ECONOMIC ANALYSIS FOR BUSINESS DECISIONS

MBA, SEMESTER-I

According to the New Syllabus of **'Savitribai Phule Pune University, Pune'**

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Swami Prabhupada,

Founder Acharya, International Society of

Krishna Consciousness"

- Dr. Padmalochana Bisoyi

"Dedicated to my Father

Retd. Principal ***Mushtak Ahmed Nazir Ahmed Shaikh***

A Teacher par Excellence

&

my Mother

Mrs. Nasim Shaikh

A Greatest Mother

-Dr. Rijwan Ahmed Mushtak Ahmed Shaikh

Preface

We feel very happy in bringing this book **Economics Analysis for Business Decisions** for the students of MBA First Semester of the **Savitribai Phule Pune University, Pune**. This book is specially designed to cover the syllabus of Savitribai Phule Pune University.

We sincerely believe that students should acquire balanced knowledge of theory as well as practical aspects of the subject. We have structured the fundamentals in concise and accurate form.

We have tried to remain faithful in two aspects:

1) Making the text easy to understand.
2) Keeping the material according to syllabus and relevant.

This book is a straight and humble attempt to sketch the various aspects of Economics Analysis for Business Decisions. The book is divided into five units.

1st Unit of this book is concern with the Basic Concepts of Managerial Economics.

2nd Unit covers the Utility & Demand Analysis.

3rd Unit focuses on the Supply & Market Equilibrium.

4th Unit is concerned with the Revenue Analysis & Pricing Policies.

5th Unit deals with the Consumption Function and Investment Function.

We are expecting valuable suggestions for improvements from our dear students and lecturers, which will be useful for our next edition.

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Acknowledgement

I express my deep gratitude to everyone who had helped me to complete my book on time. To begin with, I am indebted to **Dr. Danial Penkar**, Director SBPIM, **Dr. H. D. Thorat**, Head Research, DPU and **Dr. Kirti Dharadwkar**, Head (Research & Academics), who helped me and guided me to complete this book. I am sincerely thankful to my teachers **Dr. Nirmal Chandra Sahu**, Retired Professor and Head, Berhampur University and **Dr. Mrutyunjay Mishra**, Professor in Economics, BHU who taught me economics and how to write books. His contributions to this work is beyond description for which I am deeply thankful to him.

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It is my opportunity to express my sincere and cordial gratitude to my elder brother, **Mr. Trilochana Bisoyi**, and **Smt. Sujata Bisoyi**, father **Somonath Bisoyi**, mother **Madanavati Bisoyi**, Sister **Urmila** and brother in-law **Gadadhar Panigrahy**, for their blessings to complete this work. I am grateful to **Monoranjan**, **Vinod**, **Ranubala Nilesh**, **Jatin, Kalpana**, **Amarnath** and **Achyutanand**, for their well-wishes. I am also indebted to all **Bisoyi family** members. I feel it is my sincere duty to extend my respect and gratitude to my **Grandfather** and **Grandmother**, for their affectionate and blessing which has instilled moral force to develop my career.

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Last but not least, I pay respect and thank **Lord Jagannath**, for instilling spiritual force to proceed my work and enabled me to go in right direction for completion of this work.

- Dr. Padmalochana Bisoyi

First of all, I am thankful to Almighty for his immeasurable blessings upon my life. I would thank my parents, Retd. Principal **Mushtak Ahmed Nazir Ahmed Shaikh**, my father and my mother **Mrs. Nasim Shaikh**, who every time nurtures me to make progress in life. I also thank my wife **Mrs. Farhad** and son **Md. Ridan** for their continuous encouragement.

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- Dr. Rijwan Ahmed Mushtak Ahmed Shaikh

Syllabus

103 – ECONOMIC ANALYSIS FOR BUSINESS DECISIONS

Unit 1

Managerial Economics: Concept of Economy, Economics, Microeconomics, Macroeconomics. Nature and Scope of Managerial Economics, Managerial Economics and decision-making. Concept of Firm, Market, Objectives of Firm: Profit Maximization Model, Economist Theory of the Firm, Cyert and March's Behavior Theory, Marris' Growth Maximisation Model, Baumol's Static and Dynamic Models, Williamson's Managerial Discretionary Theory. (6+1)

Unit 2

Utility & Demand Analysis: Utility – Meaning, Utility analysis, Measurement of utility, Law of diminishing marginal utility, Indifference curve, Consumer's equilibrium - Budget line and Consumer surplus. Demand - Concept of Demand, Types of Demand, Determinants of Demand, Law of Demand, Elasticity of Demand, Exceptions to Law of Demand. Uses of the concept of elasticity. Forecasting: Introduction, Meaning and Forecasting, Level of Demand Forecasting, Criteria for Good Demand Forecasting, Methods of Demand Forecasting, Survey Methods, Statistical Methods, Qualitative Methods, Demand Forecasting for a New Products. (Demand Forecasting methods – Conceptual treatment only numericals not expected) (8+1)

Unit3

Supply & Market Equilibrium: Introduction, Meaning of Supply and Law of Supply, Exceptions to the Law of Supply, Changes or Shifts in Supply. Elasticity of supply, Factors Determining Elasticity of Supply, Practical Importance, Market Equilibrium and Changes in Market Equilibrium. Production Analysis: Introduction, Meaning of Production and Production Function, Cost of Production. Cost Analysis: Private costs and Social Costs, Accounting Costs and Economic costs, Short run and Long Run costs, Economies of scale, Cost-Output Relationship - Cost Function, Cost-Output Relationships in the Short Run, and Cost-Output Relationships in the Long Run. (8+1)

Unit 4

Revenue Analysis and Pricing Policies: Introduction, Revenue: Meaning and Types, Relationship between Revenues and Price Elasticity of Demand, Pricing Policies, Objectives of Pricing Policies, Cost plus pricing. Marginal cost pricing. Cyclical pricing. Penetration Pricing. Price Leadership, Price Skimming. Transfer pricing. Price Determination under Perfect Competition- Introduction, Market and Market Structure, Perfect Competition, Price-Output Determination under Perfect Competition, Short-run Industry Equilibrium under Perfect Competition, Short-run Firm Equilibrium under Perfect Competition, Long-run Industry Equilibrium under Perfect Competition, Long-run Firm Equilibrium under Perfect Competition. Pricing Under Imperfect Competition- Introduction, Monopoly, Price Discrimination under Monopoly, Bilateral Monopoly, Monopolistic Competition, Oligopoly, Collusive Oligopoly and Price Leadership, Pricing Power, Duopoly, Industry Analysis. Profit Policy: Break Even analysis. Profit Forecasting. Need for Government Intervention in Markets. Price Controls. Support Price. Preventions and Control of Monopolies. System of Dual Price. (11+1)

Unit 5

Consumption Function and Investment Function: Introduction, Consumption Function, Investment Function, Marginal efficiency of capital and business expectations, Multiplier, Accelerator. Business Cycle: Introduction, Meaning and Features, Theories of Business Cycles, Measures to Control Business Cycles, Business Cycles and Business Decisions. (7+1)

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UNIT 1

INTRODUCTION TO MANAGERIAL ECONOMICS

1.1. ECONOMICS

1.1.1. Introduction

Economics, being a social science, has the basic function of studying the ways in which people, households, firms and nations maximise their scarce resources and opportunities. This is known as maximising or optimising behaviour in economic terms. Thus, optimisation means to select among the best available options with the sole objective of maximising profits from the available resources. Economics, therefore can be termed as a social science which studies human behaviour with respect to optimisation of available resources to accomplish the given desires and wants.

For example, there are certain ways in which households allocate their income which is a limited resource, among various goods, services and luxuries consumed by them in order to optimise their satisfaction. Economics studies this and analyses how households try to maximise total utility by deciding 'what to consume' and 'how much to consume' from different goods/services.

1.1.2. Concept of Economy

The term economy means the whole structure of consumers, producers and distributers of goods and services in a national, regional or local community. In other words, an economy means an activity of producing and consuming goods and services which indicates the way in which scarce resources are distributed in different areas. Here, economy contains all the activities related with the production and consumption of all kind of tangible and intangible things. The reason behind the existence of an economy is to satisfy the needs of all its participants. There are different types of economy existing across the world, such as command economy and traditional economy.

In command economy, the control of all the activities related to production and consumption of goods and services is in the hands of the government. Whereas, in traditional economy, the trends of works, food, clothes, etc., is transferred from generation to generation. Further, another type of economy is followed in countries, like United States, here both the producer and consumer decides what product will be produced and what will be sold. As producers will decide which goods or services they will produce and what price they will charge whereas, the consumers will decide which goods or service they will purchase and what price they will pay.

The analysis of production, consumption and distribution of goods and services is done in economics. The micro and macro-economics are the major branches of economics. In microeconomics, the analysis of behaviour of a firm or a consumer is done. On the other hand, in macroeconomics, the analysis of the aggregate is done, like GDP. The GDP or Gross Domestic Product indicates the total monetary value of goods and services produced in an economy. Further, the increase of GDP from last year to next year indicates the growth of the economy.

1.1.3. Concept of Economics

The word 'Economics' in English has been originated from two Greek words namely; ***Oikos (household and Nemein (management)*** which derive a combined meaning as **'management of household'.** This is a normal phenomenon that the wants of each household are infinite but resources like food, cloth etc., have limitations to satisfy them. People are forced to make few choices when they face such limitations. They will have to pick wisely among the given resources as they cannot consume everything they want. Economics is often referred to as the science of choice due to the fact that scarcity forces people to choose.

Thus, it is the science which justifies the choices of people and forecasts the changes which can take place in their choices because of the dynamic circumstances.

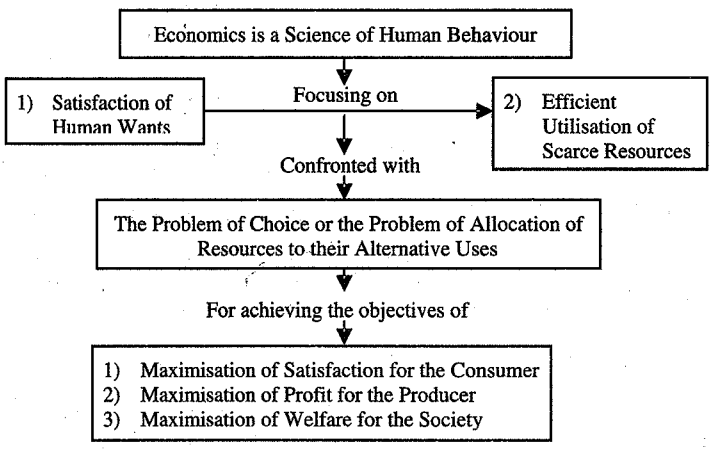


Figure 1.1: Concept of Economics

Knowledge is a growing phenomenon where economics being its branch. From time to time, various known economic experts have provided different definitions of economics. These definitions have been classified under the following categories due to their diversified meaning:

1) **Wealth Definition: According to Adam Smith,** "Economics is an enquiry into the nature and causes of wealth of nations".

   This definition is not accurate as it pertains to wealth despite the welfare of man.

2) **Material Welfare Definition: According to Dr. Marshall,** "Economics is a study of mankind in the ordinary business of life. It examines that part of individual and social action which is most closely connected with the attainment and use of material requisites of well-being".

   This definition was criticised by **Robbins** on the following grounds:
   i) Economics is the study of all economic activities whether material or non-material.
   ii) Economics studies all human beings and not just the social beings.

3) **Scarcity Definition: According to Lord Robbins,** "Economics is a science that studies human behaviour as a relationship between ends and scarce means which have alternative uses". **Lord Robbins** definition limits the scope of economics to just being a science of making choices and a mere determinant of values. It does not lay down emphasis on man's welfare and therefore, is impractical.

4) **Growth-Oriented Definition:** According to modern economists, "Economics is a science that studies those activities of man which he undertakes to maximise his satisfaction by making proper use of his scarce means". These modern experts have derived their definition from the important components of the famous definitions put forward by **Marshall and Robbins.** Going by their definitions, we arrive at the conclusion that the study and analysis of efficient allocation and consumption of limited resources which triggers economic growth and social welfare is called economics.

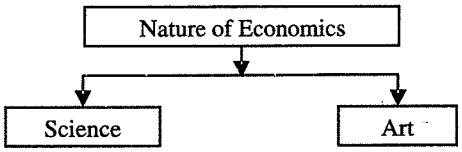
According to Samuelson, "Economics is the study of how people and society end-up choosing, with or without the use of money to employ scarce productive resources that could have alternative uses to produce various commodities overtime and distribute them for consumption now or in the future, among various persons and groups in society. It analyses costs and benefits of improving patterns of resource allocation". An analysis of how people maximise their satisfaction, producers optimise their gains and society enhances its welfare by taking decisions in respect of allocation of limited means to meet infinite needs, is called economics.

As a social science, economics deals with:
1) Studying scarce means;
2) Allocating scarce means for distinguished purposes which comprise the problem of choice.
3) Maximising welfare at both individual and social level.

1.1.4. Nature of Economics

The primary objective of the study of nature of economics is to decide whether economics is a science or an art or both. Thus, if it is a science then it is positive or normative science or both.

According to Samuelson, "Economics is the oldest of the arts, the newest of sciences – indeed the queen of all the social sciences".



1.1.4.1. Economics as a Science

The term 'Science' has its origin in term '*Scientia*' of Latin language which means "to know". The knowledge of the subject makes us understand and able to describe its causes and effects. There are various definitions where one says; "Science is a systematic body of knowledge concerning the relationship between causes and effects of a particular phenomenon". The collection of facts affects all economic activities in the economy.

For example, the observations done by an economist are related to the consumers buying large quantities and thus retard the utility of its usage and when there is a rise in prices, the demands diminishes. These are the facts which are collected by him relating to economic activities which are compiled does not signify it being a science.

According to Poincare, "Science is built-up of facts as a house is built of stones but an accumulation of facts is no more a science than a heap of stones is a house".

Thus, the significance of it being a science requires collection, classification and analysis of the facts in a systematic manner. Economics is the systematic collection, classification and analysis of the economic facts.

Collection of facts, measurement, explanation, and verification are the prominent features of science. A detailed study of these features will testify the economics as a science is as follows:

1) **Collection of Facts:** In science, facts relating to the subject are collected, similarly in economics facts pertaining to the economic activities are collected. **For example,** the observations done by an economist are related to the consumer buying large quantities and thus retard the utility of its usage and when there is a rise in prices, the demand diminishes. Thus, an economist accumulates facts relating to various economic activities.

2) **Measurement:** In science the facts are measured, thus efforts are made to properly classify the facts, and then proper presentation is done. Similarly, in economics, facts are measured where an economist will observe the contraction of demand in proportion to the rise in prices. It is a basic question that if there is unemployment in the country then what is the number of unemployed persons? What is the growth rate of national income of the country? To measure these facts the help is provided to an economist by econometrics, statistics and mathematics and the collection of facts in written form which is known as data.

3) **Explanation:** There is a need of explanation of the facts after being collected and measured. The economic laws are framed on the basis of an established relationship between cause and effect of a fact. **For example,** the formulation of the law of demand is based on the study of mutual relationship between changes in price and subsequent changes in demand and vice-versa.

4) **Verification or Validity of the Laws:** The validity of scientific laws is subject to verifications by their applications to real life situations which being a final feature of science. In other words, whether they adapt to the facts or not? There are two ways of verification as every science look for verification of the validity of its laws;
   i) Laws based on real assumption should be considered valid.
   ii) If the prediction of any law happens to be true, then it is valid.

   All the economic laws hold true when they are subject to verification and experiments are conducted in this respect. **For example,** there are various laws like law of diminishing returns, law of diminishing marginal utility etc., which are valid as they are applicable to real life situations.

From the above explanation it is clear that economics is a science but this is contradicted by some scholars who do not consider it to be science as it is not being exact and universal as compared to different laws of physics and chemistry. Practically their disagreement is unjustified. **According to Seligman,** the two vast branches of science are:

1) Social science, and
2) Natural science.

Economics is a social science as it deals with human beings. Other sciences like physics, chemistry, etc., which deals with lifeless matter are natural sciences.

1.1.4.2. Economics as an Art

Whether economics is an art or not, is yet another controversial subject among the economists. In this respect **J.N. Keynes** observed – "An art is a system of rules for the attainment of given ends".

The term "applied economics" is used by J.M. Keynes against "art". Whereas modern economists have used the term "Economic Policy" for "art". Thus economics as an art facilitates in solving the practical problems of an individual. The best methods can enable us in achieving economic ends by its study.

1.1.5. Scope of Economics

Study of scope of a subject refers to its subject matter. **According to Samuelson,** "It is a difficult task to give precisely the subject matter of economics in few lines and to delimit it from other subjects as also to apprise to a new student what economics is all about". The study of the subject matter of economics is divided into the following parts to study it conveniently:

1) **Economic Activities:** Subject matter of economics includes the following economic activities which forms the main basis of the subject:
   i) **Production:** This is the economic activity that generates resources or creates utility or value addition in goods and services. For producing biscuits, the activity involved is of using labour, sugar and maida. To transport a commodity like sand from the river bank to the required place is also production. The following are the five factors or means of production:
      a) Land,
      b) Labour,
      c) Capital,
      d) Entrepreneur, and
      e) Organisation.

      Consumption is the result of production which means that it is the direct satisfaction of wants and investment is used for indirect satisfaction of wants.
   ii) **Consumption:** The usage of goods and services for the direct satisfaction of wants is the activity related to consumption. Wearing of clothes, eating of fruits, drinking of coffee, etc., are some consumption activities.
   iii) **Investment:** The activity that results in increasing physical and human capital is called investment. Thus, it is the formation of capital to be used for the production of other goods and services.
   iv) **Exchange:** The activity relating to the sale and purchase of produced goods as well as services of the factors of production is referred to as exchange. Money is the medium of sale and purchase. Price determination is the another name of this activity which is divided into the following two parts:
      a) **Product Pricing:** It is related with the price determination of goods with respect to different conditions of market, *viz.*, perfect competition, monopoly, and imperfect competition.
      b) **Factor Pricing:** It is related with the determination of prices of the factors, concerning how prices of the services of land, labour, capital, and entrepreneur are determined in the form of rent, wages, interest, and profit respectively. Distribution is another name of this activity.

2) **Economic Systems:** Following are the division of the economic system into three parts which is based on the economic activities relating to organisations and institutions:
   i) **Capitalism:** Capitalism is the system where people freely consume, produce, and exchange goods and services. Government does not interfere in the economic activities and people can accumulate private property without any interference of the government.

ii) **Socialism:** This is the system where the government fully control the activities like consumption, production, exchange, etc., and people are not allowed to keep private property.

iii) **Mixed Economy:** This is the system where people are free to perform some kind of economic activities under the control of the government. This is considered as a mid-way between capitalism and socialism. To solve economic problems various economic policies are adopted under various economic systems. Hence, the study of economic policies is the integral part of the subject matter of economics.

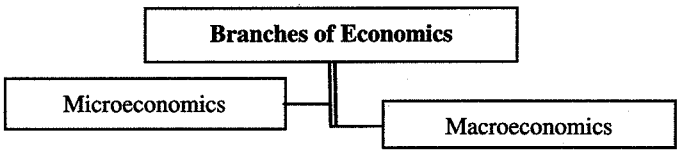
3) **Economic Policies:** Several economic problems such as unemployment, inflation, poverty, depression, etc., are the result of operation of economic systems. Policies which are enclosed to solve these problems are as follows:
   i) Monetary Policy,
   ii) Fiscal Policy,
   iii) Price policy,
   iv) Economic planning, and
   v) International liquidity, etc.

4) **Econometrics:** This not only involves the analysis of micro and macro-economic problems but also includes the methods of measurement which being the subject matter of economics. The branch of economics where there is simultaneous involvement of mathematics and statistics to measure and analyse economic activities is known as Econometrics. The study of econometrics is included in the subject matter of economics.

5) **Applied Economics:** The application of economics is not limited to theories and policies of economics but it pertains to the analysis of applied subjects like, agriculture, industry, transport, etc. Thus, industrial economic, agriculture economics, and transport economics are included in the subject matter of economics.

1.1.6. Branches of Economics

There are two main branches of economics – Micro Economics and Macro Economics.



1.1.6.1. Microeconomics

The meaning of micro is small. The behaviour of an individual is a micro-economic analysis. The study of an individual consumer, producer, price of a particular commodity, household, etc., is done under micro analysis. The micro-economic theory gives rise to the roots of business economics. In theory of price, demand concepts, elasticity of demand, marginal cost, marginal revenue, the short and long runs and theories of market structure are sources of the elements of micro-economics which solicit the business economics. The sole objective of micro-economics is to maximize utility on the demand side and to maximise profit at minimum cost on the supply side.

According to Watson, "Micro-economics is the theory of the small, of the behaviour of the consumers, producers and markets".

According to Prof. K.E. Boulding, "Micro-economics is the study of particular firm, particular household, individual prices, wages, incomes, individual industries and particular commodities."
Thus, micro-economics studies:
i) The allocation of resources to the production of particular goods and services,
ii) The distribution of goods and services among the people, and
iii) How efficiently they are distributed.

Micro-economics is the study of price theory; how to determine the price of a particular commodity like wheat, wages of a particular type of labour, interest on a particular type of capital asset, rent on a particular land and profits of a particular entrepreneur and how various resources are efficiently allocated to individual consumers and producers.

1.1.6.2. Macroeconomics

The term 'Macro' in English language has its origin in the Greek language term *'Makros'* which mean 'large'. Therefore, macro-economics studies economic problems of the entire economy. Macro-economics involves the study of aggregates or averages which cover the entire economy, such as total employment, national income, national output, total investment, total consumption, total savings, aggregate supply, aggregate demand, and general price level, wage level and cost structure.

Alternatively, it is cumulative economics which examines the inter-relations among the various aggregates, their determination and causes of fluctuations in them. **Theory of income and employment** is another name of macro-economics, or simply income analysis. It includes the problems of unemployment, economic fluctuations, inflation or deflation, international trade and economic growth.

According to Boulding, "Macro-economics theory is that part of economics which studies the overall averages and aggregates of the system".

According to Edward Shapiro, "Macro-economics deals with the functioning of the economy as a whole".

1.1.7. Limitations of Economics

Following are the main limitations of economics:

1) **Study of Human Activities Only:** Economics studies only the human activities and does not study birds, animals or lifeless matter.

2) **Study of Rational Man:** The study done by economics is based on the man who is of sound mind. A consumer is willing to maximise his satisfaction and if he is a producer then he will always be willing to maximise his profit.

3) **Assumption-Based Principles:** The laws and principles of economics are based on the various assumptions. The main assumption of micro-economics is- "Other things remaining constant". This signifies that in the analysis of any economic phenomenon, all other elements are assumed to be constant while some of the elements are taken to be variable. Thus, the significance of the term "Other things remaining constant" is for the constant elements. **For example,** to understand the relationship between the price and demand for a commodity, price is a variable factor and all the other factors that affect demand are income of the consumer, price of the substitutes, etc., are taken to be constant.

4) **Economic Laws are not Exact and Universal:** The laws of economics are based on trends which are neither exact nor universal as the laws of natural sciences such as physics and chemistry. These are not considered to be valid in all situations of existence. Since, they are based on the assumption then changes occur accordingly if there is any change in the assumption. The main reason is that these laws are based on the behaviour of men, as man being virtuous and cannot act according to the law of demand. It is not necessary that everybody will demand more quantity of a commodity when there is a fall in price.

5) **Economics is not Non-Controversial:** Economics is controversial, as being a normative science whenever suggestions are offered as solutions to economic problems by economist; there are different opinions to the same problem by different economists.

1.2. MANAGERIAL ECONOMICS

1.2.1. Meaning and Definition of Managerial Economics

The synthesis of economic theory with business practice is known as Managerial economics. Various tools which are provided by economics, applied to the business management. Alternatively, the application of economic theory to the problems of management is referred to as managerial economics. The solving of problems at the level of the firm by applying economics is known as managerial economics. The business executive is capable of assuming and analysis of things and thus profit maximisation is the emphasis of economics resulting in getting satisfactory profits.

According to Spencer and Siegelman, "Managerial economics is the integration of economic theory with business practice for the purpose of facilitating decision-making and forward planning by management".

According to Mc Nair and Meriam, "Managerial economics is the use of economic modes of thought to analyse business situation".

According to Jeal Dean, "The purpose of managerial economics is to how economic analysis can be used in formulating policies".

According to Mansfield, "Managerial economics is concerned with application of economic concepts and economic analysis to the problems of formulating rational managerial decision".

The other names for managerial economics are, **Business Economics** or **Economics for Firms**.

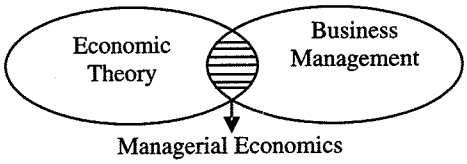


Figure 1.2: Managerial Economics

Managerial economics is a venture to use economics and economic logic in formulating the policies of business. Thus, managerial economics is that form of economic knowledge which is used in the analysis of business problems for taking the relevant business decisions and devising forward plans.

1.2.2. Nature of Managerial Economics

Following is the nature of managerial economics:

1) **Micro-Economic in Nature:** The branch of economics which deals with individual units of an economy is called micro-economics. These individual units may be either a firm or a person or a group of firms or a group of persons.

2) **Pragmatic:** Managerial economics is practical in nature. Rigid and abstract theoretical frameworks are provided by managerial economics to managers. It is said to be pragmatic in nature because it avoids complex abstract issues of economic theories at one end, while at another end it incorporates complications which is ignored by economic theory in order to analyse the overall situation where managerial decisions are to be taken.

3) **Related to Normative Economics:** Economics can also be classified as positive or normative. Positive economics states the economic phenomenon that is observed and normative economics determines what should be done which means discriminating the ideal from the actual.

4) **Conceptual in Nature:** Managerial Economics is based on the solid conceptual foundation of economics. The subject-matter of managerial economics is not irrational. Analysis of business problems on the basis of established concepts is the objective of managerial economics.

5) **Utilises Some Theories of Macro-Economics:** Macro-economics envisages all the individual matters that are integrated to be the part of analysis of the problems of the economy as of a nation not pertaining to an individual that being operated in the environment affects and gets affected. Thus, majority of the aspects that are related in such a manner becomes the subject-matter of micro-economics.

6) **Problem-Solving in Nature:** Managerial economics not only analyses the managerial problems of business units but it aims to resolve the business problems also which provides optimal solutions. Hence, it is problem-solving in nature.

7) **Managerial Economics Deals with the Application of Economics:** Managerial economics studies and deals with economic theory that are applied to business management. The relevant aspects of economic theory like demand, supply, production, pricing, markets etc., must be carefully understood by the managers which facilitate them to manage the business in an organised manner by applying the principles of economics in making managerial decisions. Following are some of the important aspects of economic theory which can be applied to business management:
   i) **Demand Analysis:** It includes Laws of demand, elasticity of demand, determinants of demand etc.
   ii) **Production Analysis:** It includes Laws of variable proportions, Law of returns to scale, economies and diseconomies of scale etc.
   iii) **Market Analysis:** It includes nature of product market like perfect competition, monopoly, oligopoly, monopolistic competition. Product pricing and output decisions in different forms of market etc.

8) **Managerial Economics is the Study of Allocation of Resources:** Proper allocation of resources is fundamental in managing business. The theory of economics deals with the problems of resource allocation such as what to produce, how to produce, and for whom to produce. Thus, theory of economics, which is the fundamental of managerial economics, specifies the significance of the allocation of resources to the application of the principles of managerial economics. Allocation of resources includes the following concepts:
   i) **Input Allocation:** It includes raw materials, labour hours, machine hours, etc.
   ii) **Output Allocation:** It includes meeting customer's demand in different segments of market at different locations.
   iii) **Allocation of Funds:** It includes efficiency of usage of cash and other resources. Ratio analysis and financial statement analysis helps in making decisions about fund allocation.

9) **Inter-disciplinary:** Managerial economics being a new discipline is the integrated form of multi-disciplines. Its techniques, tools, theories and contents emerge from different subjects, such as Economics, Management, Statistics, Mathematics, Accountancy, Sociology and Psychology.

1.2.3. Scope of Managerial Economics

Following topics constitute the scope or subject matter of managerial economics:

1) **Demand Analysis and Forecasting:** A business firm is transforming the productive resources into goods which are sold in the market. Accurate estimation of demand is an important basis of managerial decision-making. Estimation of future sales facilitates management for scheduling the production and employing resources. It will be helpful for management to maintain or strengthen its market position and profit-base.

2) **Cost and Production Analysis:** Cost estimation is very useful for taking management decisions and various factors which affects the cost estimation should be given due consideration for planning purpose. Cost and production analysis includes various concepts such as cost concepts and classifications, cost-output relationships, economies and diseconomies of scale, production.

3) **Pricing Decision, Policies and Practices:** Price gives income to the firm and it constitutes the most important field of managerial economics. It covers various concepts such as price determination in various market forms, pricing policies, pricing method, differential pricing, productive and price forecasting.

4) **Profit Management:** Earning maximum profit is the main objective of a business firm. Due to variation in costs and revenue an element of uncertainty about profits always exist. It covers various aspects such as nature and measurement of profit, profit policies and techniques of profit planning like break-even analysis.

5) **Capital Management:** For any firm, the most complex problem is firm's capital investments. Capital management implies planning and control of capital expenditure because it involves a large sum and moreover the problems in disposing the capital assets are so complex that they require considerable time and labour. The main topics dealt with under capital management are cost of capital, rate of return and selection of projects.

6) **Analysis of Business Environment:** The working and performance of any business firm is affected by the various environmental factors. Therefore, in decision-making process environmental factors are considered by the managers. If manager does not consider the environmental factors then it will be harmful for the firm. Therefore, the management must be aware about the economic environment, particularly those economic factors which creates the business climate.

7) **Allied Disciplines:** The concepts which are helpful for the management in decision-making are qualitative in nature. Therefore, for determining the relationships between economic variables, mathematical tools are generally used. The linear programming technique is used by the firms to maximise or minimise their objective function. Similarly, for taking business decision, statistical and accounting principles are used.

1.2.4. Uses of Managerial Economics

Following are some important uses of managerial economics:

1) **Used for Integration of Economic Theory:** Economic theory and business practice is integrated with the help of managerial economics.

2) **Used as Solution to Practical Business Problems:** It acts as a mean to apply economic concepts and principles to solve practical business problems in real life. Managerial economics helps in employing the most modern instruments and tools to find solutions to business problems.

3) **Optimum Use of Scarce Resources:** It helps in making optimum use of scarce resources of a firm to maximise profits.

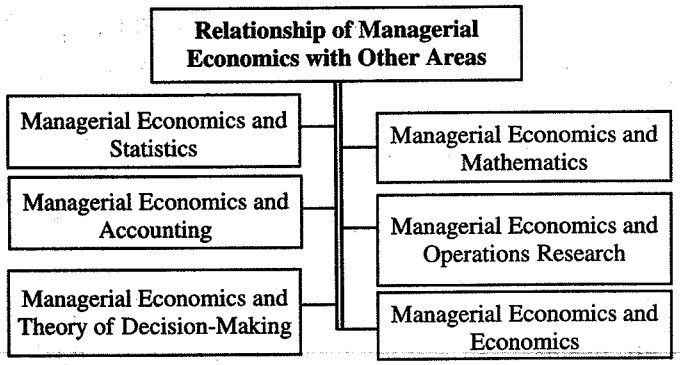
4) **Used for other objectives:** The various objectives such as attaining industry leadership, expansion of market share, etc., are achieved with the help of managerial economics.

5) **Used for overall development:** Managerial economics facilitates in making overall development of a firm.

6) **Used in making right decisions:** A manager understands the details of business problems and taking a right decision at the right time with the help of managerial economics.

1.2.5. Relationship of Managerial Economics with Other Areas

The relationships of Managerial Economics with other subjects are as follows:



1) **Managerial Economics and Statistics:** Statistical tools are playing very important role in business decision-making. Statistical techniques are used in collecting, processing and analysing data, testing the validity of the economic laws with the real economic phenomenon before they are applied to business analysis. Probable economic events are the basis of a good business decision. Various statistical tools such as theory of probability, forecasting techniques etc., help the decision-makers in the prediction of future economic events.

2) **Managerial Economics and Mathematics:** The main challenge of a businessman is how to minimise cost or how to maximise profit or how to optimise sales. To find the answers of these questions, various mathematical concepts and techniques are widely used in economic logic. The knowledge of geometry, trigonometry and algebra is not only important but various mathematical tools and techniques such as logarithms and exponentials, vectors, matrix, calculus, differential and integral are also necessary for managerial economics.

3) **Managerial Economics and Accounting:** Various data are required by a managerial economist for the decision-making purpose. Accounting details are included in data. **For example,** the profit and loss statement of a firm gives details about the performance of the firm and guides the managerial economist to prepare the future course of action–whether it should improve or close down.

4) **Managerial Economics and Operations Research:** Models and tools of operations research or quantitative techniques are affecting the managerial economics. Operations research is a subject that consists of a number of models and analytical tools which are developed on the basis of inter-disciplinary research for solving complex problems of planning and allocation of scarce resources, primarily in defence industries. Managerial economics has generalised and developed the models and tools of operations research for the purpose of business decision-making. Linear programming models, inventory models, game theory, etc. are a few tools that have originated in the works of operation researchers.

5) **Managerial Economics and Theory of Decision-Making:** Decision theory has been developed to deal with problems of choice or decision- making under uncertainty, where the applicability of figures required for the utility calculus are not available. Economic theory is based on the assumptions of a single goal whereas decision theory breaks new grounds by recognising multiplicity of goals and persuasiveness of uncertainty in the real world of management.

6) **Managerial Economics and Economics:** Managerial economics has been described as economics applied to decision-making. Managerial economics has been studied as a special branch of economics, bridging the gap between pure economic theory and managerial practice. Economics has two main branches – micro-economics and macro-economics.

1.2.6. Fundamental Concepts of Managerial Economics

Various principles or fundamental concepts and analytical tools which are offered by managerial economics, are used by modern business organisations in decision-making process. These can be explained as under:

1.2.6.1. Opportunity Cost Principle

The sacrifice of alternative courses of action for any decision is referred to as opportunity cost. Opportunity cost may be defined as, 'the revenue foregone or opportunity lost by not using the resources in second best alternative use'. It is also called **imputed cost**. Measurement of sacrifice is done by opportunity cost. The sacrifice which is made for taking a decision is measured by opportunity cost.

This concept can be explained by following points:

1) The opportunity cost of the funds employed in one's own business is the interest that could be earned on those funds had they been employed in other ventures.
2) The opportunity cost is the time an entrepreneur devotes to him.
3) Business is the salary he could earn in other occupations.
4) The opportunity cost of using a machine to produce one product is the earnings forgone which would have been possible from other products.
5) The opportunity cost of using a machine that is useless for any other purpose is Zero since its use requires no sacrifice of other opportunities.

Thus, it should be remembered that opportunity costs require ascertainment of sacrifices. If a decision involves no sacrifices, its opportunity cost is nil. For decision-making, opportunity costs are the only relevant costs. The opportunity cost principle may be stated as, "The cost involved in any decision consists of the sacrifices of alternatives required by that decision. If there are no sacrifices, there is no cost."

1.2.6.2. Incremental Principle

Marginal costs and marginal revenues of economic theory are related with Incremental principle concept. Estimation of the impact of decision alternatives on costs and revenues, emphasising the changes in total cost and total revenue resulting from changes in prices, products, procedures, investments or whatever may be at stake in the decision are involved in the incremental concept.

The two basic components of incremental reasoning are:

1) **Incremental Cost:** The change in total cost resulting from a particular decision may be referred to as the incremental cost.
2) **Incremental Revenue:** The change in total revenue resulting from a particular decision is referred to as the incremental revenue.

According to incremental principle, a decision is profitable if:

1) It increases revenue more than it increases cost,
2) It reduces some cost more than it increases others,
3) It increases some resources more than it decreases others, and
4) It decreases costs more than it decreases revenues.

Some businessmen consider that for making an overall profit they must make a profit on every work. The orders are refused which do not cover full cost (labour, materials and overhead) plus a predefined minimum profit. Incremental reasoning shows that this principle may be varied with profit maximisation in short-run. A refusal to accept business below full cost may mean rejection of a possibility of adding more to revenue than to cost. The relevant cost is not the full cost but rather the incremental cost.

1.2.6.3. Marginal Principle

The application of marginal concepts in economic theory is referred to as the marginal principle or marginalism. Marginalism is associated with arguments concerning changes in the quantity used of a good or of a service, as opposed to some notion of the overall significance of that class of good or service, or of some total quantity thereof. If the resources are scare then the manager has to be very careful about the full utilisation of each and

every additional unit of resources (inputs). For taking decision about the use of an additional man-hour or machine-hour, manager is required to know what the additional output is expected there from. Similarly, for taking decision about additional investment, manager required to know that what the additional return from that investment is. For all such additional amount of output or return, the term 'marginal' is related.

The marginal concept is a very important concept in Economics which measures the rate of change in the dependent variable. However, it is very difficult to apply the concept of marginalism in the real world. There is a problem that the independent variable may be subject to 'chunk changes' rather than 'unit change'.

Difference between Marginal and Incremental Concept

Marginal Concept	Incremental Concept
1) It focuses on the function of single variable or it considers independent variable. For example, revenue depends upon output.	It considers more than one independent variable. It is multi-variable function.
2) It is more specific.	It is general in sense.
3) All marginal concepts are somehow incremental concepts.	But all incremental concepts are not essentially marginal.
4) It is expressed in terms of unit change.	It is expressed in terms of bulk change.

1.2.6.4. Equi-Marginal Principle

The allocation of the available resources among the alternative activities is deal with by the Equi-Marginal Principle. According to this principle, an input should be allocated in a way that the value added by the last unit is the same in all cases. This generalisation is called the equi-marginal principle. Let us consider that a firm is involved in three activities such as A, B and C activity. All the activities require the services of labour and the firm should allocate the available labour in such a way that the value of Marginal Product of labour is equal in all the three activities.

In symbols:

$$VMP_{La} = VMP_{Lb} = VMP_{Lc}$$

Where, VMP = Value of Marginal Product
L = Labour
a, b, c = Activities i.e.,

the value of the marginal product of labour employed in a is equal to the value of the marginal product of the labour employed in B and so on.

For example, if in activity 'A', the value of marginal product of labour is ₹30 while that in activity 'B', it is ₹40. Hence, it is profitable to shift labour from activity 'A' to activity 'B' thereby expanding activity 'B' and reducing activity 'A'. When the value of marginal product is equal in all the three activities then the optimum condition will be achieved.

1.2.6.5. Principle of Time Perspective

A business firm should take any decision only after considering the effects of decisions on costs and revenues in short-run and long-run. The short-run refers to a time period which is long enough to allow using the variable factors of production in different amounts. The long-run refers to a time period which is long enough to allow using all the factors of production in different amounts. It is required that proper balance should be maintained between short-run and long-run effects.

Hence, it is very important to give proper consideration to the time perspective. The principle of time perspective may be stated as: "A decision should take into account both the short-run and long-run effects on revenues and costs and maintain the right balance between the long-run and short-run perspectives".

For example, suppose, there is excess productive capacity in a firm. An order of 3000 units comes to management notice. The customer is willing to pay ₹6 per unit or ₹18000 for the whole lot but no more. Suppose, the short-run incremental cost (ignoring Fixed Cost) is ₹5. Therefore, contribution to overhead and profit is ₹1 per unit (₹3000). But there may be long-run effects and these must be kept in mind.

1.2.6.6. Discounting Principle

Discounting is a process of reducing the future values to their present values. In this context, a discount rate is referred by the interest rate which is used in present value problem. For making investment decisions, it is very important concept in managerial economics. The origin of this principle is valuation of the money received at different point of time. Various transactions which involve making or receiving cash payments at numerous future dates used the discounting principle. When a buyer takes home loan from any bank then he promise to make monthly payments for twenty years.

When a person injured in an automobile accident then he accepts a settlement of ₹4,000 per month as compensation for the damage from the insurance company for the life time. The concept of time value of money explains that the money which is received at different future dates will not be same today. Thus, it is necessary to understand the methods which are used for measuring the present value of money to be received or paid at different future dates. After one year the value of ₹1000 is not equal to ₹1000, but less than that. But it is required to know that how much money today is equal to ₹1000 of one year hence. The rate of interest is required to find out this. So, we will discount ₹1000 at that rate of interest to ascertain the value of 1000 rupees one year hence or two years hence.

This discounting principle is:

$$FV_n = PV(1+r)^n$$

Where, FV_n = Future value in years;
PV = Present value
r = Discount interest rate;
n = Time between the future value and present value

1.2.6.7. Scarcity Principle

'Excess demand' of any commodity or service is referred to as scarcity. At any time, if demand (requirement) for anything exceeds its supply (availability) then that thing is called scarce. Amount of demand in relation to supply determines the scarcity and hence it is called relative concept. The scarcity required managerial attention because it lies at the root of any problem. **For example,** scarcity of jobs is known as unemployment. Scarcity of goods is known as inflation. Unsold stock of inventory is essentially the scarcity of buyers. Under-utilised capacity at the plant level may be primarily due to scarcity of power or other supporting facilities.

Had these scarcities not been there, there would have been no managerial problems. It is because of this scarcity that a manager has to decide on optimum allocation of scarce resources of men, materials, machines, money, time and energy. This is one of the fundamental concepts of economics relevant for business economics. This concept lies at the heart of resource allocation problem of a business enterprise. The essence of any economic problem, micro or macro, is the scarcity of resources. The managers who decide on behalf of the corporate unit or the national economy always face the economic problem of scarcity of one kind or the other.

1.2.6.8. Principle of Risk and Uncertainty

While taking decision a business executive has to face these important situations: risk and uncertainty. Therefore, the business executive should understand the meaning and implications of situations so that he can take business decisions easily and effectively.

Risk is the probable measurement of uncertainty. When uncertainty is reduced to a number of possible results to alternative courses of action, it is called risk. Thus, risk is a state of knowledge in which the result of every alternative is expected to be one of the certain possible results. Thus, risk is a state of knowledge in which it is possible to calculate the possibilities of the result of every decision and such results can be forecasted. In comparison to the situation of certainty, risk is more uncomfortable because under this situation, the results depend upon possibility.

Uncertainty has been defined as, "a state of knowledge in which one or more alternatives result in a set of specific outcomes but where the probabilities of the outcomes are neither known nor meaningful". Thus, uncertainty is a situation in which the decision-maker does not know probable results of the alternatives. It is opposite to the situation of certainty. It is a situation in which the results of various decisions cannot be predicted and their probability cannot be measured also. Uncertainty is also termed as immeasurable risk due to imperfect knowledge about each alternative. In this situation, all the decisions are taken in the atmosphere of uncertainty.

1.2.7. Significance of Managerial Economics in Decision-Making

Managerial economic plays an important role for firms in taking the following decisions:

1) **Production Decisions:** Supply of goods and services for sale in a market to satisfy consumer wants is done through the production which is an economic activity and this is made possible to get maximum profit. It is necessary that the business executive has to make the rational allocation of available resources. He may face problems that how to use best combination of the factors to gain maximum profit or how to use different machine hours for maximum production advantage, etc.
2) **Inventory Decisions:** The amount of goods, raw-material or other resources that are idle at any given point of time which is held by any firm is referred to as inventory. The decision about how much inventory is required for meeting the demand is very important. In certain situation, production planning is affected by the level of inventories and hence it is a strategic management variable. When any firm maintain large inventory of raw-materials, intermediate goods and finished goods then it means blocking of capital.
3) **Cost Decisions:** The ability of any firm to produce goods at the lowest cost is influenced by the competitive ability of the firm. In business decisions cost structures, reduction of cost and cost control plays an important role. If cost control is not used then profits would come down because of increasing cost. It is required to know the cost involved with any resource because it affects the future business decisions about which alternatives are selected from available resources.
4) **Marketing Decisions:** For marketing planning, a marketing executive must take decisions about target market, market positioning, product development, pricing channels of distribution, physical distribution, communication and promotion. In marketing, a businessman has to take mainly two different but interrelated decisions namely sales decision and purchase decision. Sales decision includes the decision that how much to produce and sell to get maximum profit. The purchase decision includes the decision that how to acquire various resources at the lowest possible prices so as to maximise profit.
5) **Investment Decisions:** For any investment decision, the problems of risks and wrong estimation are very essential. In real business situation, there is seldom an investment which does not involve uncertainties. Investment decision covers issues like the decisions regarding the amount of money for capital investment, the source of financing this investment, allocation of this investment among different projects over time. These decisions are of immense significance for ensuring the growth of an enterprise on sound lines.
6) **Personnel Decisions:** The services of a large number of personnel are required in an organisation. These personnel are appointed at various posts. For achieving organisational objectives, every position of the organisation should have specific duties. Manpower planning, recruitment, selection, training and development, performance appraisal, promotion, transfer etc., are covered by the personnel decisions.

1.2.8. Limitations of Managerial Economics

The main limitations of managerial economics are as follows:

1) Managerial economics has led to the emergence of monopolies in the free market economies for the production of some important products or services, e.g., electricity producing and distributing companies, telephone service, railways, etc., have been exploiting the consumers by charging high prices and making excessive profits.
2) It has also led to the emergence of oligopoly, whereby few producers or firms formally collide with each other or form a cartel, and thus charge high price and restrict output.
3) There is an exploitation of workers by the undesirable activity of the private business, due to unequal bargaining power of employers and workers. Child labour and women are employed at a very low wages while a lot of work is obtained from them.
4) The limitations of micro-economics and managerial economics can be classified into two broad categories:
   i) Those requiring use of optimisation techniques, and
   ii) Those requiring supply-demand analyses to arrive at equilibrium solutions.
5) Multinational corporations have given rise to cut-throat competition whereby closing the future prospects of small business enterprises.
6) Focus is only on firm's internal management.
7) There could be some other goals or criterion than efficiency or profit:
   i) Environment protection,
   ii) Humanism, charity,
   iii) Nationalism or patriotism.

1.2.9. Managerial Economics and Decision-Making

Managerial economics is the application of economic theory and quantitative methods (mathematics and statistics) to the managerial decision-making process. It is a combination of two terms 'managerial and economies'. As the term 'managerial' refers to the functions of the management, the main functions of management are decision-making and forward planning. 'Decision-making' means the process of choosing from among available alternatives to achieve the firm's established goals. Forward planning refers to making the plans for the future.

Managerial economics takes the help of decision-making sciences or quantitative methods. Decision-making sciences or quantitative methods refer to the tools and techniques of analysis, optimisation analysis, statistical methods, game theory and capital budgeting. Decision-making sciences make special use of mathematical economics and econometrics to derive optimal solutions to managerial decision-making problems and other optimisation and forecasting.

Economics is the study of the optimal use of scarce resources to satisfy human needs and wants. It is the study of how human beings make choices to allocate scarce resources to satisfy their unlimited wants to maximise their satisfaction. Managerial economics applies economic theory mainly micro-economics to business and administrative decision-making. Because it uses the tools and techniques of economic analysis to solve managerial problems. Managerial economics links traditional economics with the decision sciences to develop vital tools/or managerial decision-making.

It tells managerial managers how things should be done to achieve organisational objectives efficiently. It identifies ways and means to achieve virtually any of the managerial organisation's goals. The managerial economics formulates principles which provide a framework for evaluating whether resources are being allocated efficiently within a firm. **For example,** managerial economics can help the manager to determine if profit can be increased by reallocating labour from a marketing activity to the production line.

In short, managerial economics arrives at a set of operating rules that aid in the efficient utilisation of scarce human and capital resources. Managerial economics is just as relevant to the management of non-managerial and non-profit organisations, such as government agencies, cooperatives, schools, hospitals and similar institutions as it is to the management of profit-oriented managerial. Managerial economics uses economic concepts and decision science techniques to solve managerial problems. It becomes clear from the following **figure 1.3**.

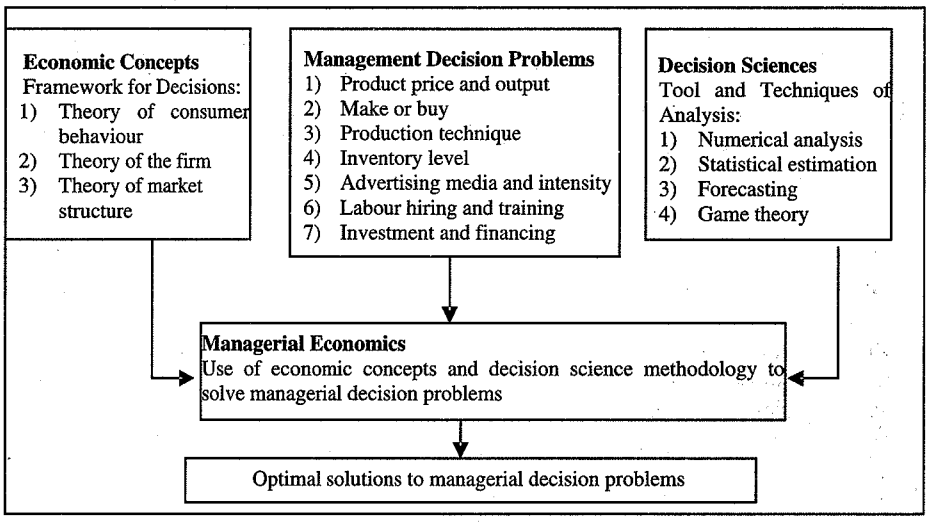


Figure 1.3: Managerial Economics and Decision-Making

Here, it may be mentioned that managerial economics does not provide readymade solutions to all the problems faced by a firm. It provides only the logic and methodology to find-out answers and the answers as such. It all depends on the manager's ability, experience and intelligence to use different tools of economic analysis to find-out the correct answer to business problems. Managerial economics is also referred to as business economics, economics of the firm, economics of the business management, economics of the enterprise, applied economics and so on.

1.2.10. Difference between Managerial Economics and Economics

Basis of Difference	Managerial Economics	Economics
1) **Meaning**	It is the study of proper allocation of resources in a particular firm	It is the study of allocation of resources in a society as a whole.
2) **Presence of Principle**	The proper application of economic principles to the problems of the firm.	The body of the principles are being dealt itself.
3) **Nature**	It is individualistic.	It is comprehensive.
4) **Type of Economics**	It is micro-economic in character.	It is both macro and micro-economic in character.
5) **Deals in**	It deals only with a firm.	It deals with a firm and its industry.
6) **Scope**	It is narrow in scope.	It is wider in scope.
7) **Type of Work**	The modification and enlargement of models is done by adopting and revaluating of economic models to match specific conditions and to help in the process of problem-solving.	It speculates economic relationships and builds simplified economic models.
8) **Factors Used**	Variables such as objectives of the firm, multi-product nature of manufacture, constraints on resource availability, environmental aspects, legal constraints, etc., are utilised.	Theoretical assumptions are used only.

1.2.11. Difference between Micro, Macro and Managerial Economics

Basis of Difference	Micro-Economics	Macro-Economics	Managerial Economics
1) **Objective**	The main objective of micro-economics is to increase the utility on demand side while to reduce the cost and profits on the supply side.	The situation of full employment, price stability economic growth and favourable balance of payment are the essential objective of macro-economics.	The main objective of managerial economics is to analyze the way of distributing the resources to a specified firm.
2) **Driving Forces**	The price mechanism is one of the most important driving forces in micro-economics which is regulated by the forces of demand and supply. The equilibrium price is decided by these forces in the market.	The national income, output and employment are the main driving forces in macro-economics which is regulated by the aggregate demand and aggregate supply of the market.	The intention of managerial economics is to fulfill the particular situation and to help in the process of problem solving; the managerial economics chooses and redesigns the economic models. Therefore, the economic models are altered and expanded.
3) **Assumptions**	The main assumption of micro-economics theory is that the other factors are remaining constant i.e. a phrase used to explain this assumption is that "ceteris paribus". This assumption is related to the behaviour of the consumer.	The assumption of macro-economics depends on the factors like aggregate volume of the output of an economy, the amount to which the resources are engaged, with the size of the national income and with the general price level.	Managerial economics is based on the assumptions like objectives of the firm, nature of different commodities, obtainability of resources, environmental factors, legal factors, etc.
4) **Subject-matter**	The theory of micro-economics does not consider the element of time but the equilibrium condition is aroused at a specified time period. Hence, the analysis of micro-economics is regarded as a stable approach.	The theory of macro-economics focuses on the time perspective like time delays, changes in rates, previous and anticipated value of the variables.	The managerial economics has the element of micro-economic character.

1.3. FIRM

1.3.1. Concept of Firm

The proper combination and organisation of resources to produce goods and services for the purpose of sale in an organisation is known as firm. This firm is capable to transform the hired inputs into saleable outputs. The factors of production which are called inputs are divided into two broad categories, namely; human resources and capital resources.

Further, human resources are sub-divided into labour resource and entrepreneurial resource inputs whereas land, man-made capital, forests, rivers, etc., are capital resources. Land, labour, capital and entrepreneur (organisation) are the four factors of production and the remuneration which are given to these factors are wage, rent, interest and profit respectively in economics.

According to Professor S.E. Landsbury, "Firm is an organisation that produces and sells goods with the goal of maximising its profits."

According to Professor R.L. Miller, "Firm is an organisation that buys and hires resources and sells goods and services."

The firms are classified on the basis of investment limit, number of owners and government participation in business.

Thus, the classification of firms is as follows:

1) Small, medium and large,
2) Proprietorships (owned individually), partnerships (owned by two or more individuals) and corporations (owned by stockholders), and
3) Public sector, private sector and joint sector.

Business and non-business are the two categories of firm. Profit is the sole objective of the business firm whereas non-business firms like universities, hospitals, libraries and museums operate as 'not-for-profit' organisations.

1.3.2. Rationale for the Firm

Firms exist because they are useful. Firms survive by public consent to serve social needs. If social welfare could be measured, business firms might be expected to operate in a manner that would maximise some index of social well-being. Maximization of social welfare requires answering the following important questions: What combination of goods and services (including negative by-products, such as pollution) should be produced? How should goods and services be provided? How should goods and services be distributed? These are the most vital questions faced in a free-enterprise system, and they are key issues in microeconomics.

The reasons for the existence of the firm are manifold, such as:

1) **Concept of Transaction Costs:** Any market transaction between autonomous individuals required time and negotiation and therefore had a cost. The firm is able to coordinate production and exchange at less cost than the market. Long-term contract with labour saves the transaction costs because no new contract has to be negotiated every time a labour is to be hired or given new assignment.
2) **Government Interference:** Government interference in the marketplace applies to transactions among firms rather than within firms. Many government regulations like price-control and sales taxes are also saved by dealing of transaction within the firm. (Sales tax is levied for transaction between firms). By internalizing some transactions within the firm that would otherwise be subject to those interferences, production costs are reduced.
3) **Diminishing Returns to the Management:** Although the production costs are reduced by organizing production factors into firms, this process cannot continue to grow into just one large firm, thereby,

limiting the size of the firm. Production cost per unit of output will tend to rise as firms grow larger. There are two reasons:

i) **Rise in Cost of Organizing Transactions:** The cost of organizing transactions within the firm tends to rise as the firm gets larger. The firm will internalize the lower-cost transactions first and then the higher-cost transactions. At some point, these internal transactions cost will equal the costs of transacting in the market. At that point, the firm will cease to grow.

ii) **Entrepreneurial Organizational Skill:** resources within the company may not be efficiently allocated if the firm's size exceeds the manager's ability to control the operation. To overcome this problem, profit centers are developed, dividing the organization into small groups, managed somewhat independently.

4) **Public Good of Information:** The public good of information has become increasingly important in complex modern economies, so the advantage of firm over market has increased in many sectors. Information, once created, may very easily disperse across a large population without any diminution of its utility to each additional consumer.

Because it is so easy to come by, unless protected, no individual has an incentive to declare and part with its true value. But if all refuse to pay, the incentive to create information disappears. The firm overcomes this problem by using authority to help ensure a return to those who acquire title to information, whether it be the location of a good, a technique for refining it, or an innovative and effective administrative system for bringing it to market.

1.3.3. Objectives of Firm

In managerial economics the entire operations of every firm which is considered as a single unit is carried-out by an entrepreneur. The main objective of the entrepreneur is to organise and coordinate various factors of production, supervise and manage them for producing goods and services for the firm in order to fulfil the objective of the firm. There are various theories regarding the objectives of a firm. The important ones are as follows:

1.3.4. Profit Maximisation Model

The sole objective of business firms is profit maximisation as stated by traditional economic theory where profit maximisation is the largest complete amount of money profits accrued by given demand and supply conditions. Profit maximisation hypothesis is the basis of common price theory. This hypothesis helps in the prediction of the behaviour of the business firms as well as in the price-output performance under different market conditions as no other hypothesis is capable in explaining and forecasting the performance of the firms. Thus, a firm will always maximise its profit at that output level where the difference between total revenue and total cost is maximum.

The two profit maximising conditions are:

1) MC = MR, and
2) MC must be rising at the point of equality between MC and MR. Let us explain these two conditions. We take first condition.
   i) If MC < MR, total profits are not maximised because firm will earn more profits by increasing output.
   ii) If MC > MR, the level of total profit is being reduced and firm can increase profit by decreasing production.
   iii) If MC = MR, the profits could not increase either by increasing or decreasing output and hence profits are maximised.

Assumptions of Profit Maximising Theory

The assumptions of profit maximisation theory are as follows:

1) The basic objective of the organisation is to gain maximum profit. Here, profit means the excess of revenue over cost.
2) The firm is owned by an entrepreneur.
3) The consumer's habit and preferences remains same.
4) The technique used for production of the goods and services are given.
5) The organisation deals in the production of single, standard, and perfectly divisible goods and services.
6) The organisation possesses complete knowledge of quantity of goods and services produced which can be sold at a certain price.

7) The organisation confidently knows the demand and cost of goods and services produced by it.
8) No new firm can enter the market in short-term period. For entering the market, the new firm needs long period of time.
9) The organisation can earn maximum level of profit over a fixed period of time.
10) The profit maximisation is possible in both the long run and short run.

The profit maximisation model can be estimated by these assumptions under perfect and monopoly competition.

Criticisms of Profit Maximisation

The objective of profit maximisation has been extremely criticised by economists on the following grounds:

1) **Profits Uncertain:** The principle of profit maximisation is based on the assumption that the firms are sure about the maximum level of their profit but conversely, the profits are not sure as they accumulate from the difference between the receipts of revenue and the costs incurred in future. Therefore, it is impossible for the firms to maximise their profits under uncertain conditions.
2) **No Perfect Knowledge:** The basis of the hypothesis of profit maximisation is that all the firms are fully knowledgeable about their own costs and revenues and about other firms as well. But, actually these firms have insufficient and inaccurate knowledge about the conditions of their operation.
3) **Firms do not Bother about MC and MR:** In reality, the firms are unconcerned about the calculation of marginal revenue and marginal cost and are even unaware of the two terms. There are some firms having no knowledge about the demand and marginal revenue curves and most of them have no adequate information about cost structure.
4) **Static Theory:** The neo-classical theory of the firm is static in nature which does not tell the duration of either the short period or the long period. The extent of time of the neo-classical firm consists of identical and independent time periods. Decisions are considered as temporarily independent. This is a serious weakness of the profit maximisation theory.
5) **Not Applicable to Oligopoly Firm:** In economic theory, the objective of profit-maximisation has been kept for perfectly competitive or monopolistic competitive firms. Whereas, due to criticism against oligopoly firms they are discarded and thus economists have put forth different objectives in the theory of firm which are either duopoly or oligopoly.
6) **Doubtful Validity:** There is no specific reason to believe that all the businessmen decide for same goal and thus aim at sales maximisation, expansion of market share, etc. Thus in this context of real business the profit-maximisation assumption holds doubtful validity.
7) **Not Applicable in Modern Business World:** In modern business world, the traditional theory assumes that firms managed by their owners is not valid and these firms are complex organisations managed by managers who are salaried employees whose interest may or may not often align but differ from those of the shareholders who want maximum profits.
8) **Lack of Predictive Power of Managers:** The firms settle with less-than-maximum profit as their goal because of lack of predictive power of the managers who are risk-hesitant. Due to lack of proper intra-firm communication, firms are restricted to maximise their profits.

1.3.5. Economist Theory of the Firm

The economist theory of the firm is explained below:

1) The firm is referred to as transformation unit in which the factors (inputs) are changed into the products. During the process of this transformation, organisation aims toward creating surplus values. Here, surplus value means the excess of value of product produced over the value of factors used in the production. It includes good understanding of:
   i) Least cost factor combination, and
   ii) Production function.
2) The firm always tries to gain more and more surplus values. These surplus values will create a greater amount of profit for the firm. The focus of the organisation remains over maximising the level of profit. For earning this maximum profit it either raises its revenue through increased sale, or it reduces its expenditure through cost cutting or sometimes firm follows both the techniques.

3) The firm achieves its goal of maximising profit in an efficient way. The economic efficiency of firm on the basis of behaviour indicates that:
   i) The state of equilibrium of the firm is profit maximisation,
   ii) Firm takes decisions and follow such actions which will help to reach the level of equilibrium, and
   iii) Firm takes every possible measure to return back on the track to achieve the goal of equilibrium.
4) The firm can increase the profits and will achieve the state of equilibrium by equalising marginal revenue (MR) with marginal cost (MC) *i.e.,* MC=MR. The firm decides to produce output (Q) in order to gain addition to total revenue (R) to cover the any addition of total cost (C). Here, the correct position of organisational equilibrium will be:

   $$\frac{dR}{dQ}=\frac{dC}{dQ}$$

   Or

   $$\frac{dR}{dQ}=\frac{dC}{dQ}=0$$

   In order to make equilibrium stable, these conditions must be satisfied:
   i) The firm must necessarily maintain the condition of MC=MR.
   ii) At the point of equilibrium, MC must be raising.
5) The market environment where the firm works is given. The nature of market competition is defined in terms of:
   i) Number of firm in market,
   ii) Nature of product (whether homogeneous or heterogeneous) which organisation is producing, and
   iii) Movement *i.e.,* entry and exit of organisations in the market.

   The amount of maximum profit is estimated based on whether it is perfect or imperfect market but the MR=MC condition for maximum profit will remain same.
6) The firm may be determined in term of economic concept of time *i.e.,* short term or long term. The condition for profit maximisation will remain unchanged. The time perspective is an important from the perspective of organisational structure, size, scale of operation, level of output, etc. These perspectives have a direct or indirect effect on organisational profit and hence organisation desires to maximise profit for long term.
7) There are various decision variables which are affected by the organisation such as, rice, quality and quantity of raw materials, and finished goods, etc. Various behavioural theories neglectes the perspective of quantity factor, as it is not easily measurable by the economic measurement. In an implicit way, it is assumed that firm possess a complete knowledge of all the important factors while taking decisions. Further, the firm need not to pay any money in order to acquire this information.

   Basically, 'Economist Theory of the Firm' moves around the assumptions of concept of profit maximisation and assumptions that the behaviour of organisation is rational in order to attain this objective because of the constraints imposed on the firm by production technology, market environment, factor endowment, etc.

1.3.6. Cyert and March's Behaviour Theory

In the year **1963, Cyert and March** introduced **'Behavioural Theory of the Firm'** which is an extended detail of 'behavioural model of the firm' given by Simon. The focus of the behavioural theory is on the way- the big and modern firm take decision under the uncertain environment of imperfect market. The internal structure of these firm acts as the base for this theory. Thus, the theory tries to evaluate the issues created by the internal structure of firm and the effect of these issues is on the process of decision making. This theory also gives a detailed explanation of how firms react to similar set of external factors as per the difference in internal factors of the organisation.

The main four research commitments made by Cyert and March are as follows:
1) This theory focuses on various small but important decisions.
2) This theory helps to develop such model which is process oriented.
3) It connects the firm's model with empirical observations.
4) This theory helps to develop general theories which are beyond the specific study of the firm.

The behaviour theory suggested by Cyert and March in the book 'A Behavioural Theory of the Firm' is stated below:

1) **Firm as Coalition of Groups with Conflicting Goals:** The focus of this theory is on how the big and modern firms take decisions under the uncertain environment of imperfect market. In traditional theories, firms were considered as having sole objective and as a single unit of decision making but under this theory firms are considered as focusing on multiple objectives and decisions.

2) **Process of Goal Formation – the Aspiration Level:** Each member of the firm or group of firms has variety of demands from the firm. These demand of a member usually conflict with the demand of other members and the overall objective of the firm. Both the past achievements and demands are directly connected with each other. Demand motivates the firm and these changes in demand rely on the past achievements and changes of the firm and its environment.

3) **Goals of the Firm:** The top executives of the firm set the goals. The production, sales, raise in value of share, inventory, and profit are the five main objectives of the firm. For setting goals, the top executives continuously discuss among the associated groups. During the process of goal setting the top executives tries to satisfy the demand of as many members as possible. Further, Cyert and March states that organisational satisfying behaviour is rational within the boundaries of the firm where it operates. Hence, firm is satisfying organisation instead of maximising organisation.

4) **Means for Resolution of the Conflict:** Since organisation has limited resources thus it is unable to meet the demand of each and every member. This inability to meet demand gives raise to conflict. Therefore, the different groups continuously bargain and struggle so that their demands can be satisfied with the available resource. Further, this conflict among the members do not turns up to develop instability in firm, whereas the firm still maintains its stability. For the smooth functioning of the organisation, top executives uses several tools some of them are as follows:
   i) The sharing and use of the budget by different groups.
   ii) Firms are adaptive in nature. It decides its current objectives, decisions and behaviour on the basis of its past experiences.
   iii) There is allocation of activities which restricts the decisions of different associated groups. Hence, it helps in minimising the causes of disputes.
   iv) Giving the financial benefits to employees is one of the sources to satisfy the desires of associated groups but unlike traditional theory it is not the only source for this goal.
   v) The side payments given to the associated members are a good source to satisfy the demand and it often takes the form of policy commitment from the top management.
   vi) The slack payments are the payment made to the workers other than payment for efficient working. **For example,** in order to retain the associated members, management may pay high wages to the workers, high perks to the managers and high dividend to the shareholders.
   vii) The demand of each member in a sequence should be given proper care. For this, top executives lists out all the demands and first they satisfy the most urgent demands.
   viii) The decision making in the organisation must be decentralised. The top management of the firm makes such departments to which authorities or decision making powers are delegated.

5) **Decision Making Process:** The goals or objectives which are set by the top management and approved by board's members need to be implemented through the decision making. The decisions are taken at different levels of control. The two levels of decision making can be differentiate by the help of:
   i) Top level decision making, and
   ii) Lower level (administration) decision making.

   Usually, the top level decisions are neither taken after a comprehensive study of all the alternatives nor any significant study of marginalistic rule or cost-benefit are done. Instead of that, top management goes through a quick review of only important alternatives and after that it advices to study about these alternatives and make further study of the most suitable alternatives.

   On the other hand, the process of decision making at the lower level enjoys much freedom to take actions. The administrative staffs working at lower level possess knowledge and experience. It also helps in implementing the decision through rules and blue prints provided to them.

6) **Uncertainty and the Environment of the Firm:** There are two kinds of uncertainties suggested by Cyert and March. They are uncertainty of competitor's reactions and market uncertainty.

i) **Market Uncertainty:** It means the probable alteration in technology used in production, changes in the tastes and preferences of the customers, etc. This uncertainty can be controlled to some extent through market research.

ii) **Uncertainty of Competitor's Reactions:**The uncertainty due to the competitor's action and reaction are ignored by this theory by believing that the present business has been united to certain type of associated work.

The inappropriate attention is emphasised by this theory on the environment and its impact on the process of goal formation and on the decisions related to price and output at top level management. Further, it analyses the allocation of internal resources while assuming the collusion.

7) **Simple Model for Illustration of Theory**

The simple model given by Cyert and March states the case of duopoly market structure. The process of decision making includes the output determination of homogeneous products, so that only the single price will survive in the market. Definitely, each organisation decides its level of output which brings price change in market. Further, when both the firms decide the level of output, then the price market will determine the price. In this model, no changes are allowed to be made in inventories. The various steps of simple model for illustration theory is as follows:

Step 1: Forecast of Competitors' Reactions: Basically, the forecast of competitor's reaction is a prediction made on the basis of the past behaviour or trend which the competitor follows.

Step 2: Forecast of Firms Demand: After forecasting the competitor reactions, the demand of the firm is forecasted. Therefore, the future demand is an estimation of firm's sale in past years.

Step 3: Estimation of Costs: After the estimation of demand, firm needs to estimate its cost. Here, it is assumed that cost will remain same as last year. Though, if the goal of profit is being achieved over last few years then the average cost per unit will raise due to slack payment made by the firm.

Step 4: Specification of Goals of the Firm: Under this model, profit is the ultimate goal of the firm and estimation of profit will be done at the approximate average level of last few years.

Step 5: Evaluation of Results by comparing them to the Goals: With the help of information provided from prior steps a solution is obtained. It gains an estimated level of production, price, and profit. Thus, these estimations are compared with the firm's target. In case after this comparison, firm analyses that, it will be able to achieve its goal, then it adopts them, but if the firm finds that it will be unable to reach its set targets, then it will move to next stages.

Step 6: Re-Examines the Estimate of Costs: Here in this step, re-examination begins with cost as this comes under direct control of the firm. Generally, this cost cutting starts with reduction in expenses and slacks.

Step 7: Evaluation of the New Solution by Comparing it to Goals: In case this reduction in cost leads to firm at the decided level of profit, then this measure will be adopted otherwise the firm will move to next step.

Step 8: Re-examine the Estimate of Demand: In this step, re-examination includes the consideration of changes in sales strategies, like, changes in the market research, advertisement, recruiting more salesmen, etc. Hence, it leads to adjustment of demand estimation which is done earlier.

Step 9: Evaluation of New Solution by comparing it to Goals: In case after the new estimation of the demand and cost, the target profit is achieved, then the firm will adopt this strategy otherwise the firm will move to next step.

Step 10: Readjusts Downwards its Aspiration Levels: In this step, the firm readjusts to downward aspiration level, if with the revised cost and demand, the goals are not achieved by the firms.

Cyert and March explained this model on the four actual case studies and two experimental studies done with the hypothetical firms. It is clear that the theory is based on very less number of cases which will make it possible only in general working. Thus, the part which shows the decision making procedure and the proper utilisation of resources in big companies should be formed and should include other theories of the firms.

1.3.7. Marris' Growth Maximisation Model

Marris' hypothesis is that executive actions are limited by the need for management to protect itself from dismissal or takeover raids in the event of failure. The theory of Marris states that modern firms are controlled by the managers and shareholders. Managers will have the aim to maximise the rate of growth of the company, whereas the shareholders will try to attain maximum return in the form of dividend and increased value of share price. Marris approach illustrates that the ownership and control are lying with shareholders and managers respectively. The manager is more concern on variables such as salary, status, power and security. On the other side, shareholders are more concern towards companies profit, market share, output, capital and public esteem. In precise, the role of managers and the shareholders are completely different, but pushing company for betterment.

The utility function of managers (U_m) and that of the owners (U_0) may, therefore, be defined as:

U_m = f (salaries, power, status, job security)

U_o = f (profits, market share, output, capital, public esteem).

According to Robin Marris, there is a strong correlation between managers, owners and size of the firm. Further, he states that manager aims to maximise the rate of growth for the firm than absolute growth. Marris assumes that managers are mostly concerned towards the rate of growth of the firm which is defined in terms of variables such as capital, output, revenue and market share. While the absolute growth is the total book value of assets, inventories and short-term assets. Thereby, the rate of growth is the concern of most of the managers and shareholders than the absolute growth. Thus, the growth rate has a positive effect on promotion of the manager as well as satisfies the shareholder. The two major constraints recognised by Marries for maximisation of rate of growth are:

1) **Managerial Constraints:** The capacity of the managerial team determines the upper limit of the rate of growth of the firm. There is a high possibility of management losing control over the firm which is experiencing rapid growth. The rapidly increasing firm needs accommodation of more managers, for which managers are hired. Due to this, there will be a lag in time for managers to get adapted to culture, know their responsibilities clearly, developing a new product for launch and so on. Similarly, the research and development department cannot be expected to continuously produce new products, even if they are capable. There will exist a lag in time for getting the requirement from the manager who is involved in interaction with the customer to know their need and want.

2) **Financial Constraints:** Finance is an important ingredient for growth; lump-sum money mostly cannot be generated by the firm for expansion for the rapidly growing company. Thereby, borrowing the same involves a potential risk of repaying as debt with interest, hence, the management has to involve more tediously in generating revenue and further growth of the firm which would give them job security. If the borrowing is made from the capital market, the case is serious because there is a possibility of takeover if the company is not functioning well; thereby the job security to manager therein is further pushed to perform better. Hence, most of the companies are discouraged to maximise the rate of growth despite their opportunity in the market due to financial constraint.

   To judge the prudence of a financial policy, Marris proposes the concept of financial constraint (a), which is mainly determined by the risk attitude of the top management. A risk-loving management would prefer a high value of 'a', while a risk averting management would prefer a low value of 'a'. Marris defines 'a' as the weighed average of the following three security ratios:

   i) $\text{Liquidity Ratio } (a_1) = \frac{\text{Liquid Assets}}{\text{Total Assets}}$;

   ii) $\text{Leverage Ratio } (a_2) = \frac{\text{Value of Debts}}{\text{Total Assets}}$; and

   iii) $\text{Profit - Retention Ratio } (a_3) = \frac{\text{Retained Profits}}{\text{Total Profits}}$

1.3.8. Baumol's Static and Dynamic Models

The ultimate objective of the firm is to maximise the sales, through which profit maximisation can be attained by the company. Due to the separation of ownership and control in the modern organisation, the managers are forced to achieve sales target for prestige and better salary through incentives. Baumol who was a consultant to various firms expressed that, when he asked various managers that how their business went last year then they

replied that the sales value touched three million dollars. Baumol states that maximisation of sales to attain maximisation in profit is the actual behavior of every firm. According to Baumol every firm is more focused on sales maximisation because it tends to achieve both short-run and long-run objective of the management, whereas short-run revenue maximisation will not support long-run profit maximisation.

Number of arguments has also been passed by Baumol to support his theory. Those are as follows:

1) Every firm has sensitivity towards sales; even slightest declines in sales are much concerned by firms.
2) The sales of the firm are directly tend to influence the bank credit, creditors and the capital market.
3) Decline in sales will also cover the way for its own distributors and dealers to stop taking an interest in the particular product.
4) The unpopularity of the product may stop the consumer to buy the particular product.
5) Sales are the revenue generators for the firm and if it is declined then the firm needs to cut down its expenses through cutting down its staff and other operating expenses.
6) A large sale of the firm is called economies of sales. Through economies of sales the firm expands and earns a large profit.
7) The extent of sales is directly proportional to the salaries of the worker and the management.

Baumol model for revenue maximisation through sales is depicted in **figure 1.4** where TC- total cost curve, TR- Total Revenue Curve, TP- Total profit curve and MP- Minimum profit. It can be seen from the diagram that a firm attains maximum profit for the output level OQ, the corresponding point is denoted by B in the TP curve. But the objective of the firm is to maximise the sales rather than its profit; thereby sales maximisation output is denoted by OK where the total revenue generated correspondingly is denoted by KL in the TR curve.

From the curve, it can be interpreted that sales maximisation output OK is higher than profit maximisation OK. But the sales maximisation is subject to minimum profit constraint. For sales maximisation, the firm produces output to a certain level to cover the minimum profit and to generate highest total revenue, whose point is OD, the DC here represents the minimum profit generated which is equivalent to OM for the output level DC. DE represents the total revenue generated for the output level OD.

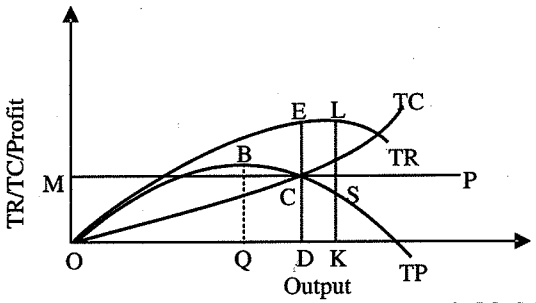


Figure 1.4: Baumol's Static and Dynamic Model

In Baumol's model, the point of profit maximisation output, OQ is smaller than sale maximisation output OD. In the model, lower price under sales maximisation is that both total revenue and total output are equally higher while under profit maximisation, total output is much less as compared to total revenue. Imagine if QB is joined to TR in **figure 1.4**. "If at the point of maximum profit," writes Baumol, "the firm earns more profit than the required minimum, it will pay the sales maximiser to lower his price and increase his physical output".

Assumptions

The assumptions of Baumol's Static and Dynamic Models are as follows:

1) The firm has a single period time horizon.
2) In order to gain profit, the firm aims toward maximisation of sales revenue.
3) The minimum level of profit is decided according to the present market value of shares in the competitive market.
4) The firm is operating in oligopolistic market structure where the cost curve is U-shaped and demand curve is sloping downward. The revenue and total cost curve of the firm is also conventional in nature.
5) Under oligopolistic market, advertisement acts as a major tool because of the existence of non-price competition.
6) The cost of advertisement is independent from production cost.
7) The price of product will remain same.

Criticisms of Sales Maximisation

Following are the limitations of the Baumol model:

1) Rosernberg has illustrated that it is difficult to specify relevant profit of the firm.
2) According to Hawkins, sales maximisation will involve more production and advertisements than a profit maximiser.
3) According to Baumol model, multi-product yield same value of revenue and profit maximisation, but Williamson showed that there is the difference in sales maximisation with respect to profit maximisation for multi-product.
4) The model does not give an explanation for the interdependence of prices, especially with oligopolistic firms.
5) The model fails to explain that the products possess inelastic demand whose price is kept constant for a considerable period of time.
6) The model ignores actual competition as well as threat of potential competition from rival oligopolistic firms.
7) Baumol model does not establish relationship between the industry and the firm. It made assumption that every firm is involved in sales maximisation.
8) Professor Hall uses 500 companies in his analysis to come to the conclusion that most of the companies are not operating with sales maximisation objective.

Types of Baumol's Sales Revenue Maximization Models

There are two important models developed by the Prof. Boumol. They are static model and dynamic model. The brief introduction of these models are given below:

The Static Model of Sales Maximization

The assumptions of static model of sales maximisation is given as follows:

1) This model is applicable only on the specific time period, whereas at different time period this model is not applicable.
2) The objective of the firm is maximising the sales revenue subject to minimum profit constraint.
3) The firm demand curve will slope downward from left to right.
4) The firm average cost curve is unshaped.

The Dynamic Model of Sales Maximization

Practically, in the firm various changes take place from time to time and these changes impacts greatly on the business decision. Therefore, to give consideration to these changes, the another model of dynamic of sales maximisation was developed by the Boumol. This model gives a clear explanation of how changes made in the expenditure of advertisement will affect the sale revenue under high competition in market. The assumptions of dynamic model are as follows:

1) With high cost of advertisement the sale revenue of the firm will increase.
2) The market price will be same.
3) The firm's cost and demand curve are conventional in nature.

Sometime, to face high degree of competition in market, the firm undertakes the strategy of aggressive advertisement. This shifts the demand curve to the right. This forward shift implies the increase in the demand due to high advertisement. Generally, sales increase as a result of decrease in cost but raise in sales usually depends on whether the demand is elastic or inelastic.

When demand is elastic then any reduction in price will lead to increase in sales but in case demand is inelastic, then reduction in price may affect sales negatively. Therefore, advertisement plays a very significant role in sales. Mostly, the firm experiences that whenever they increases the cost of the advertisement their sales volume will increase. The profit maximise will incur lesser amount of advertisement expenses as compared to the sales maximise. Further, it is very crucial to remember that the expenditure made on promoting the sales must bring sales and profit in an amount more than proportionate if it is not done then it will impact the business decisions negatively.

Therefore, a successful attempt was made by Boumol in order to understand the competitive behaviour of the firm under oligopoly with the help of introduction of a non-price variable in the model. Under the oligopoly market, there will be only one big firm which will be competing with other small firms by either producing same or different product. Thus, high advertisement will help organisation to increase its sale and as a result increase in profit.

1.3.9. Williamson's Managerial Discretionary Theory

A very crucial model of managerial utility popularly known as the 'Model of Managerial Discretion' was introduced by O. Williamson. Williamson argues that managers after attaining a minimum level of profit, pursues policies to maximise their utility. The managerial utility function includes variables such as salary, security, status, power, prestige and professional excellence. Out of all variables in the utility function, it is the only salary that is quantitative or measurable. Other variables are associated with an increase in the satisfaction level of the managers. Expenditure on staff will generate a positive satisfaction (utility) because these expenditures are the source laying security and in turn reflecting the power, prestige and status of the individual manager for achievement.

The managerial utility function is expressed as:

$$U = f(S, M, I_D)$$

Where, S = Staff expenditure (including managerial salaries, the administrative and selling expenses)

M = Depicts managerial emoluments

I_D = Discretionary investment

Staff increment indicates the span of increase in the power, status and prestige of the manager within the firm. Progressive increase in the staff provides better platform to managers for achieving personal and professional development alongside firm growth through high output resulting out of more staff employed.

Managerial Emoluments or slack refers to the payments apart from giving minimum needs such as expense account, luxurious offices, company cars and so on will provide a comfortable zone for the managers to attach with the development of the firm. Also, expense, before mentioned, has tax benefit and better visibility that is capable if attracting more and more talented employees and employers towards the firm.

Discretionary Investment refers to the investment made by the discretion managers for better operation of the firm. This gives satisfaction to the managers to fulfil their project as well as provides additional responsibility for getting good revenue out of the project to acquire maximum dividend out of the investment made. Discretional investments are investments beyond the requirement, thereby they help in smooth functioning of the project as well as it is included in the minimum profit for lending satisfactory dividend policy.

1.3.10. Alternate Objectives of the Firm

Besides maximisation of profits and sales, a firm may keep certain other economic and non-economic goals. Some of these goals are as under:

1) **Economic Objectives:** Economic objectives of the firm are as follows:
   i) **Maximum Growth Rate:** This is the primary objective of the firm and it is associated with the profit constraint. In order to achieve a maximum rate of growth, the firm has to attain optimal net profit. **Kenneth E. Boulding** identified various stages in the life cycle of the firm. In early stage or initial stage, the firm will aim to establish itself and its product in the market. During the second stage, which is called growth, when the firm is well established, it focuses on improving its efficiency to attain high growth rate. The third stage is called maturity stage.

      At this period of time, the competition becomes numerous and the price penetration becomes hard for the firm, thereby the demand becomes stagnant and profit value will tend to decline. Finally during the fourth stage or decline stage, the demand declines and heavy competition makes the firm lose its profit, eventually the firm will exit out of the industry and the market or introduce new product in the same or different stream of line that has market value and once again start the cyclic process of initial, growth, maturity and decline stage.

   ii) **Desire for Liquidity:** According to Professor Joel Dean, liquidity is important for smooth functioning of operations and profit maximisation. He refers liquidity as the adequate amount of cash that is needed to avoid crisis such as opportunity cost.

      The fear of bankruptcy and financial crisis known as **"Banker's Mentality"** are the major factors influencing the firm to maintain sufficient amount of liquidity cash for utilising opportunity cost during strikes and attain profit.

2) **Non-Economic Objectives:** Non-economic objectives of the firm are as follows:
   i) **Survival: Peter F. Ducker** says that it is important that the firm sustain in the market rather than profit maximisation or growth. Survival is an essential component for the firm to produce better profit and growth for which customer demanding quality goods and services are required from the producer end to its customers. Goodwill earned by the firm through its quality goods and services with respect to time helps the firm to sustain in the industry and market over longer period of time. Hence, survival is considered as the long-term objective of the most of the firm.
   ii) **Building-Up Public Confidence for the Product:** This is the secondary objective next to survival. Firm induces public confidence through strong advertising. Once the firm establish itself in the industry and market, it attracts more public and installs confidence when it produces better quality products in the market and sell at considerable price that its customer can afford with satisfying service.
   iii) **Welfare:** Business firm's responsibility includes providing fair wages, good working conditions and motivational benefits such as transportation, subsidised canteen, medical care, schools and housing for the workers. This not only enhances the productivity but also helps in retaining the skilled employees and employers within the firm. As the business grows there lays the responsibility of the firm towards the society. Hence, some social welfare programs such as construction of schools and hospital should be run by the firm.
   iv) **Sound Business Practices:** Business ethics strictly provides sound business practices that help in gaining goodwill for the firm from society. Business practices such as disclosing necessary information, price lists, and replacement or refund for defective products will make a platform for increasing the customer's confidence on the product, service, and the firm.
   v) **Progressive Management:** Management should frame certain policies for dynamic growth and expansion of the firm by involving workers in management and training program to compensate fast changes occurring due to dynamic growth in the firm.

1.4. MARKETS

1.4.1. Concept of Market

Market is defined as a group of individuals or organisations that is interested in purchasing products and services, is capable of doing it, and is having permission of law to purchase that product.

According to Cournot, "Economists understand by the term market, not any particular market place in which things are bought and sold, but the whole of any region in which buyers and sellers are in such free intercourse with one another that the prices of the same goods tend to be at equality easily and quickly".

According to Philip Kotler, "A market consists of all the potential customers sharing a particular need or wants who might be willing and able to engage in exchange to satisfy that need or want".

According to Pyle, "Market includes both place & region in which buyer & seller are in free competition with one another".

The number of persons, whose needs are not satisfied but they have the ability to perform the transaction for fulfilling their requirements, determines the size of the market.

1.4.2. Importance of Market

Following points reflect the importance of a market:

1) **Reciprocal Benefits:** Market creates a reciprocal benefit in a way that consumers can fulfil their requirements by purchasing the products, whereas the sellers can acquire the market for continuing their marketing activities.
2) **Incentive to Producers:** The main aim of producing goods is to sell them to consumers. But the goods cannot be sold if market does not exist. So, the existence of market for certain product plays a very important role in motivating the producers to continue their production activities.

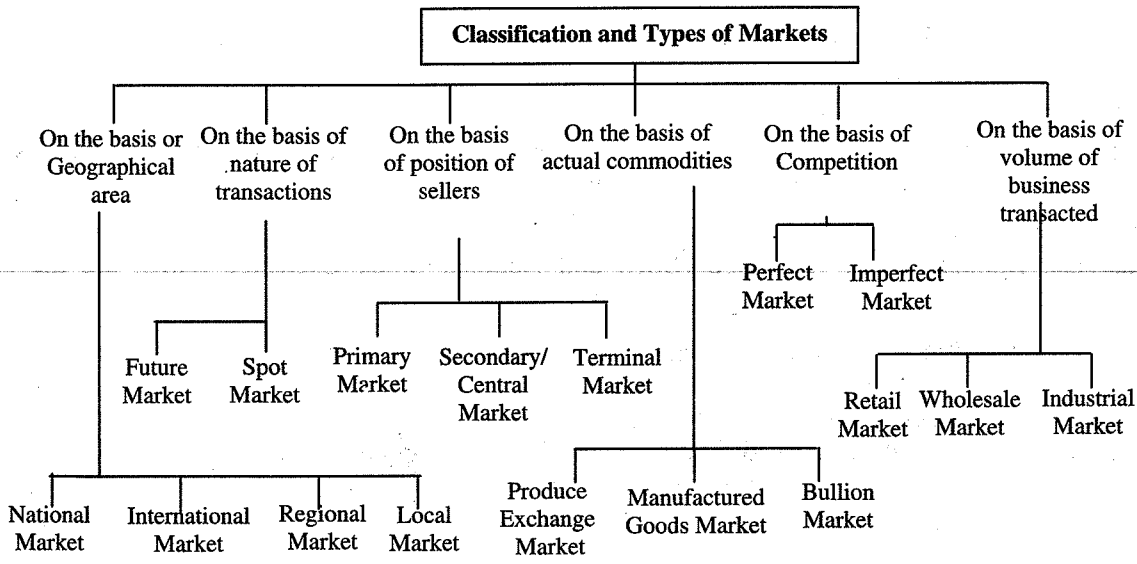
3) **Generation of Employment:** Market also plays a vital role in creating new opportunities of employment because day to day selling and purchasing of goods and services require manpower for performing various operations.

4) **Index of Economic Situation:** Market condition of any country helps in determining its economic situation. If a country is performing its business operations in worldwide market then it is considered to be an advanced country as far as business is concerned.

5) **Supply *vs.* Demand Adjustment:** In the absence of market there will be no demand for product and services. Some products such as, sugarcane, cotton, jute, etc., are produced seasonally but their demands persist throughout the year. So over a long period of time, a well-managed market helps in maintaining a balance between supply and demand and also plays a vital role in controlling the price fluctuations.

1.4.3. Classification and Types of Markets

On the basis of different approaches, markets can be classified in various ways, which are mentioned below:



1) **Classification According to Geographical Areas:** On the basis of geographical areas, market can be of following types:
   i) **National Market:** A country is considered to be a national market for a particular commodity when all the facilities of production, distribution, communication, etc., are available in that country. Currently in India, goods or services can be delivered quickly in any corner of the country because of effective transportation and communication system. That is why India is national market for near about all the products and brands.
   ii) **International Market:** A market is referred to as an international market when the buyers and sellers of products and services are involved in the process of exchange beyond the geographical boundaries of a country.
   iii) **Regional Market:** When a market is confined to a specific region or area of a country, then that market is termed as a regional market. **For example,** India is distributed in four regions namely Eastern India, Western India, Northern India, and Southern India for most of the purposes.
   iv) **Local Market:** A market that is restricted to a small area for doing business operations is called local market. Local market involves markets of consumable goods of daily requirement such as, milk, vegetables, etc.

2) **On the Basis of Nature of Transaction:** On the basis of nature of transactions, market can be of following types:
   i) **Spot Market:** Spot market is that type of market where commodities are delivered immediately as soon as their transaction is settled.
   ii) **Future Market:** In case of future markets, contracts are signed for the transaction of a commodity which is to be settled on any future date, i.e., payment and delivery of product will take place in future. Contracts are signed generally for price fixation.

3) **Classification According to Position of Sellers:** On the basis of position of sellers, market can be of following types:
   i) **Primary Markets:** In primary markets, organisations sell their products to some middlemen like, distributor or wholesaler.
   ii) **Secondary Market:** In this type of market, the middlemen like wholesalers, further sell their products to another middleman like retailers.
   iii) **Terminal Market:** Terminal market is that market where goods are finally sold to the end users of the product.

4) **On the Basis of Commodities/Goods:** On the basis of commodities, market can be of following types:
   i) **Produce Exchange Market:** Sale and purchase of only one commodity takes place in 'produce exchange markets' and these markets are set-up and managed by the buyers and sellers of that commodity. These markets can only be seen in cities which are industrially developed. Cotton Exchange Market of Mumbai and Wheat Exchange Market of Hapur are some of the examples of such markets.
   ii) **Manufactured Goods Market:** Manufactured goods are bought and sold in these types of markets. Some examples of these markets are Piece Goods Exchange of Mumbai, Leather Exchange Market of Kanpur, etc.
   iii) **Bullion Market:** Commodities like gold and silver are bought and sold in these types of markets. **For example**, Bullion markets of Kolkata, Kanpur, Mumbai, etc.
   iv) **Stock Market:** Mutual funds, equity shares, bonds, debentures, etc., are bought and sold in these types of markets. Bombay Stock Exchange (BSE) and National Stock Exchange (NSE) are the stock exchanges that regulate the stock markets of India.

5) **Classification According to Nature and Degree of Competition:** On the basis of nature and degree of competition, market can be of following types:
   i) **Perfect Market/Competition:** Many sellers selling same type of products to many buyers is the characteristic of perfectly competitive market. Production of goods by using productive techniques and minimum utilisation of the factors of production is the feature of efficient market. This type of market is considered to be unrealistic but, it is of special importance for hypothetical and theoretical reasons.
   ii) **Imperfect Market/ Competition:** The situations pertaining to perfect market are different form imperfect market. In any market the conditions being necessary for perfect competition are unsatisfied is the competitive situation of imperfect competition. The market having large number of producers offering closed substitute goods which are not identical as regards to perfect competition. Various forms of imperfect competition are:
      a) **Monopoly:** The origin of the word monopoly is from the Greek word *mónos*, single, and *polein*, to sell. This is a form of imperfect competitive market structure described by a single seller and large number of buyers where entry and exit barriers are normally associated with monopolistic situation pertaining to single seller producing for many buyers. The product is greatly discriminated due to the absence of competitive sellers producing closed substitute products.
      b) **Monopolistic Competition:** With respect to large number of sellers contributing discriminated product to large number of buyers makes this competition different.
      c) **Oligopoly:** Selling of competing products by some sellers to many buyers is called Oligopoly which is originated from the Greek word *oligos*, few, and *polein*, to sell. The characteristic of this type of competition is having limited firms but more than one which are producing homogeneous good. The interdependent strategy between the firms as they are fewer in number in the market clearly states that oligopolistic firm's profit totally depends on the actions of their competitors.
      d) **Duopoly:** Duopoly type of oligopoly originates from the Greek word, *duo*, two, and *polein*, to sell. There are only two firms in the market producing a homogeneous goods characterises imperfect competition. Normally to avoid any complications oligopolies are studied by analysing duopolies. Such that, in a market A, there are only two dominant suppliers which means a market where prices are being jointly controlled by them

6) **On the Basis of Volume of Business Transacted:** On the basis of volume of business, market can be of following types:
   i) **Retail Market:** In these types of markets goods are directly sold to end consumers. End consumers buy goods for personal consumption and not for making profit from it.
   ii) **Wholesale Market:** In these markets, dealers or retailers buy goods in bulk quantity for the purpose of re-sale.
   iii) **Industrial Market:** In industrial markets, goods are bought in bulk either for consumption or for the purpose of further processing on it, **e.g.**, raw material.

7) **On the Basis of Sale:** On the basis of sales, the market can be divided into two market:
   i) **Product Market:** Product market is a market where the final goods and services are bought and sold.
   ii) **Factor Market:** The factor market is a place where factors of production (land, labour, capital) are bought and sold.

Participation and Interaction between Firms and Individuals in Product Market

Market is a place where sales and purchase of finished goods and services takes place. A place where household sector purchases consumer goods, business sector purchases capital investment goods and foreign or government sector purchases goods by paying a price are known as product market. Generally, this sale of goods and services is made by the business sector.

In economics, a large portion of macroeconomics describes the ways and the reasons why there is an increase and decrease in the GDP when exchanged through product market takes place. The product market is also known as the output market or goods market. Product market is one of three sets of macroeconomic market whereas financial markets and resource markets are the other two sets.

Further, exchange of scare resources, factors of production, raw materials or any form of intermediate goods are not included in product market. The gross domestic product is used to calculate the gross value of goods exchanged in product market within a year. The components of demand side of the product market consist of government purchases, investment expenditure, consumption expenditure and net amount of export whereas the components of its supply side include the total production made by business sector.

Participation and Interaction between Firms and Individuals in Factor Market

In terms of economics, factor market means such a market where sale and purchase of factors of productions (i.e. land, labour, capital and entrepreneurship) takes place. In other words, factors market is a place where an organisation purchases such resources which it will need for producing goods and services. Labour markets, capital markets, raw material markets are some of the factor markets.

The transformation of raw material into finished goods is known as production. For any organisation, factors market is the source of both raw material and factors of production. The product market is the place where the goods are sold. In a general aspect, both factor market and product market are same because in both the markets equilibrium of demand and supply determines the price of the product, in both markets firms try to earn maximum profit, their factors possess the capacity to effect the price and both law of demand and supply are applicable in these markets.

The participants of the market purchases and sales the factors of production which are required to perform the production process. **For example,** if a person seeks job then he makes himself available for the producers as a factors known as labour. The person sells his talent and skill and the producer pays back to him in the form of wages or salary. Therefore, organisation purchases different productive resources against the factor price. Further, the principle of derived demand is involved when product market and factors market interacts.

Here, derived demand means such demand which is derived out of the demand of finished goods and services. **For example,** in case there is an increase in demand of bikes in the market the producer will meet this increasing demand through raising the demand of raw material and factors used for production of bike. The interaction of the demand and the supply of various factors determine the prices of the factors. The market helps in the proper allocation of money among the various production factors.

1.5. EXERCISE

1.5.1. Very Short Answer Type Questions

1) What is economy?
2) Explain the meaning and definition of managerial economics.
3) Highlight the meaning of firm.
4) Discuss the rationale for the firm.

1.5.2. Short Answer Type Questions

1) Is economic an art or science?
2) What is the nature of managerial economics?
3) Explain the opportunity cost principle.
4) Differentiate between economics and managerial economics.
5) What are the objectives of firm?
6) Discuss some major differences among micro, macro and managerial economics.
7) Highlight the economist theory of the firm.

1.5.3. Long Answer Type Questions

1) What is economics? Explain its scope and limitations.
2) Explain the micro and macro branch of economics.
3) Highlight the scope and use of managerial economics.
4) Discuss the incremental and marginal principle.
5) What are the significance and limitations of managerial economics?
6) Write note on managerial economics and decision making.
7) "Is profit the only objective of a firm?" Explain.
8) Explain the profit maximisation model of firm.
9) Give a brief introduction of Cyert and March model, Marris' growth maximisation model and Williamson's managerial discretionary theory.
10) Describe the concept market. Also highlight its types and importance.
11) Explain how firms and individuals participate and interact in the product market and in the factor market.

UNIT 2

Utility & Demand Analysis

2.1. CONSUMER DEMAND

2.1.1. Introduction

Why, when, how and where should people buy or do not buy product is the study of consumer behaviour which is the mix of essential features from psychology, sociology, social anthropology and economics. The decision-making process of the buyer is attempted to comprehend, both individually and in groups. To understand people's wants, characteristics of individual consumers like demographics and behavioural variables are studied by consumer behaviour. Groups such as family, friends, reference groups and society in general which affect the consumers are also assessed.

The three different roles of user, payer and buyer being played by the customer are studied as behaviour of the customer on its buying behaviour. The powerful asset for analysing customer behaviour is relationship marketing as it is enthusiastic to re-discover the true meaning of marketing by asserting the importance of the customer or buyer. Consumer retention, customer relationship management, personalisation, customisation and one-to-one marketing are also being given great importance.

2.1.2. Meaning of Utility

The power to satisfy one's want or desire is the utility of the commodity. The literal meaning of the term 'utility' is 'usefulness' but in economics, it implies the capacity to satisfy the want of a good. Simply to say, utility is value-in- use of a commodity. Utility is the derivative of consumption which implies subjective satisfaction. Simply, 'utility' describes consumer preferences according to modern theory of consumer behaviour. Satisfaction and utility are different to each other as satisfaction is derived whereas utility is in the thing. Thus, a thing possessing utility leads to or provides satisfaction. Satisfaction being the ultimate end, while utility is the channel. Utility analysis provides awareness in understanding market demand and forms a foundation of modern micro-economics. Investigation of consumer behaviour , especially market purchases is the analysis in particular which depends upon the satisfaction of wants and needs(utility) generating from the consumption of a good.

2.1.3. Measurement of Utility

The concept of measuring utility in units is **utils**, which helps in understanding the consumer's behaviour. It is used analytically to distinguish between the following two utility concepts:

1) **Initial Utility:** After consuming first unit of a commodity the utility derived is the initial utility. Satisfaction of the thirst by taking first glass of water is the utility gained which is described as initial utility that is always positive.

2) **Total Utility (TU):** After consuming various units of the commodity, the sum total of satisfaction that is received by a consumer is the total utility. The total utility or satisfaction of a consumer to a certain point will be greater on consumption of more units of a commodity. The point of saturation that represents maximum total utility is achieved by the consumer as he keeps on increasing the consumption of the commodity. The total utility of the consumer starts declining as he further consumes the units of commodity.

3) **Marginal Utility (MU):** When a single unit is added to total utility in the rate of consumption or the stock of goods possessed is defined as marginal utility which is the net additional unit. Thus, for a given unit of time, marginal utility is the increment or decrement in total utility that is proportionately caused by increment or decrement of one unit in the consumption of a commodity.

2.1.4. Relation between Total Utility and Marginal Utility

Following table and diagram illustrates the relationship between total utility and marginal utility:

Table 2.1

Quantity	Total Utility	Marginal Utility	
1	10	10	Initial utility
2	18	18 – 10 = 8	
3	24	24 – 18 = 6	
4	28	28 – 24 = 4	Positive
5	30	30 – 28 = 2	
6	30	30 – 30 = 0	Zero
7	28	28 – 30 = –2	Negative

It is clear from **table 2.1** that total utility increases as every successive unit is consumed by the consumer. At the first unit total utility is equal to 10 which is initial utility and further, it increases to 18, 24, 28, 30 and so on. Here, one must remember that this increase is at a diminishing rate, but still it is positive utility. At 6th unit, total utility is constant, i.e., 30 units and later on it start declining at an increasing rate. At 7th unit, total utility is recorded as 28 units. Therefore, one can say that total utility goes on increasing as long as the marginal utility of a commodity remains positive. On the other side, at the first unit of consumption marginal utility is equal to 10 and on successive units, it continues to fall till 5th unit of commodity is consumed. At 6th unit, marginal utility becomes zero, this is the point of full satisfaction or point of saturation. Here, consumer gets maximum satisfaction. At 7th unit, marginal utility is –2, i.e., negative. This means, the point of over satisfaction.

Diagrammatic Representation of Total Utility and Marginal Utility

Figure 2.1, shows the relationship between total utility and marginal utility. Here, units of commodity are measured on X-axis while marginal utility and price are measured on Y-axis. According to the figure initial utility is depicted by point 'A' i.e., utility derived from the first unit. From point 'A' to 'M', total utility increases upto 5th unit. Equivalent to this, there is a fall in marginal unit (MU) at 6th unit and becomes zero at point P. Further to this, MU turns negative (-2). Thus at point M total utility is maximum and marginal utility is zero. There is a decline in total utility after point M and marginal utility becomes negative below X-axis. Thus, the conclusion is that both total utility (TU) and marginal utility begins from the same point but, as total utility declines, marginal utility turns negative.

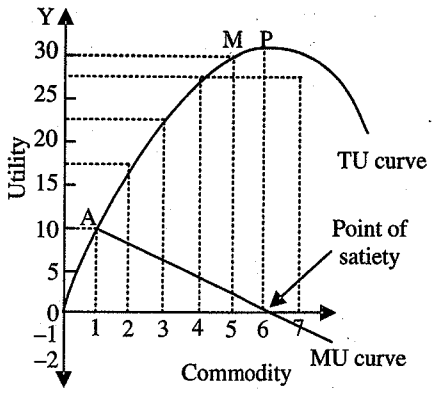
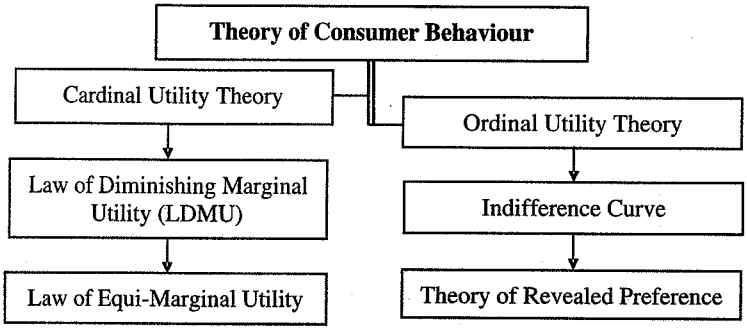


Figure 2.1: Relationship between TU and MU

2.1.5. Theory of Consumer Demand

There are two theories or approaches of consumer behaviour. The terms 'Cardinal' and 'Ordinal' are borrowed from the vocabulary of mathematics. The number 1, 2, 3, 4, 5, etc., are cardinal numbers. The number, 2, e.g., is twice the size of number 1. In contrast, the number 1st, 2nd, 3rd, 4th, 5th, etc., are ordinal numbers. Such ordinal numbers are ordered or ranked and there is no way of know, just from the ranking, what is the size relation of the numbers. The second one might be or might not be twice or thrice as big as the first one. The ordinal numbers 1st, 2nd, 3rd, 4th and 5th, could be 50, 100, 150, 200 and 250 or they could be 50, 51, 52, 53 and 60. In case of ordinal number, the second number is greater than the first, the fourth is greater than the third, etc. They are as follows:



2.2. CARDINAL UTILITY ANALYSIS

2.2.1. Introduction

Neo-classical economists, namely, **Gossen** of Germany, **William Stanley Jevons** of England, **Leon Walras** of France and **Karl Menger** of Austria developed the theory of cardinal utility. Meaningful clarifications were made in the cardinal utility theory by **Alfred Marshal**. According to this background, the other name of cardinal utility theory is **Marshallian theory**, **Neo-classical Utility analysis** and more.

Cardinal measurement of utility is contributed by Marshall where the consumer pays the higher price for a commodity; the higher is the utility and *vice-versa.* The utility of a commodity can be measured by price. Therefore under the assumption, a consumer is capable to assign a number to every commodity that represents the measure of utility associated with it. Likewise weights, utility numbers can also be manipulated. Law of diminishing marginal utility and law of equi-marginal utility have an access to cardinal utility.

2.2.2. Assumptions of Cardinal Utility Analysis

The assumptions of cardinal utility approach are as follows:

1) The consumer is a rational person.
2) Consumers are perfectly aware of the price in the market.
3) There is variety of choices of the goods in the market.
4) There are various commodities whose prices are not affected by variations in their supply.
5) Utilities are measured in terms of money as there are no substitutes.
6) Due to price of the commodity being very low or very high, the consumer is unable to purchase it.
7) Law of Marginal Diminishing Utility is being assumed.
8) The units of the commodity must be suitable.
9) There is no change in the tastes of the consumer.
10) The utilities of the commodities do not depend upon one another.

2.2.3. Limitations of Cardinal Utility Analysis

The cardinal utility approach suffers from many shortcomings:

1) It is assumed that utility is cardinal. In reality, it is subjective and it is difficult to put a specific value on the utility that one gets from the consumption of a good. One can certainly compare the utilities from two goods but one cannot give a definite value to the utility derived from the two goods.
2) Utility is assumed to be additive and thus to arrive at total utility, the utility from different goods was added together. This is not true because utility is not cardinal and also because one derives different types of utility or satisfaction from different types of goods. One cannot add different types of utility or satisfaction from different goods.
3) The utility derived from the consumption of a good was assumed to be independent of the utility from the consumption of another good. However, this is not true as the different goods are interrelated and the utility from one good often depends upon the utility from another good.
4) The marginal utility of money was assumed to be constant. This is not true and it is often a subject ot much debate as to whether the law of diminishing marginal utility applies to money.
5) The utility analysis is unable to explain the existence of the Giffen goods.
6) The utility theory docs not analyse the effect of a change in the price in terms of (he income effect and the substitution effect.

2.2.4. Criticisms of Cardinal Utility Analysis

Following are the criticisms of the cardinal utility approach:

1) **Utility is Subjective:** Though the analysis of consumer's demand is based on its objective, concept of utility is subjective relating to man's psychology. Therefore, it is not possible to be objective about it.

2) **Every Commodity is not an Independent Commodity:** According to the assumption that the basis of the analysis is that every commodity being an independent commodity. But actually, utility of a commodity depends upon the utility of other commodities as no commodity is independent. Therefore, absolute measurement of consumer's behaviour through utility analysis is not possible.

3) **Marginal Utility cannot be Estimated for all Commodities:** Concept of marginal utility is the basis of utility analysis. Those commodities which are divisible, their marginal utility can be measured but division of commodities like, television set, refrigerator, etc., is not possible.

4) **Marginal Utility of Money does not Remain Constant:** The marginal utility of money remains constant is the other assumption of utility analysis. This is not a realistic assumption as the quantity of money with a person increases; there is a decrease in its marginal utility and *vice-versa.*

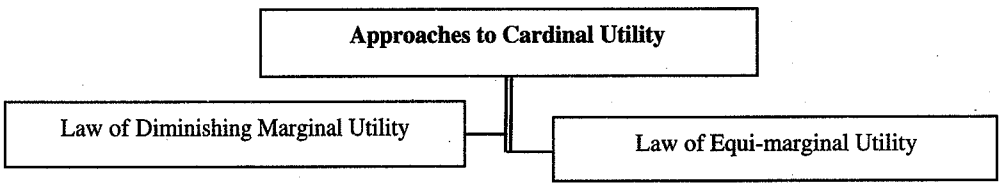
5) **Income Effect and Substitution Effect not separated:** Price effect is not divided into income effect and substitution effect by utility analysis. Neither it indicates any change in demand with respect to change in price nor what amount of changed demand is the result of change in real income (income effect) and how much the demand has changed by substituting cheap good for the expensive good (substitution effect).

6) **Consumer is regarded as a Computer:** This analysis states that consumer is a computer who while spending his money will always compare the amount of gain accrued to him by the utility of the commodity purchased with the loss incurred to him by sacrificing the money spent. But no consumer is so much calculative in real life.

7) **Does not Explain Giffen Paradox:** Giffen paradox cannot be explained by cardinal utility analysis and there is no explanation to why do the demand curve of many inferior goods slopes upward from left to right. In simple words, why does the demand expand with rise in price and contracts with fall in price?

2.2.5. Approaches to Cardinal Utility

The following approaches of cardinal utility are as follows:



2.2.6. Law of Diminishing Marginal Utility (LDMU)

The law of Diminishing Marginal Utility (LDMU) being the fundamental laws of economics is also the fundamental hypothesis of the theory of cardinal utility. Wants are satiable, therefore it is an important characteristic of human wants. Though there are many wants but only one particular want is satisfied at a time. Hence the other name of this law is the law of satiable wants. According to this law, as the consumer consumes more and more units of a commodity, there is a steady decline in the additional utility derived.

Thus, more the consumer consumes, there is an increase in his total utility but at a diminishing rate. It being very natural that on consuming additional units of a particular good at a point of time by a consumer, his intensity of desire diminishes for every additional unit resulting in the decrease in utility derived from each additional unit.

2.2.6.1. Explanation of LDMU

The Law of Diminishing Marginal Utility is explained with the help of the following **table 2.2.** From the **table 2.2,** total utility increases whereas marginal utility decreases. When the 6^{th} apple is consumed total utility is 82 and marginal utility is 2. But the consumption of the 7^{th} unit did not add to the total utility and hence, marginal utility is reduced to zero. The 8^{th} unit actually reduced total utility to 80 and marginal utility to –2. This shows that the consumer clearly dislikes the 8^{th} apple.

Table 2.2: Diminishing Marginal Utility

Number of Apples	Total Utility	Marginal Utility
1	30	30
2	50	20
3	65	15
4	75	10
5	80	5
6	82	2
7	82	0
8	80	–2

$MU_n = TU_n - TU_{n-1}$

Where; MU_n = Marginal utility of n units,
TU_n = Total utility of n units, and
TU_{n-1} = Total utility of n – 1 units

On analysing the above mentioned situation, it is clear that if there is a continuous consumption of some units, there is a partial satisfaction of our demands. Hence, the intensity of want and satisfaction diminishes accordingly.

2.2.6.2. Diagrammatical Representation

Figure 2.2 explains the law of diminishing marginal utility.

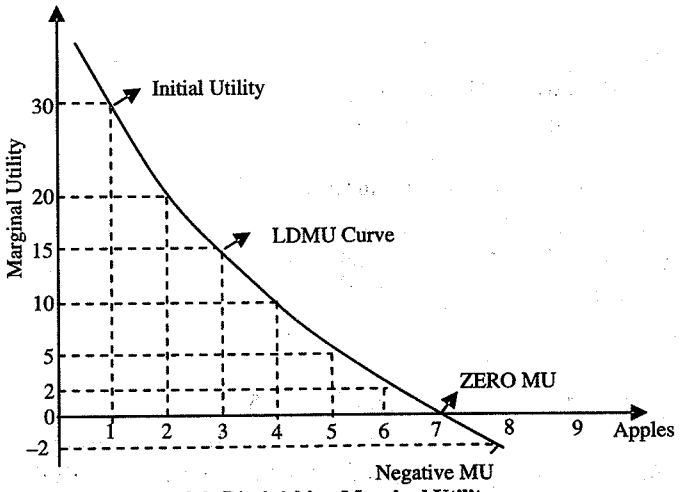


Figure 2.2: Diminishing Marginal Utility

According to the **figure 2.2** the units of apple is shown on the horizontal axis and the vertical axis measures the marginal utility derived from the units of apple. The utility curve displays the units gained from each additional apple. The falling of the curve down from left to right states that on every additional unit of apple, marginal utility is decreasing. Thus, the marginal utility of first 30 apples is the initial utility that reduces to zero for the 7^{th} apple. This point is termed as satiety point. Further, the MU of the 8^{th} apple is –2 and the marginal utility curve lies below the horizontal axis.

2.2.7. Law of Equi-Marginal Utility

On attaining maximum utility or satisfaction from his expenditure, a consumer is in the state of equilibrium. The law of maximum satisfaction explains the principle of consumer's equilibrium. The following have been mentioned by this law; the Law of Equi-marginal Utility, the Law of Substitution, the Law of Consumer's Equilibrium, etc.

According to Samuelson, "A consumer gets maximum satisfaction when the ratio of marginal utilities of all commodities and their prices is equal".

According to Hicks, "Utility will be maximised when the marginal unit of expenditure in each direction brings the same increment of utility". It means:

$$MU_1 = MU_2 = MU_3$$

The behaviour of the consumer is described by the law of Equi-marginal Utility on spending his limited income on various commodities and services. To derive maximum satisfaction and equilibrium every consumer properly allocates his resources on various items of expenditure. On buying one unit of commodity the consumer is in equilibrium whose marginal utility is at par with the marginal utility of the rupee that is represented by the price. The consumer is in the state of equilibrium when the marginal utility of the good consumed is zero. On the consumption of an economic commodity by the consumer, he is in equilibrium when the price paid by him is equal to the marginal utility of the commodity.

2.2.7.1. Assumptions of Law of Equi-Marginal Utility

The following assumptions form the basis of the law of Equi-marginal Utility:

1) It is a desire of every consumer to maximise his satisfaction.
2) There is no change in the prices of substitutes or complementary goods and the price of the commodity remains unchanged.
3) There is no change in the income of the consumer.
4) There can be further divisions and sub-divisions of the commodities into necessary commodities.
5) The behaviour of the consumer is logical.
6) There is no change in the marginal utility of money.
7) The income, taste, fashion, habits and customs of the consumer are unchanged.

2.2.7.2. Functional Representation of Law of Equi-Marginal Utility

Practically, the consumer spends his income on many available goods and services but he can only achieve equilibrium of maximum satisfaction or utility if he acts according to the equi-marginal utility as he is having a fixed income and is facing the market prices of different goods.

There should be a correlation between the marginal utilities of different goods and their prices according to the requirement of the principle of equi-marginal utility. The basic condition of consumer equilibrium that can be written in terms of marginal utilities and price of different goods is as follows:

$$\frac{\text{MU of A}}{\text{Price of A}} = \frac{\text{MU of B}}{\text{Price of B}} = \frac{\text{MU of C}}{\text{Price of C}}$$

It holds true for the consumer of sound mind with a given money income.

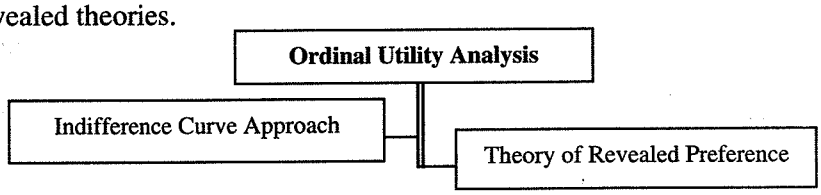
2.3. ORDINAL UTILITY ANALYSIS

2.3.1. Introduction

The number of pairs of two commodities that a consumer keeps in his mind to achieve equal level of satisfaction is the fundamental idea behind ordinal utility approach. This means that the utility can be ranked qualitatively. Thus, both the approaches of utility whether ordinal or cardinal differs from one another as the satisfaction derived from various commodities cannot be measured objectively. There are various names given to ordinal theory, namely, Neo-classical theory of consumer equilibrium, Hicksian theory of consumer behaviour, indifference curve theory and optimal choice theory. The consumer's equilibrium is explained by this approach that is challenged by multiple objectives and scarcity of money income.

According to the assumption of cardinal theory, utility is measurable on the contrary; utility is not measurable according to the ordinal theory postulate. The limitations of the usefulness of this approach is because, in real life utility is subjective and cannot be measured. The limitations of the cardinal approach that offers valuable awareness to consumer behaviour motivated economists to develop ordinal utility theory whose approach is completely different that states about consumer's behaviour on the assumption that utility from different units

of a good or between different goods can only be rankable but not measureable. Supposing a consumer gains more utility from good X than from good Y then he will rank good X above good Y irrespective of the knowledge by how much X has been given preference. The important ordinal theories are indifference curve analysis and revealed theories.



2.3.2. Indifference Curve

The utilisation of the analytical approach by modern demand theory is known as indifference curve analysis to understand the economic choices being made by the household. It is based on ordinal measurement and the name indifference states the fact that the consumer is not bothered to know about the combination of goods that he or she gets if they are all along the curve and are yielding the same utility. These curves are sometimes known as **"utility curves"**. The representation of different combinations of two goods on a curve is done by indifference curve that provides a consumer equal level of satisfaction. Between these curves the consumer is indifferent as he gets equal satisfaction from all the combinations. The locus of the points representing different combinations of two goods yielding equal satisfaction to the consumer is an indifference curve.

According to Prof. Leftwitch, "The different combinations of X and Y is shown by a single indifference curve that yields equal satisfaction to the consumer".

2.3.2.1. Assumptions of Indifference Curve

The assumptions of indifference curve are as follows:

1) **Rationality:** Consumer is an individual of sound mind who wants to maximise his satisfaction.
2) **Ordinal Utility:** Preferences are ranked by the consumer in accordance to the satisfaction of each combination.
3) **Non satiety:** Preference is given to goods of larger amount than to a smaller amount by the consumer where other things are same.
4) **Consistency:** If the consumer chooses combination A over B at a given time then he is not choose the combination B over A at some other time despite the availability of both the combinations. This is under the assumption that the behaviour of the consumer is consistent over the period of time.
5) **Transit of Choice:** It means if A = B, B = C, then A = C
6) **Diminishing Marginal Rate of Substitution:** Satisfaction being constant and the quantity of one good is given-up in exchange of one unit of the other good, it is termed as marginal rate of substitution. Hence, the marginal rate of substitution of X for Y is as follows:

$$MRS_{xy} = \frac{-\Delta Y}{\Delta X} = \frac{\text{Decrease in Units of Y}}{\text{Increase in Units of X}}$$

MRS_{xy} diminishes; when the consumer consumes an additional unit of X the units of goods Y are discarded at a diminishing rate.

7) **Scale of Preferences is Independent of the Market Prices:** Under the assumption, that the consumer is not affected by his preferences and being indifferent between the combinations by the market prices of other goods.
8) **Weak Ordering:** The ordering form of choice hypothesis is weak, which is the basis of the indifference curve analysis.
9) **Assumption of Transitivity:** This assumption being fundamental to indifference curve analysis is that the consumer's position of indifference between combinations taken as a whole and taken separately is not contradicted by the consumer's choice or indifference relations.
10) **Assumption of Continuity:** The meaning of continuity is that the consumer can rank all probable combinations of the required goods according to his choice of indifference. This assumption of continuity is the basis of the indifference curve analysis given by **Hicks** and **Allen.**

2.3.2.2. Properties of Indifference Curve

The properties of indifference curve are as follows:

1) **Negative Slope:** The slope of the indifference curve is always downward from left to right because the consumer consumes additional units of a commodity which results in discarding some units of other goods by the consumer so that the level of satisfaction is maintained on an indifference curve. This is clearly depicted by the given **figure 2.3**.

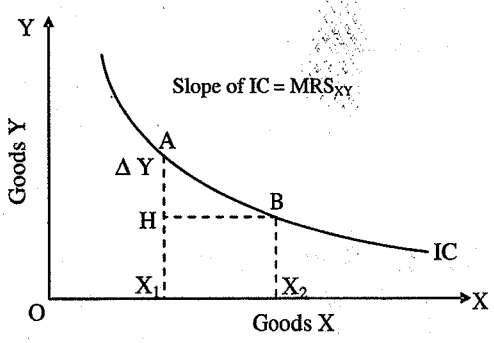


Figure 2.3: Negative Slope

According to the figure, IC is an indifference curve where a consumer moves to combination B from A he sacrifices HA units of goods Y to consume additional units of goods X i.e., HB (X_1X_2). Supposing the slope of the IC is upward to the right or it is parallel to the axis-X or axis-Y then a single IC is unable to yield equal satisfaction from different combinations on the curve.

2) **Indifference Curves are always Convex to the Origin:** This property of an indifference curve is based on diminishing marginal rate of substitution which is the assumption of indifference curve analysis. This property is made clear by the given figure. When the consumer goes from A to B, B to C and C to D combination, each time he takes one additional unit of goods X and gives EA, FB and GC units of goods Y respectively. Therefore, MRS_{xy} is diminishing when:

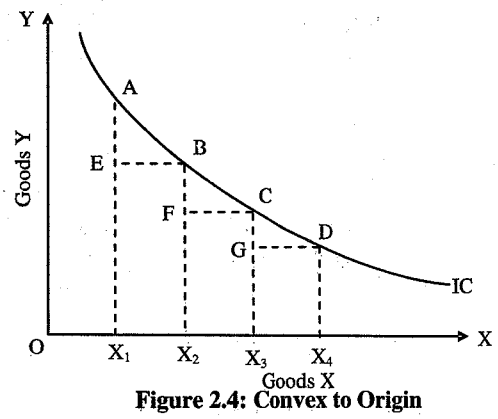


Figure 2.4: Convex to Origin

He goes from A to B, B to C and C to D combination because EA > FB > GC. An indifference curve cannot be a straight line or concave to the origin because marginal rate of substitution will be same or increasing respectively.

3) **Higher Indifference Curve represents Higher level of Satisfaction:** In an indifference map, higher indifference curve represents those combinations which give more satisfaction than the combinations on a lower indifference curve. It is exhibited by the following **figure 2.5**:

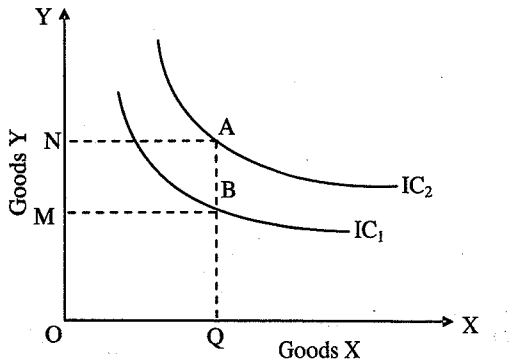


Figure 2.5: Higher Indifference Curve Represents Higher Level of Satisfaction

In both A and B combinations the quantity of goods-X is equal (OQ) but the quantity of goods-Y in combination A is ON which is greater than quantity of Y (OM) in combination B. Therefore, the satisfaction of combination 'A' (which is an higher IC) is greater than satisfaction of combination B.

4) **Two Indifference Curves can Never cut Each Other:** The reason is that different indifference curves represent different levels of satisfaction. If two IC cut each other at a point then it means that at that point, the level of satisfaction is the same on the two indifference curves, but it is impossible. It is made clear by the **figure 2.6**:

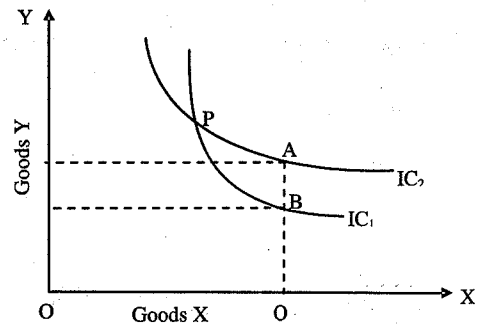


Figure 2.6: Two Indifference Curves can Never Cut Each Other

Taking IC_1, by definition the satisfaction of:

P = B (Both on IC, curve)

Taking IC_2, by definition the satisfaction of:

P = B (Both are on IC_2 Curve)

∴ Satisfaction of A and B combinations must be equal (A = B)

But satisfaction of A ≠ B, because the quantity of both goods is equal in the two combinations but quantity of goods Y in combination A is greater than in combination B. Therefore, two indifference curves can never intersect each other.

5) **Indifference Curve either Touches X-axis or Y-axis:** Indifference curves have the basic assumption that the consumer purchases combinations of different commodities. Therefore, he is not supposed to purchase only one commodity because in that case, the indifference curve will touch that axis. Purchasing only one commodity means monomania that is consumer's lack of interest in the other commodity or his insistence on purchasing only one commodity.

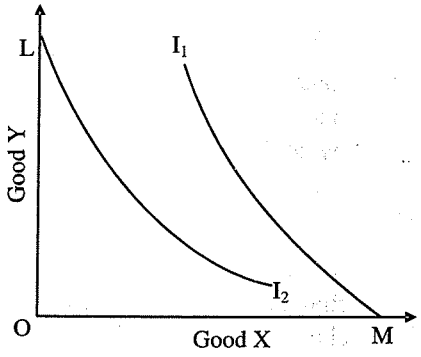


Figure 2.7: Indifference Curve either Touches X-axis or Y-axis

The **figure 2.7** shows an indifference curve meeting the horizontal axis at C and vertical axis at D. This violates the basic assumption of indifference curves because at point C, the consumer is purchasing OC of the commodity A. Similarly, at the point D the consumer is purchasing only commodity B, nothing of commodity A. This is against the basic assumption that the consumer always purchases the two commodities in a combination.

6) **Indifference Curve Need not be Parallel to Each Other:** Indifference curves need not be parallel. They can be parallel to each other only if the two commodities being shown on the horizontal axis and the vertical axis are independent of each other in the sense that they are neither substitute nor complementary goods, and further that none of them is 'inferior' or 'superior' in relation to each other whatever the amount of each with the consumer.

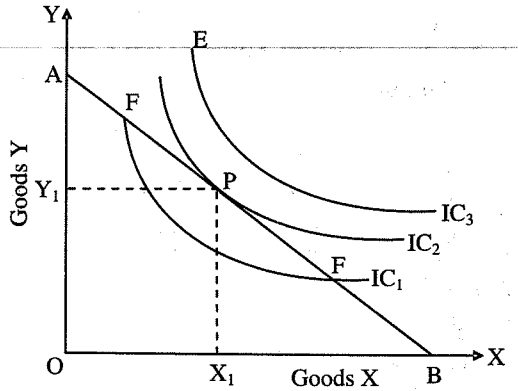


Figure 2.8: Indifference Curve need not be Parallel to Each other

The assumption of independence and absence of any income effects by **Dr. Marshall**, if introduced in the indifference curves, would give us a set of indifference curves parallel to each other and always equidistant from one another. It is obvious that this is an extreme case. Normally, a consumer demands goods which are related for which the income effect is positive.

Therefore, the normal position of indifference curves is that they converge on both sides to one another as we go left upwards or right downwards. They do not meet. The indifference map in **figure 2.8** shows normal indifference curves.

2.3.2.3. Marginal Rate of Substitution (MRS)

When total utility remains constant and the quantity of one good is given-up as the consumption of other good increases by one unit, it is marginal rate of substitution (MRS).

According to Professor Bilas, "The marginal rate of substitution of X for Y is defined as the amount of Y the consumer is just willing to give up getting one more unit of X and maintaining the same level of satisfaction."

Hence the marginal rate of substitution is:

$$\text{Marginal Rate of Substitution} = \frac{\text{Change in the Consumption of One Good}}{\text{Change in the Consumption of Another Good}}$$

The slope of indifference curves is determined by Marginal Rate of Substitution (MRS). It is the rate at which one good is substituted for another good by a consumer without changing the level of satisfaction.

2.3.2.4. Indifference Map

"Indifference map" is the family of indifference curves. The drawing of the indifference curves on the assumption that consumer having only one indifference curve where a particular level of satisfaction is assigned to the various combinations of two commodities. The indifference curve drawn will either be higher or lower than the one having been drawn already. Hence, the higher level of satisfaction is shown by higher indifference curves since more good is being preferred by the consumer. Whereas, the lower level of satisfaction is indicated by the lower indifference curves that include less of at least one of the two commodities for the consumer.

In the **figure 2.9**, there are four indifference curves that are numbered to show that higher number is assigned to the higher indifference curve in the scales of preference. Higher level of satisfaction is shown by the curve IC_2 than IC_1 and with IC_3 and IC_4 where, the utility of bundle of goods is displayed by each indifference curve. In ordinal sense, the significance of the number assigned to indifference curves is to indicate different levels of satisfaction. Following are the features of indifference curve:

1) The combination of two goods providing the same utility is represented as an indifference curve.
2) The pattern of preference is represented by Indifference map. The higher the indifference curve, higher is the level of satisfaction.

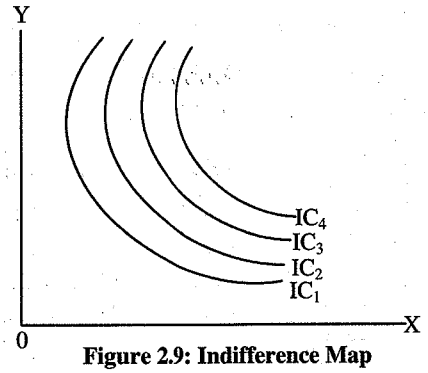


Figure 2.9: Indifference Map

2.3.2.5. Budget Line/Price Line

Exclusively for the consumers, preferences are indicated on an indifference map. Depending upon his income and the prices of the goods, the consumer actually selects two goods. There are several assumptions of price income or budget line. The following table explains a price or budget line. Let us consider that a consumer's monetary income is ₹40 which he wants to spend on two goods X and Y where the price of good X is ₹2 per unit and ₹1 per unit of good Y.

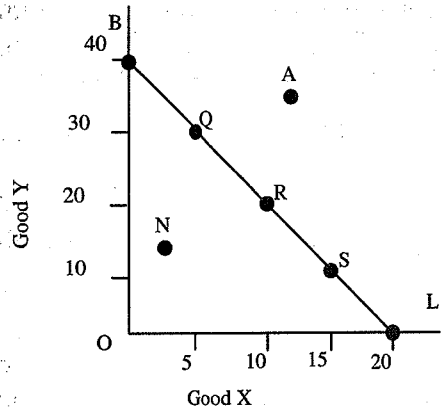


Figure 2.10: Budget Line/Price Line

The alternative consumption or expenditure potential of a consumer is shown in the following table where the income of the consumer and the prices of the two goods are given. According to the table, on any of the alternative combinations of goods X and Y, the consumer can spend but, if he spends his total income on either of the good X or Y, accordingly he can buy only 20 or 40 units. This not being possible as he has to buy only a combination of two goods, thus he has to buy any other combination.

Alternative Consumption or Expenditure Possibilities

Combination	Units of Good X (₹ 2 Per Unit)	Units of good Y (₹1 Per Unit)
B	0	40
Q	5	30
R	10	20
S	15	10
L	20	0

The consumer's budget line which is his budget line or budget constraint line is shown in the given **figure 2.10** where the various combinations are shown by the BL line. The consumer having ₹40 can buy any one of the combination Q, R, or S of goods X and Y but he is restricted to obtain any combination A beyond this line neither he can obtain the combination N inside the BL line due to the reason that he cannot spend all his income (₹40) on both the goods X and Y.

2.3.2.6. Consumer Equilibrium

If the consumer does not intend to make any changes in the purchases of goods then he is said to be in equilibrium position. At the given income and prices, a consumer who is in equilibrium position purchases that basket of goods that provides him maximum satisfaction. Thus a consumer is said to be in equilibrium position by purchasing that combination of goods that is to his budgetary constraints and on the highest indifference curve.

According to Scitovosky, "A consumer is in equilibrium when he regards his actual behaviour as the best possible under the circumstances and feels no urge to change his behaviour as long as circumstances remain unchanged".

According to Marshall, "If a person has a thing which he can put to several uses, he will distribute it among these uses in such a way that it has the same marginal utility in all."

Assumptions of Consumer Equilibrium

The assumptions of Consumer Equilibrium are as follows:

1) Logical action of the consumer.
2) Cardinal number like 1,2,3,4 etc., are the measures of utility.
3) Consumer is completely knowledgeable.
4) Marginal utility of money remains constant.
5) The price of the commodity and the income of the consumer are fixed.
6) There is no change in the tastes of the consumer.

Conditions of Consumer Equilibrium

For the consumer to be in equilibrium, following two **conditions** are to be satisfied:

1) At the point of equilibrium on

$$MRS_{xy} = \frac{P_x}{P_y}$$ Price line must be tangent to the indifference curve.

2) At the point of equilibrium indifference curve must be convex to the origin, i.e., MRS_{xy} is diminished. Explanation of consumer's equilibrium is given by **figure 2.11.**

In **figure 2.11,** price line AB shows all the possible combinations of goods X and Y that can be purchased by the consumer at given prices and with given money income. He cannot purchase any combinations on IC_3 as these combinations are beyond the power of the consumer to purchase. But any combination of two goods that are on the price line AB can be purchased by an imaginary consumer.

P combination is the point where consumer is at equilibrium as well as both the conditions of equilibrium are satisfied. The price line AB being tangent to the indifference curve IC_2, the slope of the budget line (P_x/P_y) and the slope of the indifference curve IC_2 are equal at equilibrium point P. Therefore,

$$MRS_{xy} = \frac{P_x}{P_y}$$

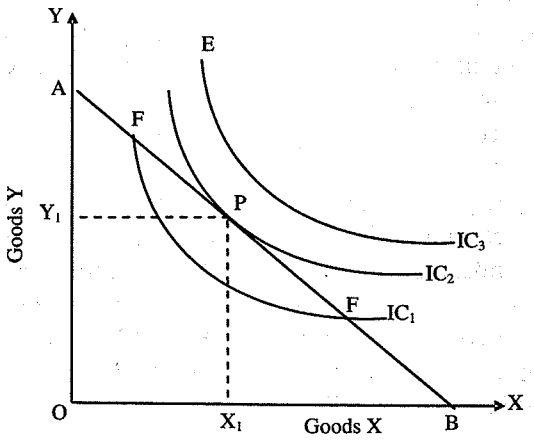


Figure 2.11: Consumer Equilibrium

The indifference curve IC_2 is convex to the origin being point tangent to the price line at P satisfying the second condition of equilibrium. Thus, at this point of equilibrium P, marginal rate of substitution falls. At E and F, consumer cannot be in equilibrium because at both the point $MRS_{xy} \neq P_x/P_y$ .Therefore, at point E, $MRS_{xy} > P_x/P_y$. The consumer will substitute X for Y to equalise MRS_{xy} and P_x/P_y and at point F, $MRS_{xy} < P_x/P_y$. He will substitute Y for X to equalise MRS_{xy} and P_x/P_y. Equilibrium position of the consumer changes as the income of the consumer or price of a commodity change accordingly.

2.3.2.7. Price Effect

The change in the equilibrium position of the consumer as well as in his purchases of goods corresponding to a change in price of one good and no change in the income and price of the other good is referred by Price Effect (PE). Depending upon the changes in the price of goods whether rising or falling, the consumer will be either at better-off or worse-off position than before. Illustration of the price effect is given by the **figure 2.12**:

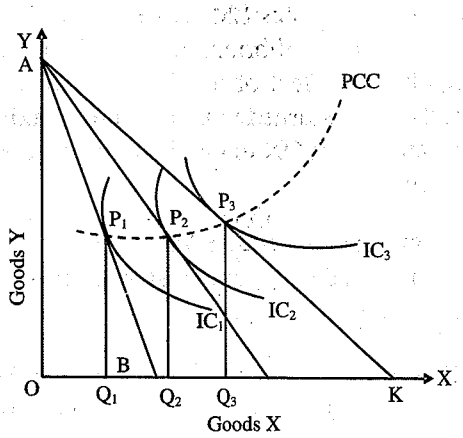


Figure 2.12: Price Effect

Let AB be the initial price line and P_1 is the equilibrium point of the consumer where he purchases OQ_1 quantity of goods X and Q_1P_1 quantity of goods Y. On the fall of the price of goods X and money income and price of goods Y remains the same, the new price line shifts to AL on the right of the X-axis. Corresponding to this, consumer's equilibrium also shifts from P_1 to P_2 resulting in the increase in demand of goods X to Q_1Q_2 which is the price effect on the demand of goods X due to fall in its price.

Similarly if there is a further fall in price of X, the new price line will be AK with a corresponding change in the equilibrium of the consumer from point P_2 to P_3. The price effect is shown by the price consumption curve PCC after joining the points P_1, P_2, and P_3. The PCC reveals the same information as being revealed by a demand curve showing different quantities purchased of goods X at its different prices.

2.3.2.8. Income Effect

A change in the position of equilibrium of the consumer or change in the purchase of goods refers to income effect when there is a change in the money income of the consumer as other things remains the same. It is illustrated by **figure 2.13**.

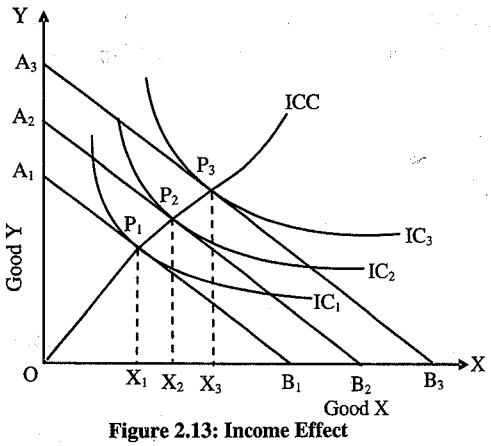


Figure 2.13: Income Effect

Let the money income and the prices of goods X and Y are given and A_1B_1 is the initial price line. At an equilibrium point P_1, the consumer purchases OX_1 quantity of goods X and X_1P_1 quantity of goods Y. Supposing the money income of the consumer increases and there is no increase in the prices of the goods X and Y, a new price line A_2B_2 which is parallel to A_1B_1 shifts to the right.

Accordingly the consumer equilibrium position changes from P_1 to P_2 due to rise in his income and his purchases of goods X increase from OX_1 to OX_2. Similarly due to further increase in the income of the consumer, the price line shifts to A_3B_3 and the new equilibrium position is P_3 on the curve IC_3. A new curve (ICC) Income Consumption Curve showing income effect will be formed by joining the equilibrium points P_1, P_2 and P_3. The nature and type of goods determine the position of ICC. If both the goods are normal goods, the slope of ICC will be upward but if the goods X is superior and goods Y being inferior, there will be a bend in ICC towards X-axis and ***vice versa.***

2.3.2.9. Substitution Effect

Substitution effect refers to change in purchases of goods (change in equilibrium) by a consumer when one goods, say X, becomes relatively cheaper, real income remaining constant. Substitution effect is shown in **figure 2.14**. In this figure, AB is the price line and the consumer is in equilibrium at point P_1, and consumer purchases OQ_1 quantity of goods X. Now suppose the price of goods X falls. With the fall in the price of goods X, the real income of the consumer increases.

Now a compensatory variation in the money income of the consumer is made in such a way that the consumer is neither better-off nor worse-off, i.e., the consumer remains on the same indifference curve. Here, compensatory variation in income means that the consumer's money income is reduced so much that consumer remains on the same indifference curve. After reducing the money income equal to compensatory variation, the

price line becomes LM and consumer's equilibrium shifts P_1 to H point on the same IC_1 curve. Due to shift in consumer's equilibrium from P_1 to H point, the demand of goods X is increased by Q_1H quantity. Q_1H is substitution effect of relative change in price of goods. Now of the money income reduced by compensatory variation is returned to the consumer, the equilibrium position shifts from H to P_2 point and demand of goods X increases from ON to OQ_2, i.e., by NQ_2 .

Change in equilibrium position from point H to P_2 shows income effect of change in price. Here income effect of change in price is NQ_2. In the given figure shift in equilibrium position from P_1 to P_2 point shows the price effect and due to price effect demand of X goods increase from OQ_1 to OQ_2, i.e., as a result of fall in price of X its demand increase by Q_1Q_2 (other thing remaining constant). Thus Q_1Q_2 is price effect. It is also proved by the figure (d) that price effect is the sum total of substitution effect and income effect because $Q_1Q_2 = Q_1H + NQ_2$.

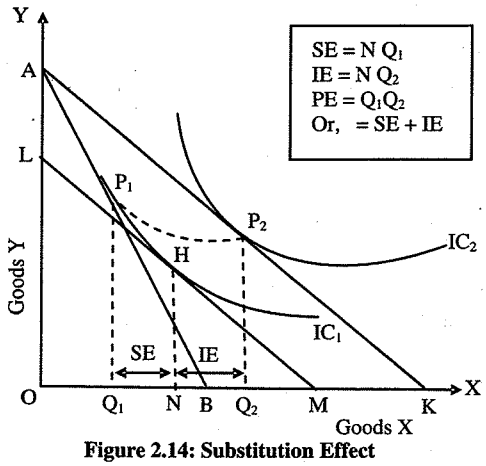


Figure 2.14: Substitution Effect

Where Q_1H is substitution effect and NQ_2 is income effect. Thus, substitution effect refers to the substitution of those goods which has become relatively cheaper for those goods which is now relatively dearer leaving the consumer neither better-off nor worse-off. It is also clear that the direction of price effect depends upon the nature of income effect and degree of substitution effect.

2.3.2.10. Separation of Price Effect into Income Effect and Substitution Effect

When the price of a commodity changes, the money income of consumer held constant, two separate and different forces are simultaneously altered to affect his demand behaviour:

1) **Income Effect:** The change in the real income or the purchasing power of consumer's money income either makes him better-off or worse-off.

2) **Substitution Effect:** When the price of a commodity falls, it becomes relatively cheaper, so the consumer is induced to buy more of it, and when its price rises, the commodity become relatively dearer, so the consumer tends to buy less of it as he will replace it by buying more of other cheaper goods.

Evidently, the price effect can be interpreted as the sum of income effect plus substitution effect.

Price Effect (Pe) = Income Effect (Ie) + Substitution Effect (Se)

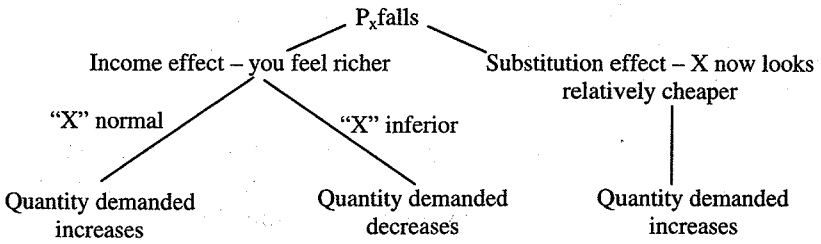


Figure 2.15: General Effect of a Price Fall

The movement of the consumer along the price consumption curve such as from equilibrium point P_1 to P_2 as a result of a fall in the price of X, is in reality a resultant of two forces. The first of these components is the feeling of better offenses that a consumer experiences when the price of X falls. There is an increase in the potential purchasing power of the consumer's income following a relative fall in the price of X. It is income effect. It is as if his income has increased and the price of both goods has remained the same. This income effect is the first component of the price effect. The consumer is operating along the income consumption curve.

There is also the second component of the price effect, *viz.*, the substitution effect. Now that X is cheaper than before while the price of Y has remained unchanged, there will naturally be a tendency on the part of the consumer to buy more of the cheaper goods and less of the relatively dearer one. In other words, he will substitute cheaper goods for the dearer one. This second component is called the substitution effect and can be viewed as operating along the price consumption curve in the given **figure 2.15**.

These two components of the movement from P_1 to P_2 are shown in **figure 2.16**. Initially, with the price-income line ML_1, the consumer is in equilibrium at point P_1. With a fall in price of X, so that the new relevant price-income line is ML_2 he moves to a new equilibrium position P_2 on a higher indifference curve C_2. RS is a hypothetical price-income line drawn parallel to ML_1 and touching the higher indifference curve C_2 at point T.

RS shows as if the price of X had remained the same (as represented by ML_1), but instead consumer's income had increased by an amount just sufficient to make the consumer as well-off as he is at point P_2, when his money income remains the same but price of X is lower than at P_1. As RS touches the higher indifference curve at T, the line passing through P_1T is the income consumption curve of the consumer.

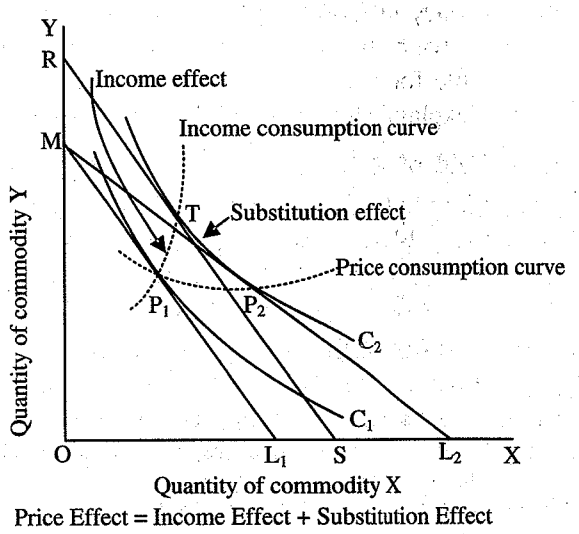


Figure 2.16: Separation of Price Effect into Income and Substitution Effect

The line P T shows both the magnitude and the direction of the income effect. The portion TP_2 of the higher indifference curve C_2 shows the direction and magnitude of the substitution effect. The substitution effect means that the cheaper good X is substituted for the dearer good Y.

The splitting-up of the price effect into substitution effect and income effect is very useful in bringing-out clearly the response of a consumer to a change in price of a commodity. This response will reveal itself through substitution effect and the income effect.

The difference between the two may be clearly understood. The substitution effect is always positive. That is, if the price of goods falls, more of it will be purchased and substituted for other goods whose prices have not fallen. Thus, the direction of the substitution effect is very clear and certain. But we cannot be so sure of the income effect.

The income effect may be positive (i.e., more of the goods may be purchased if the income goes up) or it may be negative (i.e., increase in income may lead to less being purchased of a goods). This will happen if the consumer regards it as inferior goods. Thus, the substitution effect and the income effect may move in the same direction, i.e., they may both be positive. In that case, the positive income effect will reinforce the positive substitution effect in increasing the demand for goods the price of which has fallen.

But in some cases, the substitution effect and the income effect pull in opposite directions. That is, the substitution effect is positive, as it generally is but the income effect may be negative. In that case, the negative income effect may dilute or negative altogether the positive substitution effect.

That is, the demand for the commodity, whose price has fallen, may not much increase or it may diminish instead of increasing as in the Giffen case. Hence, the direction in which the quantity demanded of a goods will change as a result of a fall in it price will depend upon the direction and strength of the income effect on the one hand and the strength of the substitution effect on the other.

2.3.2.11. Significance of Indifference Curve Analysis

Indifference curves technique has affected special significance through its application in all the spheres of economic activity. A few such applications can be mentioned as follows:

1) **In the Theory of Production:** Basically, a producer intends to attain a cost combination that is low. To realize this objective, indifference curves are useful and on using these curves in the theory of production, they are known as iso-product curves. When one of the iso-product curves on the map is tangent to the producer's budget line at the point at which low cost combination is obtained i.e., Producer's equilibrium.

2) **In the Theory of Exchange:** The exchange of two goods between two consumers results in mutual gains that are shown by the technique of indifference curves being used by **Prof. Edgeworth.** Due to exchange, it was possible for both the consumers to attain a higher level of satisfaction. Therefore 'Contract curves' helped in explaining the process of shifting to the higher level of satisfaction.

3) **In the Field of Rationing:** Indifference curve technique is also useful in the field of rationing. Usually there is a rationing of two commodities to different individuals without regarding their preferences but if the preference of these individuals is considered and the amounts of the two commodities is distributed in accordance to their scale of preferences among consumers then each of the consumer will be in a position to search for a higher indifference curve and satisfaction.

4) **In the Measurement of Consumer's Surplus:** The old Marshall's concept of consumers' surplus that has been buried for a decade under the weight of the assumptions that were unrealistic and illusory has been rehabilitated by the technique of indifference curve. Thus, without any need of making unrealistic assumptions this technique helps in the measurement of consumer's surplus.

5) **In the Field of Taxation:** The application of this technique in choice testing between a direct and indirect tax from where it is shown that a direct tax is more preferable than indirect tax regarding its effects on consumption and satisfaction of the tax payer with the help of indifference curves.

Though, it has been declared by certain writers that the technique of indifference curve is 'an old wine in a new bottle'. **For example** and according to the view of **Prof. Robertson** that new concepts and equations have been substituted by this analysis in place of the old ones. Thus, as regards to the above application of this technique, it forms an integral part of the modern welfare economics that has been maintained.

2.3.2.12. Criticisms of Indifference Curve Analysis

Due to number of uses and merits, indifference technique is a definite improvement over utility analysis. Despite having merits, there are the following shortcomings in the analysis of indifference curve:

1) **Unrealistic Assumptions:** The basis of this indifference approach is in the assumptions that are unreal logic, imperfect competition, having perfect knowledge of scale of preference to the consumer and that the goods are divisible.

   Because habits, customs and fashion affect the purchases of the consumer to a greater extent. Hence, he is not rational in his actions and one cannot expect a consumer having knowledge of his indifference map due to goods not being divisible and perfect competition is a myth.

2) **No Novelty: Prof. D.H. Robertson** remarked that the technique of indifference curve is merely "An old wine in new bottle". As this being similar to the utility analysis and has changed the existing names to new names like utility concept is replaced by scale of preference, diminishing marginal utility has been replaced by diminishing marginal rate of substitution & cardinal numbers are replaced by ordinal numbers.

3) **Indifference Curve is Non-Transitive:** According to the **Prof. W.E. Armstrong** argument, a consumer not being able to perceive the difference between close alternative combinations that leads to his indifference between the two. The reason why, the different combinations on the same indifference curve do not yield equal satisfaction is due to the increase in difference of combinations as the difference in the satisfaction of different combinations is evident.

   Hence, considering the acceptance of Armstrong's argument different points giving different satisfaction on an indifference curve resulting in, indifference curve becoming non-transitive.

4) **Fails to Explain Risky Choice:** To choose among those alternatives involving risk or uncertainty of expectation and unable to explain consumer behaviour, indifference curve is criticised. There emerges a need of quantitative measurement of utility to decide whether to take the risk in choosing among uncertain alternatives. Thus, consumer behaviour could be explained by coordinal system of utility in such situations.

5) **Absurd and Unrealistic Combinations:** Assumed combinations are the basis of indifference curve analysis. Considering these different combinations of two goods where some of them are meaningless and impossible in real life.

6) **Does not Provide Behaviouristic Explanation of Consumer Behaviour:** The character of the indifference map is hypothetical as well as subjective and does not depend upon observed market behaviour due to the fact that functions and curves are not set in pure objective terms. The possibility of obtaining quantitative data leads to the possibility of pure objective indifference curves. It is being difficult to quantity indifference curves due to the logical structure of the theory of indifference curve. However, despite number of attempts to quantity indifference curves success is still limited.

7) **Based on Weak Ordering:** The hypothesis of weak ordering which being subjective is the basis of indifference curve analysis where a consumer can be indifferent between a large numbers of combinations. Thus according to **Prof. Samuelson,** finding of many situations of indifference in real world is not possible.

2.3.3. Theory of Revealed Preference

On the basis of actual observations, **Samuelson** realistically examines how a consumer reacts to changes in price and income. He takes an objective approach. As such, the revealed preference theory is supposed to offer more realistic and scientific explanation of consumer demand. Like **Hicks, Samuelson** also assumes ordinal measurement of utility.

But, unlike **Hicks**, his approach is objectively based on actual observations – rather than imagination of consumer behaviour. His theory is, thus, regarded as the **'behaviouristic-ordinal utility approach'** to the theory of demand. This theory is fundamentally based on 'axioms of preference', where the axioms are expressed not in terms of what a consumer prefers but what he chooses. The major postulate of the theory is that – choice reveals preference.

2.3.3.1. Assumptions of Revealed Preference Theory

Samuelson develops his theory of revealed preference on the basis of the following assumptions:

1) **Two-commodity Model:** It is assumed that the consumer wants to buy two commodities (say, commodity X and commodity Y) per unit of time.

2) **Given Price-Income Situation:** The consumer has a given money income which is wholly to be spent on these two goods under consideration, i.e., X and Y. And their prices (Px and Py) are given.

3) **Constancy of Taste:** The habit and taste of the consumer remain unchanged over the period of time involved in the process of analysis.

4) **Rationality:** The consumer is rational in his behaviour. Rationality implies that he prefers a combination of larger stock of the two goods (X and Y) to a smaller one, as he experiences greater satisfaction from the larger quantities in comparison to the satisfaction derived from the smaller quantities of the same goods. (The original measurement of utility is thus implied here.)

5) **Revealed Preference Axiom:** The consumer's choice for a particular combination of the two goods within his budget reveals his definite preference. That is to say, choice reveals preference. It is not a statistical concept but a single act of choice.

6) **Strong Ordering:** It means that strong form of preference hypothesis is presumed. That is to say, the consumer reveals his positive and definite preference for a particular combination of the two goods against all other possibilities encountered by him under the given price-income situation. Strong ordering, thus, rules-out any position of indifference on the part of the consumer in making his choice. Thus, he makes definitely one and only one choice and thereby reveals his specific preference.

7) **Transitivity Condition: Samuelson** assumes transitivity in choice. It implies three term consistency in the sense that in a particular situation, if the consumer prefers A to B, and B to C, then he must prefer A over C.

   To express in symbolic terms, transitivity means – when $A > B$ and $B > C$, then $A > C$.

8) **Positive Income-Elasticity of Demand:** The theory assumes that the income-elasticity of demand of the consumer is always positive. That means when his income increases, the consumer always tends to buy more of the given commodity. Thus, zero income-elasticity or negative income-elasticity of demand is ruled out.

2.3.3.2. Diagrammatic Representation

Professor Samuelson put forward the Revealed Preference Hypothesis in the following words:

"Though any observed equilibrium point A, draw the budget equation straight line with arithmetical slope given by the observed price ratio. Then all combinations of goods on or within the budget line could have been bought in preference to what was actually bought. But they were not. Hence, they are all 'revealed' to be inferior to A".

A graphical description of the Samuelson's idea of the revealed preference hypothesis, as in **figure 2.17:**

1) There are two commodities X and Y.
2) Prices of X and Y are given.
3) Consumer has a given money income.

ΔODM represents the area of consumer's choice. It suggests that the consumer can choose any point such as A, B, etc., on the price-line DM, or below it, such as C, E, etc. Any combination like point K which lies outside the choice triangle (ΔOPM) is obviously beyond the reach of the consumer, because it is not within his budget. Suppose, the consumer chooses point A combination of the two goods (i.e., OP amount of Y and OQ amount of X). This means that he has revealed his preference for A against all other possibilities like B, C, and E. The consumer has a definite choice of point A based on his strong ordering of preference. Thus, in his choice position – the revealed preference – any possibility of indifference is ruled out.

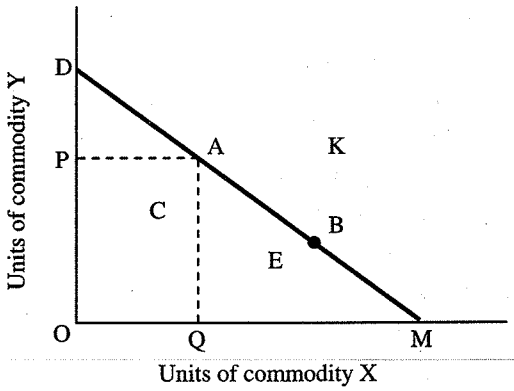


Figure 2.17: Revealed Preference Hypothesis

When a consumer expresses his choice for a particular combination, there may be either of the following two reasons behind his choice:

1) He likes it more as compared to all other possible choices. In our illustration, the consumer likes the combination A more than the combinations B, C, and E.
2) The chosen combination is found to be cheaper than other alternatives.

Strong ordering and consistency in choice behaviour of the consumer are the distinguishing features of the revealed preference hypothesis.

According to Professor Hicks, there are following two types of ordering:

1) **Strong Ordering:** In a strong ordering, each combination in a consumer's scheme of purchases is assigned a definite rank or definite order of preference, which enables him to make a definite specific choice and clearly reveal his preference. Thus, in illustration when the consumer chooses A, it is chosen in preference to all other possibilities like B, C, E, etc. In other words, his choice of A implies rejection of the rest. Thus, by choosing one definite combination and rejecting all others in a given situation, the consumer expresses his definite preference for it. In this way, choice reveals preference, under strong ordering.
2) **Weak Ordering:** On the other hand, there may be some combinations to which no definite order of preference is given, as they may be equally preferred by the consumer. Weak ordering is the case of indifference curve. In weak ordering, thus, the consumer cannot reveal his preference. **Samuelson** considers this to be unrealistic in observed behaviour of the consumer. When the behaviour of the consumer reflects his preference, there is a definite choice involved, and the possibility of his indifference is apparently out of question.

For methodological reason rather than for the sake of convenience, thus, **Samuelson** rules out the possibility of indifference or weak ordering and asserts strong ordering in consumer choice. Further, the consumer is always consistent in his choice. His behaviour is never self-contradictory. **For example,** if it prefers A combination against B in one situation, he cannot consistently choose B against A in another situation, when A is available. This is described by **Professor Hicks** as two-term consistency. The consistency condition of the revealed preference hypothesis relates to each single act of the consumer's choice.

2.3.3.3. Significance of Revealed Preference Theory

Samuelson's revealed preference theory is superior to the Marshallian cardinal utility analysis and Hicksian indifference curve approach to the theory of demand on the following counts:

1) **It is Behaviouristic in Approach:** Marshallian and Hicksian approaches are introspective and provide psychological explanations of consumer behaviour, having no basis for the empirical test. The revealed preference theory, on the other hand, is based on observed behaviour of the consumer in the actual market.
2) **It is More Realistic, Objective, and Scientific:** Samuelson's behaviouristic approach is certainly better, being objective, more realistic and scientific in comparison to the introspective method of analysing consumer behaviour.
3) **It Establishes the Law of Demand Directly:** The revealed preference theory does not require the use of the concept of utility to derive the demand curve. It proves the law of demand directly on the basis of revealed preference axiom.
4) **It Requires no Restrictive Assumptions:** The revealed preference theory does not make any dubious hypothesis like the law of diminishing marginal utility and the law of diminishing marginal are of substitution. It does not require to assume that the consumer always maximises his satisfaction.
5) **It does not Require the Assumption of Continuity:** Unlike indifference curve, the revealed preference hypothesis does not assume continuity. It assumes consistency. **Samuelson,** in fact, emphasises discontinuity so there is no indifference position in consumer's choice and a definite preference is revealed.
6) **It is a Simpler Analysis:** Samuelson's theory explains income effect in a much simpler way than the Hicksian concept of income-consumption curve.
7) **It Provides a New Vista in Welfare Economics:** The revealed preference hypothesis provides the basis for welfare economics in terms of observed behaviour.

2.3.3.4. Criticism of Revealed Preference Theory

There are many 'gaps' and defects in Samuelson's revealed preference theory. Its major drawbacks are as follows:

1) **It is not a General Theory of Demand:** The revealed preference theory is conditional. It is based on the assumption of positive income elasticity of demand. It, thus, explains the case of normal goods only. It does not consider the negative income elasticity and the negative income effect. It, thus, fails to explain the cases of inferior goods and Giffen paradox.
2) **It has Bias Towards Strong Ordering:** The revealed preference theory over-stresses the strong ordering, as such, takes no note of the possibility of indifference position in consumer's choice. **Hicks** opines that infinite cases of choice ordering can be strong. But, there may be infinite alternatives, and then ordering may be weak. Thus, the possibility of indifference position in consumer's choice cannot be just ruled out. It is the serious drawback of the theory to neglect indifference in the consumer behaviour altogether.
3) **It Over-Stresses the Consistency Condition of Rationality:** The theory observes demand behaviour of the consumer in current market situation under consistency condition. But, there can be influences of other considerations in his demand behaviour which may force him not to act so rationally. Then his behaviour may not be consistent.
4) **It is an Imperfect Theory:** To some critics, the revealed preference theory is not only destructive of the contemporary theory of welfare, by disregarding the phenomenon of maximum satisfaction it has also failed to serve welfare purpose. It serves only economic purpose to the demand theory. A perfect theory of demand should serve both the economic as well as the welfare purposes.
5) **It is More an Academic Exercise: Samuelson** gives no empirical proof of his theory. His demand theorem is not based on observed consumer behaviour in actual market. Because, in reality it is not necessary that choice always should reveal preference. If **Samuelson** would have examined actual market behaviour, he might have come across such a phenomenon. In short, like his predecessors **Samuelson** also presents an imaginary logical reasoning without empirical support.
6) **It has a Limited Scope of Applicability:** The revealed preference theory is not applicable to the situation of consumer's choices under the condition of uncertainty and risks.

2.4. CONSUMER SURPLUS

2.4.1. Meaning and Definition of Consumer Surplus

The concept of consumer surplus was evolved by Alfred Marshall. This concept occupies an important place not only in economic theory but also in economic policies of government and decision-making of monopolists. It has been seen that consumers generally are ready to pay more for the goods than they actually pay for them. This extra satisfaction which consumers get from their purchase of goods is called by Marshall as consumer's surplus.

According to Marshall, the concept of consumer's surplus is defined as "excess of the price which a consumer would be willing to pay rather than go without a thing over that which he actually does pay, is the economic measure of this surplus satisfaction ..... "It may be called consumer's surplus".

Thus,

Consumer's Surplus = What a Consumer is Ready to Pay – What He Actually Pays

The concept of consumer's surplus is derived from the law of diminishing marginal utility. As we know from the law of diminishing marginal utility, the more of a thing we have, the lesser marginal utility it has. In other words, as we purchase more of a good, its marginal utility goes on diminishing. The consumer is in equilibrium when marginal utility is equal to given price i.e. he purchases that many number of units of a good at which marginal utility is equal to price (It is assumed that perfect competition prevails in the market). Since the price is fixed for all the units of the good he purchases except for the one at margin, he gets extra utility; this extra utility or extra surplus for the consumer is called consumer's surplus.

2.4.2. Assumptions of Consumer's Surplus

In explaining the concept of consumer's surplus, the following assumptions are made:

1) Utility can be measured.
2) The marginal utility of money remains constant.
3) The income of the consumer, taste and preferences remain constant.
4) There is no substitute for the commodity.
5) There exists independent marginal utility of each unit to all commodities.
6) Working of the Law of Diminishing Marginal Utility.

2.4.3. Uses of the Concept of Consumer Surplus

Inspite of criticisms, the concept of consumer's surplus has been applied in a number of cases. Its uses or importance are as follows:

1) **Choice of Taxes:** Concept of consumer's surplus makes a case for the imposition of direct tax in place of indirect tax. Both taxes reduce Consumer satisfaction, but the sacrifice of satisfaction is less in case of direct tax. When direct tax is imposed the consumer will sacrifice the marginal utility of some goods. But when an indirect tax is imposed then the consumer will have to sacrifice not only the marginal unit of the commodity but also the units preceding marginal unit. Therefore, the total amount of sacrifice will be more in case of indirect tax than the direct tax.

2) **Importance to Finance Minister:** Indirect tax imposed on a commodity increases government revenue and reduces consumer's surplus. The concept of consumer's surplus supports the imposition of tax on commodities that are subject to the law of diminishing returns (increasing cost). In such cases, all in consumer's surplus will be less than government revenue. The concept of consumer's surplus also guides the government to provide subsidies to that industry which is subject to increasing returns.

3) **Difference between Value-in-use and Value-in-Exchange:** The goods which passes more consumer's surplus, such as newspaper, salt, match-box, etc., have mere value-in-use and less value-in-exchange. Value-in-exchange of such goods is less because their marginal utilities are low. On the other hand, the goods which possess less consumer's surplus have more value-in-exchange because of high marginal utility.

4) **Importance to Monopolist:** The concept of consumer's surplus helps a monopolist in pricing its product. A monopolist will charge high price for its product if the product yields more consumer's surplus to the buyers.

5) **Importance in International Trade:** The domestic consumers are ready to pay higher prices for those goods which are either scarce or not available within the country. Therefore, a country should arrange these goods through imports to increase consumer's surplus. On the contrary, the goods that are available in abundance be exported to earn foreign exchange. Similarly, the government can tax relatively cheap imports to extract some part of consumer's surplus.

6) **Economic Welfare:** The economic policies of the government can be evaluated on the basis of their impact on consumer's surplus and economic welfare. Consumer's surplus is an index of the level of economic welfare of the people of a country.

7) **Pricing of Public Utilities:** The prices of public utilities cannot be fixed and should not be fixed on the basis of cost of these utilities. Therefore, it is suggested that the government while fixing their prices for different classes of consumers, the amount of consumer's surplus of the public utilities for different consumers be kept in mind.

2.4.4. Criticisms of Consumer's Surplus

The concept of consumer's surplus is criticised on the following grounds:

1) **Cardinal Measurement of Utility is Impossible:** Marshallian concept of consumer's surplus is impractical because utility cannot be measured in cardinal numbers.

2) **Unrealistic Assumptions:** The concept of consumer's surplus is based on unrealistic assumptions such as constant marginal utility of money, independent utilities. Therefore, it is not possible to measure consumer's surplus with the help of utility approach.

3) **Ignores Income Effect:** Assumption of constant income is unrealistic. The price which a consumer is willing to pay depends on the level of income. At higher income a consumer may be ready to pay more and vice-versa. Thus, increase in income increases consumer's surplus.

4) **Assumption of Constant Marginal Utility of Money:** Consumer's surplus cannot be measured if we assume constant marginal utility of money. When a consumer spends some part of his income on a commodity, then he is left with less amount of money to purchase other goods. Marginal utility of money will increase with decrease in money income and this makes the measurement of consumer's surplus difficult.

5) **Articles of Distinction and Giffen Goods:** Consumer's surplus cannot be measured in case of articles of distinction because the law of demand is not applicable to these articles. Similarly, since the Marshallian demand curve does not include Giffen goods in it, the consumer surplus is not measurable for these goods also.

6) **For Necessities of life Consumer Surplus is Infinite:** Many a times a consumer may be ready to forego even his entire income to satisfy his necessary wants. Life saving drugs is an example to quote. Therefore, consumer surplus is infinite in case of necessities of life.

7) **Substitutes:** Marshall has assumed that a commodity has no substitutes. Availability of substitutes cannot be ruled out. In the absence of substitutes a consumer may be willing to pay high price for a commodity. But if substitutes are available then the consumer will be ready to pay some lower price. Thus, measurement of consumer's surplus becomes difficult due to the presence of substitutes.

8) **Vague Concept:** Consumer's surplus is no doubt a theoretical concept. There is no meaning, according to Prof. Nicholson, in saying that utility of £ 100 at one place is equal to the utility of £ 1000 at another place.

2.5. DEMAND ANALYSIS

2.5.1. Introduction

The analysis of such factors which affects the demand of any commodity or service is called 'demand analysis'. The demand can be forecasted with the help of such factors only. The important phenomenon which are studied under demand analysis are individual and market demand, demand function, demand distinctions, demand estimation, elasticity of demand and methods of its measurement, sales forecasts and different methods to forecast sales or demand, manipulating demand and appropriate change in allocation of resources. The demand analysis helps the producer to modify his production with respect to the demand to minimise the objective function.

2.5.2. Concept of Demand

The demand refers to a consumer's willingness for such quantity of goods and services and ability to pay for same at various prices dealing within a period of time. In Economics, demand is something more than desire to purchase. **For example,** if any person is feeling hunger but he does not have money to pay for it, in that case his demand is ineffective. Effective demand involves three things:

1) Desire,
2) Means to purchase, and
3) Willingness to use those means for that purchase.

According to Benham, "The demand for anything, at a given price, is amount of it, which will be bought per unit of time, at that price".

According to Bobber, "By demand we mean the various quantities of a given commodity or service which consumers would buy in one market in a given period of time at various prices".

The demand for any commodity at a given price is that amount which is paid for that commodity at that price. Therefore, demand refers to demand curve which is the combination of different quantities of commodity demanded at different prices.

2.5.3. Objectives of Demand Analysis

Various objectives of demand analysis are as follows:

1) **Demand Forecasting:** Demand forecasting is the process of prediction of future demand of any commodity with the help of present and past behaviour patterns of some related events.
2) **Production Planning:** For any business firm demand analysis is essential criterion which is necessary for the production planning. Firm's output is fully based on such estimation of future demand. In the absence of proper estimation firm has to confront with overproduction and subsequent losses.
3) **Sales Forecasting:** Effecting sales forecasting also depends on demand analysis because advertising function of firm cannot occur without sales forecasting.
4) **Control of Business:** For making effective control on business, it is required to have a proper budget of costs as well as profit which is based on the assessment of annual demand/sales and prices.
5) **Inventory Control:** For any business there must be adequate control on its inventories such as raw materials, intermediate goods, semi-finished product, finished product, spare parts, etc. Such control on future necessities can be done only through demand analysis.
6) **Growth and Long-Term Investment Programmes:** For significant growth rate and long-term investment planning programme for any firm, demand analysis is quite necessary.
7) **Economic Planning and Policy Making:** For a better planning and lucid allocation of country's productional resources, demand analysis helps the economic planners a lot. The long-term demand forecasting and estimation for various goods facilitate the government to establish import and export policies in the country.

2.5.4. Types of Demand

The types of demand are as follows:

1) **Individual and Market Demand:** Individual demand refers to such demand of individual for quantity of a commodity which he is eager to buy at a particular price during a specific time period, given his money income, his taste, and prices of other commodities (particularly, substitutes and complements). On the contrary, market demand is the sum of all individual demands for a commodity over a specific period of time, and at a given price, other factors remaining the same.
2) **Demand for Firm's Products and Industry's Products:** The demand for the quantity which a firm produce at a given price over a time period is demand for firm's product. Whether the cumulative demand for the product of all the firms of an industry is known as the market demand for 'industry's product'.

3) **Autonomous and Derived Demand:** The demand for a commodity that is not related with the demand of other commodities is known as autonomous or direct demand. **For example,** the demand for goods such as food, shelter, clothes, and vehicles, etc., is autonomous because it arises due to biological, physical, and other personal needs of consumers. The demand for a commodity that is related with the demand of other commodities is known as derived demand. **For example,** demand for petrol, diesel and other lubricants are related with the demand of vehicles.

4) **Demand for Durable and Non-Durable Goods:** Goods can be classified into two categories such as durable and non-durable goods. On the basis of these two categories the demand can be classified into two categories such as demand for durable and non-durable goods. Durable goods are those which can be used for indefinitely long period of time or continuously over a period. Durable goods may be consumer as well as producer goods.

   Durable consumer goods include clothes, shoes, owner's-occupied residential house, furniture, utensils, refrigerators, scooters, cars, etc. The durable producer goods include mainly the items under 'fixed assets', such as building, plant, machinery, etc. On the other hand, non-durable goods are those which can be used only once. **For example,** food items.

5) **Short-Term and Long-Term Demand: Short-term demand** refers to the demand for products that are used for a very short time period or for current period. **For example,** demand for umbrellas, raincoats, sweaters, etc., is short-term in nature.

6) **Long-term demand** refers to the demand for products over a longer period of time. **For example,** demand for durable goods, demand for producer goods, etc., is long-term in nature.

7) **Joint Demand and Composite Demand:** When demand for two or more than two goods are mutually demanded at the same time to satisfy a single demand then it is called **joint demand**. Demand for bike and petrol, coffee and sugar, pen and ink etc., are the example of joint demand. When a commodity is demanded for two or more distinct uses then it is said to be composite demand. **For example,** steel is demanded for the purpose of making car bodies, making utensils, etc.

8) **Direct and Indirect Demand: Direct demand** refers to such demand which is directly used by the end consumer. The demand of such goods is also called consumer's goods demand because these are used for final consumption. Demand for all consumers' goods such as sweets, oil, readymade clothes, motorcycles, house is direct demand. **Indirect demand** is such demand for those goods which are not used by end consumers. These goods are used for producing other goods. Due to this, indirect demand is also referred as producer's goods demand.

9) **Total Market and Market Segment Demand:** The total market demand is referred to as summative demand for the product from all the segments whereas market segment demand would refer to demand for the product in that specific market segment. For analysis of demand both total market as well as market segment type of demand is necessary. The market may be segmented on the basis of age, gender, geographical area, etc. **For example,** demand for Vadilal ice-cream in India is total market demand, demand for Vadilal ice-cream in Rajasthan or demand for Vadilal ice-cream by women is a market segment demand.

2.5.5. Determinants of Demand

Following are the determinants of demand:

1) **Price of the Commodity:** According to law of demand, there is an inverse relationship between price and quantity demanded of the commodity. It describes that if price increases then the demand of the commodity decreases and if price decreases then the demand of the commodity increases. It happens because of income and substitution effects.

2) **Income of the Consumer:** Demand is also influenced by the income of the consumer. The purchasing power is affected by the consumer income. If consumer has more income he will buy more and if he has less income he will prefer less to buy. The high income consumer has more purchasing power.

3) **Tastes and Preferences:** Consumer's tastes and preferences are also considered as an important determinant of the demand. If consumers have more preference for any commodity then automatically his demand will increase and *vice versa.*

4) **Advertisement and Sales Propaganda:** In present scenario, the tastes and preferences of consumers can be changed by advertisement and sales propaganda. Information about the product such as availability of the product, superiority of the product by the competitor's products, provided to the potential consumers with the help of advertisement increase the demand of the product. The demand for FMCG products is largely triggered by advertisement.

5) **Consumer's Expectations:** Consumer's expectation is considered to be an important determinant of demand. If consumer expects a hike in prices he may prefer to buy more quantity of that particular commodity and *vice versa*. Similarly, expectation of rising income may force him to increase his current consumption.

6) **Growth of Population:** Growth of population is an important determinant of demand. If population increases the demand of the goods also increases naturally.

7) **Weather Conditions:** Climatic conditions affect the demand most. The demand for certain items purely depends on climatic and weather conditions. **For example,** the demands for cold drinks, fans and coolers during the summer season and the demand for umbrellas and raincoats during the rainy season.

8) **Tax Rate:** The demand is affected by the tax rate. If there is a high tax rate then there will be a less demand for the goods. The tax is used as a weapon by the government for restricting the consumption of a particular commodity at a certain time. The demand of that commodity will be lower if the imposed tax is high.

9) **Availability of Credit:** One of the most considerable determinants of demand is credit availability. The purchasing power of the consumer is highly affected by the credit availability if there is availability of cheap credit, the consumers try to buy more products.

10) **Pattern of Saving:** Saving pattern also influence the demand. In the case when consumer wants to save more money than his demand will become less for the product and *vice versa*.

11) **Circulation of Money:** Circulation of money affects the demand greatly. As more money circulates among the people it increases the purchasing power of the consumer and they will buy more products.

2.5.6. Demand Equation

An algebraic expression which shows the relationship between demand for a commodity and its various determinants that affect this quantity is known as demand function. There are two types of demand functions:

2.5.6.1. Individual Demand Equation

The quantities of a commodity which are demanded by an individual at various prices, his income, prices of related goods and tastes, is called individual's demand function.

It is expressed as: $D = f(P)$

2.5.6.2. Market Demand Equation

An individual demand function is the basis of demand theory. But it is the market demand function that is main interest to managers. It refers to the total demand for a good or service of all the buyers taken together. The market demand function may be expressed mathematically thus,

$$Dx = f(Px, Pr, M, T, A, U)$$

Where,
Dx = Quantity demanded for commodity x
f = Functional relation
Px = Price of commodity x
Pr = Prices of related commodities, i.e., substitutes and complementaries
M = Money income of the consumer
T = Taste of the consumer
A = Advertisement effect
U = Unknown variables

By demand function, economists mean the entire functional relationship. This means the whole range of price quantity relationship and not just the quantity demanded at a given price per unit of time. The demand function expressed above is really just a listing of variables that affect the demand. The demand function must be made explicit and clear for use in managerial decision-making. The industry must have reasonably good knowledge and information about its demand function to formulate effective long-run planning decisions and short-run operating decisions.

2.6. LAW OF DEMAND

2.6.1. Meaning and Definition of Law of Demand

Law of demand explains the inverse relationship between the change in quantity demanded of the commodity as well as the change in the price of the commodity respectively.

This law states that if price will be high then quantity demanded of the commodity will become less whereas when price gets down the quantity demanded of the commodity increases. In other words, we can say that both price and quantity demanded have inverse relationship, all other things remains constant.

According to Marshall, "The amount demanded increases with a fall in price, and diminishes with a rise in price". Thus, it expresses an inverse relation between price and demand. The law refers to the direction in which quantity demanded changes with a change in price.

2.6.2. Assumptions of Law of Demand

Following points are to be considered as assumptions of law of demand:

1) **Income Level should Remain Constant:** If the income level of the buyer remains constant the law of demand works better. If the income rises while the price of the commodity does not fall then the demand may increase with the effect of income.
2) **Tastes of the Buyer should not Change:** Tastes and preferences of the consumer creates an obstacle in the functioning of law of demand. Mostly it happens in the situation when tastes or fashions change and people revise their preferences.
3) **Prices of Other Goods should Remain Constant:** The demand of a commodity is generally affected by the change in the prices of other goods. If prices of commodities for which demand is inelastic rise, the demand for a commodity other than these in all probability will decline even though there may not be any change in its price. Thus, it is necessary for operating the law of demand that prices of other goods do not change.
4) **No New Substitutes for the Commodity:** If any new substitutes for a commodity is present itself in the market, it generally declines the demand. It is usual that the demand of existing product will fall even though price remains unchanged due to the reason that some buyers will purchase new available substitutes. Thus, the law of demand only apply in the situation when new substitutes do not exist in the market.
5) **Price Rise in Future should not be Expected:** If the buyer expects that the price of any commodity will increase in future then they raise the demand of that commodity in response to an initial increase in the price. The law of demand is violated by this type of behaviour. Thus, for applying law of demand it is required that there should not be any expectation of price rise in future.

2.6.3. Demand Schedule

Demand schedule is defined as a table or a chart which shows the inverse relationship between price and quantity demanded of a commodity.

Types of Demand Schedules

Demand schedule is of two types:

1) **Individual Demand Schedule:** Individual demand schedule is a tabular presentation which shows the different quantities of a commodity demanded by a consumer at different level of prices within a given period of time. **For example,** let us consider that a household purchases 500 grams of oranges per day at a price of ₹20/kilogram. When the price falls to ₹16/kilogram, it increases its demand to 700 grams. As the

price keeps decreasing, the household continues to increase its demand. When the price falls to ₹5/kilogram, its demand shoots upto 6 kilograms.

Daily Demand Schedule of a Household for Oranges

Price per kg	Oranges Demanded (in kg)
20	0.50
16	0.70
13	2.00
11	2.6
8	3.5
7	5
6	6

2) **Market Demand Schedule:** After added the individual demand of different buyer at each price for any commodity, one determines the market demand schedule. Different quantities of oranges will be purchased by every individual buyer at different prices. The slope of this demand curve will be negative, i.e., it will slope downward from left to right. If there are only two buyers in the market and their demand schedules are identical, then the demand for the commodity in the market will be exactly double the quantity demanded by a single buyer.

In practice, however, the demand schedules of all the buyers for a product are not identical. Therefore, we have to add up the quantities demanded by various buyers at different prices. This gives us the market demand schedule.

Weekly Market Demand Schedule for Oranges

Price per kg	Qty Demanded by Household 1	Qty Demanded by Household 2	Market Demand (in kg)
120	2	4	6
110	3	4	7
100	5	5	10
90	8	7	15
80	8	9	17
70	10	11	21
60	13	12	25
50	17	15	32
40	21	16	37
30	22	20	42

2.6.4. Demand Curve

Demand curve is of two types:

1) **Individual Demand Curve:** Individual demand curve is a graphical presentation that shows different quantities of a commodity demanded by an individual per period of time at various prices of commodity, keeping all the other factors constant which affect the demand. Hence, Individual demand curve shows the different quantities of a commodity demanded by an individual at different prices.

In **figure 2.18**, quantity demanded are shown on X axis and different prices are shown on Y axis. The figure represents that when the price of the commodity was ₹20 per unit then quantity demanded by an individual was 2 units. When the price of the commodity was reduced to ₹16 per unit then quantity demanded by an individual was increased to 3 units and similarly when the price of commodity was further reduced to ₹10 per unit then quantity demanded by an individual was increased to 5 units.

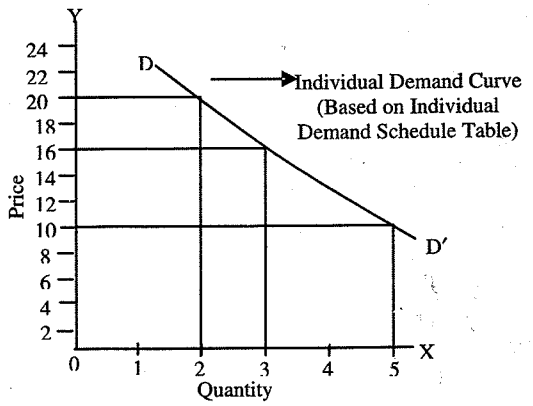


Figure 2.18: Individual Demand Curve

2) **Market Demand Curve: Market demand curve** is a graphical presentation that shows different quantities of a commodity demanded by all the existing consumers per period of time at various prices of commodity, keeping all the other factors constant which affect the demand. Hence, market demand curve shows the different quantities of a commodity demanded by all the consumers at different prices. Any point on the graph represents the quantity demanded at respective price. The demand curve DD_1 shows the different quantities of a commodity demanded by all consumers at various prices.

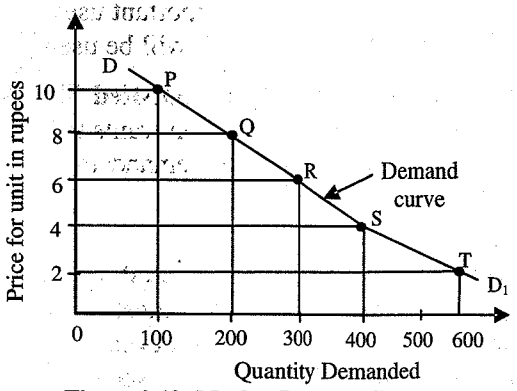


Figure 2.19: Market Demand Curve

In the **figure 2.19**, quantity demanded are shown on X axis and different prices are shown on Y axis. The figure represents that when the price of the commodity was ₹10 per unit then quantity demanded by all consumers was 100 units. When the price of the commodity was reduced to ₹8, ₹6, ₹4 and ₹2 per unit then quantity demanded by all consumers was increased to 200, 300, 400 and 600 units respectively. Thus, the demand curve DD_1 shows that the increase in demand of commodity when its price falls. This curve shows the inverse relation between price and demand.

2.6.5. Why does Demand Curve Slope Downwards?

The inverse relationship between quantity demanded and price of a commodity is shown by the downward slope of the demand curve. Following are the reasons for the downward or negative slope of the demand curve:

1) **Income Effect:** Income effect says that when there is a fall in price of commodity, then the consumer can buy the same quantity of the commodity with lesser money or he can buy more amount of same commodity with the same money. In other words, the purchasing power of the consumer increases when the price of the commodity decreases. Thus, demand for that commodity increases.

2) **Substitution Effect:** When the price of any commodity decreases then it becomes cheaper in comparison to other similar commodities. This condition influences the consumer to substitute the commodity by the commodity whose price has decreased. In this condition the total demand increases for the commodity whose price has fallen increases.

3) **New Consumer Creating Demand:** When price of any commodity decreases then various consumers again start buying it who could not buy it previously due to its higher price. The number of consumers increases and hence the demand of that commodity also increases at a lower price.

4) **Based on the Law of Diminishing Marginal Utility:** The law of Diminishing Marginal Utility is the basis of the law of demand. According to this law, the marginal utility of a commodity continues to decline when a consumer buys more units of that commodity. Therefore, in the condition when the price of the commodity decreases then the consumer will buy more units of that commodity. When fewer units are available, utility will be high and the consumer will be ready to pay more for the commodity. This justifies that the demand of a commodity will be high at a lower price and it will be less at a higher price. This is the reason that why the demand curve is downward sloping.

5) **Price Effect:** Price creates a drastic effect on demand of the commodity. As one knows that each product has certain consumers, but when its price falls, new consumers start consuming it, as a result demand increases but at the same time when increase in the price of the product, many consumers will either reduce or stop its consumption and the demand will be reduced. Thus, due to the price effect when consumers consume more or less of the commodity, the demand curve slopes downward.

6) **Different Income Groups:** Different income groups are available in every society but the low income group is available in majority. This group influences the downward sloping demand curve. Generally people buy more units when the price falls and buy fewer units when price increases. The rich class of society does not affect the demand curve because they are able of buying the equal quantity even at a higher price.

7) **Different Uses:** Various commodities and services are used for different purposes which are responsible for the negative slope of the demand curve. In the case, when the prices of such products increases then it will be used only for more important uses and their demand will fall. On the other hand, when the price of such products decreases then it will be used for various purposes and their demand will rise.

8) **Tendency to Satisfy Unsatisfied Wants:** Each person has a tendency to satisfy unsatisfied wants which were not fulfilled. A person wants to satisfy his demand when the price of any good such as pomegranate is increases and hence its demand increases. The demand curve slopes downwards to the right due to this human tendency.

2.6.6. Exceptions to Law of Demand

The law of demand is valid in most of the cases but there are certain cases where this law does not hold true. The following are the important exceptions to the law of demand:

1) **Conspicuous Goods:** In our society, there are some consumers who measure the utility of any commodity only by its price. They think that if the commodity is expensive then it gives more utilities. As such, they buy less of this commodity at low price and more of it at high price.

2) **Giffen Paradox:** An economist, named **Sir Robert Giffen,** was surprised to find out that the British workers purchased more bread while the price of bread increased. This was against the law of demand. He wanted to know the reason behind this. The reason was given for this incident is that when the price of bread increases, the purchasing power of poor people decline and they were forced to cut down the consumption of meat and other more expensive foods. Since the bread was a cheaper food at higher price in comparison to the other expensive food articles such as meat, ice-cream, etc. Hence the consumption increases in the condition when price went up.

3) **Necessities of Life:** Generally, the necessities of life such as food, cloth, etc., are not ruled by the law of demand. The demand of such items is not reduced when the prices are increased. These items were purchased by reducing the demand for comfortable goods even the price is very high. This is the reason that the demand curve slopes upward to the right.

4) **Conspicuous Necessities:** The demonstration effect of the consumption pattern of a social group affect the demand of various goods. Due to the regular use, these goods become necessity of life. **For example,** when the prices of television sets, refrigerators, coolers, cooking gas, etc., is rising but the demand of these goods does not show any tendency to fall.

5) **Future Expectations about Prices:** It has been observed that when the household expect any rise in the price of a commodity in future then they tend to buy larger quantities of the commodities. **For example,** in case of man-made and natural disasters there is a shortage of industrial and agricultural goods. This creates uncertainty and fear in the economy. Resulting this, common man willing to pay higher prices and demanded greater quantities of foodgrains.

6) **Impulsive Purchases:** The law of demand fails in the condition when consumers tend to make impulsive purchases without any proper calculation about price and usefulness of the product.

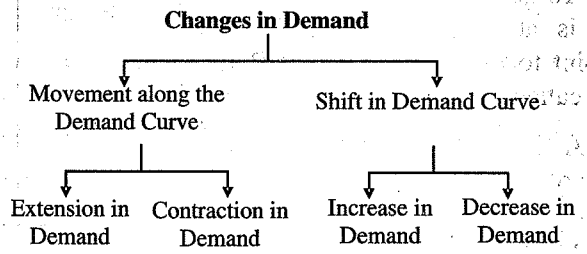
7) **Ignorance Effect:** It is generally assumed that the household have perfect knowledge about price and quality of goods. A consumer's ignorance is another factor which encourages him to purchase larger quantity of a commodity even at a higher price.

8) **Outdated Goods:** Goods which are going out of use due to technological advancements are known as outdated goods. Durable goods such as radio, telephone, etc., are included in this category. When the touchscreen mobile phones are introduced then it reduces the demand of key used mobile phones even though they may be available at lower prices. Seasonal goods which are not use during the off-season shows the similar demand pattern.

2.6.7. Changes in Demand

If the other factors are remaining same, then a movement alongwith the demand curve due to the price change refers the changes in the quantity demanded. If the other factors such as income, tastes or the price of some other goods have changed, then the change in demand at each possible price referred as shift of the demand curve. Thus, when an economist tells about an increase or a decrease in demand then it means to a shift of the

whole demand curve because other factors which were assumed to remain constant earlier have changed. When the economist tells about the change in quantity demanded then it means a movement along the same curve (i.e., expansion or contraction of demand) which occurs due to fall or rise in price of the commodity.



Shift of the demand curve to right or left due to the changes in other factors such as income, tastes, prices of other goods, etc., is referred as 'change in demand' and movement upwards or downwards on the same demand curve due to a change in price of the commodity is referred as 'change in quantity demanded'.

1) **Movement Along the Demand Curve:** A movement alongwith the demand curve is due to the change in the price of the good or service. When the quantity demanded of a commodity increases with a fall in its price then it is called extension of demand. When the quantity demanded of a commodity decreases with a rise in its price then it is called contraction of demand.

   i) **Extension of Demand:** If other things remain same, demand of a commodity tends to increase due to fall in its price. This can be shown with the help of **table 2.3** and **figure 2.20**:

Table 2.3: Extension of Demand

Price	Quantity
10	1
8	2
6	3
4	4
2	5

In the above **table 2.3** when price of a commodity is ₹10, its quantity demanded is 1 unit. As the price of the commodity falls ₹8, quantity demanded increases to 2 units. Again when price falls to ₹2, its quantity demanded further increases to 5 units. In **figure 2.20** quantity of commodity has been measured on X-axis and price on Y-axis. AB is the demand curve.

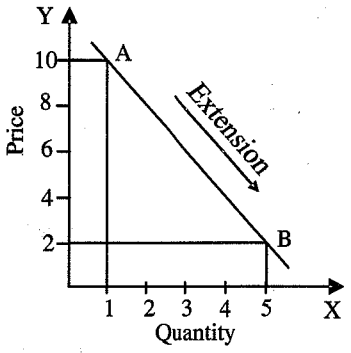


Figure 2.20: Extension of Demand

When price of the commodity is ₹10, its quantity demanded is one unit and consumer rests upon point A of the demand curve. As the price falls to ₹2, quantity demanded increases to 5 units. The consumer is at point B of the demand curve. Thus, movement from higher point to lower point (from A to B) along the demand curve is called the **extension of demand.**

   ii) **Contraction of Demand:** If other things remain same, demand of a commodity tends to decrease due to rise in its price. This can be shown with the help of a **table 2.4**:

Table 2.4: Contraction of Demand

Price	Demand
2	5
4	4
6	3
8	2
10	1

In the **table 2.4** when price of a commodity is ₹2, its quantity demanded is 5 units. When the price of the commodity increases to ₹6, quantity demanded falls to 3 units. Again as the price increases to ₹10, quantity demanded falls to one unit.

In **figure 2.21**, AB is the demand curve. When price is ₹2, quantity demanded is five units and consumer in at point B. Now as the price rises to ₹10 quantity demanded falls to one unit, and the consumer is at point A. In other words, movement from lower point to higher point (from B to A) along the demand curve is called **contraction of demand**.

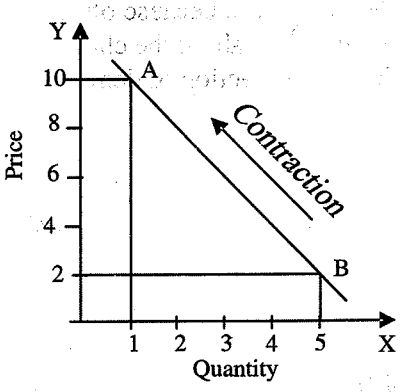


Figure 2.21: Contraction of Demand

2) **Shift in the Demand Curve:** A shift in the demand curve due to the change in non-price factors of demand such as income, rise in the price of substitutes, fall in the price of complements, change in tastes and technological change, etc.

i) **Increase in Demand:** An increase in demand means that at the same price, the consumers are willing to buy more of a quantity. The demand curve shifted to the right when there is an increase in demand. It must increase in two ways:

a) **Same Price More Demand:** In the **table 2.5** when price of a commodity is ₹10 per unit then the demand is 2 units. Now, if price remains same, i.e., ₹10, still demand increases to 4 units and further to 6 units.

Table 2.5: Increase in Demand

Price	Demand
10	2
10	4
10	6

b) **More Price Same Demand:** In the **table 2.6** when price of a commodity is ₹2 per unit, quantity demanded is 8 units:

Table 2.6

Price	Demand
2	8
4	8
6	8

Now, as the price increases to ₹4 per unit, its quantity demanded remains same, i.e., 8 units. Again, when price further increases to ₹6 per unit, quantity demanded remains 8 units.

In **figure 2.22** increase in demand has been shown. DD is the original demand curve. When the price of a commodity is OP_1 the consumer is at point X_1 and demanded OX_1 units. The demand curve shifts to D_1D_1 with the effect of changing in any factor other than price. New demand curve represents that the demand for commodity increases from OX_1 to OX_2 without changing the price of the commodity. Similarly, if the price increases from OP_1 to OP_2 then the demand remains same, i.e., X_1.

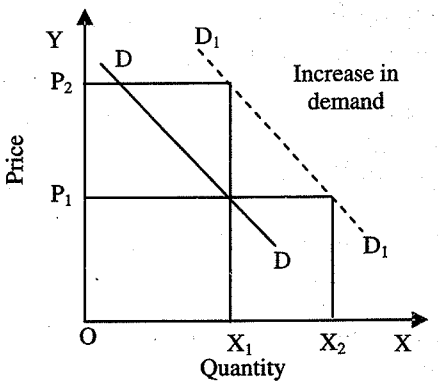


Figure 2.22: Increase in Demand

ii) **Decrease in Demand:** An increase in demand means that at the same price, the consumers are willing to buy less of a quantity. The demand curve shifted to the left when there is a decrease in demand. It must decreases in two ways:

a) **Same Price Less Demand:** In **table 2.7** when price of commodity is ₹6, its quantity demanded is 4 units:

Table 2.7: Decrease in Demand

Price	Demand
6	4
6	3
6	2

The demand tends to decline to 3 units and further to 2 units in the condition when price remains same.

b) **Less Price Same Demand:** In the **table 2.8** when price is ₹6 per unit, quantity demanded is 4 units:

Table 2.8

Price	Demand
6	4
4	4
2	4

If price falls to ₹4 per unit, quantity demanded remains same, i.e., 4 units. Even at price ₹2, quantity demanded is 4 units. In **figure 2.23**, DD is the original demand curve. When the price of a commodity is OP_2 the demand for a commodity is OX_1. If demand curve shifts to D′D′ (downwards) due to the change in the value of other factors then it indicates the situation of decrease in price. The demand curve shows the fact that the demand decreases to OX_2 at the same price of OP_2.

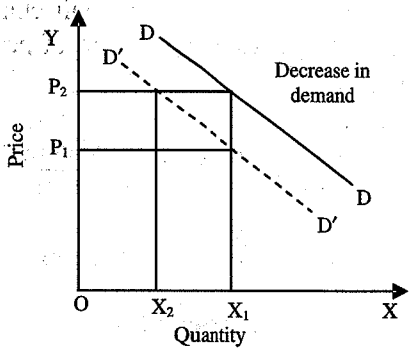


Figure 2.23: Decrease in Demand

2.7. ELASTICITY OF DEMAND

2.7.1. Meaning and Definition of Elasticity of Demand

The relative responsiveness of a supply or a demand curve in respect to price is referred as elasticity. If the curve is more elastic then it means the more quantity will be changed in respect to price change. On the other hand, if the curve is more inelastic then it means the less quantity will be changed in respect to more price change. The goods with elastic demand are not very important for the consumers because consumers can easily find the substitutes. Goods with inelastic demands are very important and necessary for the consumers because consumers cannot easily find the substitutes.

If the percentage change in quantity demanded is divided by the percentage change in one of the variable which affect the demand, then it is called 'elasticity of demand'. In other words, price elasticity of demand is generally referred as elasticity of demand. The rate at which the demand changes with respect to the change in price is denoted by the elasticity of demand.

According to Prof. Marshall, "The elasticity (or responsiveness) of demand in a market is great or small according as the amount demanded increases much or little for a given fall in price, or diminishes much or little for a given rise in price."

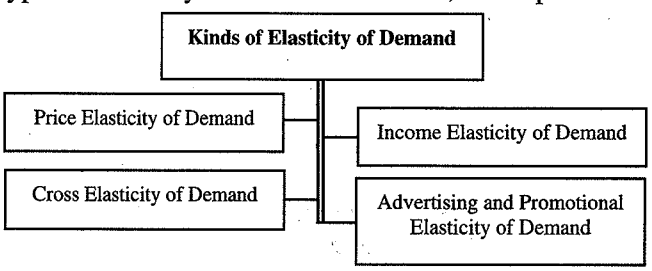
According to Prof Benha, "The concept relates to the effect of a small change in price upon the amount demanded."

According to Prof Boulding, "The elasticity of demand may be defined as the percentage change in the quantity demanded which would result from one per cent change in price."

According to Mrs. John Robinson, "The elasticity of demand at any price or at any output is the proportional change of amount purchased in response to a small change in price divided by the proportional change in price."

2.7.2. Kinds of Elasticity of Demand

There are various types of elasticity of demand. However, the important ones are given below:



2.7.3. Price Elasticity of Demand

Price elasticity of demand denotes the change in quantity demanded of goods with respect to a change in price, given that the consumer's income, his tastes and prices of all other goods.

According to Mrs. Robinson's, "The elasticity of demand at any price is the proportional change of amount purchased in response to a small change in price, divided by the proportional change of price." Thus, price elasticity of demand is the ratio of percentage change in amount demanded to a percentage change in price. It may be written as:

$$\text{Price Elasticity}(E_p) = \frac{\%\ \text{Change in quantity demanded}}{\%\ \text{Change in prices}}$$

$$\text{Or, } E_p = \frac{\frac{\text{Change in quantity}}{\text{Original quantity}} \times 100}{\frac{\text{Change in price}}{\text{Original price}} \times 100}$$

$$\text{Or, } E_P = \frac{\text{Change in Quantity}}{\text{Original Quantitiy}} \times \frac{\text{Original Price}}{\text{Change in Price}}$$

$$\text{Or in symbolic terms, } E_p = \frac{\Delta Q}{Q} \times \frac{P}{\Delta P} = \frac{\Delta Q}{\Delta P} \times \frac{P}{Q}$$

Where,

E_p = Price elasticity,

Q = Quantity, P = Price,

Δ Q = Change in quantity demanded,

ΔP = Change in price.

2.7.3.1. Types/Degrees of Price Elasticity

Price elasticity of demand is classified into five categories:

1) **Perfectly Elastic Demand (E = ∞):** If the demand for a commodity changes even though there is no change in price then demand is called perfectly elastic. It also indicates that if a very small change is done in price then quantity demanded would changes indefinitely and due to this reason seller would not change the price.

   In the **figure 2.24**, straight-line parallel to X-axis denotes the demand curve. The figure shows that at the current price OD, any amount of the commodity can be sold which indicate the perfect elastic demand or E = ∞ .

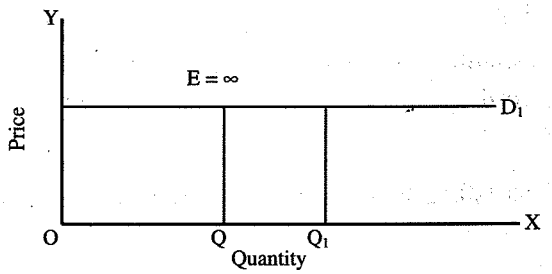


Figure 2.24: Perfectly Elastic Demand

2) **Perfectly Inelastic Demand (E = 0):** In case of perfectly inelastic demand, the demand for a commodity does not change even though there is change (increase or decrease) in price. In real life it is very difficult to find the practical example of perfectly inelastic demand. Let us consider that the price of any commodity is decreased by 40% but there is no change in quantity demand, then it is said to be a perfectly inelastic demand.

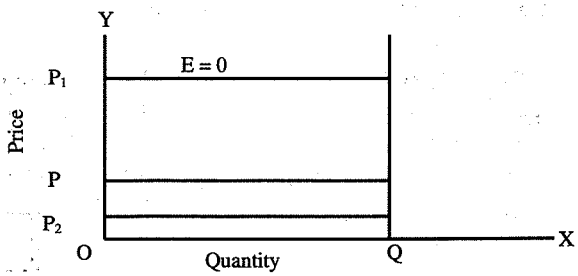


Figure 2.25: Perfectly Inelastic Demand

In the **figure 2.25**, an increase or decrease in price of a commodity is followed by absolutely no decrease or increase in the quantity demanded. Thus elasticity becomes zero i.e., E = 0.

3) **Unitary Elastic Demand (E = 1):** If the change in demand is exactly equivalent to the change in price then demand of that product is known as unitary elastic demand. In this case the demand curve is represented by a rectangular hyperbola (**figure 2.26**).

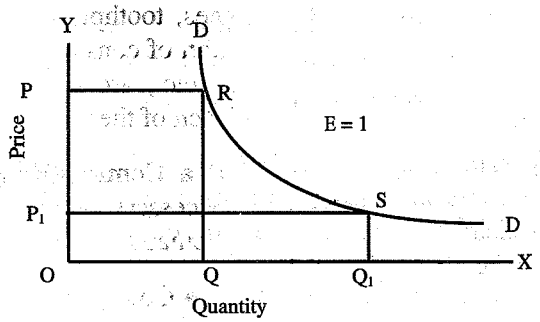


Figure 2.26: Unitary Elastic Demand

4) **Elastic Demand (E > 1):** If the percentage change in price is less than the percentage change in quantity demanded then price elasticity of demand is greater than one. This is known as elastic demand (**figure 2.27**).

For example, when price increases from ₹20 to ₹40, quantity demanded declines from 200 units to 80 units. As a result, the total outlay declines from ₹4,000 to ₹3,200. On the other hand, suppose price decreases from ₹20 to ₹10 and quantity demanded increases from 200 to 500 units, as a result the total outlay has increased from ₹4,000 to ₹5,000.

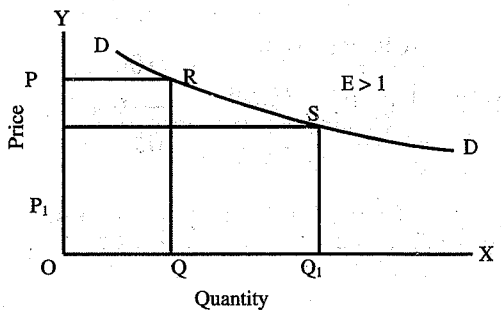


Figure 2.27: Elastic Demand

In this case, the price and total outlay are inversely related. This is due to the fact that the quantity demanded is changing comparatively faster the change in price.

5) **Inelastic Demand (E < 1):** In case of inelastic demand, percentage change in demand for a commodity is less than the percentage change (increase or decrease) in price. Price elasticity of demand is less than one. This is referred as inelastic demand (**figure 2.28**).

For example, when price increases from ₹20 to ₹40, and the quantity declines from 200 units to 150 units, the total outlay increases from ₹4,000 to ₹6,000. On the other hand, if the price declines from ₹20 to ₹10, and the quantity demanded increases from 200 units to 250 units, the total outlay declines from ₹4,000 to ₹2,500.

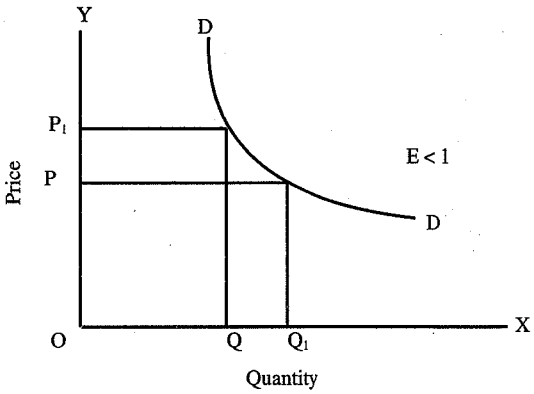


Figure 2.28: Inelastic Demand

This shows that the price and total outlay are directly related. This is because the quantity demanded changes relatively slower than the change in price.

2.7.3.2. Factors Influencing Price Elasticity of Demand

The main determinants of elasticity of demand are as follows:

1) **Availability of Substitutes:** The availability of substitutes of a commodity is the important determinant of the demand elasticity for that commodity. If the substitutes are very close then the elasticity of demand would be greater for the commodity.

For example, Coke and Pepsi may be considered as a close substitute for each other. If the price of one good increases then the other commodity becomes cheaper in comparison to first commodity. Therefore, consumer starts to buy ralatively more cheaper good, and less expensive good. The elasticity of demand will be higher for these two goods. On the other hand, soap and toothpaste do not have their close substitutes and price-elasticity is lower for these two goods.

2) **Position of a Commodity in a Consumer's Budget:** The elasticity of demand is also affected by the proportion of income which consumers spend on a particular commodity. If the proportion of income spent on a commodity is very small then its demand will be inelastic and if the proportion of income spent on a commodity is very high then its demand will be elastic.

For example, salt, matches, toothpastes, etc., are demand inelastic products because a household spend only a very small proportion of consumer's income on each of them. On the other hand, demand for goods like vegetables, clothing, etc., are demand elastic products because a households spend a very good proportion of income on each of them.

3) **Nature of the Need that a Commodity Satisfies:** Generally, the demand for luxurious goods is price elastic and demand for necessary goods is price inelastic. Therefore, the demand for air-conditioner is relatively elastic, and the demand for food and housing is inelastic.

4) **Number of Uses to which a Commodity can be Put:** The wider the range of alternative uses of a product, the higher the elasticity of its demand and *vice versa.* **For example,** milk has several uses. If the price of milk decreases then it can be used for various purposes such as preparation of curd, cream, ghee and sweets. But if the price of milk increases then it will be used only for few purposes such as feeding the children and sick persons.

5) **Possibilities of Postponing the Consumption:** There are various commodities to which consumption cannot be postponed even for any period of time. **For example,** foodgrains, salt, sugar, and medicines, etc., are included in this category. The demand for these commodities is generally inelastic because there is no option for the consumer to postpone the consumption. Consumers have to maintain their consumption level even in case of price rise.

6) **Joint Demand:** If the demand of two goods are combined then the demand of one good varies with the demand of other goods. In such case, the elasticity of demand of the good determine the elasticity of the other good which is more important. **For example**, the elasticity of demand for car will depend considerably on the elasticity of demand for fuel.

7) **Consumer Habits:** If a consumer is a habitual consumer of a commodity then the demand for this type of commodity will be inelastic at any price change.

8) **Tied Demand:** The demand for those goods which are tied to other goods is generally inelastic as against those whose demand is of autonomous nature.

9) **Price Range:** The demand of the goods have inelastic which are very costly or high range but those goods which lies in the middle range have elastic demand.

10) **Brand:** The price elasticity of demand may be elastic for a given brand of product. If the price increase then people start to buy products of other brand and it is called substitution effect. **For example,** if the price of Bajaj bikes increases, the consumer will buy the Honda bikes.

11) **Distribution of Income:** The demand for various goods is elastic in the case when a country has equal distribution of income and wealth because there is a huge number of middle class person whose purchasing power is almost equal.

12) **Time:** If the time period is longer in which the demand is considered then the demand for a product becomes more elastic. Consumers have more time to compare the existing product with another product and develop a habit to use another product.

2.7.3.3. Measurement of Price Elasticity of Demand

There are five methods to measure the price elasticity of demand. These methods are:

1) **Total Outlay Method:** Total expenditure method was developed by **Dr. Marshall** to measure the price elasticity of demand. According to this method, measurement of price elasticity of demand is based on the change in total outlay or total expenditure in respect to a change in the price of the commodity.

Total Outlay = Price × Quantity Demanded

Dr. Marshall maintains that elasticity of demand can be of three types:

i) **Unitary Elasticity:** Price elasticity of demand is unitary in the condition when total expenditure does not change with change in price of the commodity. In other words, if change in quantity is equivalent to the change in the price then price elasticity of demand is called unitary.

ii) **Elastic Demand:** If a small change in price leads to a relatively large change in the quantity demanded then demand is elastic.

iii) **Inelastic Demand:** If a small change in price leads to a relatively smaller change in the quantity demanded then demand is inelastic.

2) **Proportionate Method:** This method is also given by **Dr. Marshall**. According to this method, "The ratio of percentage change in the amount demanded and percentage change in price of the commodity is referred as price elasticity of demand". It is also known as the Percentage Method, Flux Method, Ratio Method, and Arithmetic Method.

3) **Point Method or Geometric Method:** This method was also developed by **Marshall** and a straight line demand curve taking into consideration to measure the elasticity of demand on different points of demand curve. It is opposite of proportionate method. To measure the elasticity of demand we take a straight line demand curve as shown in **figure 2.29**. In **figure 2.29**, AB is the demand curve.

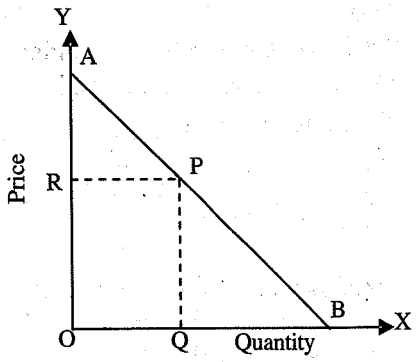


Figure 2.29: Point or Geometric Method

We know that, $E_p = \frac{P.\Delta Q}{Q.\Delta P}$

The term $\frac{\Delta Q}{\Delta P}$ is the reciprocal of the slope of demand curve. The slope of AB demand curve is PQ/QB in the figure price is PQ and quantity demanded at PQ price is OQ and $\frac{\Delta Q}{\Delta P} = \frac{QB}{PQ}$.

Therefore,

$$E_p = \frac{P.\Delta Q}{Q.\Delta P} = \frac{QB}{PQ} \times \frac{PQ}{OQ} = \frac{QB}{OQ} \text{ and } \frac{QB}{OQ} = \frac{QB}{RP} \text{ Because, } OQ = RP$$

Now ΔARP and ΔPQB are similar right angle triangles. Therefore, part PB is the lower section and part PA is the upper section of the AB demand curve. Thus, the elasticity of demand at point P is equal to $\frac{PB}{PA}$.

$$\text{i.e., Point Elasticity of Demand} = \frac{\text{Lower Segment of Demand Curve}}{\text{Upper Segment of Demand Curve}}$$

4) **Arc Method: According to Professor Baumol,** "Arc elasticity is a measure of the average responsible to price change exhibited by a demand curve over some finite stretch of the curve".

This method of measuring elasticity of demand is also known as "Average Elasticity". In this method, we use $\frac{P_1 + P_2}{2}$ rather than P. Thus, we apply $\frac{Q_1 + Q_2}{2}$ rather than the formula for Arc elasticity of demand is as follows.

$$\text{Arc Elasticity of Demand } (E_A) = \frac{\text{Change in Demand or Original Demand + New Demand}}{\text{Change in Price or Original Price + New Price}}$$

Arc elasticity of demand in notational form can be expressed as:

$$E = \frac{Q_2 - Q_1}{Q_2 + Q_1} \div \frac{P_1 - P_2}{P_1 + P_2}$$

Where,
Q_1 = Original quantity demanded
Q_2 = New quantity demanded
P_1 = Original price

When change in price is very large then Arc elasticity method should be used.

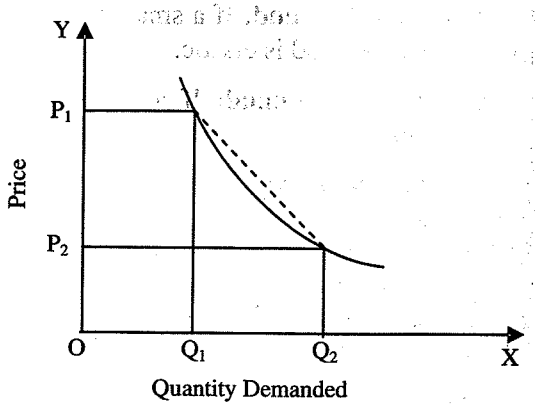


Figure 2.30: Arc Elasticity

In **figure 2.30** we can measure arc elasticity between points A and B on the demand curve; we will have to take the average prices of OP_1 and OP_2 and average of the two quantities demanded (original and the new).

5) **Revenue Method:** This method was given by **Mrs. Joan Robinson**. She suggested that elasticity of demand can be measured on the basis of average revenue and marginal revenue. Therefore, the income that a firm obtains by selling its products is called its revenue. When total revenue is divided by the number of units sold then it is called **average revenue**. **Marginal revenue** is referred as the additional **revenue** that will be generated by increasing product sales by one unit. Therefore, the formula to measure elasticity of demand can be written as:

$$E_d = \frac{A}{A - M}$$

Where, E_d = Demand; A = Average revenue; M = Marginal revenue

2.7.4. Income Elasticity of Demand

Income elasticity of demand is defined as, "the percentage change in the quantity demanded of a good divided by the percentage change in the income of the consumer". In symbolic form:

$$E_y = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q}$$

Where, E_y = Income elasticity,
Q = Quantity demanded,
Y = Income,
ΔQ = Change in quantity demanded,
ΔY = Change in income

Types of Income Elasticity of Demand

Types of income elasticity of demand are as follows:

1) **High-Income Elasticity:** Income elasticity of demand is said to be high when the quantity demanded for a good increases by a larger percentage as compared with the income of the consumer.
2) **Unitary Income Elasticity:** Income elasticity of demand is said to be unitary when the percentage change in quantity demanded for a product is equal to the percentage change in income.
3) **Low-Income Elasticity:** Income elasticity of demand is said to be low when the quantity demanded for a good increases by a smaller percentage as compared with the income of consumer.
4) **Zero Income Elasticity:** Income elasticity of demand is said to be zero when the quantity demanded for a good remains unchanged upon the change of income.
5) **Negative Income Elasticity:** The income elasticity of demand is said to be negative when the quantity demanded for a good falls in response to an increase in income.

Example 7: When the income is ₹100, demand is 25 units; now suppose that the income increases to ₹150, as a result of this demand increases to 30 units. Calculate the elasticity of income.

Solution: Elasticity of Income

Y_1= ₹100, Y_2= ₹150

Q_1= ₹25, Q_2= ₹30

$\Delta Y = Y_2 - Y_1 = 150 - 100 = 50$

$\Delta Q = Q_2 - Q_1 = 30 - 25 = 5$

$$E_y = \frac{\Delta Q}{\Delta Y} \times \frac{Y}{Q} = 5/50 \times 100/25 = 2/5$$

The coefficient 2/5 shows that the demand is inelastic.

2.7.5. Cross-Elasticity of Demand

Cross-elasticity of demand for a good is defined as a change in demand of that good due to change in price of another good. It is defined as:

$$E_c = \frac{\text{Percentage change in quantity demanded of good A}}{\text{Percentage change in price of good B}} = \frac{\Delta Qx}{\Delta Py} \times \frac{Py}{Qx}$$

Where, E_c = Cross elasticity

Qx = Original quantity demanded of x

Py = Original price of good y

ΔQx = Change in quantity demanded

Δpy = Small change in the price of y

The elasticity coefficients give significant results about the type of goods:

1) **Substitute Goods:** Let us consider two goods X and Y. Goods X and Y are called substitute goods if demand of X move in the same direction as a change in the price of Y. **For example**, Pepsi (X) and Coca-Cola (Y). An increase in the price of Pepsi causes consumers to buy more Coca-Cola, resulting in a positive cross-elasticity. The larger the positive cross-elasticity coefficient, the greater is the substitutability between the two products.

2) **Complementary Goods:** Let us consider two goods X and Y. Goods X and Y are called complementary goods if demand of X move in the opposite direction as a change in the price of Y. **For example**, an increase in the price of printers will decrease the amount of ink cartridges purchased. The larger the negative cross-elasticity coefficient, the greater is the complementarity between the two goods.

3) **Independent Goods:** Let us consider two goods X and Y. Goods X and Y are called independent goods if demand of X is unchanged as a change in the price of Y. **For example,** movie ticket and bread; we would not expect a change in the price of movie ticket to have any effect on purchases of bread, and *vice versa.*

2.7.6. Difference between Price and Cross Elasticity

Basis of Difference	Price Elasticity	Cross Elasticity
1) **Meaning**	Price elasticity of demand denotes the change in quantity demanded of goods with respect to a change in price, given, that the consumer's income, his tastes and prices of all other goods.	Cross-elasticity of demand for a good is defined as a change in demand of that good due to change in price of another good.
2) **Coefficient**	The coefficient of price elasticity of demand is always negative due to inverse relation between price and quantities demanded.	The positive coefficient of cross elasticity shows that the given commodities are substitutes. Negative cross elasticity shows that the given commodities are complementary to each other and when it is zero, then the given commodities are unrelated to each other.

3) **Degree of Elasticity**	The coefficient of price elasticity shows different degrees of price elasticity like elastic, inelastic and unitary demand.	The coefficient of cross elasticity shows the relation between the given set of goods.
4) **Formula**	Price Elasticity (E_p) $= \frac{\%\text{ Change in quantity demande}}{\%\text{ Change in prices}}$	$E_c = \frac{\text{Percentage change in quantity demanded of good A}}{\text{Percentage change in price of good B}}$

2.7.7. Uses/Importance of Concept of Elasticity

The price elasticity of demand generally called elasticity of demand is very important for the producers or sellers, workers and government to formulate their policies. It has the great importance in the managerial decision-making. Following applications clear the practical importance of the elasticity of demand:

1) **Determination of Price Policy:** For deciding the price of any product, a businessman considers the elasticity of demand for the product. He should consider for a product that if price of the product decreases then demand of the product increases or not? If demand of the product increases due to decrease in price then he should consider that up to what extent the price should decrease and resulting this profit will be increase or not. If the increased sales are proportionally more than the price reduction then the total revenue will increase and his profit will be also larger. Generally, for a product having inelastic demand, producer will select a high price whereas the products which are having elastic demand the producer select a lower price.

2) **Price Discrimination:** The process of selling technically same product at different prices to the different section of consumers or selling in the different markets, referred as price discrimination. If the elasticity of demand is different for any product in different markets then the policy of price discrimination is profitable for monopolist. Higher price can be charged for any product if the demand of that product is inelastic and charged lower price if the demand of the product is elastic.

3) **Shifting of Tax Burden:** The producer can shift the burden of indirect tax on the consumer and at what extent it is shifted to the consumer by increasing the product price, is decided on the basis of the elasticity of demand.

4) **Importance in International Trade:** The concept of elasticity of demand is very important in various aspects of international trade. The elasticity of demand for exports and imports of the country is the basis of the success of the devaluation policy to correct the adverse balance of payment.

5) **Importance in the Determination of Factors Prices:** A factor of production charge higher price in comparison to other factor if the elasticity of demand is relatively high. This helps trade unions to know the reasons for which they can get the higher wage rate. The elasticity of demand is the basis for the bargaining capacity of trade unions for the workers services.

6) **Pricing of Joint Supply Products:** The goods which are produced by a single production process are joint supply products. The cost of production is also joint for these goods. The elasticity of demand is considered when the prices of these products are determined. There is fixed high price for that product whose elasticity of demand is low, and fixed low price for that product whose elasticity of demand is high.

7) **Effect on Use of Machines on Employment:** Generally, labour think that use of machines reduces their demand. Due to this, trade unions generally oppose the use of machinery because they fear about unemployment. But this is not always true that the demand of labour reduces due to the use of machines. Price elasticity of demand for the products is the basis of demand of labour.

8) **Public Utilities:** The elasticity of demand also justified the nationals of public utility services. Generally, the demand of various public utilities such as electricity, water supply, post and telegraph, public transportation, etc., is inelastic in nature. Public enterprises determine the prices of the public utilities on the basis of elasticity of demand.

9) **Output Decisions:** The decision of the production is based on the elasticity of demand. With the help of elasticity of demand, optimum production-mix for various products is selected by the businessman.

2.8. FORECASTING

2.8.1. Introduction

Forecasting is the process of estimating the relevant events of future, based on the analysis of their past and present behaviour. The future cannot be probed unless one knows how the events have occurred in the past and how are they occurring presently. Thus, the past and present analysis of events provides information about their future occurrences. Since forecasting may require the use of various statistical techniques, some persons equate this analysis with statistical analysis.

2.8.2. Meaning and Definition of Demand Forecasting

Forecasting is a technique that facilitates the process of planning and enabling efficient decision-making. By this estimates, projections are made for the future. Combining the aspect of forecasting with demand, it serves with the future projections regarding the quantity demanded in precise. Certain questions like which product will be in demand? What will be the quantity demanded?

Where will be the demand? are answered by demand forecasting. It also helps in developing relation between the demand and price alongwith the price of the substitutes that are available. The change in demand as a result of change in any of these factors is also revealed. So, primarily demand forecasting serves as a mechanism to forecast and estimate with precision the future demand of the product.

According to Pynes, Demand forecasting is "the anticipation of the leadership workforce that will be needed to accomplish future functional requirements and carry out the mission of the organisation".

According to Cundiff and Still, "Demand forecast is an estimate of sales during a specified future period which is tied a proposed marketing plan and which assumes a particular set of uncontrollable and competitive forces".

According to American Marketing Association, "Demand forecasting is an estimate of sales in rupees or physical units for a specified future period under a proposed marketing plan".

2.8.3. Level of Demand Forecasting

The demand forecasting has a great effect on the firms. In an industry, the level of demand forecasting is carried out at three levels. These levels are as follows:

1) **Macroeconomic Forecasting:** At macroeconomic level, the demand forecasts are assumed for the general economic condition. This forecast focus on the condition of business over the whole economy. Usually, it is measured through suitable index or through the estimation of production of the industry, national income or expenditure, etc.

2) **Industry Forecasting:** The industry forecasting is done through Chamber of Commerce or trade association and is made available to its members. These forecasts are based on the preference of the consumers and analysis of trends of the past data. Hence, firm can use this type of data in order to compare the various industries.

3) **Micro/Firm Level Forecasting:** At this level, the forecasting is done in order to predict the demand of those products whose sale depends on the policies of the firm. To forecast the demand, the firms give due consideration to various factors like – technology, competitive strategy, changes in income of the consumer, taste, and preference of the customer, etc.

2.8.4. Criteria for Good Demand Forecasting

Good forecasting is generally based on the following criteria:

1) **Simplicity and Ease of Comprehension:** Management must be able to understand and have confidence in the techniques used. Complicated mathematical and statistical techniques may be avoided.

2) **Economy:** Cost must be weighed against the importance of the forecast to the operations of the business. The criticism should be the economic consideration of balancing the benefits from increased accuracy against the extra cost of providing the improved forecasting.

3) **Availability:** Techniques should give quick results and useful information.

4) **Durability:** Durability of the estimating power of a demand and functions depends on reasonableness and simplicity of functions fitted.

5) **Accuracy:** Market planning is based on the sales forecasting and, hence, sales forecasts should be accurate.

6) **Flexibility:** Flexibility of method is another attribute of a good forecasting method which means we should be able to make some changes in the variable which determine the demand. If the model for demand forecasting is very rigid, then its predictive power is considerably reduced.

7) **Availability of Data:** Any forecasting is based on the availability of data, which is fitted in a model to estimate future demand. If the data, whether primary or secondary, is easily available, then only it can be used. If the data requirement is clumsy or unavailable, then obviously the forecasting method is of little use.

2.8.5. Factors Affecting Demand Forecasting

Demand forecasting depends upon the following factors:

1) **Nature of Goods:** Demand forecasting differs on the basis of the fact that the goods may be producer goods, consumer goods or services, etc. Apart from this, demand forecasting also depends on the basis of durable and non-durable commodities.

2) **Level of Competition:** Market competition affects the process of demand forecasting. In a highly competitive market, the forecast is conservative taking into account the actions of competitors. It also depends on the future outlook of the industry due to entry of new firms.

3) **Price:** Demand forecasting is deeply influenced by a firms pricing policy. Higher prices in future may influence future demand of the product or service unless it is supplemented by high quality of product.

4) **Level of Technology:** Rapid changes in technology have the potential of rendering the existing products obsolete in future. **For example,** the drastic fall in demand of Black and White T.V. due to introduction of colour T.V. in the last 15 years in India. Technological factor plays an important role in arriving at a reliable demand forecast in future.

5) **Economic Outlook:** Future economic development plays an important role to forecast demand in future. Positive economic development due to high level of investment, liberalisation, globalisation, etc., gives enough reason to have an optimistic demand forecast for future.

2.8.6. Significance of Demand Forecasting

The importance/ significance of demand forecasting is as follows:

1) **Price Control:** Demand forecasting helps in controlling prices by matching the output with future expected demand. **For example,** if demand is expected to go up in the future the steps may be taken to increase output and keep the prices in control.

2) **Business Planning:** Demand forecasting helps in business planning based on future activities to be taken up. **For example,** entrepreneurs may plan their export, sales, and production targets on the basis of future demand.

3) **Competitive Strategy:** Demand forecasting helps business to effectively formulate their competitive strategy in terms of manpower, finances, advertising and other overheads. Future demand stimulates entrepreneurs to plan for future resource mobilisation.

2.8.7. Limitations of Demand Forecasting

The limitations of demand forecasting can be explained as under:

1) **Change in Fashion:** Change in fashion is an inevitable consequence of advancement of civilisation. Results of demand forecasting have short lasting impacts especially in a dynamic business environment.

2) **Consumer's Psychology:** Results of forecasting depend largely on psychology of consumers, understanding which itself is difficult. Thus, demand forecasting becomes an uphill task.

3) **Uneconomical:** Forecasting requires collection of data in huge volumes and their analysis, which may be too expensive for small firms to afford. Moreover, estimation process may take a lot of time, which may not always be affordable by firms.

4) **Lack of Experts:** Accurate forecasting necessitates experienced experts, who may not be easily available. Forecasting by less experienced individuals may lead to erroneous estimates.

5) **Lack of Past Data:** Demand forecasting requires past sales data which may not be easily available. This may be a typical problem in case of forecasting for a new product.

2.9. METHODS OF DEMAND FORECASTING

After analysing and interpreting the data and subsequently the information that is obtained from various primary and the secondary sources, the next step is to adequately predict the future demand for the underlying product or the service. For facilitating these projections, numerous methods are available which can be taken up by the market analyst. Different methods that are available are categorised in:

1) Qualitative Techniques
2) Quantitative Techniques

2.9.1. Qualitative Techniques/Opinion Polling Method

This method is characterised by the opinion of different forces that drives the demand i.e. buyers, experts, retailers, etc. The information provided by them serves the base for demand forecasting. The opinion polling methods of demand forecasting are of following kinds:

2.9.1.1. Survey Methods

Survey method is the most commonly used direct method in short run for the projection of the demand. Under this, different surveys are organised in routine to gather the information about the likes and the dislikes of the buyers and the probable future purchases. Survey method include:

1) **Complete Enumeration Survey:** The survey is done by visiting all the houses during the forecast and all the household opinions are taken into the consideration.

Advantages of Complete Enumeration Survey

i) Quite accurate as it surveys all the consumers of a product.
ii) It is simple to use.
iii) It is not affected by personal biases.
iv) It is based on collected data.

Disadvantages of Complete Enumeration Survey

i) It is costly.
ii) It is time consuming.
iii) It is difficult and practically impossible to survey all the consumers.
iv) The size of the data increases the chances of faulty recording and wrong interpretation.
v) Useful only for products with limited consumers.

2) **Market Experiment Method and Sample Survey:** Rather than contacting all the households, some households are selected randomly and their opinions are sought. One underlying assumption is considered that the sample is the true indicator of the population. As a result of the sample survey, test marketing has evolved. Under this, the product is handed over to the different users on random basis for specified time and their responses are noted down. As an outcome of their response the future demand is projected.

- **Advantages of Sample Survey**

i) An important tool especially for short-term projections.
ii) It is simple and does not cost much.
iii) Since only a few consumers are to be approached, the method works quickly.
iv) The risk of erroneous data is reduced.
v) This method gives excellent results, if used carefully.

Disadvantages of Sample Survey

i) The conclusions are based on the view of only a few consumers and not all of them.
ii) The sample may not be a true representation of the entire population.

3) **End-Use Method:** Demand survey is conducted for the industries that are actively using the product as an intermediate. The consumption of the intermediate product is considered as the base for the future projections of the final product.

Advantages of End-Use Method

i) The method yields accurate predictions.
ii) It provides sector wise demand forecast for different industries.
iii) It is especially useful for producer's goods.

Disadvantages of End-Use Method

i) It requires complex & diverse calculations.
ii) It is costlier as compared to the other survey methods & is more time consuming.
iii) Industry data may not be readily available.

2.9.1.2. Sales Force Opinion Method

It is also termed as **Collective Opinion Method**. Under this, rather than approaching the consumers for their opinions, the verdict if those are affecting the sales are given the priority i.e., the shopkeepers and the salesmen. The salespersons who are engaged with the company are to make spate projections of their sales to the respective regions. These individual forecasts are put forth for further study and final data for the organisation as whole is prepared.

Advantages of Collective Opinion Method

1) Perhaps the simplest of the forecasting methods.
2) It is less costly.
3) Collecting data from its own employees is easier for a firm than to do it from external parties.

Disadvantages of Collective Opinion Method

1) Consumer's tastes and preferences keep changing with time. What held well in the past may not necessarily continue to do so in the future as well. The opinion of the sales force may thus be erroneous.
2) The sales force may give biased views as the projected demand affects their future job prospects.

2.9.1.3. Delphi Technique

It comprises of the certain rounds of the structural surveys and those who participate in the survey are the experts of their own field. This survey is conducted in different rounds and the results of the initial rounds are given as feedback in the later round for the purpose of follow-ups. It can be observed that the expert's responses in the second round are influenced by the opinion of the others. It is basically a group communication system is structured for survey, so that the incomplete information or knowledge is utilised to derive the most appropriate outcomes.

Advantages of Delphi Technique

1) It is simple to conduct.
2) Can be used where quantitative data is not possible.
3) The forecast is reliable as it is based on the opinion of people who know the product very well.
4) It is inexpensive.
5) It takes little time.

Disadvantages of Delphi Technique

1) The results are based on mere hunch of one or more persons and not on scientific analysis.
2) The experts may be biased.
3) The method is subjective and the forecast could be unfavourably influenced by persons with vested interests.

2.9.2. Quantitative Techniques/Statistical or Analytical Methods

The statistical methods, which are frequently used, for making demand projections are:

2.9.2.1. Trend Projection Method

The data of the past can be used a stepping stone for the future demand projections. A trend can be established on the basis of the past figures. Such trend can be graphically or equationally established. The trend can be estimated by using any one of the following methods:

1) **Graphical Method:** Past data of sales in different regions are plotted on the graph and the sales trend line is drawn by joining all the points. The line shows the most likely trend which may be upward or downward. As only the direction of the trend is reflected but the actual figures cannot be predicted precisely.

2) **Least Square Method:** This method assumes consumption that the previous rate of change continues to prevail in future. A trend line is drawn mathematically considering the past data in a manner that the difference between calculated and that of observed value is minimum. A trend line is formed which is the best fit for the data that is available.

   The trend line is based on two variables, where one variable is dependent and the other variable is independent. The value of the independent variable is put forth to estimate the value of the dependent variable. This method is simple to use and the cost associated with this method is also very less.

3) **Time Series Methods:** These methods (graphical and least square) are dependent upon analysis of the past data. Just like the other methods, this method also assumes that the past data can be effectively used to predict the future demand of the product. Usage of different methods like simple moving average and weighted moving average is adopted on the same assumption that the future events are based on the past figures.

   Time Series Demand has the following components:
   i) **Average:** It is the arithmetic mean of all observations.
   ii) **Trend:** Timely increase/decrease in the outcome over different periods.
   iii) **Seasonal Influence:** Other cyclic behaviour that also remains constant over different time periods i.e. daily, monthly, seasonally, yearly, etc.
   iv) **Cyclical Movement:** It refers to the unpredictability as a result of the long-term cyclical behaviour because of lifecycle of a product or a service.
   v) **Random Error:** Other aspects that was not covered by the above four components.

4) **Exponential Smoothing:** It is the fourth method in drawing the trends. Under this method, the past data of demand is used considering the weights. The weights used are however exponential in nature. Higher weights are allotted to the most recent period and it goes on decreasing exponentially as we go down.

   It can also said that the weights used are not linear rather they are exponential and the magnitude of the weights decrease for later years.

Advantages of Trend Projection Method

1) It is a very simple method.
2) The method provides reasonably accurate forecasts.
3) It is quick and inexpensive.

Disadvantages of Trend Projection Method

1) Can be used only if past data is available.
2) It is not necessary that past trends may continue to hold good in the future as well.
3) There is no analysis of causal relations between the demand and time series explaining the whys of it.

2.9.2.2. Barometric Method

This method is also termed as 'leading indicators forecasting'. National Bureau of Economic Research of U.S.A. has spotted three different indicators under this method i.e. leading indicator, co-incidental indicator and lastly the lagging indicator.

To have better result a relation between sales of product and economic indicators need to be studied to arrive at accurate projection of sales and study the effect of the indicators on the sales. In case, where the product is new to the market and no past data is available, establishing the relation may be difficult.

Advantages of Barometric Method

1) It is a simple method.
2) It predicts directional changes quite accurately.

Disadvantages of Barometric Method

1) It does not predict the magnitude of changes very well.
2) Finding out a leading indicator for any series is not always feasible.
3) The lead time is maintained consistently by a very few time series.
4) The method can be used for short-term forecasts only.

2.9.2.3. Regression Method

This is one of the most commonly used methods in forecasting of demand. Here, relation between quantity demanded being dependent variable and other independent variables like price, income, price of substitutes, etc. is established and a regression equation is formulated as $Y = A + BX$. Such relation between the dependent and the independent variables is often a linear one but sometimes it may also be curvy-linear.

After formulating the regression equation the value of variable Y can be calculated by using the value of dependent variable X and the two constants i.e. A and B.

Advantages of Regression Method

1) As the method is based on causal relationships, it produces reliable and accurate results.
2) Besides generating the forecast, it also explains the economic phenomenon.
3) It is neither as subjective as the qualitative techniques nor as mechanistic as the quantitative ones.
4) This method not only forecasts the direction but also the magnitude of the change.
5) The method is quite consistent.

Disadvantages of Regression Method

1) The method uses complex calculations.
2) It is costly and time consuming.
3) It requires the use of some other forecasting technique for estimating the value of the causal variables.

2.9.2.4. Econometric Method

The econometric model forecasting involves estimating several simultaneous equations, which are, generally, behavioural equations, mathematical identities and market-clearing equations. The econometric model technique is known also as simultaneous equations method and complete system approach to forecasting. This technique uses sophisticated mathematical and statistical tools.

Advantages of Econometric Method

1) This method possesses all the advantages of the regression method.
2) Furthermore, if one is not worried about the size of the model, it can almost eliminate the regression method's major problem of forecasting the values of the independent variables in the prediction period.

3) Provided the data are available, one should not really be concerned with the model's size, for the computer is there to help the forecasters to estimate and solve the model.

Disadvantages of Econometric Method

1) To the extent the structural changes have taken place, the past relationship between independent and dependent variables will not continue in the future.
2) The forecasts under this method assume that the forecast equation holds exactly in the prediction period. To the extent it is stochastic, i.e., the disturbance term is non-zero, the forecasts will be wrong.

2.9.3. Demand Forecasting for A New Product

According to **Joel Dean,** there are six methods of demand forecasting which are mentioned below:

1) **Evolutionary Approach in Forecasting Demand:** The main idea for the development of this approach is that the demand of the new products depends upon the performance of the existing product. Hence, in order to obtain the demand of the new product, it is necessary that the demand conditions of the existing product are to be considered.

   However, this approach is beneficial in the condition when the existing product is closely associated with the new product. **For example,** keypad mobile to touch screen mobile; bulb to LED lights, etc.

2) **Substitute Approach in Forecasting Demand:** This approach of demand forecasting helps to know the substitute of the existing product or services.

3) **Growth Curve Approach in Forecasting Demand:** According to the growth pattern of the existing product, the growth rate and the level of demand of the new product is assessed. Hence, an estimate of the growth curve of the new product is obtained based on the growth of existing product. **For example,** the growth in the demand of new cosmetics can be assessed from the average sale of the already existing powder such as Talcum powder in the market.

4) **Opinion Poll Approach in Forecasting Demand:** This is an essential method of demand forecasting in the consumer responses regarding the new product is gathered by directly interacting with the consumers. The sample survey method is adopted for evaluating the demand of the new product. The main advantage of this method is that the accuracy of the data is obtained. The disadvantage of this method is that it is expensive, complex process, etc.

5) **Sales Experience Approach in Forecasting Demand:** Under the method of sales experience, the sample of the new product is send to the sample market in order to predict the demand about that product and this activity is performed by the channels of distribution like departmental stores or cooperative society, etc., or by direct mailing. Hence, with the help of the sample market total demand of the new product is forecasted. The major problem is in estimating the allowance for making immaturity of the sample market and full-fledged market.

6) **Vicarious Approach in Forecasting Demand:** Under this approach, a mediator is appointed i.e. specialised dealers who establish a connection between the producer and consumers and is capable to predict the requirements or needs of the ultimate consumer. The dealer can also assess the consumer's reaction towards the new product as per the taste and preferences of the consumer. Hence, the organization is able to know the response of the consumer regarding the new product.

2.10. EXERCISE

2.10.1. Very Short Answer Type Questions

1) What is consumer behaviour?
2) Explain the concept of consumer equilibrium.
3) Discuss the meaning of budget line.
4) Give any two important assumptions of consumer surplus.
5) Explain the objectives of demand analysis.
6) What is law of demand?
7) Discuss the elasticity of demand.
8) What are the levels of demand forecasting?

2.10.2. Short Answer Type Questions

1) Explain the meaning of utility and also discuss the measurement of utility.
2) Elucidate the limitations of cardinal utility theory.
3) What is the assumption of indifference curve?
4) Highlight the practical usefulness of consumer surplus.
5) Explain individual and market demand equation.
6) Discuss the demand curve and demand schedule.
7) Highlight the exceptions to law of demand.
8) Why does demand curve slopes downward?
9) Explain income elasticity of demand with suitable examples.
10) Explain market experiment method and regression technique of demand forecasting.

2.10.3. Long Answer Type Questions

1) Highlight the relationship between total utility and marginal utility.
2) What is cardinal utility theory? Explain its assumptions.
3) Explain the law of diminishing marginal utility and law of equi-marginal utility.
4) What are the assumptions, significance and criticism of revealed preference theory?
5) Explain the concept and various determinants of market demand.
6) Explain the concept of "Elasticity of Demand". Differentiate between Price and Cross Elasticity of Demand.
7) What is price elasticity of demand? Explain various kinds of price elasticity with suitable examples.
8) What is survey method, sales force opinion methods and Delphi technique?
9) Highlight the Statistical methods of demand forecasting.

UNIT 3

Supply, Market Equilibrium & Cost Analysis

3.1. SUPPLY ANALYSIS

3.1.1. Introduction

One of the basic concepts of economics is supply. Supply states the quantity of goods and services which is present for the consumers. In order to present supply on the graph, it can be related with the quantity of goods or services available at a specific price. Further, the supply has a direct relation with the demand of goods or services at a specific price. In case, the supply and demand of goods and services equalises, then producer will increase the price of goods because every producer wants to attain the level of maximum profit.

3.1.2. Meaning and Definition of Supply

Supply refers to the schedule of the quantities of a good or service during a specific period of time that will be offered for sale at various prices. Thus, supply is defined always at a price and at a specific period of time. **For example,** the price of crude oil is changes every day and hence, the supply of crude oil is defined on the basis of daily prices. When the price of a commodity is high then the producer or seller will supply greater amount of the commodity. On the other hand when the price of a commodity is low then the producer or seller will supply less amount of the commodity. Hence, the price and amount supplied is positively correlated.

According to Thomas, "The supply of goods is the quantity offered for sale in a given market at a given time at various prices".

According to Meyers, "Supply means the amount offered for sale at a given price".

Factor of production is supplied by the individuals to the firms. The factor of production is converted into consumable goods by the firms which is the group of individuals. In the production process, the supply of factor of production depends not only on individual's decisions but also it is based on the firms' capability to convert these factors into consumable goods. The supply of goods which is categorised as non-produced is directly associated with the goods market. **For example,** banks provides various services such as taking deposits, locker facility and so on, to the customers.

3.1.3. Supply Function

Supply of a good or service is defined under the assumption that if other factor remaining same then the amount of goods or services which is offered by the seller at various prices at a specific time period. The amount of goods and services supplied by the seller or producer is not only affected by the price but also affected by the various factors such as the prices of other commodities, the price of factors of production, the objectives of producers or sellers and the level of technology used. The equation which is formed with the use of these factors is called **supply function**. Supply function can be represented as follows:

$S_x = f(P_x, P_y, C, T, O, F, W, N, T)$

3.1.4. Determinants of Supply

Following are the factors that determine the supply:

1) **Product Price (P_x):** If the other factors remaining constant then the amount of goods which is supplied by seller is directly related to the price. Usually, producer increases the supply of the goods and services to get more profit in the situation when prices are increases. The producers are encouraged to supply of a goods and services because of earn higher revenue from sales.

2) **Prices of Related Products (P_y):** Prices of goods related in production influences firm's to supply goods and services. **For example,** suppose the sellers of pizza notice that the price of hot dogs increases substantially. They may reduce the amount of resources devoted to the selling of pizza in favour of hot dogs. This will decrease the supply of pizza. If the sellers were already selling two (or more) goods, the change in market conditions would prompt them to reallocate their resources towards the more profitable ones. The price of a complementary good is expected to affect the supply of the good under consideration in a direct manner. **For example,** if the price of computer hardware increases, other things remaining constant, the supply of software would increase and *vice versa*. This is due to the reason that, an increase in price of computer hardware increases its supply that results in an increase in supply of software.

3) **Costs and Technology (C and T):** If two factors are closely related then they can be treated as one. The cost of the factors which are used in production process is known as costs. If other things are remains constant then any increase in factor price increases the cost of the production. In this situation the firm decreases the supply. **For example,** an increase in the price of labour will increase the cost of production. As a result, the supply of the product is decreases because sellers may charge more than before for any quantity supplied. For reducing the unit cost of production or increasing factor productivity technological improvement or innovations are introduced. Technological improvement reduces the total cost and it increases the supply of the product or services.

4) **Objectives of the Firm (O):** Firms set various goals to achieve. The objective of the firm influences the quantity of the goods or services which are supplied. **For example,** if the objective of the firm is to maximisation of sales then firm supply more amount of goods or services in comparison to firm whose objective if maximisation of profit.

5) **Future Expectations (F):** As the demand is affected by consumer's expectation, similarly, supply is influenced by seller's expectation. There are various factors such as expectations of prices, costs, sales and the general macroeconomic conditions which affect the supply. **For example,** if a seller expects that the price of the goods or services will be increase in future then he decreases the market supply to gain more profit. Similarly if the seller expects that the production cost will be increased in the future then he increases the market supply to avoid the decrease in profit margin.

6) **Weather Conditions and Other Short-Term Factors (W):** The supply of goods or services is adversely affected by the various short-term factors such as floods, droughts, strikes, lockouts etc. The supply of refrigerator and air-conditioner has a tendency to increase in the summer because there are limited buyers in other months and there are additional carrying costs of inventory. The supply of agricultural goods is reduced in the condition of floods and droughts. Supply of industrial goods is negatively affected in the situation of strikes and lock-outs.

7) **Number of Sellers (N):** The supply of goods or services is directly affected by the number of sellers. Market supply is directly correlated with the number of sellers. If the number of sellers is high then the market supply will be greater. In the condition when sellers make a mutual agreement then they restrict the supply to obtain more profit. In the condition when there are competition among the seller then they increase the supply to capture larger market share.

8) **Taxation Policy (T):** If heavy taxes are imposed on a commodity then the production will be discourages and resulting from this the supply will also be reduces. On the other hand if there are various tax discounts are available then it encourages the producer to increases the supply.

3.2. LAW OF SUPPLY

3.2.1. Meaning and Definition of Law of Supply

The **law of supply** derives the relationship between price of the commodity and quantity supplied. According to this law, other thing remaining the same, the supply of commodity is directly related with the price of the commodity.

According to Dooley, "Higher the price the greater the quantity supplied or the lower the price, the smaller the quantity supplied".

In the other words, other thing remaining the same, when the price of commodity increases then its quantity supplied increases and when its price decreases, quantity supplied also decreases. Symbolically, the law of supply can be expressed as follows:

$S_X = f(P_X)$

3.2.2. Assumptions of the Law of Supply

Just, as similar to the law of demand, the law of supply also assumes, other things remains constant. In order to have the direct relationship between the price and the quantity supplied, the factors which are taken into consideration are as follows:

1) The prices of factors of production remain constant.
2) The prices of related goods like substitute goods etc., remains unchanged.
3) The objective of the firms does not change.
4) There is no expectation of the producer to change the price of the commodity in the forthcoming future.
5) The technology remains unchanged.
6) The number of producer do not changes.
7) There is no change in the Government policy in relation to payment of taxes and subsidies.

3.2.3. Exceptions/Criticisms to the Law of Supply

There are various exceptions to the law of supply which are as follows:

1) **Future Expectation About The Change In Prices:** In case the producer assumes that the prices in future will reduce then he will increase the supply even though there is fall in the prices of the commodity. Hence, to eliminate the future losses the supplier sells the product even at reduced prices.
2) **Increase of Agricultural Goods:** The supply of agricultural goods is not affected by the prices of the goods rather it is controlled by the rainfall, fertility of the land and weather conditions.
3) **Perishable Goods:** Perishable goods refer to the goods which must be used within a shorter period of time like fruits, vegetables, milk etc. Hence, as these goods cannot be stored for long period of time. Thus, the producer supplies the goods even when there is fall in the price of the commodity.
4) **Goods of Action:** The goods which are smaller in quantity are purchased during a given period of time. In this case, the law of supply does not hold true.
5) **Disposal of Old Stock:** In order to clear the reserved stock the seller sells the goods without concerning about the prices of the commodity. There is no profit from the money which is invested in the reserved stock.
6) **Rare Goods:** The goods which are rare like old coins, artistic goods and rare poems have unusual prices inspite of having less supply. The consumer who purchases these goods is not affected by the change in prices.
7) **Backward Countries:** Backward countries do not have adequate capital and the required inputs which are used for the production process i.e., with increase in the prices the capability of the producer and the quantity supply remains same.

3.2.4. Supply Schedule

Supply schedule refers to a tabular representation of the quantity supplied and the prices. It shows the different quantities of a product that the seller is willing to sell at different prices.

Supply schedule is divided into two types as follows:

1) **Individual Supply Schedule: Individual supply schedule** is a schedule showing various quantities of a commodity that an individual producer is willing to sell to the consumer at various prices in the market during a given period of time.
2) **Market Supply Schedule: Market supply schedule** refers to a schedule which shows various quantities of the commodity that each and every producer and the firms are willing to sell to the consumer at various prices during a specified period of time.

   Market supply schedule is also said as the summation of all the individual supply schedules at each price.

   For example, there are three individual supply schedules of the firms A, B and C and by adding these we obtain the market supply schedule.

Market Supply Schedule

P_x (₹)	Supply by the Firm 'A' (Units)	Supply by the Firm 'B' (Units)	Supply by the Firm 'C' (Units)	Market Supply (Units)
1	5	3	2	5 + 3 + 2 = 10
2	9	6	4	9 + 6 + 4 = 19
3	14	8	5	14 + 8 + 5 = 27
4	16	10	7	16 + 10 + 7 = 33
5	20	12	11	20 + 12 + 11 = 43

3.2.5. Supply Curve

Supply curve refers to the graphical representation of the supply schedule as depicted in the **figure 3.1:**

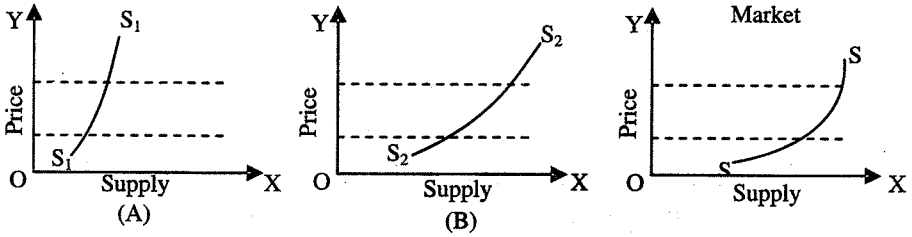


Figure 3.1: Supply Curve

It shows the various quantities of a commodity that the producer is willing to sell at different prices in the market. Supply curve is also divided into two types i.e. individual supply curve and market supply curve. The aggregate of all the individual supply curves gives the market supply curve as shown in the **figure 3.1,** the addition of two individual supply curve of firm 'A' and firm 'B' gives the market supply curve SS. As a result, the market supply curve depicts the direct relationship between the quantity supplied and the price of the commodity. As the price increases, the quantity supplied by every firm also increases hence, the supply curve is upward sloping from left to right.

3.2.6. Movement Along Supply Curve

When the price of a commodity changes, other factors kept constant, the quantity supplied of a commodity changes suitably. A change in price either causes supply curves to expand or contract.

3.2.6.1. Expansion of Supply

Expansion of supply, like that of demand, refers to a movement along the supply curve in response to changes in price. A rise in price, other things remaining same, leads to a rise in supply.

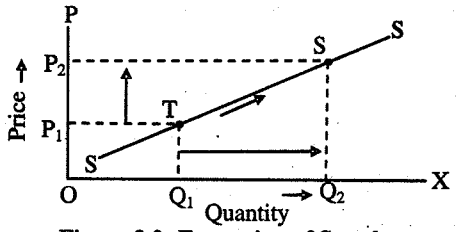


Figure 3.2: Expansion of Supply

3.2.6.2. Contraction of Supply

Contraction of supply is just opposite of its expansion. A fall in price offered leads to a fall in supply. It results in a downward movement along the supply curve.

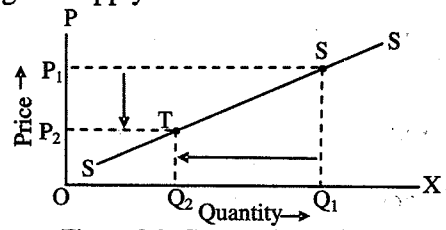


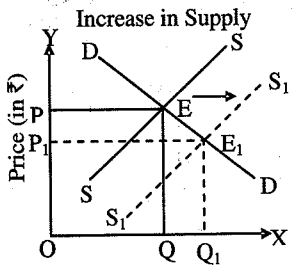
Figure 3.3: Contraction of Supply

3.2.7. Change or Shift in Supply

Under the law of supply, the supply curve shifts or change in supply arises because of the change in other factors which was assumed to be constant. There can be two types of changes in supply curve *i.e.,* 'Increase in Supply' or 'Decrease in Supply'. The increase in demand or change in price would cause no effect on supply curve. **For example,** supply curve will shift when technology changes.

3.2.7.1. Increase in Supply

In case of increase in supply, the supply curve moves rightwards from SS to S_1S_1, whereas there is no change in demand.



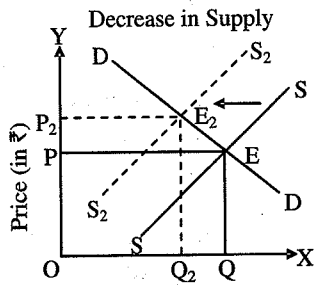
1) New equilibrium is determined at E_1
2) Equilibrium price falls from OP to OP_1
3) Equilibrium quantity rises from OQ to OQ_1.

Figure 3.4: Quantity demanded and supplied (in units)

When supply of a commodity increases without change in price it is called rise in supply. In the above **figure 3.4**, an excess supply is created at the old equilibrium price OP in case when supply curve increases from SS to S_1S_1 resulting into the decrease in price due to rise in competition among the seller. When price reduces, the demand increases and supply decreases. Till the period when a new equilibrium is not established at point E_1 these changes continue to happen. When supply curve increases, quantity demanded increases from OQ to OQ_1 and price falls from OP to OP_1 a new equilibrium is reached from E to E_1.

3.2.7.2. Decrease in Supply

In case of decrease in supply, the supply curve shifts towards left from SS to S_2S_2 while there is no change in demand as shown in **figure 3.5.**



1) New equilibrium is determined at E_2
2) Equilibrium price rises from OP to OQ_2
3) Equilibrium quantity falls from OQ to OQ_2.

Figure 3.5: Quantity demanded and supplied (in units)

In the given **figure 3.5**, an excess demand is created at the old equilibrium price of OP in case of reduction in supply curve from SS to S_2S_2 resulting into the increase in price due to rise in competition among buyer. When price increases, the demand decreases and supply increases. Unless the new equilibrium point is not reached to E_2, these changes continue to happen. When supply curve decreases, the quantity demanded reduces from OQ to OQ_2 and equilibrium price rises from OP to OP_2 and a new equilibrium is achieved from E to E_2.

3.2.8. Why does the Supply Curve Slopes upwards to the Right?

The increase in prices allows the producer to make further profits for the firm. When the other things remains constant, the direct relationship between price and quantity supplied leads to emergence of some situations. There are various causes for the upward sloping of the supply curve which are as follows:

1) **Change in Stock:** As there is positive relationship between supply and the price, therefore, with increase in price, the quantity supply of the old stock also increases and vice-versa. This result into the increase in inventory with the rise in prices.

2) **Profit and Loss:** Normally, with increase in price the profits and the production of the firm also increases. Alternatively, with reduction in the prices the supply of the firm reduces and it results in losses for the firm and loss of confidence among the producer who makes the decision to produce smaller quantities. Hence, the cost of production is reduced considerably.

3) **Entry and Exit of the Firms:** The increased price with increased profits encourages new firms to join the industry which results into rise in supply. In the situation of loss, the existing firms try to move out from the industry which leads to reduction in supply because the producer who are not efficient do not expect to earn any profit in the market at reduced prices. Hence, the supply curve slopes upward from left to right.

4) **Incentive for Innovation:** Nowadays, the scientific researcher uses modern technology for the production process which results into the positive effects.

3.3. ELASTICITY OF SUPPLY

3.3.1. Meaning of Elasticity of Supply

Elasticity of supply illustrates the tendency of change in the quantity of supply with respect to change in the price. This means that with the increase or decrease in price, the quantity of supply also increases or decreases. This change in the quantity of supply can be measured using price elasticity of supply. However, this amount of change in the supply due to the change in price has no accurate measurement.

According to Samuelson, "Elasticity of supply is defined as a measure of the degree of responsiveness of supply to the change in price".

According to Prof. Bilas, "Elasticity of supply is defined as the percentage change in quantity divided by percentage change in price". Elasticity of supply may be described as the ratio of proportional change in quantity supplied to the proportional change in price.

$$\text{Elasticity of Supply} = \frac{\text{Proportional change in supply}}{\text{Proportional change in price}} = \frac{\text{Percentage change in quantity supplied}}{\text{Percentage change in price}}$$

$$= \frac{\frac{\Delta Q_s}{Q_s}}{\frac{\Delta P}{P}} \text{ or } \frac{\frac{P - P_1}{P}}{\frac{Q_s - Q_{S1}}{Q_s}} = \frac{\Delta Q_s}{\Delta P} \times \frac{P}{Q_s}$$

where, E. S. = Elasticity of supply;

S = Original/Initial supply;

P = Original/Initial price;

Q_{S1} = New quantity of supplied

ΔQ_s = Change in supply

ΔP = Change in price

P_1 = New price

When comparison is done, one can find that elasticity of demand and elasticity of supply are quite similar. The only difference is that elasticity of supply will show a positive sign since it holds a direct relationship between price and supplied quantity.

3.3.2. Factors Determining Elasticity of Supply

Following are determinants of elasticity of supply:

1) **Nature of Inputs Used:** Elasticity of supply greatly depends upon the nature of inputs used during the production of the product. In case of elements of non-specialised nature, their volume of supply can be increased easily, which means the elasticity of supply of such products is high. However, in case of elements of specialised nature, their volume of supply cannot be increased; hence their elasticity of supply is low.

2) **Natural Constraints:** In case of the product which is used as agriculture based raw materials, the elasticity of supply will be low because production of such raw materials is itself dependent on the climatic conditions of nature. Hence, elasticity of supply is also determined by the natural constraints.

3) **Risk-Taking:** When the producer of the products has high risk-taking ability regarding the bulk production of products, than the elasticity of supply of such products is high. Whereas if the producer is not willing to take risk, then elasticity of supply of such products will be low.

4) **Nature of the Commodity:** The nature of product which is being supplied also determines the elasticity of supply. If the products are of perishable nature, than storing it for long time period is not possible which decreases its elasticity of supply. However, products of durable nature can be stored for quite a longer time period, hence they have more elasticity.

5) **Laws of Production:** The various laws of production play an important role in determining the elasticity of supply. Wherever Law of increasing returns is being applied, there the elasticity of supply will be high. However, in case of Law of Diminishing Returns, the elasticity of supply will be limited.

6) **Time:** If involved time period is longer in supply of product then more will be its elasticity. More elaborately it means, in case of long-run production, the elasticity of supply will be more whereas in case of short-run, the elasticity of supply will be less.

7) **Techniques of Production:** In case where complex techniques of production involving specialised labour, more capital, etc. are used, there the elasticity of supply will be quite low. Whereas, in case of simple production techniques which uses minimum capital and unskilled labours, will have better chances of higher elasticity of supply.

8) **Nature of Indusfry:** Elasticity of supply is also determined by the nature of industry, that is the products manufactured by the cottage industries or small firms will have comparatively less elasticity of supply. However, products that have been manufactured by the big firms or industries will have more elasticity of supply.

3.3.3. Degrees of Price Elasticity of Supply

There are five degrees of price elasticity of supply which are as follows:

1) **Perfectly Elastic Supply:** In case of perfectly elastic supply, the supply of the product increases or decreases, the price remaining the same or undergoing nominal change. **For example,** the below table shows the changes in quantity of supply, price remaining constant.

Supply Schedule

Price (₹)	Supply
10	8
10	10

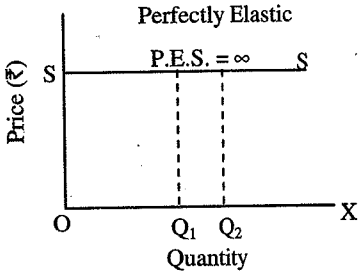


Figure 3.6: Perfectly Elastic Supply

Figure 3.6, shows supply curve SS which is parallel to X-axis (OX). This is applicable in productions where Law of constant returns is being operative. Here, the quantity supplied is zero as well as OQ_1 and OQ_2 at the price OS and since, the price remains constant, therefore, P.E.S. = ∞.

2) **Perfectly Inelastic Supply:** In case of perfectly inelastic supply there is no change in the quantity of supply with respect to any change in price of the product. There are very few products which falls under this category. The table given below can be used as an example,

Supply Schedule

Price (₹)	Supply
0	8
10	8

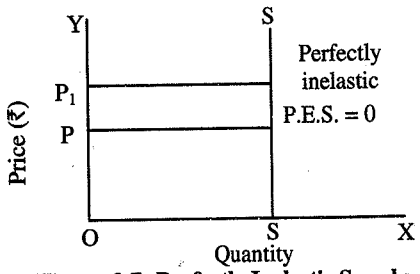


Figure 3.7: Perfectly Inelastic Supply

Figure 3.7 shows supply curve SS which is parallel to Y-axis (OY) and is touching point S on X-axis which means there is no change in the supply of quantity at this point. Here, the price may be zero or OP or OP_1but the quantity of supply will remain OS, hence, P.E.S. = 0.

3) **Unitary Elasticity:** In Unitary Elasticity, change in the quantity of supply is directly proportional to the change in the price of the product. This can be illustrated with the example that, if there is increase in price of a product by 30% then its supply will also increase by 30%.

Supply Schedule

Price (₹)	Supply (Unit)
10	20
15	30

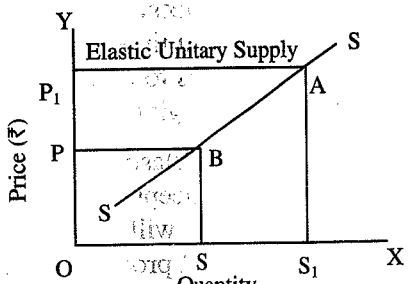


Figure 3.8: Elastic Unitary Supply

Figure 3.8 shows Supply curve SS which illustrates proportionate change in price PP_1 is equal to proportionate change in quantity supplied SS_1.

4) **More than Unitary (Relatively Elastic):** In case of relatively elastic, the proportionate change in the quantity of supply is comparatively more than the proportionate change in the price. Example for this can be illustrated by the table given below.

Supply Schedule

Price (₹)	Supply (Unit)
10	10
15	12

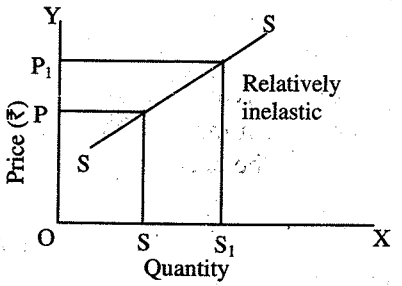


Figure 3.9: Relatively Inelastic Supply

Figure 3.9 shows supply curve SS which illustrates that the proportionate change in price PP_1 is less than the proportionate change in quantity supplied SS_1.

5) **Relatively Inelastic (Less than Unitary) Supply:** In case of relatively Inelastic, the proportionate change in the quantity of supply is comparatively less than the proportionate change in the price. Example for this can be illustrated by the table given below.

Supply Schedule

Price (₹)	Supply (Unit)
10	10
15	20

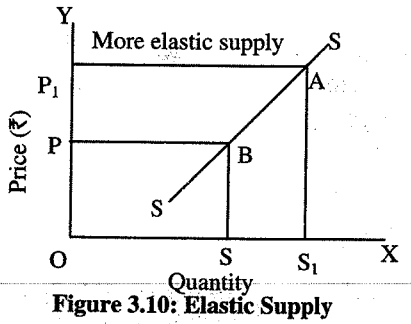


Figure 3.10: Elastic Supply

In **figure 3.10**, the change in price, P to P_1, is more than the change in supply (S, S_1). The supply curve SS has more steepness showing less elasticity of supply.

Different Price Elasticities of Supply

Value or Elasticity of Supply	Degrees of Elasticity	Relation
$E_S = 0$	Perfectly inelastic	Change in price has no effect on supply
$E_S < 1$	Less than unitary	$\frac{\Delta S}{S} < \frac{\Delta P}{P}$
$E_S = 1$	Unitary elastic	$\frac{\Delta S}{S} = \frac{\Delta P}{P}$
$E_S > 1$	More elastic	$\frac{\Delta S}{S} > \frac{\Delta P}{P}$
$E_S = \infty$	(Infinite) Perfectly elastic	Supply changes without change in price

3.3.4. Practical Importance/Use of Elasticity of Supply in Managerial Decision-Making

Elasticity of supply is useful in managerial decision-making for various businessmen and producers in following ways:

1) **Determining Prices:** Elasticity of supply helps in determination of prices, however, price is greatly influenced by total time involved in the production. In case of short-run production, the elasticity of supply is less which leads to increase in the prices. Whereas in case of long-run production, the chance of change in supply is more, giving chances to decrease the prices.

2) **Factor Pricing:** Elasticity of supply also helps in determining factor pricing. Modern economists have developed the concept of rent which depends on the elasticity of factors of production. It illustrates that the specialised factor will get more rent since they have inelastic or limited supply. However, in case of common factors of production having more elasticity of supply, rent paid will be comparatively less.

3) **Taxation:** It can be used in determining the amount of tax to be charged, i.e., products having comparatively low Elasticity of Supply can be allocated high taxation charges because supply of such products will undergo less changes.

3.4. EQUILIBRIUM ANALYSIS

3.4.1. Meaning and Definition of Equilibrium

Equilibrium originated from Latin words ***acqui*** and ***libra*** which mean 'equal' and 'balance' respectively, the term 'equilibrium' denotes 'equal balance'. Mechanically speaking, it may also be called a condition where factors are stationary or immobile, i.e., 'state of rest'. Equilibrium is achieved when there is a perfect balance between interlinked factors which work in contrary directions and they do no tend to move away or diverge from the balanced position. **For example,** demand and supply.

According to Prof. J.K. Mehta, "Equilibrium denotes in economics absence of change in movement, while in the physical sciences it denotes absence of movement itself".

According to Machlup "Equilibrium is a constellation of inter-related variables so adjusted to one another that no inherent tendency to change prevails in the model which they constitute".

Micro- as well as macro-economic studies encompasses intensive use of concepts like equilibrium and disequilibrium. Generally, micro-economics involves analysis of partial equilibrium. Whereas, macro-economics mainly involves general equilibrium analysis. The concept of partial equilibrium is limited merely to sectoral analysis in macro-economics. The situation where 'aggregate demand' becomes equivalent to 'aggregate supply' signifies equilibrium in an economy. The overall demand of goods and services of both consumer and capital nature, having been given the money's aggregate demand, is called 'aggregate demand'. While the sum total of supply of all goods and services of both consumer and capital nature, having been given the money's aggregate supply is termed as 'aggregate supply'. The economy maintains its equilibrium state till it is not affected internally or externally by any unbalancing factors.

3.4.2. Market Equilibrium: Competitive Equilibrium

Equilibrium is a market situation where demand forces and supply forces comes to a balance. If supply of goods and services increases then it will disturb the equilibrium situation and as a result price of goods and services will decrease. In this type of situation, mostly market develops new equilibrium. Hence, at this new equilibrium, both the price and quantity supply will remain equal till the demand or supply force brings any change. In order to analyse the behaviour of buyers or sellers, an analytical structure is provided by the demand and supply. Here, the demand indicates the way buyer responses to the changes in the price and other factors which affect their purchasing capacity. On the other hand, the supply indicates the way seller responses to the changes in price and factors that affects their selling capacity. Hence, when the buyers and sellers meet in market they develop a competitive equilibrium.

Therefore, this equilibrium is a situation where buyers can buy and seller can sell all the goods and services they desire. It is assumed that perfect competition exists in market. In this perfect competition, there will be large number of buyers and sellers and each buyer will accept the price and will demand the quantity of goods on that price. Therefore, each consumer possesses their own demand curve and by combining the demand curve of each consumer, market demand curve is derived. The market demand curve provides an overview of entire quantity demanded by the consumers at different level of price.

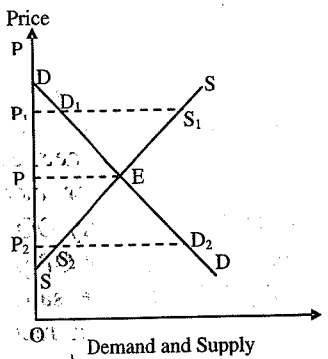


Figure 3.11: Equilibrium of Supply and Demand

On the other hand, the seller also takes price and decides the quantity which must be offered for sale. Hence, each seller possesses their own supply curve and by combining the supply curve of each supplier, market supply curve is derived. Further, the market supply curve provides the overview of entire quantity of goods and services supplied by the seller at different level of price.

In **figure 3.11,** the horizontal axis indicates the market demand and supply, whereas the vertical axis indicates the level of price. Here, the curve DD shows the market demand and curve SS shows the market supply. The curve DD slopes downward and curve SS slopes upward. Both the curve DD and SS intersect each other at a point E and at this point OP is the price. Hence, PE is the equilibrium price where total supply and demand are equal.

Further, OP is the equilibrium price where the total demand is same as total supply in the market. Thus, in this case the price is greater than or less than OP then the total demand are not the same as total supply. At price OP_1 total demand will be P_1D_1 and the total supply will be P_1S_1. Hence, total demand will be less than total supply where D_1S_1 is the amount of excess supply. Again at price OP_2 total demand will be P_2D_2 and total supply will be P_2S_2. Here, total demand will be greater than total supply where S_2D_2 is the amount of excess demand.

At price OP, total supply and total demand will be equal and equilibrium will be attained, but if price falls below or above to OP, then demand and supply will not be equal and hence no equilibrium will be attained. Here, either demand will be in excess or supply will be in excess. In case, there is excessive demand, then there will be high degree of competition among buyers which will result in increase in demand, whereas in case of excessive supply, there will be high competition among the sellers which will result in decrease in level of price.

Therefore, even if the price is either above or below than OP, it will move toward the point OP. Say, OP_1 is the initial price and at this point consumers wants to purchase P_1D_1 unit of goods, but seller wants P_1S_1 units to sell which is greater than P_1D_1. Hence, in this case, all the seller will not be able to sell their goods. Therefore, with an aim to dispose the excess supply, seller will decrease the price. Thus, with the reduction in price, quantity demanded will increase and supply of quantity will decrease.

Further, this process will be followed until OP price is reached. In a similar way, if the initial price of good is OP_2, S_2D_2 will be excess demand, which will result in increased price. If price increases, the demand will reduce and supply will increase and this process will continue unless OP point is attained. In the given demand function, D indicates total demand, S indicates total supply and P indicates level of price. The demand function is written as:

$$D = f(P) \quad \ldots\ldots (1)$$

Similarly like demand curve, supply curve is written as:

$$S = g(P) \quad \ldots\ldots (2)$$

In the above equations, g and f are the notations for function. In order to attain equilibrium, following condition must be attained.

$$D = S \quad \ldots\ldots (3)$$

Here, D is total demand and S is total supply. Hence, from the above equations, three unknown D, S and P are derived. When these equations are solved then the value of these three unknown will be derived.

3.4.2.1. Changes in Market Equilibrium

The change which takes place in market equilibrium are mentioned below:

1) **Increase in Demand:** The DD is the actual demand curve and the SS is the actual supply curve in **figure 3.12**. Both demand and supply curve meet each other at equilibrium point OP. If income of the buyer increases, then demand curve will move to D_1D_1, but supply OQ will remain same. On new demand curve *i.e.*, D_1D_1, at OP point demand will reach to OQ_2 where supply is still the same *i.e.*, OQ. As supply is less than demand, price will increase upto point OP_1. Further, with higher price, supply will also increase and new point of equilibrium will be attained. At this equilibrium, OP_1 is price and OQ_1 is quantity demanded and supplied.

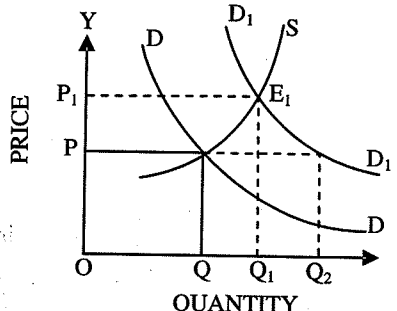


Figure 3.12: Increase in demand, causing an increase in equilibrium price and quantity

2) **Decrease in Demand:** In case, the demand decreases due to decrease in income of the buyer, then demand curve will move to left where supply curve will remain same. At new demand curve, at price OP, quantity demanded will be OQ_2 and quantity supplied will be OQ. When supply is more and demand is less, then the price will decrease and new equilibrium is developed at OP_1, where demand OQ_1 and supply OQ_1 will be equal.

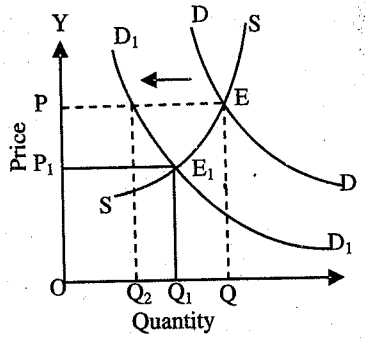


Figure 3.13: Decrease in Demand resulting in a decrease in price and quantity demanded

3) **Increase in Supply:** Let's assume that the demand remain constant while there is increase in supply due to improvement in technology. There will be shift in the supply curve from SS to S_1S_1. Further, at original point of equilibrium *i.e.,* OP, the quantity demanded will be OQ and quantity supplied will be OQ_2. As supply is more than quantity demanded, the price of equilibrium will reduce and it will become OP_1 where quantity OQ_1 will be demanded and supplied.

4) **Decrease in Supply:** In case due to any reason, the quantity supplied decreases, the equilibrium price will improve and amount purchased and sold will reduce.

3.4.2.2. Simultaneous Shifts of Supply and Demand

Practically in market, there are various forces which simultaneously affect the demand and supply. In this case, both the demand curve and supply curve can change their position simultaneously. Here, the new point of equilibrium may be either less than, more than, or equal to the initial point of equilibrium. This greatly depends on the intensity of shift of the curves. In case, demand curve and supply curve shift in equal ratio then the point of equilibrium shifts to right in same ratio. Therefore, equilibrium point remains the same.

This phenomenon is explained in **figure 3.14.** At point P_0 the initial price level will remain same because of the change in both demand and supply curves by the shift of equilibrium position from E_0 to E_1. In case, there is higher shift in demand curve then supply curve, the point of equilibrium price will rise. As shown in **figure 3.15** the shift of supply is less than shift in demand curve. Therefore, there is a change in equilibrium point from E_0 to E_2 and the resultant changes in level of price and quantity will be P_0P_1 and Q_0Q_1.

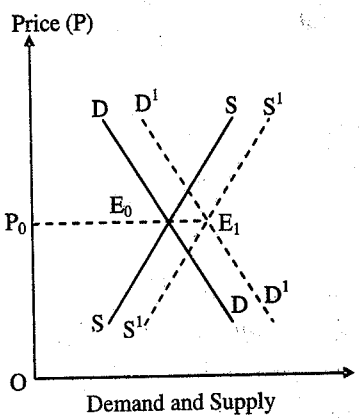


Figure 3.14: Shift in both Demand and Supply

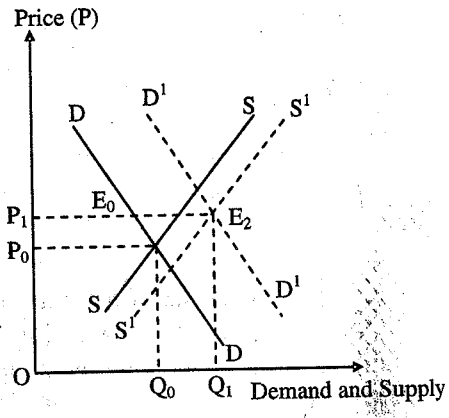


Figure 3.15: Increased Shift in Demand than Shift in Supply

There will be rise in the level of equilibrium price, if the demand curve moves to right and supply curve moves to left. Further, this rise in price may or may not result in change in quantity. If change in demand and change in supply is equal, then there will be no change in the level of price. Additionally, if there are more shifts in demand curve than in supply curve, there will be increase in equilibrium. Therefore, through the demand curve and supply curve, we not only derive the equilibrium price and equilibrium quantity, but also we derive the change in price and change in quantity demand.

3.4.3. Price of a Product under Demand and Supply Forces /Demand & Supply Equilibrium

Figure 3.16 shows a graphical representation of equilibrium price determination. Plotting the schedules of demand and supply on similar price and quantity axes gives the demand and supply curves, i.e., DD′ and SS′ respectively. The equilibrium price in **figure 3.16** lies at ₹300 as the demand curve and the supply curve intersect each other at point 'E'.

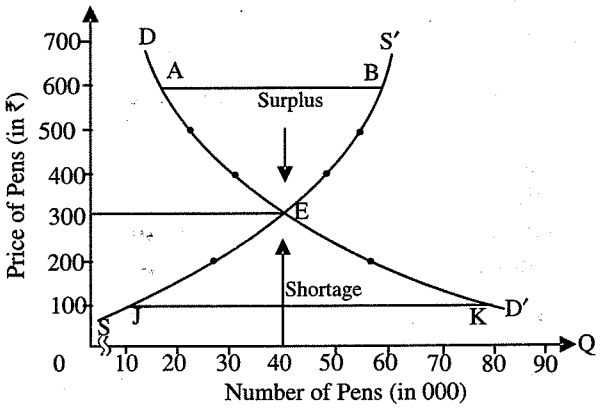


Figure 3.16: Equilibrium of Demand and Supply. Price Determination

At point 'E', the demanded quantity is equivalent to the supplied quantity, i.e., 40,000 pens. Therefore, the equilibrium quantity here would be 40,000 pens and the equilibrium price would be 300. As shown in the figure, the two curves do not cross each other at any other point except 'E', and therefore no other equilibrium condition is possible.

3.4.4. Calculation of Demand-Supply Equilibrium

After understanding the graphical determination of equilibrium through demand and supply curves, the next concept is calculation of equilibrium price and quantity in terms of algebra. This can be calculated when the demand functions as well as supply functions are available. In the following illustration, a commodity 'X' and its demand and supply functions have been assumed in order to determine the equilibrium price and quantity.

Let the demand function for commodity X be given as:

$Q_d = 150 - 5P_x$

and supply function as:

$Q_s = 10P_x$

It is known that demand and supply equilibrium is achieved when the supplied quantity is equal to the demanded quantity, i.e., where

$Q_s = Q_d$ .....(1)

On replacing supply and demand functions into equation (1), we get

$10P_x = 150 - 5P_x$ .....(2)

On obtaining the equation (2), the equilibrium price can be calculated as follows:

$10P_x = 150 - 5P_x$ .....(3)

or, $10P_x + 5P_x = 150$

$15P_x = 150$

$P_x = 10$

Thus, at equilibrium, i.e., $P_x = 10$, the equilibrium price is ₹10.

The equilibrium quantity demanded and the equilibrium quantity supplied can conveniently be calculated once the equilibrium price, i.e., $P_x = 10$ has been obtained. Equilibrium quantity demanded equals:

$150 - 5P_x = 150 - 5\ (10) = 100.$

Similarly, equilibrium quantity supplied can be calculated as:

$10\ (P_x) = 10\ (10) = 100.$

The equilibrium price and quantity determination in terms of algebra can be graphically represented as shown in **figure 3.17**. The demand function ($Q_d = 150 - 5P_x$) and the supply function ($Q_s = 10P_x$) have been used to

obtain the demand curve (DD′) and the supply curve (SS′) respectively. Point 'P' is the intersecting point of demand and supply curves. On drawing a perpendicular from 'P' to both price and quantity axes, equilibrium price at ₹10 and equilibrium quantity at 100 units can be obtained respectively. This means that at point 'P', the quantity supplied is equal to the quantity demanded.

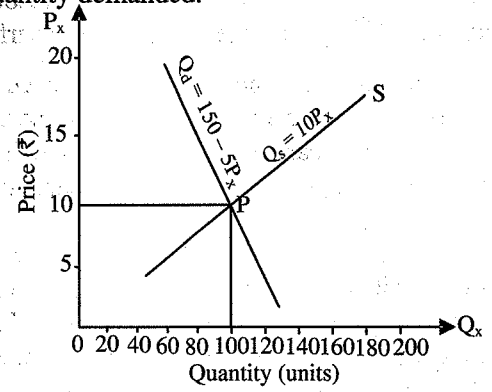


Figure 3.17: Determination of Equilibrium Price and Quantity

3.4.5. Uses of Equilibrium

The concept of equilibrium has significant uses in economic analysis. These are as follows:

1) **Visualisation of Ideal Situation:** In economic generalisations, visualising an ideal and desirable situation is a necessary step. Equilibrium helps in envisioning this ideal situation in which:
   i) All the economic resources of an individual, household, an organisation, or the entire economy are utilised to the maximum,
   ii) All the produced units are consumed, and
   iii) Production, employment and prices are stable.
2) **Study of Behaviour of Economic Units:** Equilibrium assists in behavioural analysis of economic units like producers, consumers, etc., by bringing change in one of the significant variable or by presenting a new variable which disturbs the equilibrium state. How quickly the economic variables are changing and the direction of change can be anticipated by the analyst through a dynamic analysis. He/she can also gain information regarding the equilibrium being re-claimed or not.
3) **Comparative-Static Analysis:** The future values likely to be obtained by economic variables after the equilibrium shift as a result of an independent change in one of the variables, can be predicted by comparing two different equilibrium positions.

3.5. PRODUCTION ANALYSIS

3.5.1. Introduction

The **theory of production** focuses to show the link between the input and output factors of production of the various goods and services. In order to have the best possible output, the production analysis focuses only on the operational efficiency rather than the cost of production even though it is considered as an essential factor for the production.

3.5.2. Meaning and Definition of Production

In economics, **production** means the procedure with the help of which the natural resources are transformed into effective produce which results into the satisfaction of human wants. Production can be said as an organised activity of converting raw material into finished goods and services and thereby it helps in creation or addition of the utility. Utility denoted the ability to satisfy the wants of the consumer by consuming the various amount of goods and services. In the production process, the creation of utility takes place in regard with the form utility, place utility and time utility.

According to James Bates and J.R. Parkinson, "Production is the organised activity of transforming resources into finished products in the form of goods and services; and the objective of production is to satisfy the demand of such transformed resources".

According to J.R. Hicks, "Production in any activity directed to the satisfaction of other people's want through exchange".

3.5.3. Factors of Production

The factors which are used in the process of production are as follows:

1) **Land:** In economics, the word **land** has been used in different form. The word land consists of all the natural resources which are present in the atmosphere like natural resources, fertility of soil, water, air, natural vegetation, etc., rather than considering only the soil or earth's surface itself.

2) **Labour:** The term **'labour'** means mental or physical exertion directed to produce goods or services. In other words, it refers to various types of human effort which require the use of physical exertion, skill, and intellect. Labour, to have an economic significance, must be one, which is done with the motive of some economic reward. **Division of labour** is an important feature of modern industrial organisation.

   It refers to a scheme of dividing the given activity among workers in such a way that each worker is supposed to do one activity or only a limited and narrow segment of an activity. When one worker does one activity again and again, he becomes an expert in it and specialises in it. The efficiency of individual worker is enhanced by means of greater specialisation. Division of labour increases output per worker on account of higher efficiency, specialised skill, and on account of repetitive work on a given task or a part of it.

3) **Capital: Capital** refers to the wealth of the person which helps in generating additional wealth. Capital includes both the concept i.e. stock concept and flow concept. It includes stock concept due to the reason that the initial investment is done for the production and includes flow concept because it helps in generating income from the finished goods.

   Capital is defined as "produced means of production". As both land and labour are not produced ones so it is differentiated from the capital with the help of the above definition. They are considered as the primary factors but the capital is the formed by the human being by performing the various tasks in the nature. Hence, capital is said as the man-made tool of the production.

4) **Entrepreneur: Entrepreneur** is the person who starts a business but in economics term, he utilizes the other factors of productions like land, labour and capital in the correct quantity and then starts the production process. He is also responsible to take the risk which arises during the production process. An entrepreneur is called by different names such as the organiser, the managers or the risk taker. In present scenario, with the emergence of innovative entrepreneur the work of the organiser and the manager seems different as compared to the entrepreneur.

3.5.4. Production Function

The **production function** describes the functional relationship between the input and the output. In short, production function refers to the aggregate of the quantities of output which is produced from the given set of inputs under the specified level of technological factors. Here, the inputs are the various factors of production like land, labour, capital and entrepreneur and the outputs are the quantity of the produced goods and services. Symbolically, the production function is written as:

$$Q = f(X_1, X_2, X_3 \ldots\ldots X_n)$$

Where,
Q = Quantity produced during a given period of time
$(X_1, X_2, X_3 \ldots\ldots X_n)$ = Quantities of various inputs used in production

In other words, the production function means the function which depicts the least possible quantities of various inputs that are needed to produce the specified quantity of output.

3.5.4.1. Assumptions of Production Function

The assumptions of the production function are as follows:

1) The input and output can be perfectly divided.
2) It is not necessary that one factor is substituted for the other.
3) The state of technology does not change.
4) In short-run the supply of fixed factors is inelastic.
5) Production function is associated with the given period of time.

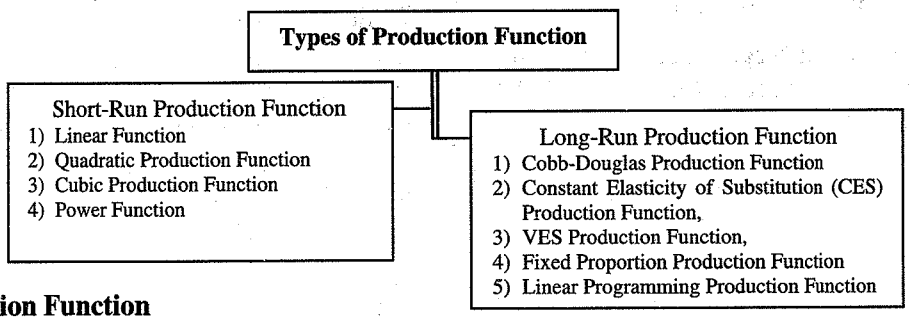
3.5.4.2. Importance/Applications of Production Functions in Managerial Decision-Making

In managerial decision-making the production function analysis is of great importance. The managerial usage of Production function analysis is as follows:

1) Production function analysis aid the manager in taking decisions related to the short period by the following two methods:
   i) The decision related to the method adopted by which best possible output can be generated from the available input.
   ii) The decision related to the method adopted to attain the specified output from the availability of less input.
2) The decisions which are taken for the long period of time makes utilization of production function analysis. In case, there is rise in returns to scale than it would be advisable to raise the level of production. It is better to reduce the level of production if there is decrease in the returns to scale and when there is no change in the returns to scale then it is difficult for the producer to take the decision that either to rise or reduce the production level when there is no restriction on the demand.
3) In order to compute the minimum and maximum cost from the combination of several factors of input from the particular output level, the production function analysis is beneficial.
4) Production function analysis is rational and practical in the approach. In case, there is reduction in the price of one input factor whereas the price of other factor input increases then it is assumed to substitute one factor for the other.
5) There is a major significance of production function analysis in taking decision related to the proper allocation of the variable input of factor in the process of production.

3.5.4.3. Types of Production Function

The production function analysis can be divided as follows:



Short-Run Production Function

Short-Run production function is divided into the following:

1) **Linear Function: Linear production function** can be expressed as mentioned below:
   Total product: $Y = a + bX$
   It is concluded from the above function that the average product can be written in the form of the following equation:
   $$\frac{Y}{X} = \frac{a}{X} + b$$
   The marginal product can be written in the following form of equation as:
   $$\frac{\Delta Y}{\Delta X} = b$$
2) **Quadratic Production Function: Quadratic production function** can be expressed as follows:
   $Y = a + bX - cX^2$

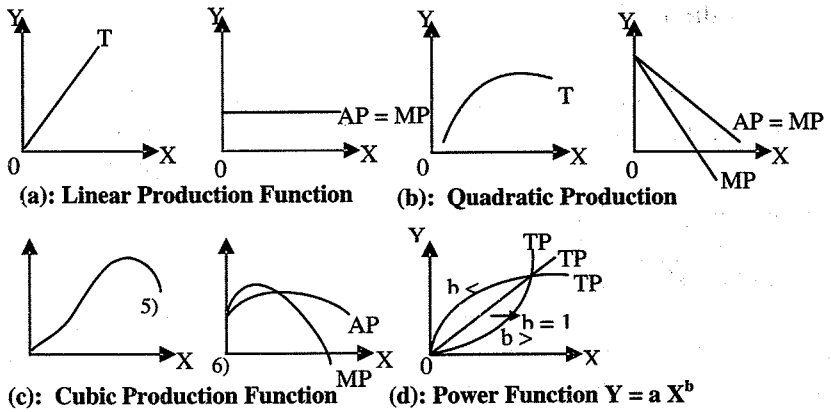
   Where,
   Y= dependent variable representing total output,
   X= independent variable representing total input, and
   a, b and c= constraints whose values are determined by statistical analysis of data.
3) **Cubic Production Function: Cubic production function** can be mathematically expressed as:
   $$Y = a + bX + cX^2 - dX^3$$

The cubic production function has essential properties which are as follows:

i) The rise and fall of marginal productivity is allowed.
ii) At every point, the elasticity of production changes along the curve.
iii) In the later stages, the marginal productivity reduces at a faster rate.

4) **Power Function:** A **power function** expresses output, Y, as a function of input X in the form

$Y = aX^b$



(a): Linear Production Function (b): Quadratic Production

(c): Cubic Production Function (d): Power Function $Y = a X^b$

Figure 3.18: Short Run Production Function

Long-Run Production Function

Long-Run production function is divided into the following:

1) **Cobb-Douglas Production Function: Cobb-Douglas production function** was developed by **Charles W. Cobb and Paul H. Douglas** who analyses the relationship between inputs and outputs. This is known as Cobb-Douglas production function. The C-D production function was initially applicable in the whole of the manufacturing production rather than the production of the individual firms.

The Cobb-Douglas production function can take the following form:

$$Q = AL^{\alpha} K^{\beta}$$

Where, Q = Output
L and K = Inputs of labour and capital respectively
A, α and β = Positive parameters where $\alpha > 0$, $\beta > 0$.

The above mentioned equation expresses that the output is directly based on the labour (L) and capital (K) and A which is the 'residual' includes the output which is not described by L and K and is normally known as technical change.

The marginal products of labour and capital are the functions of the parameters A, α and β and the ratios of labour and capital inputs, i.e.

$$MP_L = \frac{\partial Q}{\partial L} = \alpha AL^{\alpha-1} K^{\beta} \qquad MP_K = \frac{\partial Q}{\partial K} = \beta AL^{\alpha} K^{\beta-1}$$

In C-D production function the degree of the homogeneity is measured by two constraints i.e. α and β which are taken simultaneously. This function describes the returns to scale as:

$\alpha + \beta > 1$: Increasing returns to scale
$\alpha + \beta = 1$: Constant returns to scale
$\alpha + \beta < 1$: Decreasing returns to scale.

Properties of Cobb-Douglas Production Function

The Cobb-Douglas production function has the following properties:

1) There are constant returns to scale.
2) Elasticity of substitution is equal to one.
3) α and β represent the labour and capital shares of output respectively,
4) α and β are also elasticities of output with respect to labour and capital respectively.
5) If one of the inputs is zero, output will also be zero.
6) The expansion path generated by C-D function is linear and it passes through the origin.

7) The marginal product of labour is equal to the increase in output when the labour input is increased by one unit.
8) The average product of labour is equal to the ratio between output and labour input.
9) The ratio α/β measures factor intensity. The higher this ratio, the more labour intensive is the technique and the lower is this ratio and the more capital intensive is the technique of production.

Importance of Cobb-Douglas Production Function

Cobb-Douglas production function is most popular in empirical research. The reasons for this are many:

1) The Cobb-Douglas function is convenient for international and inter-industry comparisons. Since α and β (which are partial elasticity coefficients) are pure numbers (i.e., independent of units of measurement) they can be easily used for comparing results of different samples having varied units of measurement.
2) Another advantage is that this function captures the essential non-linearities of production process and also has the benefit of the simplification of calculations by transforming the function into a linear form with the help of logarithms. The log-linear function becomes linear in its parameters, which is quite useful to a managerial economist for his analysis.
3) In addition to being elasticities, the parameters of Cobb-Douglas function also possess other attributes. **For example,** the sum of $(\alpha + \beta)$ shows the returns to scale in the production process; α and β represent the labour share and capital share of output respectively, and so on.
4) This function can be used to investigate the nature of long-run production function, *viz.*, increasing, constant and decreasing returns to scale.
5) Although in its original form, Cobb-Douglas production function limits itself to handling just two inputs (e.g., L and K), it can be easily generalised for more that two inputs, like

$$Q = AX_1^a \cdot X_2^b \cdot X_3^c \ldots . X_n^P$$

Where, Q = Output
$X_1, X_2, \ldots X_n$ = Different inputs

Criticisms of Cobb-Douglas Production Function

1) The function includes only two factors and neglects other inputs.
2) The function assumes constant returns to scale.
3) There is the problem of measurement of capital which takes only the quantity of capital available for production.
4) The function assumes perfect competition in the factor market which is unrealistic.
5) It does not fit to all industries.
6) It is based on the substitutability of factors and neglects complementarily of factors.
7) The parameters cannot give proper and correct economic implication.

2) **Constant Elasticity of Substitution (CES) Production Function:** The **constant elasticity of substitution production function** was established by **Arrow, Chenery, Minhas and Solow.** This function includes three variables i.e. Q, K and L and three constraints i.e. A, α and θ. The CES production function is also called as **Homohighplagic production function**. It can be written as:

$$Q = A[\alpha C^{-\theta} + (1 - \alpha)\ L^{-\theta}]^{-1/\theta}.$$

Where,
Q = total output,
K = capital,
L = labour,
A = efficiency constraint,
α = distribution constraint or capital intensity factor coefficient, and
θ = substitution constraint.

The efficiency constraint symbolizes the state of technology and organizational aspects of production. It means that the efficiency constraint causes change in the production function with the change in the technology and organizational factor. Thus, it can also be expressed as:
$A > 0; 0 < \alpha < 1; \theta > -1.$

3) **Variable Elasticity Substitution (VES) Production Function:** In **VES production**, it is supposed that the substitution constraint (α) become variable than the variable elasticity of substitution production function come into effect. This substitution constraint arises from the CES production function and as VES production function is expressed in the same way as CES, hence, both these functions have same properties and drawbacks. It has an additional benefit that it is not required to fix a priority value of elasticity of substitution.

Properties of VES Production Function

The properties of VES production are as follows:

i) VES fulfils the conditions of neo-classical production function,
ii) The fixed co-efficient models are contained in the VES production function, and
iii) VES production function is common.

4) **Fixed Proportion (or Leontief) Production Function:** The **fixed proportion production function** depends on the requirement that in the production process, the several factors of production which are to be taken into consideration must be used in fixed proportion. In case larger quantity of input is available than it is needed in this proportion then the extra quantity will not be required and when the inputs are available in smaller quantity than required then the other inputs will not be required. **For example,** if in a process 20 labour and 8 units of capital are needed and if the firm has 24 labour and 8 units of capital then it finds that 4 labour units is not utilized whereas when there are 18 labour units and only 6 capital units are being utilized then 2 capital units will be treated as jobless. It can be expressed as follows:

$$X = \text{minimum}\left(\frac{K}{a}, \frac{L}{b}\right)$$

Where,
K and L = units of capital and labour respectively,
a and b = constants

The word 'minimum' means that output X is based on the smaller of the two ratios (given in the bracket).

5) **Linear Programming Production Function:** The inputs in terms of factors of production are not completely fixed and also that they do not change constantly. Hence, the input and output both changes in a number of way. This is seen by linear programming function and it is expressed as follows:
Maximize $X = a_1 X_1 + a_2 X_2 + ... + a_n X_n$

Where,
$X_1, X_2, ..., X_n$ = outputs
$a_1, a_2, ..., a_n$ = constants

It is concluded from the above equation that the firms availability helps in the determination of some input and capacity parameters according to the budget.

3.5.4.4. Estimation of Production Function

Example 1: Mr. Rohan is a senior professor in a university and is having an attractive collection of cards. He wants to earn money by selling his cards. Three of his friends are ready to buy his cards. He finds out the individual demand equations of three of his friends (1,2,3) as given below:

$Q_1 = 60.00 - 2.00P$
$Q_2 = 45.00 - 1.50P$
$Q_3 = 75.00 - 2.50P$

Give the market demand equation of Mr. Rohan's card and the number of cards he would be able to sell for each one rupee decrease in price if each card. If he has 120 cards, what price should he charge to sell his entire collection?

Solution: Market demand Qm, is the sum of individual demands. Thus,
$Q_m = Q_1 + Q_2 + Q_3 = (60.00 - 2.00P) + (45.00 - 1.50P) + (75.00 - 2.50P)$
Simplifying Qm = 180.00 – 6.00P

Because P is measured in rupees, a rupee decrease in price will increase quantity demanded by three. To sell the entire 60-card collection, the price must be set such that Qm = 120. That is, 120 = 180 – 6.00P.
On solving, we get
P = 10.00

Example 2: Let the production function of a firm be $Q = 25, L^{1/2} K^{1/2}$. Find out the maximum possible output that the firm can produce with 100 units of L and 100 Units of K.

Solution:

$Q = 25,$
$L = 200$ units,
$K = 200$ units,

Production Function $= 25L^{1/2} K^{1/2}$
$= 25 \times 100^{1/2} \times 100^{1/2}$
$= 25 \times \sqrt{100} \times \sqrt{100} = 2{,}500$ units

Example 3: The production function for the firm is as follows:

$Q_x = 5K + 10L$

Find the total output of K and L that a firm can produce if:

i) K= 2 and L= 4, and
ii) When the inputs are doubled

Solution:

i) K= 2
L= 4
$Q_x = 5K + 10L$
$= 5(2) + 10(4)$
$= 10 + 40 = 50$ units

ii) When the input i.e. K and L is doubled than
K= 4
L= 8,
$Q_x = 5K + 10L$
$Q_x = 5(4) + 10(8)$
$= 20 + 80 = 100$ units

3.5.4.5. Limitations of Production Function

Production function analysis has the following limitations:

1) Production function analysis has restricted itself to the case of two inputs and one output. Mathematically, there is no great difficulty in extending this analysis to multiple inputs and outputs.
2) Production function analysis has assumed smooth and continuous curves, while in the real world, discontinuities in the production function may appear.
3) Production function analysis assumes that technology remains constant. But in reality, it does not remain the same.
4) Production function analysis is also applied under perfectly competitive market situations which arc rare in the real world.
5) Production function analysis assumes that units of labour are homogeneous. In reality, labour units are not identical but heterogeneous in characters.

3.6. COST ANALYSIS/ COST OF PRODUCTION

3.6.1. Concept

The study of the behaviour of the cost with respect to several criteria of production such as size of output, scale of operations, prices of factors of production and other relevant economic variables is known as **cost analysis**. Thus, under given production relations dealing with the behaviour of cost is referred as cost analysis. In other words, the consideration of cost analysis is the behavioural aspect of cost under certain given production relations in contrast to those physical aspects being considered in production analysis.

3.6.2. Meaning and Definition of Cost

The expenses incurred in the business activity of supplying goods and services to consumers are defined as **cost**. In economics, the value of the price of an object or condition is the cost of production which is determined by the total cost of resources employed for producing it. The composition of the cost is the factors of production that includes labour, land, capital and entrepreneur as well as taxation.

According to Campbell, "Production costs are those which must be received by resource owners in order to assume that they will continue to supply them in a particular time of production".

According to Gulhrie and Wallace, "Cost of production has a special meaning. It is all of the payments or expenditures necessary to obtain the factors of production of land, labour, capital and management required to produce a commodity. It represents money costs which we want to incur in order to acquire the factors of production".

Hence, the total sum of money required for producing a specific quantity of output is known as cost.

3.6.3. Types of Costs

The important types of costs are as follows:

3.6.3.1. Private Cost and Social Cost

The costs which are incurred by a firm in producing a commodity or service are known as private cost and this includes all the actual costs-both explicit and implicit. It is a cost incurred by the producer in the process of production which includes anticipated tax and profit margins. The costs which are paid by the firm for purchasing capital equipment, hire labour and buy materials or other inputs are the private costs that are acquired by the producer of good, service or an activity. These private costs must be included in production and consumption decisions being paid by the firm or the consumer.

The involvement of **social costs** on the society during the time of production in the forestry sector taking the form of pollution; damage to the elasticity of forest ecosystems itself and other ecosystems, deforestation, natural resource depletion, etc. The possible valuation method and other methods which are used to value non-market goods and services is the source of estimating the value of this social cost.

3.6.3.2. Accounting Cost and Economic Cost

Those costs which are recorded into the books of accounts when they are actually incurred are known as **accounting cost**. The procedure of accounting is based on the growth concept. Thus, the actual incurred costs are recorded into the books of accounts. The records of the accounts are based on the legal purity of the business transaction. These costs are explicit and must be paid. Assets which are the parts of the accounting cost are valued at book value.

Explicit and implicit costs are the components of the **Economic cost**. In other words, recorded and unrecorded costs are included in the economic cost. Assets are valued at replacement cost, i.e., market value in case of economic cost.

3.6.3.3. Short-Run Cost and Long-Run Cost

The size of the firm being the same, **short-run cost** varies with the variation in the output. Thus, shot-run cost is same as variable cost. When atleast one of the factors of production is completely fixed then the time period is the short-run as defined by the economists. **For example,** for a particular company this might mean that they have reached full capacity in a warehouse or at a factory site. These short-run costs consist of both fixed and variable costs.

Throughout the process of production, short-run costs increases in real time, whereas, fixed cost is not affected by short-run cost. The short-run production is affected only by variable costs and revenues. The change in the output affects the change in the variable costs. **For example,** the increase or decrease of short-run costs is based on variable cost as well as the rate of production. If a firm manages its short-run costs well over time, it will be more likely to succeed in reaching the desired long-run costs and goals.

Besides, **Long-run costs** are the costs which are incurred on the fixed assets like plant, building, machinery, etc. The implication of these costs is for long-run period as they are not being used-up in the unit batch of production. In contrast, it is defined by the economist that the long-run is the period where all the factors of production can be changed and hence the above example clearly depicts that without any problem, the company can look forward to expand its warehouse or the capacity of the factory. In response to expected economic profits or losses there is an accumulation of long-run costs when the production levels of the firm change over time.

Land, labour, capital goods, and entrepreneurship all differ to reach the long-run cost of producing a good or service. This time period is the planning and implementation stage for producers who analyse the current and the estimated state of the market to make relevant production decisions. Efficient long-run costs are maintained when the combination of outputs that a firm produces results in the expected quantity of the goods at the lowest possible cost. The long-run decision examples by which the firm's cost is affected including the change in the quantity of production, decreasing or expanding a company and entering or leaving a market.

3.6.3.4. Opportunity Cost and Actual Cost

The loss of earnings due to lost opportunities is known as **opportunity cost**. These opportunities are abandoned due to lack of resources. Unlimited resources lead to no opportunity. Thus, the opportunity cost may be defined as the loss of expected returns from the second use of the resources foregone for availing the gains from their best possible use. This is also known as **alternative cost** because opportunity cost is the expected income from the second best alternative use of resources.

The economic significances or importance of opportunity cost are mentioned below:

1) **Estimation of Relative Price of Goods:** The concept of opportunity cost helps to determine the relative price of various goods. **For example,** in case a given factors produces one sofa and two chairs in a given period of time then the price of one sofa will be equal to the twice the price of chair.

2) **Fixation of Remuneration to a Factor:** The concept of opportunity also helps to determine the price of one factor. **For example,** in case an insurance agent gets an opportunity to work as an electronic sale executive at ₹20,000 per month then insurance company needs to pay atleast ₹20,000 to the agent in order to retain the employee.

3) **Proper Allocation of Resources:** The concept of opportunity also helps to allocate resources effectively and efficiently. **For example,** in case an organisation can produces either one dining table or two sofas with the available resources during a period of time. The price of dining table is ₹12,000 whereas price of one sofa is ₹4,000. Therefore, organisation will decide to produce the dining table because the price of one dining table is more than the combined price of two sofas.

   Hence, by producing one dining table organisation will get ₹12,000 whereas by producing two sofas it will only get ₹8,000. The expenses **incurred actually** on labour, material, plant, building, machinery, equipment, travelling and transport, etc., are known as **Actual cost.** The actual cost is recorded for all practical purpose in the books of accounts as total money expenses and thus, comes under the purview of concept of accounting.

3.6.3.5. Other Types of Cost

1) **Business Cost and Full Cost:** The total expenses incurred in carrying out the business are known as **business cost.** This concept of business cost is synonymous to actual cost or real cost. The payments and contractual obligations that are made by the firm in addition to the book cost of depreciation on plant and equipment are included in the business cost.

   The total of business cost, opportunity cost and normal profit are included in the concept of **full cost.** Whereas the expected earnings from the best available resources, the market rate of interest on the total money capital as well as the value of the entrepreneur's own services being unchangeable in the current business are included in the opportunity cost. Thus, in addition to the opportunity cost, normal profit is the least necessary earning for a firm to remain in its present occupation.

2) **Direct Cost and Indirect Cost:** The unit of operation having a direct relationship with cost incurred is known as **direct cost** which is directly and easily identifiable or are associated to a particular product, operation or plant.

For example, for a branch being a costing unit and the salary of its branch manager are the direct cost. The direct cost are entered directly into the cost of production but are identified separately. The cost that are incurred on electricity, stationery and other office expenses like depreciation on building, decoration expenses, etc., whose source cannot be easily located and definitely traced to a plant, a product, a process or a department are known as **Indirect cost.**

3) **Incremental Cost and Sunk Cost:** The **incremental cost** adds up to the existing cost and are associated with marginal batch of output. They are the additional costs resulting from a transition in nature and level of business activity. **For example,** change in product line or output level, adding or replacing a machine, changes in distribution channels, etc. The costs which are unaltered by change in the level or nature of business activity are called **Sunk cost.** These costs cannot be decreased or increased by altering the activity level or the output rate and all past or actual costs are termed as sunk cost.

4) **Explicit Cost and Implicit Cost:** The payments that are made to the factors hired from outside, the command of the firm are termed as **explicit cost**. The entrepreneur who makes monetary payment for the various productive factors being purchased or hired which do not belong to him. Rent, wages, interest, salaries, payment for raw materials, fuel, power, insurance premium, etc., are the examples of explicit costs.

   The self-owned resource used in production for which the payment is made refers to the **implicit cost** which is the earnings of the owner from self-employed resources in their best alternative utilisation. **For example**, a businessman utilises his services in his own business leaving his job as a manager in a company. Thus, he foregoes his salary as a manager. This loss of salary becomes an implicit cost of its own business.

5) **Historical Cost & Replacement Cost:** The **historical cost** is the actual cost of an asset incurred at the time when asset was acquired. It means the cost of a plant at a price originally paid for it. In contrast, **replacement cost** means the price that would have to be paid currently for acquiring the same plant. So, historical costs are the past costs and replacement costs are the present costs. Price change over time cause a difference between historical cost and replacement cost. **For example**, suppose that the price of a machine in 1995 was ₹1,00,000 and its present price is ₹2,50,000, the actual cost of ₹1,00,000 is the historical cost while ₹2,50,000 is the replacement cost.

6) **Past Cost and Future Cost:** The cost that is out of the control of the management as being incurred already is described as **past cost**. The evaluation of these costs can be done with re-collective effect.

   Besides, **future cost** are those that are expected to be incurred in future. They are involved in forecasting the control of expenses, appraisal of capital expenditure, decisions on new projects as well as expansion programs and profit-loss projections through proper costing under assumed cost conditions.

7) **Shut-down Cost and Abandonment Cost:** The costs which are incurred in the situation of closure of the plant operations are known as **shut-down cost**. These costs can be saved if the operations are continued. All type of fixed costs, the costs of sheltering plant and equipment, lay-off expenses, employment and training of workers when the operation is re-started are included in the shutdown cost.

   On the other hand, the costs which are incurred due to the retiring of a plant from use are known as **abandonment cost.** These costs are associated with the problem of disposal of assets. **For example**, the costs are related to the discontinuance of diesel auto in Delhi.

8) **Real Cost and Prime Cost:** The costs which are actually incurred by the payment of labour, material, plant building, machinery, equipments, travelling and transport, etc., are called **real or actual cost**. The actual costs are the total money expenses which are recorded in the books of accounts. Actual cost comes under the accounting concept. On the other hand, the expenses which are directly associated to the production of a commodity are called **prime cost**. It consists of direct material cost, direct labour cost and other direct expenses.

   Prime Cost = Direct Material + Direct Wages + Direct Expenses

9) **Urgent Cost and Postponable Cost:** The costs which are required for the continuation of the firm's activities are called urgent costs. The cost of raw materials, labour, fuel, etc., is the suitable example for the urgent cost because they must be incurred if production is to take place. If it is possible that the costs can be postponed for sometime and this postponement does not affect the operational efficiency then it is called **postponable cost. For example**, maintenance costs which can be postponed for the time-being. This distinction of cost is very useful during war and inflation.

10) **Escapable Cost & Unavoidable Cost:** The costs which can be reduced by contraction in business activities are called escapable cost. Indirect effects are very difficult to estimate such as the closure of an unprofitable business unit which reduces the costs but increase the other related expenses such as transportation charges, etc. On the other hand, the costs which do not vary with change in the level of production are called **unavoidable cost** and they are unavoidable such as fixed costs.

11) **Out-of-Pocket Cost and Book Cost:** The items of expenditure which involve cash payments or cash transfers of both recurring and non-recurring nature are known as out-of-pocket costs. All the explicit costs (e.g., wages, rent, interest, transport expenditure, etc.) fall in this category. Besides, there are those actual definite business costs not including cash payments. Hence, an arrangement is made in the books of accounts and is therefore, accounted when finalising profit and loss account. These kinds of expenses are termed as **book cost.** Thus, alternatively, they are the payments made by the firm to self. The example of book costs are; depreciation allowances and the interest that is unpaid on the owner's fund.

12) **Fixed Cost and Variable Cost:** For a certain given output, the costs which are fixed in volume are called fixed costs. If there are variation in the output between zero and the given level of output then fixed cost do not vary. Hence, the costs which do not vary for a certain level of output are called **fixed cost.** Fixed costs include:
   i) Cost of managerial and administrative staff,
   ii) Depreciation of machinery, building and other fixed assets, and
   iii) Maintenance of land, etc.

   The costs that change with the change in the total output are known as **variable cost**. The cost of raw materials, running cost of fixed capital, such as fuel, ordinary repairs, routine maintenance expenditure, direct labour charges associated with the level of output, and the costs of all other inputs that vary with output are included in **variable cost**.

Difference between Fixed Cost and Variable Cost

Basis of Difference	Fixed Cost	Variable Cost
1) **Meaning**	The cost which remains same, regardless of the volume produced, is known as fixed cost.	The cost which changes with the change in output is considered as a variable cost.
2) **Examples**	Depreciation, Rent, Salary, Insurance, Tax etc. are the examples of fixed cost.	Material Consumed, Wages, Commission on Sales, Packing Expenses, etc. are the examples of variable cost.

13) **Total, Average, and Marginal Cost:**
   i) **Total Cost (TC):** For the production of goods and services the required total value of resources represents the **total cost.**
   ii) **Average Cost (AC):** The **average cost** is obtained by dividing the Total Cost (TC) by the total output (Q), i.e.,

$$AC = \frac{TC}{Q}$$

   iii) **Marginal Cost (MC):** The **marginal cost** is the cost which is associated with the production of additional one unit. It may be defined as:

$$MC = \frac{\Delta TC}{\Delta Q}$$

14) **Money Cost and Selling Cost:** The **money cost** of a certain quantity of commodity can be calculated by adding the market price paid to the factors of production employed in the production of that amount of the commodity. In other words, the amount spend to produce a commodity is called its money cost. **For example,** the cost of producing 100 mobile phones is ₹20, 000 then it will be money cost of producing 100 mobile phones.

   The money cost includes the following:
   i) Price of all kinds of raw material.
   ii) The wages and salary paid to the workers.
   iii) Interest on capital, rent and other trade expenses.
   iv) Normal profit of the enterprise.

The concept of money cost is very important because money cost of production is determined by the output of the firm or industry. In order to sell their products, firms under monopolistic competition have to spend a lot on advertisement and publicity. In Economics, the total amount of money spent on advertisement and publicity for pushing the sale of the product is called **selling cost**. When a monopolist begins the production of a product, he may spend some amount on advertisement by way of information to the potential customers. Therefore, selling Costs refer to those costs which are incurred to increase the sale of the product. **For example,** expenditure on advertisement, publicity, salesmen, commission to retailers, gifts and concessions to customers, etc.

3.6.4. Determinants of Cost

The following are the determinants of cost:

1) **Law of Returns Operating:** The law of returns operating is an important determinant of cost. The cost will tend to rise when the law of diminishing returns operate and reversely the cost tends to decrease when the law of increasing returns operates.
2) **Size of the Plant:** The size of the plant greatly affects the cost. When the size of the plant is bigger then the initial fixed costs will be high whereas variable cost will be low with a smaller size plant.
3) **Period:** The behaviour of cost is affected by the period of operation. For short period operations there will be speedy rise in the cost curve but the speed will be lower when the period is longer under consideration. The long-run cost curve engulfs several short-run curves as an envelope.
4) **Capacity Utilisation:** The level of capacity utilisation affects cost as per unit fixed cost makes a big difference as the higher capacity utilisation fixed cost per unit of output is always low.
5) **Prices of Factors of Production:** The prices of the factor of production affect the cost of product whereas the impact of the price of the factor of production is dependent upon the contribution made by that factor of production to the total product i.e., to say the value of a given input is related to the total cost of the product.
6) **Technology:** The cost of a product is greatly influenced by technology as these innovations of technology aspire to reduce the costs.
7) **Efficiency in the Use of Inputs:** The efficient utilisation of the inputs greatly affects the cost as those inputs which are cheaper as well as equally efficient regarding the quality of the product is concerned.
8) **Lot Size of the Product:** Lot size of the product affects the cost. The unit cost in producing bigger lot at one time and the total cost of operation will be lower than to process lot of smaller size of product.
9) **Output is Stable and Constant:** When the output is constant and stable over a period of time then the overall costs are generally lower and when the production is disrupted and broken suddenly the cost are higher.

3.6.5. Cost Function

The cost function is a mathematical relationship between cost of a product and the various determinants of costs where dependent variable is unit cost or total cost and the independent variables are the price of a factor, the size of the output or any other relevant phenomenon, which has a bearing on cost.
Symbolically we may state the function as:

$$C = f(O, S, T, P...)$$

Where,
C = Cost
O = Level of output
S = Size of plant
T = Time under consideration
P = Prices of factors of production

3.6.6. Cost Curves

The graphical presentation of costs of production as a function of total quantity produced is called the cost curve. The cost curves are used by the firms which are productively efficient in a free market economy to locate the optimum point of production by reducing the cost.

The cost curve is used by the firms to decide output quantities whose objective is to maximise profits. All types of costs curves are interrelated such as total and average cost curve and marginal cost curves, which are equal to the differential of the total cost curves. In **figure 3.19,** the short-run Marginal Cost (MC) curve initially decline and then goes up at some point and intersects the average total cost and average variable cost curves at their minimum points.

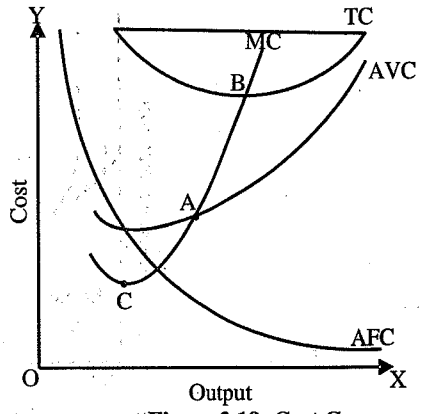


Figure 3.19: Cost Curve

The Average Variable Cost (AVC) curve goes down (but comparatively not as sharp as the marginal cost and then goes up. This does not go up as fast as the marginal cost curve. The Average Fixed Cost (AFC) curve will decline as the additional units are produced and continue to decline. The average total cost (ATC) curve initially will decline as fixed costs curves spread over a large number of units but will go up as marginal cost increases due to the law of diminishing returns.

3.6.7. Uses of Cost Analysis in Managerial Decision-Making

Following are the uses of cost analysis in managerial decision-making:

1) **Product Costing and Pricing Decisions:** For setting the prices of product or services the cost analysis is used in managerial decision-making. Several pricing strategies like cost-based pricing, market-based pricing, target pricing and others are applied by business organisations and sometimes these organisations adopt the strategy of diverse pricing. Under the situation when a firm faces a high price-sensitive customers then it can reduce the unit cost and alternatively, spreading the fixed cost over the volume of high output by adopting penetration pricing strategy for fulfilling the objective of profit maximisation.

2) **Cost Management:** The diversified problems both internal and external are being faced by the managers in running their organisations like tendency of inflexibility in the sales prices, organised employees demanding higher wages and other benefits such as increase in taxes and imposition of new regulations by the government. Thus, these situations make managers realise to control the cost by reducing it to earn regular and unhindered profits.

3) **Profit Planning:** Profit planning cannot be done without cost analysis. This requires determination of a solid operating plan of the business organisation for the future operating period and briefing its summary by presenting in the form of projected income statement. The costs are major inputs in profit planning and cost analysis. The final decision on the optimum operational activity level is based on the interrelated impact of cost, volume and profit which is to be understood by the management.

4) **Capital Investment Decisions:** The cost-benefit analysis is the simultaneous analysis of both the cost and the benefits which are basically two major items. The receiving of these benefits are often overtime. Thus, this cost-benefit analysis involves financial costs and financial benefits.

5) **Marketing Decisions:** Marketing decisions provide useful information which is the basis of cost analysis required for planning and controlling the marketing costs to evolve an appropriate marketing strategy to sell the product based on the margins contributed to the total profit of the company. Relating cost of the marketing activities to sales revenue is cost analysis by constructing a profit-loss statement for analysing any marketing component. For this, it is required to divide the cost of the firm into their functional categories which are further assigned within the appropriate classifications of marketing.

3.7. ECONOMIES OF SCALE AND DISECONOMIES OF SCALE

3.7.1. Introduction

Important information about the processes of production followed by firm for manufacturing of goods is conveyed by the shape of the long-run average total cost curve. **Economies of scale** is when output increases leading to declining of long-run average total cost. Similarly when output increases leading to rise of long-run average total cost, it is known as **diseconomies of scale.** These economies and diseconomies of scale are shown in the **figure 3.20**:

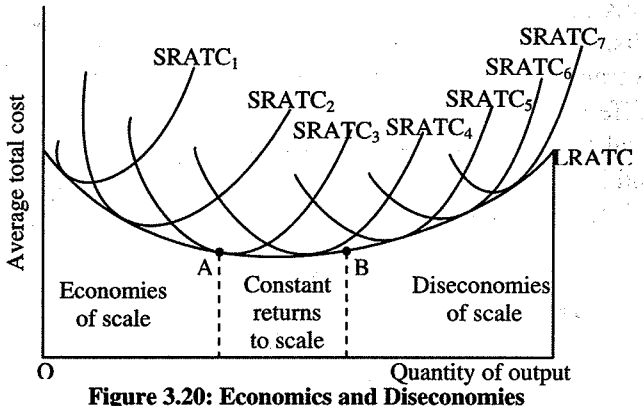


Figure 3.20: Economics and Diseconomies

U-shape of the long-run average total cost curves can be explained by this analysis. It shows that at low production levels, the firm attains benefits due to increased size as at this time greater specialisation can prove to be helpful. Meanwhile, problems regarding coordination are not yet acute. Contrary to this at high production levels, the firm have already taken advantage of specialisation and also problems about coordination become simpler with large firm. Thus, falling of long-run average total cost can be observed at low production level due to specialisation increase and rising of long-run average total cost can be seen at high production level due to coordination problems increase.

3.7.2. Meaning and Definition of Economies of Scale

Advantages arising because of production at large scale are termed as Economies of Scale which are basically economies of production at large scale. Economies of scale can be referred to the increasing efficiencies notion of goods production with increasing number of goods produced. Average production cost of goods typically diminishes alongwith additional production. This is because, fixed costs of production are shared over the additional goods produced. In micro-economics, economies of scale are termed as the advantage in terms of cost that is attained by business because of expansion.

There are certain factors that help to reduce the producer's average cost per unit with an increase in scale of output. "Economies of scale" can be explained as deduction in cost per unit as the facility size and level of usage of other inputs increase and it is a concept of long-run. Because of economies of scale, companies that are large do impart great market access in terms of media selection to access those markets, and also large companies can capture larger geographical area. Returns to scale are also determined by the economies of scale.

According to Porter, Economies of scale is the "declines in the unit's cost of production".

According to Pratten, Economies of scale is the "reduction in average unit costs attributable to increases in the scale of output".

According to Spencer, "Economies of scale is a curvilinear relationship between average cost and the number of units produced".

3.7.3. Meaning of Diseconomies of Scale

Diminishing returns to scale can be observed due to continuous expansion of the firm size. When the firm's expansion goes beyond a level, growing diseconomies can be encountered. Economies of production at large scale are cancelled by these diseconomies leading to a rise in the average production cost. Diseconomies of scale are not produced generally due to technical factors rather these technical factors more likely do 'limit' the sources of scale economies. In case, inefficiencies are caused because of overlarge size of plant, this situation needs to be avoided. Replication of plant units of smaller size may solve the problem in such cases. Most commonly diseconomies of scale generates because of problems regarding behavioural and human aspects of large enterprise management.

3.7.4. Kinds of Economies and Diseconomies of Scale

Classification of the economies of scale and diseconomies of scale can be done as follows:

1) Internal economies of scale and Internal diseconomies of scale, and
2) External economies of scale and External diseconomies of scale.

3.7.4.1. Internal Economies and Diseconomies of Scale

There is increase in the return to scale in initial stages, it remains constant for a while and finally decreases. Now the question arises that why there is an increasing return to scale which leads to fall in cost and why after a certain point there is decrease in the returns to scale which leads to cost rise.

Answer of these questions is that a firm initially enjoys internal economies of scale which after a certain limit turns into internal diseconomies of scale and company suffers from it. Major kind of internal economies and diseconomies of scale are as follows:

1) **Technical Economies and Diseconomies:** Technical economies deals with the production at large scale. With the increase in firm's scale of operations, more efficient and specialised form of all factors can be used especially machinery and capital requirement. Generally more efficient machinery is available for higher level of production that is employed so as to decrease the cost per unit of output because of larger output at once. This is result of capital equipment indivisibility.

   If there could be perfect division of capital equipments, required quantity of same would have been bought and also production of smallest output of any commodity would have been possible with large scale production machinery. However, net diseconomies of scale can be experienced by a firm beyond a certain point and this occurs because after reaching a large sized firm so as to utilise the machinery, any further size increase of the plant will lead to high long run cost due to management difficulties. Proper coordination and control becomes difficult for management when there is too large scale of operations in the firm.

2) **Managerial Economies and Diseconomies:** Managerial cost reduction is basically known as the managerial economies. With the increased output, management can be invaded with the labour division. Various departments like production can be handled by the production manager, sales can be handled by the sales manager, and finance department can be handled by the finance manager. But with further increase in the scale of production, there can be further sub-division of each department, e.g., sales department can be divided into sections for exports, customer service and advertising.

   However, with an increase in scale of production beyond a certain limit, managerial diseconomies takes place leading to difficulty in exercising control and coordination of various departments. Also there is much difficulties in the managerial structure and it gets affected by more red-tapism, more bureaucracy, lengthening of communication lines, and so on. They affect the productivity and efficiency of the management and firm.

3) **Commercial Economies and Diseconomies:** Big volume of goods production needs enough components and material. Due to this, a firm is capable of placing bulk order for components and materials that becomes available to them at low prices. The product selling can also help in achieving economies. In case, when the sales staffs do not work upto their capacity, a little extra cost can be charged for selling the additional output. Also economies of advertising can be beneficial for the large firms.

   This happens because with an increase in scale of production, costs of advertising per unit of output fall. In addition to this, by-products may also be sold by the large production which is generally unprofitable for a small firm. Finally, in a sufficiently large business, labour division introduction can be done on the commercial side, with employment of expert sellers and buyers. After an optimum scale, economies of scale become diseconomies of scale. **For example**, expenses on advertising and other marketing overheads increases after optimum scale more than a proportional relation.

4) **Financial Economies and Diseconomies:** Large firms do favour the raising finance for expansion. **For example,** better security to bankers can be offered and as per the well-known fact of raising money at lower cost when confidence of investors lies in it and prefer share that can be easily sold on the stock exchange. However, after the optimum scale of production, more proportional rise in the financial cost can be observed. This happens due to external finances dependence relatively more as compared to other factors.

5) **Risk Bearing Economies and Diseconomies:** Economies of risk bearings are enjoyed more by the large firms. This is because they have diverse multi-production capability and can easily withstand economic ups and downs. However, in case of diversification, risk may be increased if it magnifies disorder rather than providing a cover to it.

3.7.4.2. External Economies and Diseconomies of Scale

External economies and diseconomies are those economies and diseconomies that accrue to firms due to output expansion of whole industry and these aspects are independent of the individual firm's output level. They do not accrue to firms out of internal situations. Instead they deal with the outside situations and thus known as the external economies and diseconomies of scale that is the industry expansion. Available external economies to the firm are as follows:

1) **Cheaper Raw Materials and Capital Equipment:** Industry expansion results into exploration and awareness of new and cheaper raw material sources, machinery and various other capital equipments. Industry expansion demands several kinds of material and capital requirement to run smoothly. Thus, there arises need to purchase it from other industries on a large scale leading to reduction the production cost and ultimately prices. Thus, a lower price is to be paid by the firms using these materials and capital equipments.
2) **Technological External Economies:** With the expansion of whole industry, discovery of new technical knowledge is also attained and this further needs use of better and improved machinery than the prior one. This lead to change in the technical production coefficient and also firm's productivity is enhanced and reducing the production cost.
3) **Development of Skilled Labour:** Industry expansion in an area needs the well accustomed labour in that area so as to perform various productive processes. Also a good deal is learnt by the experience. This results in pooling of trained labour with the industry growth. This labour pool is equipped with the required skills that have favourable impact on the productivity level and firm's cost in that particular industry.
4) **Growth of Ancillary Industries:** A number of ancillary industries can attain specialisation in raw material production, tools and machinery production, etc., alongwith the growth of an industry. These ancillary industry tends to provide raw materials, tools, etc., at lower price to main industry. Similarly, some firms may process the industrial waste products to form the useful product from which it reduces the production cost in general.
5) **Better Transportation and Marketing Facilities:** The industry expansion that is result of a new firm's entry may possibly provide better transportation and marketing facilities to a great extent. This greatly reduces the production cost of firms. Similarly, modernisation of the communication system may result in speedy and better information.

3.8. COST-OUTPUT RELATIONSHIP

3.8.1. Introduction

In order to regulate and control production cost, one needs to properly understand the behaviour and nature of costs. Money forces are the basis of production cost along with an understanding of the functional cost relations with several forces which ultimately helps to make decisions. Another crucial cost influencing factor is Output.

In order to determine the optimum production level, there is a need to study the cost-output relationship which plays a vital role. It is beneficial and helpful to the manager with regard to the profit prediction, cost control, promotion and pricing, etc. Technically, the relation between the cost and its determinants is termed as the cost function. The cost function of the firm is determined by the combination of production function along with prices of inputs.

The cost function can further be classified based on the period of cost function as:

1) Short run cost function, and
2) Long run cost function.

As per economic theory, the period during which the firm's physical capacity is fixed and the output can be increased only by the usage of the existing capacity in more intensive way is termed as the short run cost function. Whereas, the long run cost function is the one in which firm's capacity can be increased or firm's size can be reduced, if trade is very bad.

3.8.2. Cost-Output Relationship in the Short Run

In short-run, the cost-output relationship is defined as a particular operation scale or a fixed plant. That means there is an indication towards cost variation over the plant output of a given capacity and, thus, there is variation in their relationship based on the varying capacity of the plant. In order to make decision, one needs to understand the total cost and output relationship along with knowledge of various costs and output types separately. Thus, discussion of the short-term cost-output relationship can be carried out in terms of following:

1) Marginal cost and output,
2) Average cost and output, and
3) Total cost and output.

3.8.2.1. Short-Run Total Cost

Total cost in short-run can be divided into two parts which are Total Fixed Costs (TFC) and Total Variable Costs (TVC) keeping the organisation scale fixed:

TC = TFC + TVC

1) **Total Fixed Cost or TFC:** Cost of production that does not change with respect to the output is termed as **Total Fixed Cost.** These are the costs that are totally independent of the output level. These costs infact are incurred even during temporary production stop by the firm. The costs which are included in this case such as land and building rent, insurance charges, depreciation, interest on borrowed money, property tax, wages and salaries of permanent staff, maintenance expenditure, etc. These costs are also known as the **overhead costs.**

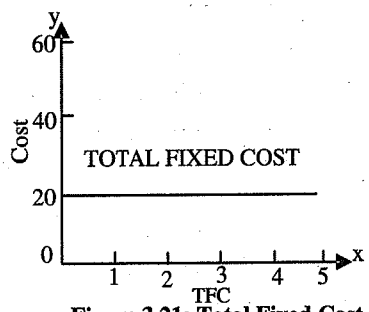


Figure 3.21: Total Fixed Cost

TFC = Quantities of the Fixed Productive Service × Factor Price

Illustration of a firm's TFC is given in the following diagram and table:

Units of Output	TFC (₹)
0	20
1	20
2	20
3	20
4	20
5	20

Total Fixed Cost curve is basically a horizontal curve which lies parallel to the X-axis representing that TFC at all output levels tends to remain same.

2) **Total Variable Cost or TVC:** The production cost that is in direct relationship with the output is termed as **Total Variable Cost.** There is rise in TVC with increase in the output and falls with decrease in the output. Costs included in this type are expenses on water, raw materials, taxes, advertising, power, labour hiring, etc. Such costs are also termed as **direct costs.**

The cost incurred on using the production's variable factors is termed as the Variable Costs.

According to Ferguson, "Total variable cost is the sum of amounts spent for each of the variable inputs used".

TVC = Quantities of the Variable Productive Service × Factor Price

Units of Output	TVC (₹)
0	0
1	18
2	30
3	40
4	52
5	65
6	82
7	106
8	140

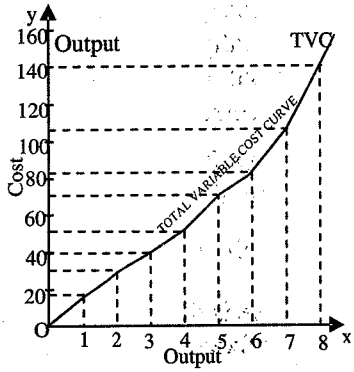


Figure 3.22: Total Variable Cost

Direct relationship of the change in TVC along with output volume is shown in the above table and diagram. Starting of the TVC curve is from the origin which moves in concave way from below upto a certain range and later it becomes convex in shape. This represents that the total variable cost rises in the beginning at a diminishing rate and later rises at an increasing rate.

3) **Total Cost or TC:** Total cost means the firm's total expenses that are made in order to produce a commodity in a given quantity. The expenses which are included are payment for wages, rent, taxes, interest and expenses on water, electricity, raw materials and advertising etc. The TFC, TVC and TC relationship can be easily illustrated by the following diagram and table:

Example

Units of Output	TFC (₹)	TVC (₹)	TC (₹)
0	20	0	20
1	20	18	38
2	20	30	50
3	20	40	60
4	20	52	72
5	20	65	85
6	20	82	102
7	20	106	126
8	20	140	160

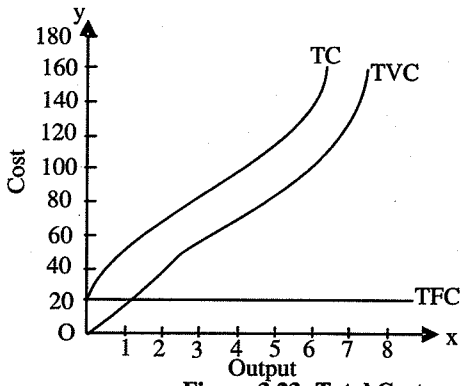


Figure 3.23: Total Cost

TFC (Total Fixed Cost) curve in the **figure 3.23** lies parallel to x-axis. The origin of this curve is from a point on y-axis which clearly indicates that fixed cost will be incurred even in case when there is no output. On the other hand, there is an increase in the total variable cost with an increase in the output as shown by the rise in TVC curve. Origin of this curve is from zero which shows that variable costs are nil when the output is zero. TC curve is obtained by simply horizontal addition of TFC curve and TVC curve.

3.8.2.2. Short-Run Average Cost Curves

Average costs are generally more important as compared to total costs in case of short-run analysis. The output units which are produced by a firm do not cost the same amount to the producing firm but their selling price must be same. Therefore, the average cost or per unit cost must be known by the firm. In a short-run, the average cost of a firm basically consists of three parts such as average variable cost, the average fixed cost and average total cost.

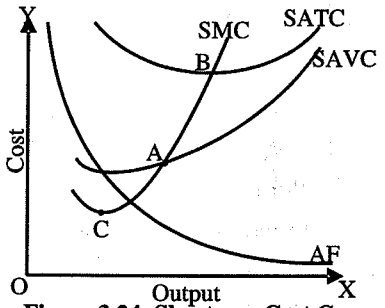


Figure 3.24: Short run Cost Curve

1) **Average Fixed Cost or AFC:** AFC can be calculated as Total Fixed Cost (TFC) at each level divided by Q i.e., the units produced. This is given as:

$$AFC=\frac{TFC}{Q}$$

There is continuous reduction in the average fixed costs along with increase in the output. This is much natural as the continuous diminishing average fixed cost is resulted by dividing the constant total fixed cost by a continuously increasing output unit. That's why the slope of AFC curve is in downward direction that approached the quantity axis but do not touch it, as clearly represented in **figure 3.24.** This curve is basically a rectangular hyperbola, as on choosing any point on the AFC curve along with corresponding output quantity produced by that, the product comes out to be same always. The reason behind this is that the corresponding output quantity leads to yield TFC which is throughout constant.

2) **Short-Run Average Variable Cost (or SAVC):** AVC at short run can be calculated by simply dividing the Total Variable Cost by the units produced:

$$SAVC=\frac{TVC}{Q}$$

There is decline in the AVC initially as output increases because larger variable factors are applied to equipment and fixed plant. But due to the law of diminishing returns, there seems to be a rise eventually. Thus, the shape of the Average Variable Cost curve at short term duration is U-shaped as represented in **figure 3.24.**

3) **Short-Run Average Total Cost (or SATC or SAC):** Short-run average total cost is basically the average cost of any output that is produced by the firm. Calculation of SATC is done by simply dividing the Total cost at each output level divided by the produced units which is given as:

$$\text{SAC or SATC} = \frac{TC}{Q} = \frac{TFC}{Q} + \frac{TVC}{Q} = AFC + AVC$$

ATC shows the influence of both parts, i.e., the average variable costs and the average fixed costs. Initially the ATC are high at low output levels. This is due to the large AFC and AVC values. But with the increase in the output, ATC leads to a sharp fall due to steady decline of both AFC and AVC till reach the minimum point. This happens because of the internal economies like optimal utilisation of labour, existing plant, etc. The lowest point in **figure 3.24** shows the optimal capacity represented by the point B. With increase in production beyond this point, there is quick rise in the ATC due to fall in AFC that is negligible in relation to the rise in AVC. From production of the above capacity, SAC curve have a rising portion which is also resulted from the management diseconomies, labour, etc. Thus, the shape of SAC curve is U-shaped.

Why is the Short-Run Average Cost Curve 'U'-Shaped?

The shape of AC curve in short-run is U-shaped which means there is initial downward fall till reach the minimum point and then there is upward rise in the curve. There are mainly two ways to explain the U-shape of the average cost curve which are as follows:

1) **Interaction of Average Fixed Cost and Average Variable Cost:** Aggregate of Average Fixed Cost and the Average Variable Cost respectively is termed as average cost (AC). AFC falls with an increase in the production. During production, initially there is fall in AVC. Thus, with AFC and AVC, aggregate of these two, i.e., AC also tends to decrease and reaches minimum point as depicted in the **figure 3.24.** There is fall in the AC curve up to point A, which is the minimum point. The firm is utilising its full capacity at this situation. The output by the firm is optimal. Optimum point means the output level corresponding to the lowest per unit production which is shown in **figure 3.24** at point A.

   While there is production by the firm beyond the point A, undoubtedly there is continuous fall in AFC but there is rise in AVC leading to ultimate rise in the Average Cost (AC). This is due to the rate of increase in AVC is much more as compared to the AFC decreasing rate after reaching the minimum point. As a result, the net effect obtained is upward rise of the AC curve. This shows clearly that the AC curve being aggregate of both AFC and AVC is a U-shaped curve that is it falls initially till reaches the minimum point and then rises rapidly.

2) **Application of the Law of Variable Proportions:** As per explanation which is already described for a short-run, if there is a combination of the variable factors with that of fixed factors then there is an increase in the output as per the law of variable proportions. In case of combining the variable and fixed factors, initially the fixed factors are used more efficiently and this leads to overall fall in the average cost. This is clearly depicted in the **figure 3.24** that before the minimum point A, output follows the law of diminishing costs or increasing returns.

   Whereas, the output is optimum and the average unit cost is minimum at the point 'A'. It is proof of the fact that the fixed factors of the production are being optimally utilised till this point. Thus, beyond this point, where the fixed factors are being optimally utilised, increasing variable factors use will further lead to fall in their ratio comparatively to the fixed factors. This results in the production's increased rate which beings to diminish. It indicates that law of increasing costs or law of diminishing returns sets in leading to upward rise in the average cost curve beyond point 'A'.

3.8.2.3. Short-Run Marginal Cost

The marginal cost is the fundamental concept that is used for exact output level determination of a firm. Marginal cost is basically variation in the total cost that is caused because of production of the additional output unit. This is given as:

$$SMC = \frac{\Delta TC}{\Delta Q}$$

Mathematically, Marginal cost is the n + 1 units Total Cost minus n units Output Cost ($MC_n = TC_{n+1} - TC_n$). Since there is no change in the TFC with respect to output, Marginal fixed cost tends to be zero at this point. So, the calculation of the marginal cost can be done with the help of TC or TVC. In both the cases, the resulted marginal cost comes same. Alike TVC or TC, Marginal cost also behaves same that is it also falls first and then rises. Thus, the shape of the SMC curve as depicted in **figure 3.24** is also U-shaped.

It is clear from the **figure 3.24** that the marginal cost curve is U-shaped. This shows that MC curve slops downward initially till reaches the minimum point and then it starts rising upward. This variation is because of the MC decreases with the low levels of the output till reaches the minimum point, and then there is increase further with increase in the output. The shape of marginal cost curve is in fact attributable to the law of variable proportions. As per the law of variable proportions, at low levels of output, the marginal product of the variable input rises and later falls with output expansion. Hence, there will be initial fall and then rise in the marginal cost curve.

Conclusion

Thus, in short-run, there are several cost curves of a firm which includes the AFC curve, SAVC curve, SMC curve and the SAC curve. Out of these four curves, the AFC curve is insignificant for determining the exact output of the firm and thus it is neglected.

Why is MC Curve U-Shaped?

Marginal cost is defined as the cost incurred due to change in variable or total cost because of producing one additional output unit. Initially on increasing the production, there is an increase in the variable and total cost at a diminishing rate. This is due to the law of increasing returns which is in operation in initial production stage where several economies are enjoyed by the firm. This results in lesser cost of each additional unit as compared to the earlier produced units.

This leads to fall in the Marginal Cost curve. Rate of increase in the variable and total costs is minimum after a given output quantity. At this point, MC is also minimum. After this, there is an increase in the variable and total cost at an increasing rate. This is due to the law of diminishing returns that operates at this stage. Many diseconomies are faced by the firm at this stage. During this stage, cost of each additionally produced unit is higher comparative to earlier produced units. This leads to pushing of the marginal cost as well. In brief, there is fall in the MC till it reaches the minimum point and later it rises. This shows why the curve is 'U' Shaped for marginal cost.

3.8.2.4. Interrelationship between Average Cost and Marginal Cost

The relationship of Average and Marginal Cost plays a vital role in price theory. The whole marginal analysis of product pricing depends upon this interrelationship. Following are some of AC and MC relationships:

1) MC is less than AC, while AC falls due to output increase.
2) MC is more than AC, while AC rises due to output increase.
3) MC is equal to AC, while AC reaches the minimum point that is the optimum point.

Table 3.1 and figure 3.25 helps to explain the relationship between AC and MC in a better way. This is described as follows:

Table 3.1: Relation between Average Cost and Marginal Cost

Unit of Output	Total Cost(₹)	Average Cost (₹)	Marginal Cost(₹)
1	150	150.0	–
2	190	95.0	40
3	220	73.3	30
4	236	59.0	16
5	270	54.0	34
6	324	54.0	54
7	415	59.3	91
8	580	72.5	165

Some points representing the AC and MC relationships are immediately clear in the **table 3.1,** which is as follows:

1) Total production cost helps to calculate both AC and MC. Derivation of both is from the common source.

$$\text{Average Cost} = \frac{\text{Total Cost}}{\text{Total Output}}$$

$$\text{Marginal Cost} = \frac{\text{Change in Total Cost}}{\text{Change in Unit of Output}}$$

Inclination of the TC curve is shown by the AC curve over the output axis. Slope of TC curve shows the MC at a particular output level. Both of these values, AC and MC can be easily obtained from the TC curve.

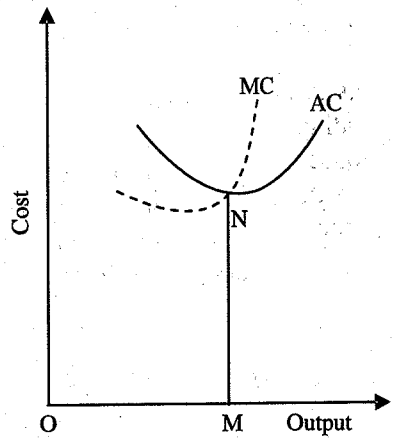


Figure 3.25: Relationship between Average Cost and Marginal Cost

2) The MC is always lower than the AC curve while there is fall in the average cost. As per a common view, there is faster fall in MC when AC falls. However, this is not the case throughout. After reaching the minimum level, there is rise in MC even if the AC is falling. The only thing to be guaranteed is that the MC lies below AC as long as AC is falling.
3) On rise of Average Cost, Marginal Cost lies above AC and there is faster rise compared to AC at that point. MC is greater than AC and also rises rapidly comparatively to AC while AC is rising.
4) There must be an intersection of MC and AC curve at AC's minimum point. The derivation of this relationship is carried out from the fact that MC = AC when AC is kept constant. This relationship is clearly depicted in the **figure 3.25.**

The AC curve is again U-shaped as shown in **figure 3.25.** This is due to the operations of the variable proportions law in short period. The dotted curve shown is the MC curve which is also U-shaped. Both curves intersect at the AC's minimum point from which a perpendicular is drawn to the x-axis giving the shortest value of minimum point. **Figure 3.25** shows that the OM is output produced at lowest average cost.

3.8.3. Cost-Output Relationship in the Long-Run

Long-run stands for the time period in which the entire productive factors of a firm's production become variable. Combination of many short-runs may be described as long-run. There are no fixed production factors in the long-run. Due to this all costs are variable and there is no fixed cost at all. A short-run may be characterised by a particular plant size. There is a short-run cost curve that exists for every plant size for the producer.

Once a particular plant size is chosen by the producer then he chooses a short-run cost curve automatically. In the short-run, producer has to operate on the short-run cost curve chosen by him instead of the output level. There are many short-run cost curves available in a long-run cost curve. From these short run curves, producer can choose one and may also move from one to another curve. For each output level, he finds that short-run cost curve which will result in the lowest production cost for that output level.

The inputs combination is used by the firm in long-run which minimises the production cost at desired output level. The firm identifies the type and size of equipment, raw materials, plant size and the labour skill, and these combinations gives the maximum output at lowest cost, considering the methods of production used and the available technology used. As choices of inputs are made for the production of desired output level, in long-run all the inputs are variable.

3.8.3.1. Long-Run Total Cost

There are two components in a short-run total cost, i.e., variable and fixed cost. As we know that fixed cost does not change with the output level and it depends on the firm structure. In case of a small structure that attains capacity at a lower level of output, fixed cost will be less but in case of large structure fixed cost comes out to be large as compared to short-run. Variable Cost follows the law of variable proportions in short-run, and thus becomes an inverted 'S' shaped curve.

There are three Short-run Total Cost Curves in the **Figure 3.26** that are $SRTC_1$, $SRTC_2$, and $SRTC_3$. A smaller plant size is represented by the $SRTC_1$ that is basically a small structure. Similarly, a relatively larger plant size is represented by the $SRTC_2$ which is comparatively smaller as compared to $SRTC_3$.

A producer has to choose any of these plants in the short-run and corresponding SRTC curve is considered accordingly. Suppose if producer chooses the curve $SRTC_2$ then production is done only on this curve and it is not affected by the output level. But he can switch from one SRTC curve to another in case of long-run. For each output level, obviously producer will select that plant which produces at lowest cost. In other words, a producer will operate on that SRTC which will give him the lowest cost of production.

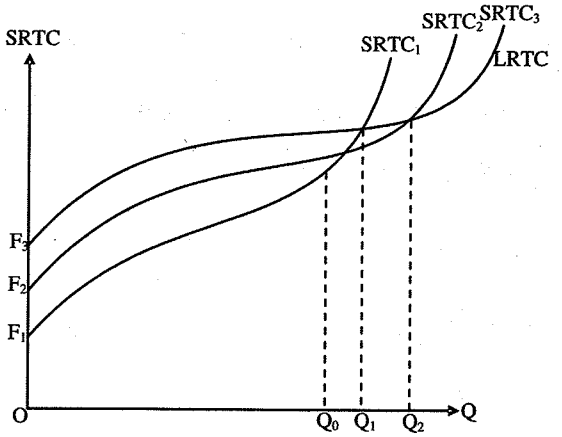


Figure 3.26: Long-Run Total Cost Curve

$SRTC_1$ gives the least production cost till the output level Q_1. Beyond that, from Q_1 to Q_2, the least production cost is given by the curve $SRTC_2$ followed by the $SRTC_3$. So, the production will be done at plant 1 upto the point Q_1, then operation on plant 2 will be done till the point Q_2 that is the production is between the points Q_1 and Q_2. If the production is more than Q_3, plant 3 will be chosen by the producer.

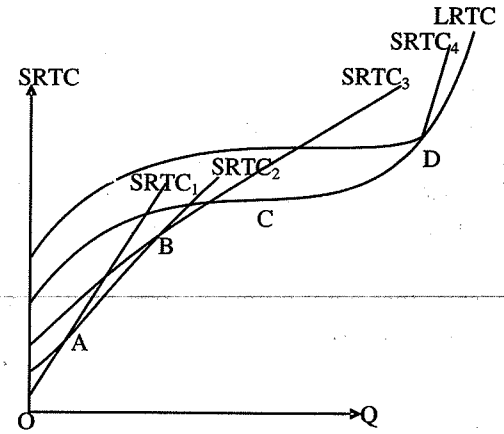


Figure 3.27: Long Run Total Cost Curve from Short-Run Cost Curves

It is assumed in the economic analysis that there are infinite sizes of the plant and thus there can be infinite SRTC curves are available to the producer. For each output level, a producer will find out the size of plant which leads to production at least cost. That producer selected that particular plant and corresponding SRTC will be followed.

Locus of all these points on SRTC curve is termed as LRTC curve i.e., Long-run cost curve. For each output level, one point of LRTC will be chosen from one SRTC curve which shows the least production cost among all of the SRTC curves. That means SRTC curves lies above the LRTC curve because in case if SRTC lies below to LRTC, for that particular level of output, lower cost is shown by the SRTC as compared to LRTC which is impossible to exist. Such a LRTC curve is drawn in the **Figure 3.27**. All relevant SRTC curves are touched by the LRTC curve from below and such a curve is known as Envelope Curve.

LRTC curve is made with the use of many SRTC curve and is flatter as compared to SRTC curve. As per observation, LRTC curve only touches relevant SRTC curve from below and lays above all the SRTC curves. That's why any tangent drawn at SRTC curve will possess same slope as that of LRTC. At all other output levels, SRTC slope is maximum in comparison to that of the LRTC. In relation to LRTC, it is applicable to SRTC curves. Therefore, LRTC curve is flatter as compared to that of SRTC curves. It is noticeable that any lower envelop curve in tends to be flatter as compared to the curve which it is enveloped.

3.8.3.2. Long-Run Average Cost

Long-run average cost curves are also known by the following names:

1) **Envelope Curve:** This curve encloses all short-run average cost curves and hence it is termed as "envelope curve". Main implication of this envelope curve is that long run average cost cannot exceed short-run average cost. Since, in the long-run, indivisible factors are fully utilised and thus the long-run average cost curve will surround the Short Run Average Cost (SAC). This will not intersect the SAC curves while rising in upward direction.
2) **Planning Curve:** Another name of the long run average cost curve is planning curve. With the use of this curve a firm can plan that which plant should be used to produce different output quantities. This helps in minimisation of the production cost.

The derivation of the long-run average cost is done by combination of the short-run average cost curves. With each short-run total cost curves, there is one SRAC in association. Long-run average cost is corresponding to the long-run total cost. As per observation, LRTC curve is usually in inverted 'S' shaped and the corresponding LRAC curve will be 'U' shape. LRAC curve can be easily obtained if one knows the LRTC curve.

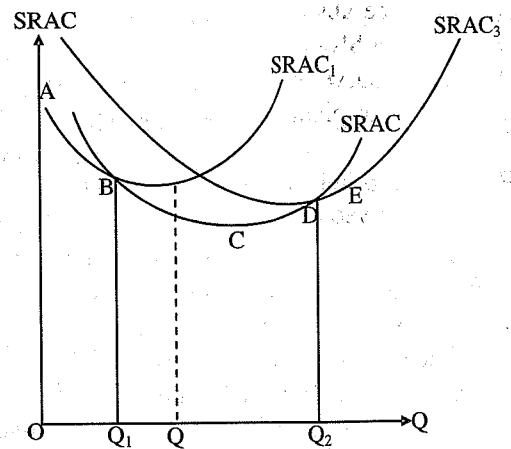


Figure 3.28: LRAC Curve from SRAC Curves

As per discussion in SRTC and LRTC curves, there is relation of SRTC curves corresponding to SRAC curve that is there is one SRAC curve for each plant size. Fixed cost is lower corresponding to the lower plant size. So, the output production capacity of a plant is at a lower volume and thus at the lower output level the SRAC curve arrives the lowest point. While deriving the LRTC curve from the SRTC curve, 3 plants are considered which are represented as $SRAC_1$, $SRAC_2$ and $SRAC_3$.

Now, $SRAC_1 < SRAC_2 < SRAC_3$ is the output up to the point Q_1. So the production is carried out on $SRAC_1$ at plant 1. Likewise, $SRAC_2$ is lowest between the points Q_1 and Q_2. And thus $SRAC_2$ is the relevant curve and after particular point Q_3, the curve that becomes relevant is $SRAC_3$. Here, ABCDE is the LRAC and also it is noticeable that LRAC may not consider the SRAC's lowest point. The lower average cost is considered for each output level. **For example,** output $SRAC_1$ reaches its lowest at point Q but $SRAC_1$ is not relevant for that output because at that point output will be $SRAC2 < SRAC_1$. So, for that output level, LRAC is made up of $SRAC_2$ as shown in the **figure 3.28**.

We move to infinite SRAC curve from a limited number of SRAC curves. Many SRAC curves combine to form a LRAC curve, those SRAC curves are merged which tends to result in lowest cost for different output levels. Therefore, LRAC curve is formed by combination of least cost resources for all output levels. For each output level, one SRAC is chosen that is lowest among all available SRACs. There are seven such SRAC curves in the **figure 3.28** which are $SRAC_1$, $SRAC_2$, ..., $SRAC_7$. Points A, B, C, D, E, F, and G are chosen by the producer on the curves $SRAC_1$, $SRAC_2$, $SRAC_3$, $SRAC_4$, $SRAC_5$, $SRAC_6$ and $SRAC_7$ respectively.

The reason behind this is that for a particular output which is Q_A, the least cost causing curve is $SRAC_1$, for output Q_B, the least cost causing curve is $SRAC_2$ and so on. As per observation, all SRACs are touched by the LRAC curves from below. Any portion of SRAC's cannot be below the LRAC curve. In such a case, some level of output arises for which LRAC curve does not taken into consideration while the SRAC that gives the lowest cost. Secondly, the making of LRAC curve is not done by combining the lowest SRAC curve's points.

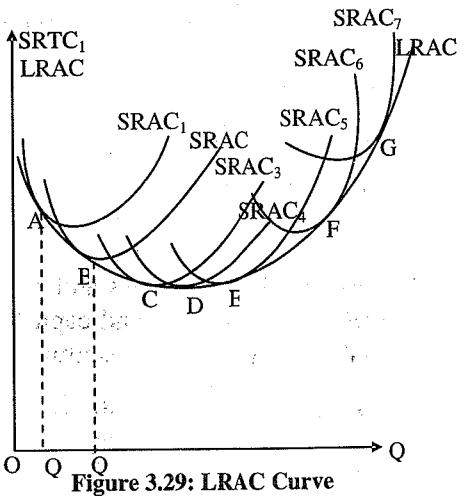


Figure 3.29: LRAC Curve from SRAC Curves

For example, the lowest output point, i.e., Q_B is reached by the curve $SRAC_1$ but at this output, the relevant curve on LRAC is $SRAC_2$. The reason is $SRAC_2 < SRAC_1$. Now, for a smooth LRAC curve as such no lowest points of SRAC curves belong to the LRAC curve except one SRAC curve. On reaching the lowest point of the LRAC curve, lowest point of one of the SRAC curve becomes the LRAC curve's point.

Point D is the lowest point of $SRAC_4$ and LRAC curve as shown in the **figure 3.29**. As LRAC curve tends to touch the SRAC curves relevantly, so it touches downward points while going downward and tends to touch upward points of the SRAC curves while going in upward direction. Thus, the lower envelope curve is the LRAC curve which is flatter as compared to SRAC curves. All relevant SRAC curves are touched by the LRAC curve from below. This means that all SRAC curves lie below that of the LRAC curve. Thus, between a SRAC and a LRAC curve, there is only one tangent point where the slopes of both curves are same. At all other output levels, a SRAC

curve's slope absolute value is higher in comparison to the LRAC curve. Due to this the LRAC curve is flat compared to SRAC curves. Normally any 'U' shaped curve of the lower envelope is flat in comparison to the 'U' shaped curve in which it is enveloped. As planning can be made based on the long-run cost curves, they are also termed as planning curves but actually one need to produce planning based on the short-run cost curves.

Relationship between the Short-Run Average Cost and Long-Run Average Cost

Relation between long-run average cost and short-run average cost curves is shown in **figure 3.30**:

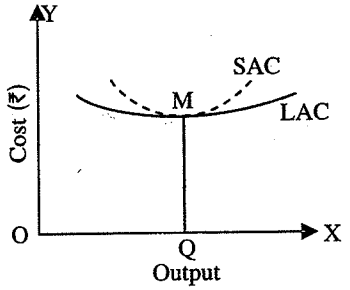


Figure 3.30 Relationship between Short Run and Long Run Average Cost

In this **figure 3.30**:

1) SAC curve represents the average costs with reference to single plant, whereas LAC curve represents the average cost with reference to several plants.
2) LAC curve like SAC curve is also U-shaped, but it is relatively flatter. The U-shape of long-run cost curve is less pronounced than that of a short-run cost curve. It means rate of increase or decrease in costs in the long run is relatively less than in the short run. The main reason being that LAC curve represents the minimum average variable cost of different quantities of output. On the contrary, SAC is the aggregate of a firm's fixed costs and variable costs. As the output of a firm increases, upto a point, both AFC and AVC fall rapidly.
3) At a given amount of output, LAC cannot be move than SAC. It is because LAC curve is tangent to SAC curve. LAC curve never cub SAC curve.
4) It may however be noted that excepting one SAC curve, LAC curve is not tangent to all other SAC curves at their minimum point. It will be tangent to that SAC curve at its minimum point, which coincides with the minimum point of LAC. This point will represent the optimum level of production.

Why is LAC Curve U-Shaped?

The shape of a long-run average cost curve is alike the English letter 'U' which clearly depicts during initial phase of production, slope of LAC is in downward direction that means with increase in production, LAC goes on falling. It becomes constant after some time and later once a given output is produced, LAC begins to rise. The reasons of long-run average cost curve being U-shaped are as follows:

1) **Economies of Scale:** Economies of scale is the main cause for downward fall of the LAC curve with increased production. The reasons of arising such economies are as follows:
   i) **Division and Specialisation of Labour:** Large number of labourers in a big plant has more opportunities as compared to small plant in terms of specialisation. Work in a large-scale production can be divided into several processes and sub-processes. Labourer can achieve specialisation in a process or sub process and attain expertise in it. This kind of specialisation enhances the efficiency of the labourer. Time and capital is also saved this way. Also new production scope is increased by the labour division. All this ultimately results in fall of per unit production cost.
   ii) **Technical Economies:** Several technical economies are faced by the firm because of increased production scale in long-run like increased automatic machine's usage. Improved techniques can help in reducing the production cost. Average cost is reduced with the help of technical economies.
   iii) **Economies of Indivisibility:** Average cost is reduced due to large scale production and the reason behind this is indivisibilities of production factors. There are some of the production factors that need to be used in a specified quantity in spite of less or more production. **For example,** the services of a production manager can be optimally utilised by the large firm whereas only one-tenth can be used by the small firm. Thus, with the increase in output, utilisation of the indivisible factors can be carried out to full capacity. This leads to reduced average production cost.

2) **Diseconomies of Scale:** Diseconomies of scale manifests themselves in the managerial constraints form. Long run average cost curve goes upward in rising part and this is due to the reason that managerial diseconomies follows large-scale production. There are management efficiency limitations in efficient and smooth coordination and working of the firm's affairs. These limitations are basically known as the scale of diseconomies. Management becomes more efficient with the firm's scale of production on account of labour specialisation and division. But these management difficulties multiply after a particular limit.

Gradually the top-management tends to lose contact with the firm's day-to-day affairs. The efficient functioning of different departments is adversely affected by this. The delegation of the decision-making responsibility may be done to other persons who are not that efficient. Various expenditures begin to increase which includes telephone bills, paperwork and travelling expenses, etc. Sometimes there is lack of coordination among different decision-making agencies. Thus, there is tendency of fall in productivity due to which consequently there is rise in the average cost.

3.8.3.3. Long-Run Marginal Cost

Derivation of the long run marginal cost is done through the short-run marginal cost curves which is shown in the following **figure 3.31.**

If LRAC's LRTC is known then it is easy to derive the LRMC. As per observation, there exists one SRAC and one SRTC for each output level. There is a short-run marginal cost corresponding to each level of output if a particular SRAC or a particular SRTC is chosen. While construction of LRAC or a LRTC curve is done, a specific point on particular SRAC or SRTC curve is chosen that is corresponding to specific output level. For that particular output level, there exists a point on the SRMC curve which is corresponding to that SRAC or SRTC curve. So, for all output levels there are points on related and different SRMC curves whose locus comes out in the form of LRMC curve.

For example, corresponding to $SRAC_1$, there is short-run marginal cost curve which is $SRMC_1$. Similarly we get $SRMC_2$, $SRMC_3$ and $SRMC_7$ which can be seen in the figure. For output level Q_A, point A or $SRAC_1$ is chosen and the output level corresponding to this is given by point A on $SRMC_1$. Likewise points B, C, D, E, F and G can be obtained on $SRMC_2$, $SRMC_3$..... $SRMC_7$ respectively. Thus, locus of all these points A, B, C.....G is LRMC. All these points are not well represented in the **figure 3.31**. It is noticeable that points D and D are same. Unlike LRTC and LRAC curves, LRMC curve is not an envelope curve.

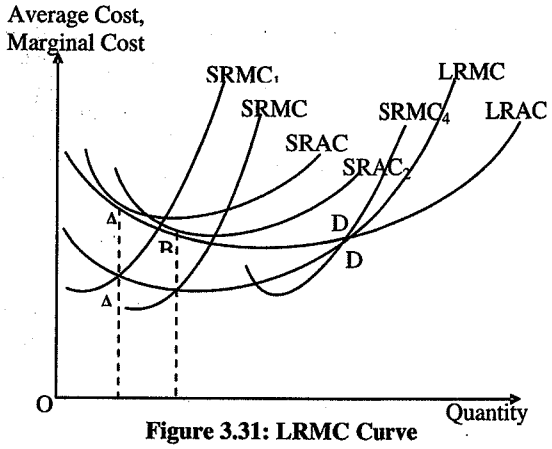


Figure 3.31: LRMC Curve

Explanation of the "U" Shape of the Long-Run Average Cost Curve

As described in the **figure 3.31**, the shape of LAC curve is "U" shaped. This U shape depends on the returns of scale. There is increase in the returns to scale with the firm expansion. The returns of scale decreases finally after a range of constant returns to scale. The long-run average cost curve on the same line turns down due to the economies of scale. The average production cost is reduced by the initial economies. However, economies of scale cannot be receipt by the firm continuously. There comes a time when diseconomies start operating and there is an increase in the average production cost. Thus, the long-run average cost curve as per the traditional economic theory will be **'U' shaped curve**.

3.8.4. Relationship Between Cost and Production Function

Some convenient features are there in cost function which helps in evaluating the number of units manufactured. In comparison to the generalized production function, calculation of cost function is much easier. By using cost function, the input demand functions can be obtained very easily rather than using other production functions. **According to Shephard's lemma,** demand for i^{th} input can be derived by differentiating the estimated cost function with respect to the price of i^{th} input.

For the derivation of demand function with the help of estimated production function the problem of constrained minimization needs to be solved. When the production function is Cobb-Douglas than it will be easy task otherwise this may be a difficult task. The derived demand functions are difficult to understand. Hence, irrespective of the reason for inspecting a production function, it must be equally helpful by the knowledge of the equivalent cost function. The association amongst physical quantities of factor inputs and the physical quantities of output is represented by the production function in a specified condition of technological knowledge for manufacturing goods and services. The quantity of input used in the manufacturing process helps in deciding the quantity of output manufactured. Hence, the association can be represented as follows:

$$Q = f(I_1, I_2 \ldots I_n)$$

Where, Q = Output of a product and I_1, I_2, etc., are quantities of the various factor inputs 1, 2, etc., used in producing that output.

The cost incurred in manufacturing the output of a product decides that whether the product that is supplied to the market is economical or not. The association between input and output of the product does not decide the economic feasibility of the product. Therefore, firm emphasis on manufacturing the product in a cost effective manner. In a firm, the association between the costs of factor inputs and the cost of output is represented by the cost function. For finding out the cost to be incurred in manufacturing the output, it is necessary to find out the quantity of input required and their prices. After adding the information about factor prices in the production function cost function can be obtained:

$$Q_C = f(p_1 I_1, p_2 I_2 \ldots p_n I_n)$$

where, Q_C is the cost of manufacturing a particular output Q, and p_1, p_2, etc., are the prices of the different factors which are used, while I_1, I_2, etc., are the quantities of factors 1, 2, etc., required. The factor prices p_1, p_2, etc., which a firm must pay in order to attract units of these factors will depend upon the interaction of the forces of demand and supply in factor markets.

With increasing returns to scale, total cost rises slower than total output. With decreasing returns to scale, total cost rises faster than total output. Total cost functions often display an S-shape, reflecting varying returns to scale at various activity levels.

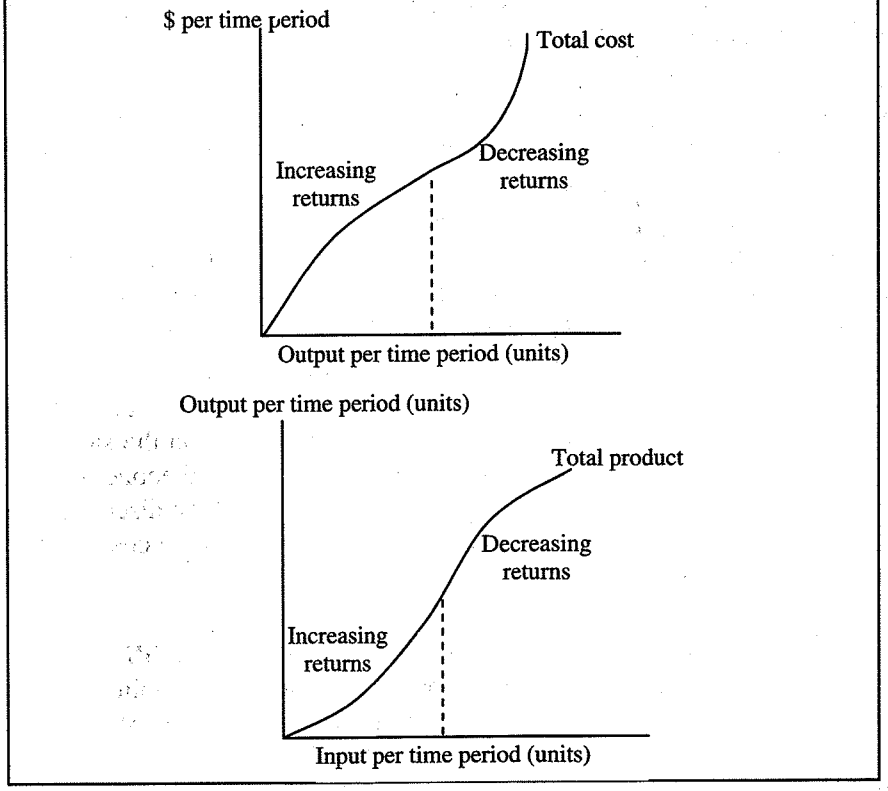


Figure 3.32: Total Cost Function for a Production System Exhibiting Increasing, then Decreasing, Returns to Scale

For example, there is direct relationship between long-run total cost and production function in case the input prices are not changed. The production function which provides constant returns to scale is linear and twice the increase in the input leads to same increase in output. The increase in input increases the total cost as the input prices remains the same. The output doubles if increasing return to scale is present but less than doubling of inputs and total cost. There will be double increase in output when the inputs and total costs will increase more than the double in case there is decreasing return to scale. The **figure 3.32** show the production function which first increases and then decreases return to scale and its implied cubic cost function shown in figure.

3.9. EXERCISE

3.9.1. Very Short Answer Type Questions

1) What is supply function?
2) Draw the supply curve.
3) Define elasticity of supply.
4) What do you understand by the term "Market Equilibrium"?
5) Explain cost function.

3.9.2. Short Answer Type Questions

1) Explain the determinants of supply.
2) What is expansion and contraction of supply?
3) Why does supply curve slopes upward?
4) Discuss the factors determining the elasticity of supply.
5) Explain the change in market equilibrium.
6) Write short notes on Opportunity Cost.
7) Discuss the explicit and implicit cost.

3.9.3. Long Answer Type Questions

1) What is law of supply? Discuss is assumptions and exceptions.
2) Explain with the help of diagram the shift and change is supply.
3) Discuss the degree of price elasticity of supply. Also, mention its practical uses.
4) What is Production analysis? Also, explain its factors.
5) Highlight the production function. Also, mention its importance and limitations.
6) Explain economies and diseconomies of scale.
7) Elaborate the meaning and various types of cost in detail.
8) Explain the concept of Average Fixed Cost, Average Variable Cost, Marginal Cost and explain their relationship.
9) With the help of suitable diagram and example explain cost-output relationship in short-run.
10) What is cost output relationship in long run? Explain.

UNIT 4

Revenue Analysis & Pricing Policies

4.1. REVENUE ANALYSIS

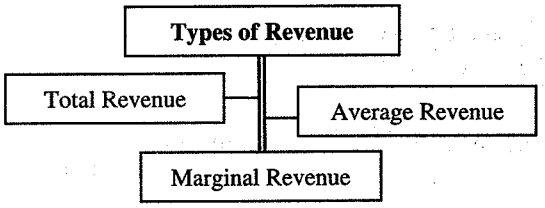
4.1.1. Introduction and Meaning of Revenue

Revenue is generally that money which any firm receives after the sale of its products. It is strictly related to demand. Basically, it is the good's sale proceeds of a firm during a particular period of a time. During a specific period of time, the amount of money which company receives includes discount and deductions for returned merchandise. The 'top line' or 'gross income' figure from which costs are subtracted to determine net income, this is also called sale proceeds. Revenue is such amount which comes out as a resultant of business activities for any firm. But on the contrary, for government it is collected in the form of taxation, fees, fines, inter-governmental grants or transfers, securities sales, mineral rights and resource rights, as well as any sales that are made. We can calculate revenue very easily by multiplying the price, at which goods are sold with the number of units of goods sold i.e., (price × quantity).

According to Dooley, "The revenue of a firm is its sale receipts or money receipts from the sale of a product".

4.1.2. Types of Revenue

There are three concepts of revenue as shown in figure below:



4.1.2.1. Total Revenue (TR)

If a firm sold 100 units for ₹10 each then its revenue is ₹1,000. Therefore, we can say that revenue refers to such amount which firm gathers by selling its certain units of commodity. Now total revenue can be expressed as:

$$TR = P \times Q$$

Where, TR = Total revenue;
P = Price;
Q = Quantity of a commodity sold.

4.1.2.2. Average Revenue (AR)

The revenue which is earned per unit of output is referred to as average revenue. Since, price is taken always at per unit of commodity, therefore, average revenue is nothing but only price of one unit of output. In symbolic form, average revenue can be represented as:

$$AR = \frac{TR}{Q}$$

Where, AR = Average revenue
TR = Total revenue
Q = Quantity of a commodity sold.

or $AR = \frac{P \times Q}{Q}$ or $AR = P$.

4.1.2.3. Marginal Revenue (MR)

Generally, marginal revenue is defined as a change in total revenue by the sale of an additional unit of commodity. It can be explained also like if seller collects ₹2,000 by selling 200 units but after selling 201 units its revenue becomes ₹2,200, then we can say that its marginal revenue is ₹200. We can say that MR is the rate of change in total revenue resulting from the sale of an additional unit.

$$\text{or } MR = \frac{\Delta TR}{\Delta Q}$$

Where, MR = Marginal revenue.
Q = Quantity of a commodity sold.
ΔTR = Rate of change.

For one unit change in output.

$$MR_n = TR_n - TR_{n-1}$$

Where, TR = Total revenue when sales are at the rate of n units per period.
TR_{n-1} = Total revenue when sales are at the rate of n-1 units per period.

4.1.3. Revenue Curves of Firm

The explanation of revenue curves of firm are as follows:

1) **Total Revenue:** The total revenue (TR) at the beginning of the stage rises at a diminishing rate with the increase in output. This curve started decreasing with the increase in output once it reaches its maximum limit. The price of a commodity increases if the competition is less in the market and it also increases the chances of earning higher super-normal profit.
2) **Average Revenue**: The slope of the average revenue curve falls with the rise in the output. The AR curve also represents the firm's demand curve.
3) **Marginal Revenue:** Marginal revenue is the increase in the revenue that results from the sale of an additional unit of output. The slope of the MR curve falls at twice a rate than the AR.

   This signifies the point at which the MR becomes zero when the total revenue (TR) reaches its maximum. MR becomes negative if the output exceeds the point where MR = 0.

For example, let us consider a producer who produces original articles and sells 20 units for ₹40,000 each. If the producer wants to sell 22 units, this can be possible only when price of the product reduces. Let us consider that 22 units can be sold at ₹39,000 each. Then,

Total revenue obtained from 18 units = ₹4,00,000
and Total revenue obtained from 20 units = ₹4,29,000

Therefore, by selling 22 units and 20 units, the difference obtained in total revenue is ₹29,000. In this example, by increasing one unit of output, the producer increases his total revenue by ₹29,000 and not by price of each unit i.e. ₹39,000 where the production level is greater. Therefore, marginal revenue earned in this case is ₹29,000

Marginal revenue refers to the additional revenue generated from the increase in the sale of one additional unit of output. In other words, marginal revenue means the proportion of variation in total revenue to a variation in selling one unit of output. It can be represented as follows:

$$MR = \frac{dTR}{dQ}$$

The calculation of total revenue, average revenue and marginal revenue can be easily done, when price and quantity schedule i.e. the demand curve of the commodity is given as presented in **table 4.1** and **figure 4.1**.

Table 4.1: Revenue Concepts

Price of Goods X	Quantity Sold of Goods X	Total Revenue = (col.1 × col.2)	Average Revenue = (col. 3/ col. 2)	Marginal Revenue = ($TR_n - TR_{n-1}$)
(1)	(2)	(3)	(4)	(5)
10	20	200	10	—
9	40	360	9	(360 – 200) = 160
8	60	480	8	(480 – 360) = 120
7	80	560	7	(560 – 480) = 80
6	100	600	6	(600 – 560) = 40
5	120	600	5	(600 – 600) = 0
4	140	560	4	600 – 560) = –40

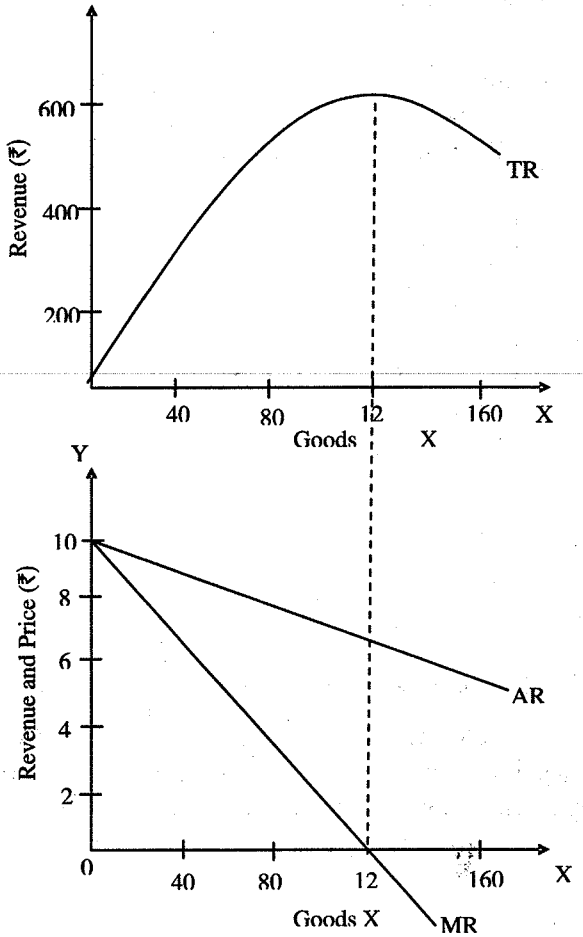


Figure 4.1: Revenue Curves

Let us assume a hypothetical **table 4.2** which involves total, average and marginal revenues.

Table 4.2: Total, Average and Marginal Revenues

Price (₹)	Quantity Sold (Units)	Total Revenue (₹)	Average Revenue (₹)	Marginal Revenue (₹)
20	1	20	20	
19	2	38	19	18
18	3	54	18	16
17	4	68	17	14
16	5	80	16	12
15	6	90	15	10
14	7	98	14	8
13	8	104	13	6
12	9	108	12	4
11	10	110	11	2

Figure 4.2 shows the graphical representation of total, average and marginal revenue curves in a graph by considering **table 4.2.**

As given in **figure 4.2**, the TR curve is used to find out AR and MR curve. AR curve is shown by the line that is drawn from the point of origin O to any point on TR curve.

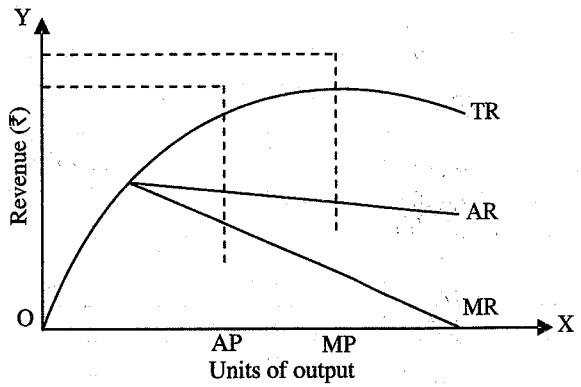


Figure 4.2: Total, Average, and Marginal Revenue Curves

Relationship between AR and MR Curve

1) The marginal revenue will overlap the average revenue, at the time when average revenue (price) and marginal revenue are same.
2) A company can sell their product in bulk when the price of the product is less resulting into the reduction in average revenue curve. Consequently, MR also reduces with the decrease in AR but MR decreases at a much faster rate than the AR. Thus, MR curve slopes downwards to the AR curve.

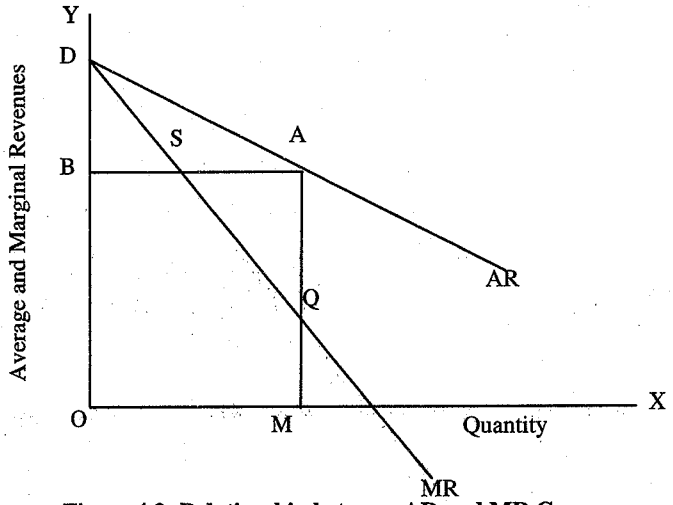


Figure 4.3: Relationship between AR and MR Curve

4.1.4. Relationship between Revenue and Price Elasticity of Demand

For business decision making, the accurate estimation of elasticity of demand (price elasticity) is of great importance. With the change in price, firm's revenue changes automatically. Total Revenue (TR) earned from sales by a firm is obtained by multiplying average unit price with the total quantity sold, i.e.,

$$TR = P \times Q$$

In **figure 4.4**, the total revenue obtained from OQ quantity sold at OP price is OPCQ:

Here, three things are clear:

1) If the demand price is elastic then, with an increase in price, there is a big fall in sales, by which total revenue also decreases and vice versa.
2) If the elasticity of demand is 1, then there will be no change in total revenue, earned from the sales even with the change in price.

   For example, with the fall in price by 5%, the sales will increase by 5% whereby the total revenue will remain unchanged.
3) Now for the case when demand price is inelastic, the sales will fall with increase in price but the total revenue will rise and vice versa.

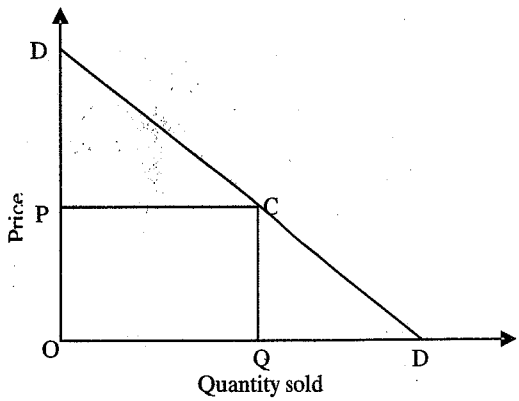


Figure 4.4: Elasticity of Demand and Revenue Relation

In general practice, unit elasticity is not in use, but when price changes in a certain ratio, the sales change in a high or low ratio automatically. Therefore, we can conclude that if management wants to increase their sales then they have to do reduction in their prices. But at same time, the reduction in price is compensated by the additional sales, the total revenue will increase or remain the same. Likewise for increasing revenue, management can increase the price of their product. But if the fall in revenue as a result of sales reduction is not compensated by the increased price, the total revenue will fall. Therefore, the effect of a change in price on the sales determines the effect of the change in price on total revenue. Additionally, the firm often remains in a fix as to whether the sales should increase or decrease and in such a situation the concept of marginal revenue is significant.

$$\text{Price Elasticity of Demand } (e_P) = \frac{\text{Average Revenue}}{\text{Average Revenue} - \text{Marginal Revenue}} = \frac{AR}{AR - MR}$$

For example, suppose that AR = ₹10, MR = ₹0. Calculate price elasticity of demand.

$$E_P = \left(\frac{10}{10-0}\right) = ₹1$$

Price elasticity of demand is unitary elastic i.e., $E_P = 1$.

4.2. PRICING ANALYSIS

4.2.1. Meaning and Definition of Pricing

There are two terms "Price" and "Pricing" commonly used in marketing. The Price is the amount which a customer pays for the product or service and pricing is an analytical process to arrive at price of the product or service per unit in terms of quantity of money currency (Rupees, Dollar, Pound etc.) at a particular time. Thus the pricing process is directly related to the value of product, service or idea in terms of money at a particular time which is accessed by the marketing manager with an aim to sell it to the customers. Precisely, pricing is the process of setting objectives, determining the available flexibility, developing strategies, setting prices and engaging in implementation and control.

According to Prof. K.C. Kite, "Pricing is a managerial task that involves establishing pricing objectives, identifying the factors governing the price, ascertaining their relevance and significance, determining the product value in monetary terms and formulation of price policies and the strategies, implementing them and controlling them for the best results".

4.2.2. Importance of Pricing

Generally, buyers and sellers try to reach an agreement for the price which is paid for a product or service. The pricing is very important to customers, industries and for promotion of business activity as described below:

1) **Importance of Pricing to the Consumers**: The importance for the consumers is as follows:
   i) **Affect Purchasing Power:** If the price of the product is high then the consumers are forced to reduce the purchase quantity because consumers have limited money to make purchases and if the product is not essential then they may even avoid purchasing it. Higher prices restrict the sale of selected commodity.
   ii) **Price Indicates Product Quality:** It is a common perception in the minds of consumers that products carrying higher price also have higher quality. In absence of other inputs about the product the higher price is the only convincing factor.
   iii) **Customer's Value Perception:** Many times the price of the product does not match the expected value by the customers and sometimes it is even lower than their expectations. In the first case, the company suffers about reputation and business in the long run and in later case the customers are delighted but company also loses its profit margin. Thus, it depends upon the right decision of the marketer to take a right decision for pricing level.

2) **Importance of Pricing to the Firms:** The importance of pricing for the firms is discussed below:
   i) **Determines Profitability:** The price of the product is directly related to the profit of the company which is set to receive. The firms receive returns in terms of money which is based on the price of product. Hence, any increase or decrease in price level of the product affects the profitability.

ii) **Determines Market Share:** The price of a commodity is a major factor that determines the demand. Price affects the competitiveness of the firm's products and its market share.

iii) **Determines Success:** The pricing policy of a company acts as double edge sword for the company. In the case when the price is set too high, the product sale will be minimum and generated revenue fails to cover up the production costs. In the other case, when the product price is too low, the company suffer loss and may be forced to leave. Thus, the price has to be sensible for the success of the company.

3) **Importance of Pricing to the Economy:** The importance of pricing for economy is described below:

i) **Influences Factors of Production:** The factors responsible for production are capital, land, machines and labour. For each factor money is spent by the producer. With increase or decrease in price of the product the profit in terms of money also increases or decreases. The increase in profit brigs the producer in better position to spend more of production factors to increase them for more profit and decrease in profit compels him to reduce the production factors to sustain the company. Thus, the pricing policy influences the production factors in both ways.

ii) **Determines Demand and Supply:** The pricing of the product also governs the demand and supply of the product in the market. The higher price leads to increase in supply but fall in the demand because of low capacity of payment of common customer. The lower price results in increase in demand and reduction in supply due to increase in number of customers and limitation of the company to match the demand. Thus, the pricing policy is the determining factor for demand and supply.

iii) **Affects Savings and Investments:** The paying capacity of the consumers remains limited and due to this reason increase in the price of product leads to reduction in their savings and investments. Any decrease in product price increases their savings and investments. This explains the pricing effect on saving and investment. In alarming stages of price rise or fall, the intervention of the government sometime becomes necessary.

4.2.3. Factors Influencing/Determinants of Pricing

The factors which influence the price can be divided into two factors:

1) **Internal Factors:** These factors are discussed below:

i) **Organisational Factors:** There are two levels in an organisation who decide the price level of the product. The top level limits itself up to planning of price range and the market segment which is related to that product. The better details of the pricing are fine-tuned by the lower level executives who go in depth of individual product to deal with the product strategies.

ii) **Marketing Mix:** The prices of the product are considered as only one factor out of many other related factors by the marketers. The increase in product cost results in increase in production, advertisement and distribution costs.

iii) **Product Differentiation:** The price of the product also depends upon the attractive characteristics of the product such as looks, colour, packaging, after sale service, matching with contemporary fashion, style, trend and the alternative usage of the product etc. These characteristics may attract the attention of the customers and prepare them to pay more.

iv) **Cost of Product:** For determining the price of a product, the cost or cost of production cannot be ignored by any company. Further, a fine tuning is required to be made for finalising the price which depends upon the market demand and price being offered by the rivals. Finally, the product has to reach the market and face the competition and any wrong calculation on pricing may make it a failed product.

v) **Objectives of the Firm:** The objective of a company can be changed with time and situations according to the financial status of a company which affect the pricing decisions.

For example, if the objective is charity then the prices are fixed at no profit and no loss basis. For increasing the market share, the prices are to be made lower than the competitors even at a loss to win over the customers. Sometimes the image or esteem of the company restricts it to increase the prices. Thus, the prices are decided as per the objectives of the company.

2) **External Factors:** These factors are as follows:
   i) **Demand:** It is a common phenomenon in the marketing that scarce commodities have increasing price which is the indication of its high demand. The demand of any product depends upon its utility, number of customers, availability of similar products in the market and their price, the paying capacity of the customers as well as their greater liking of the product due to extra features such as looks, offering schemes etc.

   ii) **Competition:** The prices of the rival products in the market make crucial impact on decision making of pricing. To remain in competition, the companies are made to make the price of the product equal to rivals or slightly lower. The higher prices can be offered only if the company brand is popular.

   iii) **Suppliers:** When the price of raw material which are used for production of product increases then the suppliers charge more money for the same quantity supplied. The producer is forced to transfer this increase to customers by increase in the price.

   iv) **Economic Conditions:** The prices are directly related to inflation or deflation conditions prevailing in the country. In inflationary situations, the prices of the input factors increases and the prices of the products are also increases. However, in deflationary situations, the prices are to be reduced to maintain the sale of the product and turnover of the company.

   v) **Buyers:** The pricing decision of the company on the product does affect the consumers. The price increase reduces the paying capacity of consumers and it develops constraints on customers. Consumers may change their plan for purchase of product. This makes a great impact on company's turn over in the case when the product is being used by large number of consumers.

4.2.4. Objective of Pricing Policies

The following objectives are to be considered:

1) **Profit Maximisation in Short- Term:** The main aim of any company is to earn maximum returns in shortest possible time. Thus, the pricing policy is designed and implemented to achieve this aim. The cost incurred in production of the product is to be recovered with respectable profit margins.

   This will certainly infuse the sense of confidence in the organisation and also boost their morale. In case when the product is meant for exclusive elite of the society, the pricing can be extremely high.

2) **Profit Optimisation in Long Term:** The commonly used philosophy to earn maximum profit is not suitable in long term. This is bound to bring in some mal-practices which are against the established code of conduct in the business and also invite criticism from the customers which is bound to harm the business in the long run. It is more practical to switchover to optimum profit instead of maximum profit.

   Thus, the optimum price is the most suitable price for adaptation in all practical purposes as strong pricing policy for industries in present times.

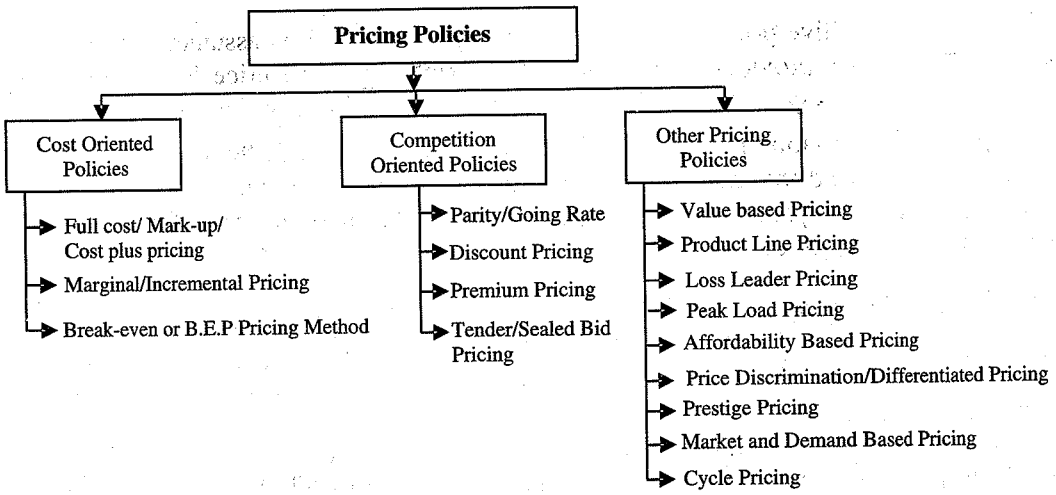
3) **Price Stabilisation:** The price stability is another important factor in the marketing field. The frequency of price variation has to be minimum because it is related to the confidence of the customers as well as the company reputation. These two factors together contribute for continuous and strong growth of the company. Therefore, a stable price is a tool to win over the customer's confidence and making company's good image.

4) **Facing Competitive Situation:** Excluding the monopolistic products, all products face competition from their rivals in setting the price of their product. The pricing policy is made to counter the existing similar products in the market and enlarge the market share. Generally, aim of the company does not end with competition but to win over the market. This strategy is a continuous process which keeps the company always busy.

5) **Entry into New Markets:** Diversification is the other way to strengthen the business potential of a company. The diversification opens up the doors of new opportunities for the company, although it has to face the risks and uncertainties in the market. In the diversified field the company has the freedom to make the price of the product more beneficial for the customers.

4.2.5. Pricing Policies

Following factors contribute for deciding the price of the products:



4.2.6. Cost Oriented Policies

The production cost of the product remains the basis of the price of the product. For arriving at any decision, an analysis of fixed cost, average cost and marginal cost is required to be made. The procedure to arrive at the price is described in following heads:

4.2.6.1. Cost Plus Pricing or Full Cost or Mark Up Pricing

In this method, the cost of production of the product is estimated based upon the calculations and profit desired is added to it to arrive at the mark up price for sale. This is the simplest method of finding the price of a product. To find the price by this method following formula is used:

$$\text{Mark up price} = \frac{\alpha}{(1-r)}$$

Where, (α) alpha = Unit cost (fixed cost + variable cost)

r = Expected return on sales expressed as a percent.

For example, if the labour cost of manufacturing 100 cars is ₹1,00,00,000

$$\text{Labour cost of 1 car is} = \frac{1,00,00,000}{100} = ₹1,00,000$$

Material Cost of 1 Car = ₹3,00,000

Rate of returns which company needs = 30%

$$\text{Hence the Mark up Price as per the above formula} = \frac{1,00,000+3,00,000}{1-0.3} = \frac{1,00,000+3,00,000}{0.7}$$

$$= ₹5,71,428.57$$

This method is based on the assumptions that the product is needed in the market and customers are ready to pay for the price. Either there are no competitors or other competitors have uniform conditions for labour and material cost and have an agreement for fixed returns of 30%. This method is a simple arithmetic addition of profit percentage to unit cost and is used by wholesalers and retailers by simply addition of percentage profit to the cost incurred by them. Due to this reason this method is also termed as **"The sum of margin method"**.

Advantages of Full-Cost Pricing

1) Full-cost pricing offers a mean by which fair and plausible prices can be found with ease and speed, no matter how many products the firm handles.
2) The prices based on full cost appear to be real and can be defended on ethical grounds.
3) Firms preferring stability use full cost as a guide to pricing in an uncertain market where knowledge is incomplete. In cases where costs of getting information are high and the process of trial and error is costly, they use it to reduce the cost of decision-making.

Disadvantages of Full Cost Pricing

1) The basic relationship of demand and price has been ignored. It is assumed that customers will pay what is being demanded.
2) It also ignores competitive price that can be put up by the rivals. It is assumed that all competitors are in agreement for price. It provides only a wishful thinking about the price by addition of profit margin. However, actual profit is gained only after the product is actually sold at the demanded price.
3) The overhead cost calculation is a very complex matter and it cannot be found as a random choice. It varies with type of product and the market in which it is to be sold. The volume of sale of product varies with variation in prices. The production cost remains a component of the price but it alone cannot help in deciding the price. This process becomes more complex for companies dealing with variety of products.

4.2.6.2. Marginal Cost Pricing or Incremental Cost pricing Method

In this method, the company gives emphasis on recovering the marginal cost and overhead expenditures. However, such procedures are useful for large organisations who intend to capture market under stiff competitions.

Advantages of Marginal Cost pricing

1) This method of pricing permits the company to develop more difficult pricing policy in comparison to cost-plus pricing. This may lead to higher sales and reduced marginal cost due to increase in productivity and reduction in input prices.
2) It has been found useful for the full life cycle of the product.
3) For such companies which deal with multi products, multi processes and multi markets, full cost pricing does not hold good because difference in impact of costs. However, in such cases marginal cost pricing is found useful.

Disadvantages of Marginal Cost Pricing

1) It is found successful for short-period and hence it can be applied on temporary basis only. It is not suitable for long term policy.
2) It does not ensure the operation of a company at zero profit.
3) Sometimes management of the companies are not aware about this method of pricing.

4.2.6.3. Break Even Point or B.E.P. Pricing Method

The situation at which the total sales revenue and total cost of the product equalises or there is no profit for the company then it is termed as **Break Even Point or B.E.P.** This is also called 'No Profit No Loss Pricing Method'. In this method, total cost of production is divided into two parts – Fixed Costs and Variable Costs. This pricing is based on the assumption that the company does not earn any profit at initial stage for a short-term but it will gradually start earning profit which will increase with growing scale of production since the fixed costs remain constant at all levels of production. The fixed costs are recovered during initial stages and company starts getting profits after crossing the break-even point. This pricing is useful for competitive products.

The B.E.P. can be calculated with the use of following formula:

$$\text{B.E.P.(InUnits)} = \frac{\text{Fixed Costs}}{\text{Selling Price per unit} - \text{Variable Costs per unit}}$$

$$\text{B.E.P.(In ₹)} = \frac{\text{Fixed Costs} \times \text{Total Sales}}{\text{Total Sales} - \text{Total Variable Costs}}$$

4.2.7. Competitive Oriented Pricing Policies

This pricing method is purely derived on the basis of demand of the product and it has no relation to production cost. This is derived from backward calculation which is based on the price that the market is ready to pay for the product. Now calculate the production cost and profit and matching it to the price which is affordable for company. The demand and price are directly proportional to each other i.e., high demand low price, low demand high price. It is the skill of the marketing manager to estimate the consumption of the product at different price levels. Generally, the companies arrive at the product price after analysing the prices of the rivals in the market. Intentionally the policies are designed to sell the product in the competitive market. There are three types of policies as given below:

4.2.7.1. Parity Pricing or Going Rate Pricing

This pricing of product is derived from the existing price of rival products in the market. It is useful for new entrepreneurs or a new product which is launched. This is based on the fact that the product is only demanded if the prices are competitive. The best way is to follow that company who is leading in the market. In this process the product price is kept equal to competitors and thus it prevents price wars in the market. This is suitable to market which is being controlled by a small group. However, this process has some serious limitations. In this process it is assumed that the company which is leading the market is efficient in their working. However, if they are inefficient then it is also reflection of others inefficiency. It is also a known fact that collective wisdom is not always the best.

4.2.7.2. Pricing below Competitive Level or Discount Pricing

This process of pricing is considered by the company to attract the customer towards its product and increase the sale and give competition to existing rivals. Such process is generally adopted by the new entrants who have limited funds and need early return of their money to continue in the business.

4.2.7.3. Pricing above Competitive Level or Premium Pricing

Premium Pricing means that the price of the product is being kept above the price of the same product by other companies in the market. The higher price always gives a hidden message in the minds of the customers for superior quality of the product. However, this pricing can only be adopted by the leading companies of repute with proven track of record.

4.2.7.4. Sealed Bid Pricing/Tender Pricing

The price quoted by a company is inclusive of production cost and consideration of competition prevailing in the market. The company may quote the price equal to production cost but it will not provide any profit to it. Therefore, it is normal practice to submit quotation of price after inclusion of profit expected in sealed bid tender. This is also a form of competitive pricing which is commonly used in government contracts, industrial marketing and in many other projects. The prices to be quoted are calculated on the basis of profits desired, profit of getting a contract. Many times after getting a contract from a reputed firm, the company finds itself on advantageous position which yields dividends for the company. The parties participating in such tender pricing have total knowledge about the company and the competition involved.

4.2.8. Other Pricing Policies

The major other pricing policies are explained in the following heads:

4.2.8.1. Cyclical Pricing

The cyclic pricing is related to periodic rise and fall of economic activities in the country. The study of past certain years reveals that there is a cyclic variation of economic activities. This has got following four phases as indicated in **figure 4.5:**

1) Trough,
2) Expansion,
3) Boom, and
4) Recession.

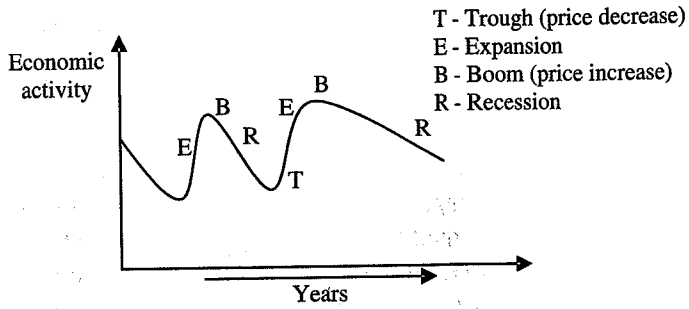


Figure 4.5: Cyclical Pricing

At point 'T' the output is at the lowest with full utilisation of available manpower. Point 'E' represents expansion of the economic activities while point 'B' represents the peak of national output. During periods of expansion and boom, the company should increase the price for business growth. The point 'R' represents the decline in national output and company has to decrease the price of the product to increase the sale and sustain the business. The company should consider the production cost and adjust the sale price according to prevailing economic situations. During recession discounts, payment on instalment may be opted to encourage sale. However in boom situation, company can afford to add costs incurred on substitutes and examine the prices put up by the rivals for finalising the sale price. In such condition apart from the cost of the product, the company has to make adjustments in the price as per market demands.

4.2.8.2. Value–Based Pricing

This is a pricing which is done after consideration of the benefit the product or service will provide to the customer and its utility for them. It is the competency of the marketer to understand the value attached by the customer and accordingly price of the product. This is termed as value-based pricing. The product designed, marketed and priced by the marketer for acceptance of customers is against the norms of this pricing. The price is never decided in isolation but price, product, promotion, and place are analysed systematically. The value based pricing never has a narrow vision of recovering the costs incurred, rather it believes that costs incurred will automatically be recovered in due course if the price offered is matching the value perception of the customers.

Following four situations in cost-value-price are described below:

1) **Value > Price > Costs:** In this situation, the company is able to recover the costs but fail to recover the value which results in loss in due course.
2) **Price > Value > Costs:** In this case, the value and costs are recovered. Since less attention is paid to customer's perceptions of price. This situation will make the customers to find other similar product which leads to decline in sale and lowering the share value of the company.
3) **Price > Costs > Value:** This is still grimmer situation for the customers. It appears that company is not much worried for customers. Sale of such product is very much doubtful. Further, the cost itself is more than the value and cost is to be recovered by still higher price. This makes the customers more negative for the product.
4) **Price = Value > Costs:** This situation can be termed as the optimum situation where customer feels comfortable which will ensure continuous sale and profit to the company. After examination of above four situations it may be noted that in all cases the price is always more than the costs and this ensures a margin of profit. However, situation four is suitable for sustained profits and therefore most suitable.

4.2.8.3. Product Line Pricing

This process is suitable for the companies which produce multiple products of same family. Different products belonging to same product family which complement each other are grouped together and priced in a group. This pricing strategies aim to maximise the sale of different products by creating more complimentary rather than competitive products. This helps the company to adopt group price which acts as cushion for fluctuating demands and maintains the maximum profits for the company by effective management of the product line.

1) **Price Bundling:** This method of pricing is adopted to maintain a continuous sale of product. It is useful for perishable products or consumer facilities like hotel rooms or seat in flight which have no substitutes. **For example,** offering off season discounts to customers for hotel rooms, flight tickets or buffet lunch in hotels at reduced prices etc. The customers are attracted and business picks up in off seasons and perishable goods also give some returns to the company.
2) **Premium Pricing:** This type of pricing is adopted by companies which manufacture different models of same products suiting to different strata of society. **For example,** a company is producing variety of television sets like normal flat screen, 3D effects with speakers giving theatre sound effects and screen size 48" or 60". Although the same purpose is served by both the models but there is no comparison between the two as far as the quality of visual and sound effects and price ratio between the two will also be more than 10 times. However, there is a class in the society who will always prefer the best available product at no consideration of cost and enjoys it as a status symbol. The company selects the premium pricing for such elite of society.
3) **Image Pricing:** This type of pricing is applicable for similar products from different companies or the same company. The consumers who expect higher quality from the product is placed at higher price. The company floats different brands of the same product suiting to different classes of the society. Such pricing can be found in cosmetics, clothes, soaps, cars etc.
4) **Complementary Pricing:** Complimentary pricing is additional price to be paid by the customers for taking possession or enjoying the product. Like purchase of flat or car is always accompanied by registration charges for legally owning it or in hotels the customer pays for sales tax and VAT in addition to original price of food.

5) **Captive Pricing Strategy:** To captivate the customers and to promote its product the company offers it as a compliment with its regular product. **For example,** toothpaste tube with a brush or shaving gel with a twin blade.

6) **Two-Part Pricing:** Many products carry two part pricing. **For example,** multi-story apartment flats have purchasing cost and monthly subscription to society running it to pay for cleanliness, round the clock power supply and security of the residents. Clubs have membership fee and separate charges for the using its facilities like swimming pool, lawn tennis and badminton etc. These are termed as fixed fee and variable usage fee.

4.2.8.4. Loss Leader Pricing

The meaning appears from this title is that it is for incurring losses but in fact it is for increasing profits. In fact, the loss leader traps the customers due to its low price who buy some other products of the company. The loss leader is also not priced at loss but at a comparatively lower price to similar products. The profit earned by loss leader is little less but the sale and profit of other products compensate it. In compliance of this pricing policy sometimes the company makes it mandatory for the customers to purchase certain items along with the popularised product. The customers feel a loss on the product which is forced to buy but they still carry a hope of profit due to lower price of the popular product.

Bob R. Holdren has given certain characteristics for the loss leader as given below:

1) The buyers must have knowledge of price of similar products at other places.
2) It should encourage the customers to make a bulk purchase to get the benefit of profit due to low price.
3) Demand of the product should not fluctuate.
4) The price reduction should be able to make the public conscious.
5) The price reduction should not give impression of inferior quality of the product or short life expiry on shelf life products.

4.2.8.5. Peak Load Pricing

The companies mark the timings, seasons or places which suit the customers the most and then accordingly design the price in order to gain maximum profit from their product or facilities. **For example,** maximum people would like to use telephones or airlines before noon to utilise the information or reach the destination comfortably to attend meeting so this becomes the peak period and company takes advantage to charge at higher rates for the facilities to accommodate who need it the most and able to pay for it. On the other hand, during night or odd hours it may be put at reduced rates to encourage the customers who are not in hurry or unable to pay higher price. This will provide higher returns for peak period and respectable returns to company during low demand periods as well. By adopting peak load pricing, the company aims at recovering the capacity costs of facilities from those who utilise it at peak hours.

The peak-load pricing can be fruitfully applied if:

1) The installed capacity of service to be used for peak-period as well as non-peak periods.
2) It is not possible to store and use it at appropriate time but it is to be used at specified time.
3) The demands vary with time.

Problems in the Application of Peak-Load Pricing

There are many problems faced in application of Peak-Load pricing. Some are illustrated below:

1) Practically accurate estimation of peak-load and lean-load are not possible.
2) Capacity costs and marginal cost measurements have certain inaccuracies. The capacity cost consists of variety of different types of equipments, the rate of utilisation is also different and the utility life of equipments also varies with use. There is certain difference of opinions on calculation of marginal costs.
3) This policy makes discriminations between the customers. Thus, it may get negative response from the customers.
4) This encourages the company to search for optimum peak price consumers as well as for off price consumers. This will give rise to serious discrimination problems. The instability of prices is not liked by the public and advance information will be needed for it.

4.2.8.6. Affordability-Based Pricing

This is applicable for essential commodities which are needed by all section of society and prices are set in such a way that all sections of society are able to consume it to the required extent. This makes the price independent of production costs; the government controls it by way of providing subsidy like in case of cooking gas. Government public distribution system is also used for the distribution of various goods.

4.2.8.7. Price Discrimination/Differential Pricing Strategy

In this pricing system company establishes different price points for different customers on the basis of different variables such as volume of purchase, payment conditions like cash down or instalments, loyalty factor of customer etc. The market consists of variety of customers and this discriminatory pricing act as a motivator for the company.

4.2.8.8. Prestige Pricing

The prestige pricing realises the psychology of the customer who needs to posses something very exclusive. Customer feels pride of owning a product which is rare. The company is aware about the class which they deal where price is not the consideration but gaining it as status symbol is the real aim.

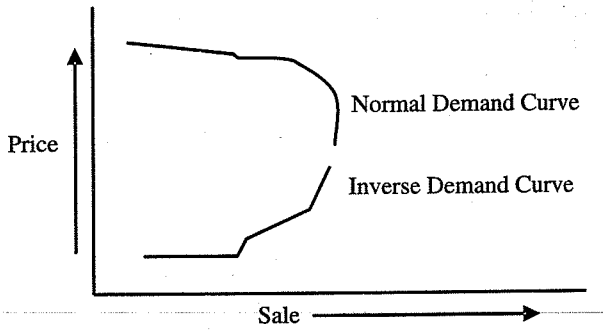


Figure 4.6: Graphical Illustration of Prestige Pricing

For example, flats constructed in a city at a location where only elite of the town reside. It is having terrace garden with fountains and separate lift attached to it along with private car parking. Certainly such products are meant for persons who are above the class and ready to pay any amount for it as a prestige. The demand curve for prestige pricing is indicated in **figure 4.6.**

4.2.8.9. Market and Demand Based Pricing

It is a practical way of pricing the product. Before finalising the price, an analysis is made on demand of the product and purchase capacity of customers. The value attached to the product by the customer is the most important factor for the marketer. It may be a customer or an industry that is willing to buy the product will certainly analyse the benefits being achieved by the product and the price being paid for it. If the analysis results justify the price only then they will go for it. Price demand relationship is based on the parameters in different markets which are discussed below:

Pricing in Different Types of Markets

There are different types of constraints on sellers in different types of market. Economists have categorised following four types of markets:

1) **Under Pure Competition:** This type of market consists of many buyers and sellers who trade in same commodity such as wheat, rice, copper or aluminium etc. There is no problem for the buyers and sellers to buy or sell any commodity at current price. No one will charge or pay more than the current price. The competition does not exist because sellers or a new entrant cannot sell the product at a little lower or higher price in order to sell his product at faster rate or gain more profits.

2) **Under Monopolistic Competition:** Practically, it is found that buyers and sellers do not follow single price line but follow a range of price line. The physical condition of the product varies according to the type of storage, time of storage and the time of purchase. The sellers wish to clear the old stock to bring the new stock. This compels the sellers to negotiate the price with buyers and sell it at different prices to each individual. For increasing the sale each seller puts different sale conditions such as discounts, instalments and gift coupons etc.

3) **Under Oligopolistic Competition:** The market is generally run by a small group of traders of the product. This group is aware about the strength and weakness of its members and make uniform decisions on pricing to break the entry of new comers in the market. **For example,** if anyone in the group increases or decreases the price of the product then the others also quickly follow the same to remain in the business.

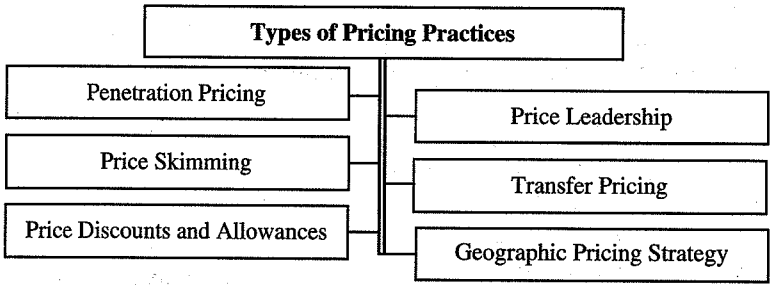
4) **In a Pure Monopoly:** This is a case of single seller in the whole market. It generally relates to essential commodities or products related to defence. These are generally government controlled or private party operating under government guidelines. **For example,** government controlled fair price shops for grain for distribution to public and the government decides the price of the grains. Power generation and distribution by private regulated body under government approval. Sometimes unregulated monopoly also emerges in the market and they are at liberty to fix the price on their own but they also do not charge at astronomical high price due to fear of government interference or take over.

4.2.9. Pricing Practices

A single price is not usually set by the companies but a price structure reflecting several variables. When an appropriate pricing method is selected then the desired price for the particular product or service is adopted by following pricing strategies or techniques that are price adaptive:

4.2.10. Types of Pricing Practices

The types of pricing practices are as follows:



4.2.10.1. Penetration Pricing

Low price of the product is being set by the entrepreneur to enter into the market in the initial stages with the objective of securing a large market share. The markets which is highly sensitive the businessman will continue to sell its product without any profit as he is interested in growth of the product instead of earning profit. When the objective is of market penetration the unit cost of production and distribution decreases as the volume of sales attains a particular target. Hence, this objective of market penetration is to obtain maximum share of the market by setting low prices deliberately.

Advantages of Penetration Pricing

1) It can result in fast diffusion and adoption. This can achieve high market penetration rates quickly. This can take the competition by surprise, not giving them time to react,
2) It creates cost control and cost reduction pressures from the start, leading to greater efficiency,
3) It discourages the entry of competitors. Low prices act as a barrier to entry, and
4) It can create high stock turnover.

Disadvantages of Penetration Pricing

1) The customer expects the prices to remain low for a long-term. They are not ready for the subsequent rise in the price and when it happens they might switch to a competitor's product. Thus, subsequent price hike leads to loss of market share gained.
2) It is believed that penetration pricing cannot create strong customer relationship and only attracts customers on the lookout for a profitable deal.

4.2.10.2. Price Leadership

Price leadership is when the leading firm of a particular sector decides the prices of product for the whole industry. The prices are decided by the price leader and are accepted by all the other firms present in the market and accordingly they change the prices of their final product equals to the price decided by the price leader. If the other competitors need to sustain in the market, they need to follow the price leader as they are left with no other choices. On the other hand, in order to attain the increased share of market, the competitors may also decide to reduce the prices. If the price leader sets the price of the product higher then, the price leadership is positive as this increase in price will removes the risk of losing their market shares from the mind of the other competitors. The profit of all the firms will increase if the prices of the products are kept high.

However, many of the market leaders like big-box retailers use their working proficiencies to reduce the prices in the market persistently. Therefore, in order to hold their market share the small competitors tend to reduce the prices. As these small competitors does not have same economies of scale as compared to the price leaders. Hence, the effort to make the prices equivalent to the leader's price may cause them to suffer huge losses over a period of time and as a result the small competitor leaves the market.

Advantages of Price Leadership

1) **Interdependence of Behaviour:** By this method most important advantage of price leadership is that the firms opt out of the uncertainty surrounding pricing decisions in oligopoly. There is interdependence between the firm's own behaviour and the behaviour of its rivals. The decision (be it price, product or promotion) which a firm takes will affect the competitors and will, encounter the countermoves from the rivals. The result of this interdependence makes predictions difficult and, therefore making the net impact of any move uncertain. When one firm is a leader and the other firms as followers, price leadership reduces the uncertainty associated with the moves and countermoves of the rivals. Hence, firms tends to choose a parallel price rather than to go for undercutting each other.
2) **Absence of Price-War:** Due to price leadership there are no chances of price-war because in the absence of price leadership a lower price that is too low is set by a low cost firm which results in price war. The effect of this will be on all the firms including the lowest cost firm leaving them worse-off than they need to be.
3) **Product Differentiation:** Due to product differentiation price war can be avoided, if there is no price leader in the oligopoly market. This non-price competition or so called product differentiation attracts firms that are oligopolistic and due to time-lag errors and before the production of the countermove in terms of another product by the rivals involving lot of time, effort and money. Thus there is only an element of understanding in price leadership between the oligopoly firms.

Disadvantages of Price Leadership

1) **Difficult for Price Leaders:** The correct assessment of the reactions of the followers is not possible by the price leader being the prime difficulty. Hence, the rival firms are unable to follow its lead.
2) **Fixation of lower prices by rivals:** There is secret charging of lower price by the rival firms when the price leader has unnecessarily fixed the higher price. Thus, they want to increase their market share without openly challenging the price leader.
3) **Non-Price Competition:** Another difficulty faced by the price leader is the indulging of the rival firms in 'non-price competition'. Their purpose is to increase their sales, yet they are charging the price set by the price leader. Advertising and other methods of sales promotion like product improvement other than the secret price-product concession are included in the mechanism of 'non-price competition'.

4.2.10.3. Price Skimming

Following are the conditions under which the pricing strategy provides their results:

1) In the target market it is believed that, if the price of the commodity is high then the quality of goods will also be high. The reason for this is that the target market has linked the commodity's quality with its price.
2) The customers are ready to purchase the commodities at increased prices because they want to become an opinion leader.
3) The customers are ready to buy the product because they believe that a particular commodity will increase their prestige in the society.
4) In case when there is no competition or any risk in the market then the existing competitors will capture the entire market due to the obstacles of low entry and exist.
5) In case when there is substantial advancement or development in the technology then the high technological product is produced in the market.

A policy is made for the commodities withdrawing from the market for discharging their part of service. In this policy it is stated that, during the initial phase of market development the prices of the commodities are kept high with the high expenditure on promotional activities while opposite condition prevails in the later phase of market development. Many commodities become successful because of the use of this policy. There are several reasons due to which this policy is successful.

Advantages of Pricing Skimming

1) The demand with respect to price is more inelastic in the initial stages in comparison to the condition when the price is grown fully. This is true about the consumer goods. A novel product such as the electric blanket is not yet accepted as a part of the expenditure pattern, consumers are still ignorant about its value compared to conventional alternatives. Moreover, at least in the early stages, the product has so few close rivals that cross elasticity of demand are low.
2) The market can be broken into segments by launching a new product with high price that differ in price elasticity of demand. This high price scoops the cream of the market being insensitive to price in the initial period of product launch.
3) More elastic sectors of the market are successively hit lightly by consecutive price reductions this strategy is demonstrated by the systematic sequence of editions of a book which are initiated by high priced limited personal edition and then culminating into low-price pocketbook.

Disadvantages of Pricing Skimming

1) The strategy of skimming of the prices can be disastrous as due to the presence of close competitors who can introduce similar products at lower prices, then the consumer will infer that the company is all the time selling the products at higher price will force them to abandon the company and its product.
2) Due to availability of strict legal and government regulations regarding the rights of the consumers the price skimming option is not viable.
3) Due to the available history of the company of price skimming the consumers will be reluctant to buy a new product; rather they would wait for some months to buy it at a lower price.

4.2.10.4. Transfer Pricing

A transfer price is a price used to measure the price of goods or services furnished by a profit centre to other responsibility centres within a company. The evaluation of managerial performance within the company through profit centres is impossible without determining transfer process in case various profit centres of the company exchange goods and services. In such situations, there is a need to determine the monetary values, called transfer price, at which the transfer should take place so that costs and revenues could be properly assigned. The implication of the transfer price is that for the transferring division/centre it will be a source of revenue, whereas for the division to which transfer is made will be an element of cost. Thus, there is a need to determine proper transfer price for the successful implementation of responsibility accounting. Some companies have the problem of pricing goods and services which are transferred to other divisions/units of the same company; such pricing is referred to as 'intracompany', 'intradivisional' or 'transfer pricing'. It is the pricing of goods and services exchanged in intra corporate purchase transactions.

Importance of Transfer Pricing

1) An inefficient transfer pricing system can lead to sub-optional decision, which may lead to inherent losses of significant propositions.
2) An error in transfer pricing may have considerable impact on divisional profits and morale of the concerned divisional manager and his staff.
3) Behavioural problems are the major problems arising out of transfer pricing system. An imposed transfer price, which is unfair either to purchasing division or the supplying division, can lead to considerable resentment, hostility and loss of motivation. This could easily counter any of the benefits claimed for divisionalised type structure. Effort should be made to avoid behavioural problem by use of system, which lends sufficient opportunity for fair arguments leading to acceptable decision in the available framework.

Limitations of Transfer Pricing

1) Inter-conflicts among different departments or divisions may spoil the spirit of transfer pricing.
2) Fixing transfer pricing by one department or division at the cost of other department or division may stand on the way of co-ordination among various departments or divisions.
3) Lack of work culture may stand on the way of objective of transfer pricing.

Transfer Pricing Methods

1) **Market Price Basis:** The suitable system of transfer of goods from one division to another under the same management to another company is the market price basis. The market price should be the transfer price. Wherever a market price exists for a product, the inter-divisional transfer price should equal the market price to avoid sub-optimisation. This method definitely avoids the possibility of passing the inefficiencies of one department to the other departments.
2) **Cost Basis:** In case the product produced by a division of the firm can be sold only to another division of the firm, the inter-divisional transfer should be priced at the level of the actual cost of production. Here transfer prices will be useful to achieve the best joint level of output. It will maximise profits.
3) **Cost Plus Basis:** Under this method the goods and services of each department are charged on the basis of actual cost plus a margin by way of profit. The major defect of this method is that the transferring department may add a high margin so as to raise the profit of the department. It may result in setting the ultimate price unduly high thereby affecting sales.

4.2.10.5. Price Discounts and Allowances

The discount on product to their customers is a valuable method with regard to the tough competition given by their competitors. The discount given to the customer must be cautiously measured and think off taking into consideration the marketing strategies. If this is not done then offering discount can be a risky practice. Nowadays, discounting is prevalent in almost every industry. It has become very common as now offering normal price without discount is pointless. The strategy of price discounting is beneficial to the seller only when they are able to earn something in exchange of the product. Nowadays, companies are suffering as they involve themselves in a difficult cash structure, quantity and other discounts. As a result, the company earns less profit.

4.2.10.6. Geographic Pricing Strategy

Geographic pricing strategy is the pricing strategies of the commodities to be sold are done according to the geographical location of the buyer. In this strategy the prices of the product are kept lower than the prices of their competitors in one market whereas, in the other market they implement penetration strategy. In this pricing the firm do premium pricing in one market, penetration pricing in other market and discounted pricing in another market. The other problem in this strategy is how they will receive payment. This problem becomes crucial at times the customer do not have currency in the physical form to give for the items they purchased.

4.3. MARKET STRUCTURE

4.3.1. Introduction of Market and Market Structure

A specific place for purchasing and selling of goods is termed as **"market".** This term has broader outlook in economics where market refers to whole area where the buyers and sellers of the products are spread and not limited to a specific place. In economics, the structure of the industry serving the market is the basis of classification of the market. The market structure variables that determine the magnitude and characteristics of competition are the basis of categorisation of the structure of the industry. For goods and services, the nature and degree of competition in the market is referred by **market structure.** The nature of competition being dominant in a particular market helps in determining the structures of goods market and service market which is also referred as features of a particular market. Various types of markets have their own particular features. Market structure affects the behaviour and performance of the firms in the industry.

4.3.2. Determinants of Market Structure

For a particular product, the numbers of determinants of market structure are:

1) **Number and Nature of Sellers:** The number and nature of sellers in the market affects the structure of market. They range from large number of sellers in perfect competition, only one seller in pure monopoly, two sellers in duopoly, and certain sellers in oligopoly and for distinctive products there are many sellers.
2) **Number and Nature of Buyers:** The structures of market are also affected by the number and nature of buyers in the market where, a monopoly market is dominated by a single buyer which is buyer's monopoly, duopoly market is characterised by two buyers acting jointly in the market and oligopoly is the market where there are certain number of organised buyers of a product.

3) **Nature of Product:** Market structure is determined by the type of the product. Monopolistic competition characterises the market where close substitutes of products are available due to product differentiation. On the other hand, in a perfect competitive market there is no product differentiation exist. Whereas, there is a complete product differentiation and no close substitutes exist; it is a pure monopolistic market.
4) **Entry and Exit Conditions:** In a market, the condition for entry and exit of the firms depends upon the profit or loss of a particular market. The entry of new firms is attracted by profits in the market and exit of weak firms is due to losses in the market. Thus, a perfect competitive market allows freedom of entry or exit of firms while there are barriers to entry of new firms in a monopolistic and oligopoly markets.
5) **Economies of Scale:** Firms which are achieving large economies of scale in production, grow larger than other firms working in an industry. These firms have a tendency to uproot other firms and leaving few firms to compete with one another resulting in the emergence of oligopoly. There exists monopoly when economies of scale are exceedingly attained by a single firm and can meet the total market demand.

4.3.3. Different Market Structures

The degree of competition is an important factor which affects the market types. Following are the two types of competition on which the types of market depend:
1) Perfect Market Structure
2) Imperfect Market Structure

4.3.4. Perfect Market Structure

Many sellers selling same type of products to many buyers is the characteristic of perfectly competitive market. Production of goods by using productive techniques and minimum utilisation of the factors of production is the feature of efficient market. This type of market is considered to be unrealistic but, it is of special importance for hypothetical and theoretical reasons.

4.3.5. Imperfect Market Structure

The situations pertaining to perfect market are different form imperfect market. In any market the conditions being necessary for perfect competition are unsatisfied is the competitive situation of imperfect competition. The market having large number of producers offering closed substitute goods which are not identical as regards to perfect competition. Various forms of imperfect competition are:
1) **Monopoly:** The origin of the word monopoly is from the Greek word *mónos*, single, and *polein*, to sell. This is a form of imperfect competitive market structure described by a single seller and large number of buyers where entry and exit barriers are normally associated with monopolistic situation pertaining to single seller producing for many buyers. The product is greatly discriminated due to the absence of competitive sellers producing closed substitute products.
2) **Monopolistic Competition:** With respect to large number of sellers contributing discriminated product to large number of buyers makes this competition different.
3) **Oligopoly:** Selling of competing products by some sellers to many buyers is called Oligopoly which is originated from the Greek word *oligos*, few, and *polein*, to sell. The characteristic of this type of competition is having limited firms but more than one which are producing homogeneous good. The interdependent strategy between the firms as they are fewer in number in the market clearly states that oligopolistic firm's profit totally depends on the actions of their competitors.
4) **Duopoly:** Duopoly type of oligopoly originates from the Greek word, *duo*, two, and *polein*, to sell. There are only two firms in the market producing a homogeneous goods characterises imperfect competition. Normally to avoid any complications oligopolies are studied by analysing duopolies. Such that, in a market A, there are only two dominant suppliers which means a market where prices are being jointly controlled by them.

4.3.6. Difference between Perfect Competition and Imperfect Competition

Reference	Perfect Competition	Imperfect Competition
Product	Homogeneous	Product Differentiation
Price	Uniform	Not Uniform
Entry of Firms	Free Entry	No Absolute Freedom
Knowledge of Market Condition	Perfect Knowledge	Imperfect Knowledge
Mobility	Perfect Mobility	Imperfect Mobility
Degree of Price Control	No Control Over Price	Control Over Price
Transport Cost	Nil	Exist

4.4. PERFECT COMPETITION

4.4.1. Meaning and Definition of Perfect Competition

A market which consists of a large number of buyers and sellers who are involved in buying and selling homogeneous products with perfect knowledge of the market is called perfect competitive market. Fruit and vegetable market is an example of perfect competitive market.

According to A. Koutsoyiannis, "Perfect competition is a market structure characterised by a complete absence of rivalry among the individual firms".

According to R.G. Lipsey, "Perfect competition is a market structure in which all firms in an industry are price-takers and in which there is freedom of entry into, and exit from, industry".

4.4.2. Features of Perfect Competition

Following are the features of perfect competition:

1) **Large Number of Buyers and Sellers:** The first condition of the perfect competition is the existence of large number of buyers and sellers or otherwise the market price cannot be affected by a single producer or purchaser who varies his supply of demand. Any single firm contributes only a small portion of its output with respect to total output whereas out of the total demand a single purchaser's demand is only a small portion of it.
2) **Homogeneous Product:** The items produced by the firms satisfying the second condition of perfect competition that it must be uniform and alike so that none of the buyers may have any choice for the product of any single seller which is superior to other sellers. The products such as salt, wheat, cotton, coal etc., are the examples of homogeneous products.
3) **Free Entry and Exit:** Under perfect competition, the firms in the industry earn common profit which can happen only if there is no restriction on the firms regarding its exit from the industry or entry into the industry. Therefore, new firms will enter if there is more profit and it will be contested for the extra profit. On the other hand, less profit will result in quitting by some firms where remaining firms will share the boosted profit.
4) **Perfect Knowledge:** There should be perfect knowledge of the market to the buyers and sellers and they must have complete awareness of the **prices** that are being offered and accepted. On the basis of this knowledge there will be an assurance of uniform prices all over the market.
5) **Absence of Transport Costs:** It is assumed that there are free transport facilities. If the prices are same, then it is necessary that no cost of transportation has to be incurred.
6) **Perfect Mobility of the Factors of Production:** For the firms, mobility of the factors of production is important to adjust their supply to demand. When demand is surplus to supply then there will be a movement of additional factors into the industry and they will be moved out into the industry in an adverse situation. The movement of factors of production is very important for the firms and the industry in achieving equilibrium position.
7) **Absence of Artificial Restrictions:** There is freedom to the sellers to sell their goods to any buyer and buyers are free to purchase from any seller. The buyers and sellers must be free from any discrimination.
8) **Absence of Selling Costs:** Due to the production of homogeneous product by the firms, the cost of advertising, sales promotion, etc., is negligible under perfect competition.

4.4.3. Assumptions of Perfect Competition

This analysis is based on the following assumptions:

1) Free entry and exit of the firms.
2) The efficiency of all the firms is same.
3) Due to homogeneity in all the factors of production they can be acquired at constant and uniform prices.
4) Uniformity in the cost curves of the firms.
5) The technology being alike, the plants of all the firms are same.
6) All the firms have perfect knowledge about price and output.

4.4.4. Advantages of Perfect Market

Following are the advantages of the perfect competition:

1) **Consumer Sovereignty:** The perfect competitive market is dominated by consumer. The consumers are rational and they are having perfect knowledge about the conditions of the market. Due to this reason, they will never purchase products at a higher price.
2) **Beneficial to Consumers:** It is beneficial for the consumer that the price is equal to the minimum average cost in the perfect competitive market.
3) **Cost-Saving:** The wastage of the resources is decreased as the producers do not incur expenditure on advertisement to promote sales. This is due to the perfectly competitive firms which are price-takers and products are homogeneous.
4) **Economic Efficiency:** The perfect competitive firm attains optimum level of functioning in the long run. This means that the firms achieve maximum economic efficiency in production. The firms actual produced output is equal to the optimum output and hence, there is no unused, or idle or surplus capacity.

4.4.5. Disadvantages of Perfect Market

Following are the disadvantages of perfect competition:

1) **No Scope for Economies of Scale**: Due to small amounts are being produced by many small firms, there is no scope for economies of scale. Industries having high fixed costs are specifically improper for perfect competition. Due to this reason perfect competition does not exist in the real world.
2) **Homogenous Products** Consumers have less of choice from boring identical products whereas different products are of great importance in the industry such as clothing and cars.
3) **Insufficient Profits:** Investment in R&D is required due to lack of supernormal profits which is important for pharmaceutical industry requiring meaningful investment.
4) **Free flow of Technology:** Due to unavailability of patents, there is no profit in developing new technology because other companies will use these technologies.
5) **Externalities:** Market would be a failure without the interference of the government due to expense in production or consumption.

4.4.6. Price Determination under Perfect Competition

A typical market structure being described by large number of buyers and sellers, free entry and exit of the firms, identical goods and price, buyers and sellers having complete knowledge of market conditions is known as perfect competitive market. Each seller supplies a fraction of the total supply and a single seller and buyer is not capable to affect the market price. The equilibrium price of the commodity under perfect competition is determined by the intersection of market demand and supply curves which are as follows:

1) **Market Demand**: It is the demand for the whole industry at different prices. It is the total of the quantity demanded by each individual consumer or user.
2) **Market Supply:** Market supply is the total quantity supplied by the individual firms in the industry. Therefore, market price is decided for the industry which is given to the each individual firm and each buyer.

In **figure 4.7** the forces of supply and demand determines the perfect competition. At the point E, price is fixed where supply and demand are at equilibrium condition. Any change in equilibrium is occurring due to change in the forces of demand and supply. Thus, in a perfectly competitive market a seller is a 'price-taker, not a 'price-maker'.

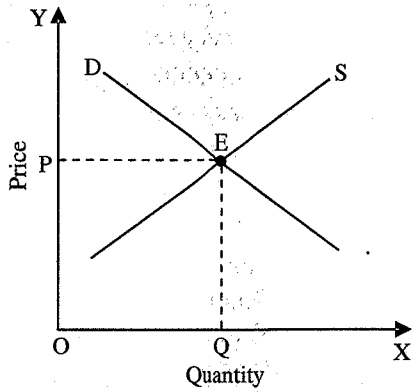


Figure 4.7: Price in Perfect Competition (market equilibrium)

The distinction between market and firm demand is illustrated by **Figure 4.8** and **4.9.** In a perfect competitive situation, considering the market price structure, at the intersection of demand and supply the price is fixed. The price is fixed by the market which has to be followed by the firm. If, the product price is fixed above 'OP' then firm will not be able to sell any product. If the price is fixed below 'OP', then all other competitors will respond to that, resulting in the fall of 'OP'.

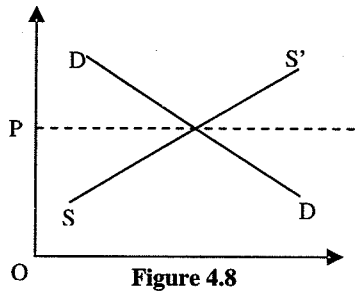


Figure 4.8

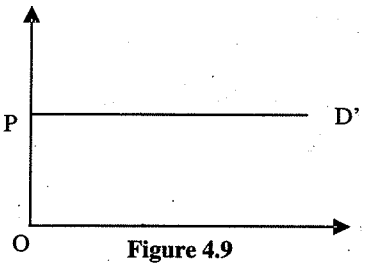


Figure 4.9

The firm's price determination and equilibrium under perfect competition is shown in **figure 4.9**. The firm's demand curve (D') is horizontal at the price determined in the market by the intersection of the demand and supply curves. The demand curve of the firm is the Average Revenue (AR) curve due to the reason that if the firm sells its output units of commodity at the same price, the revenue brought in by an average unit must be its price. when AR curve neither rising nor falling then it will be the same as the Marginal Revenue (MR) curve.

Any change in the market price is occurred due to the change in the market demand or a change in the market supply. If the demand and supply of the commodity changes in the market then the price and the quantity demanded will also be changed.

1) If the demand increases then the prices go up and vice versa. In **figure 4.10**, the demand rises and resulting demand curve shifting to the right from D to D1, whereas the supply remains the same. Due to the change in demand, price move-up from OP to OP1. Hence, the sale increases from OQ to OQ1.
2) Price goes down due to rise in supply and vice versa. This is illustrated in **figure 4.11**. Here, the supply quantity increases and resulting supply curve shifting to the right from S to S1, whereas there is no change in the demand. Due to the change in supply, price falls from OP to OP1. Hence, the sale increases from OQ to OQ1.
3) Given a shift in the demand curve, the price will rise less if the supply curve is elastic (flat). Price will rise more or fall more if the supply curve is inelastic (steep). If the rise in price is more, then the rise in sales will be less. If the rise in price is less, then the rise in sales will be more.
4) The sales will increase if both the demand and supply increase but there may or may not be an increase in price.
5) If the new demand price for the old equilibrium amount is higher than its new supply price, then there would be simultaneous increase in demand and decrease in supply raising the prices and increasing the sales. Hence, if the new supply price for the old amount is higher than its new demand price, the price will rise and will decrease the sales.

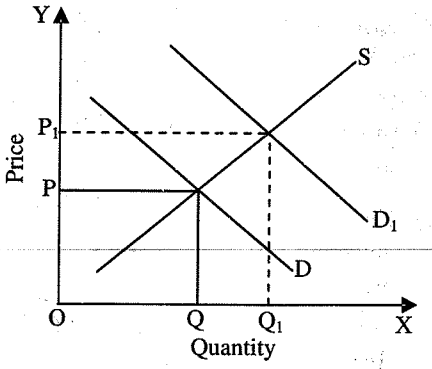


Figure 4.10: Change in Market Demand

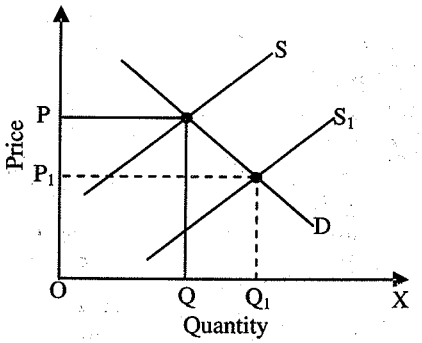


Figure 4.11: Change in Market Supply

4.4.7. Pricing–Output Determination under Perfect Competition

A firm in order to attain the equilibrium (**figure 4.12**) position has to satisfy two conditions:

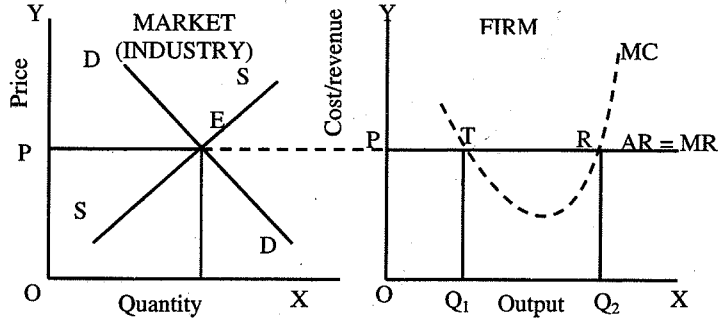


Figure 4.12: Equilibrium Position for a Firm under Perfect Competition

1) The marginal revenue should be equal to the marginal cost i.e., MR = MC. If MR is greater than MC, there is always an incentive for the firm to expand its production further and gain by sale of an additional unit adds more to cost than to reduce output since an additional unit adds more to cost than to revenue. Profits are maximum only at the point where MR = MC.
2) The MC Curve should cut MR curve from below. In other words, MC should have positive slope.

4.4.7.1. Short-Run Industry Equilibrium under Perfect Competition

When the total output remains stable and without any inclination of expanding or contracting its output then an industry is said to be in equilibrium in the short run. Hence, the industry is in equilibrium when all firms are in equilibrium that implies full equilibrium of the industry in the short run where only normal profits are being earned by all the firms. The condition for this is **SMC = MR = AR = SAC**. The full equilibrium of the industry is an abrupt accidental phenomenon as in the short run there are firms which might be earning above normal profits or some are incurring losses. But even then the industry is in short equilibrium at the price that clears the market where quantity demanded equates quantity supplied.

This is illustrated in **Figure 4.13**, where in Panel (1), the industry is in equilibrium at point E where its demand curve D and supply curve S intersect which determine OP price at which its total output OQ is cleared. But at the prevailing price OP some firms are earning supernormal profits PE1ST as shown in Panel (2), while some other firms are incurring FGE2P losses as shown in Panel (3) of the **figure 4.13**.

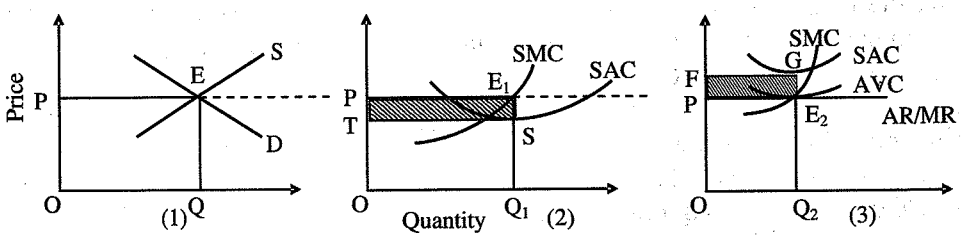


Figure 4.13: Short run Industry Equilibrium under Perfect Competition

4.4.7.2. Short-Run Firm Equilibrium under Perfect Competition

A firm is price taker rather than a price maker in the market. The difference between the role of a competitive industry and a firm in determining the price and output level can be seen through the diagram given below:

1) **Abnormal Profit/ Supernormal Profit:** A firm may get abnormal profit at the equilibrium level of output if its average revenue exceeds the average cost of production. In the **figure 4.14**, firm's equilibrium is attained at point E where the MC curve intersects the MR curve. At OP price, firm produces OQ output. At OQ output, firm's average revenue (AR) is EQ, while its average cost (AC) is BQ. Since AR is more than AC, firm gets abnormal profit. Thus, the firm gets abnormal profit EB per unit of output, total profit being PABE, i.e., per unit profit multiplied by the total output.
2) **Losses**: At equilibrium output, a firm may suffer loss. It is because of the fact that a part of the fixed cost may not be recovered in the short run. In spite of these losses the firm would decide to produce, so long as it is able to recover the average variable cost. In the **figure 4.14**, E is the equilibrium point and at this point AR = EQ and AC = BQ since BQ > EQ, firm is earning BE per unit loss and total loss is ABEP.
3) **Normal Profits or Break-Even:** When the firm just meets its average total cost, it earns normal profits. Here AR = ATC

   The **figure 4.14** shows that MR = MC at E. the equilibrium output is OQ. Since here AR = ATC or OP = EQ, the firm is just earning normal profits.

   Conclusion:
   i) if AR = AC, the firm will get normal profit (i.e., breakeven)
   ii) If AR > AC, the firm will get normal profit.
   iii) If AR < AC, the firm will suffer loss.
   iv) If AR < AVC, the firm will stop production.

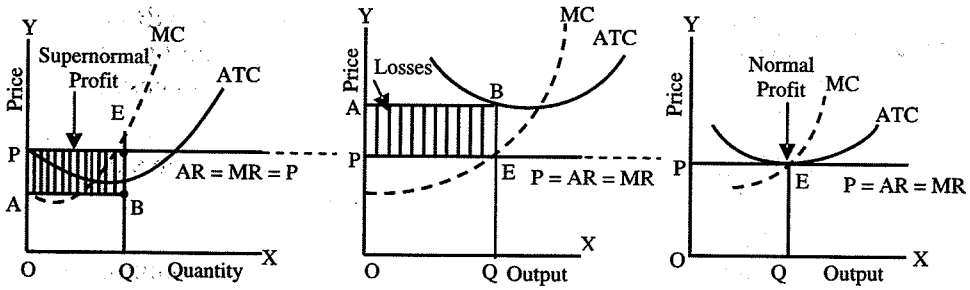


Figure 4.14: Price Determination and Equilibrium of the Firm in Short Run

4.4.7.3. Long-Run Industry Equilibrium under Perfect Competition

The following are the conditions which establish equilibrium in a perfectly competitive industry:

1) For long run equilibrium of an industry it is required that all the firms in the industry are producing an equilibrium level of output where long–run marginal cost equal to the long–run marginal revenue; (LMC = LMR). The total supply of the industry comprises of the total output of these firms.

2) There should be fixed number of firms in the industry and entry of new firms are prohibited. In this situation, all firms must be earning normal profit at the point where the price or LAR = LAC for all the firms. The industry will not attain stable equilibrium in the long run unless only normal profits are earned by all the firms, otherwise if excess of profit is earned by some firms new entrants will be motivated to enter resulting into the changes in the industry supply and market prices in the long run.

   Hence, for an industry to attain equilibrium position it becomes mandatory for all the firms to earn normal profits in the long-run.

3) The long equilibrium price is derived to attain equality between total quantities demanded and supplied in the long-run in liquidating the market. The long-run equilibrium of the industry is depicted by the **Figure 4.15.**

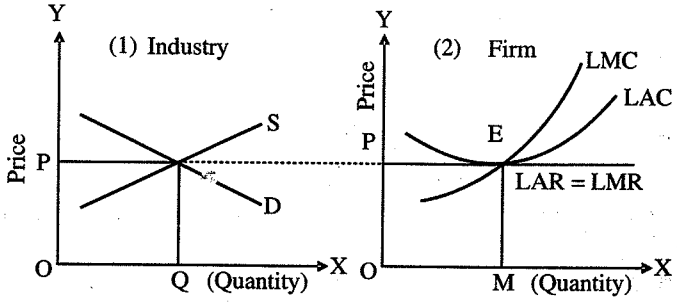


Figure 4.15: Industry's and Firm's Equilibrium in the Long-Run

In **Figure 4.15** the long-run price OP is determined by the intersection of the long-run supply curve S and demand curve D. At this price, the firm's equilibrium is determined by equating LMR = LMC.

Thus, OM is the equilibrium output of the firm in the long run. There is a full equilibrium position; Price = LAR = LMR = LAC = LMC. As such, the firm enjoys just normal profits.

Long-run equilibrium; Price = LAR = LAC = LMR = LMC.
Thus, when all the firms are in equilibrium, earning normal profits, number being stable, the market supply position becomes stable in the long-run and under the given demand condition (D in **Figure 4.15 Panel 1**), the long run equilibrium price (OP) is established making industry in the long-run equilibrium.

The firms under homogeneity conditions are identical – having identical cost functions so they must be operating at the minimum point of LAC (**Figure 4.15 Panel 2**).

There are firms quitting the industry in the long-run as they are inefficient, higher level cost functions, i.e. LAC price, fail to earn normal profits, and unable to sustain losses.

To sum up, industry and firm's equilibrium conditions in the long-run are Long-Run Equilibrium Price = LAR = LAC = LMR = LMC.

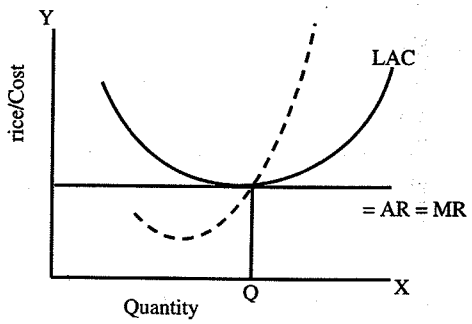


Figure 4.16: Long Run Firm Equilibrium under Perfect Competition

4.4.7.4. Long Run Firm Equilibrium under Perfect Competition

The entry or exit of the firms is not restricted in a perfectly competitive market. Those firms which are inefficient then they incur losses and then either close down the firms or enhance their efficiency. Profit earning firms attract new firms for establishment. Due to the entry of new producers, the supply of the commodity increases and the prices of the commodity decreases. All the firms are at break-even point as a result of vanishing of the profit. This situation is of long run equilibrium of firm and industry in a perfect market whereas all the firms only break-even and earn only normal profits in the long run. The firms work on no-loss-no-profit situation where firm's AR = AC which is graphically plotted in the **figure 4.16.**

In **figure 4.16**, OQ is equilibrium quantity and at this level of output average revenue and average cost both equal to QS. Due to this reason firm is making only normal profits. At this point, since the average cost is minimum, the average cost and marginal cost will be identical, and hence, the long run equilibrium conditions for a firm will be:

MR = LMC = LAC = AR = Price

4.5. MONOPOLY

4.5.1. Introduction

A market situation where there is only one seller and barriers for others to entry, product having no close substitute, cross elasticity of demand being low with every other product and no other firm produces an identical product, is called monopoly. Having control over the price, a monopolist can sell his product at the price of his choice. The price can be fixed by a monopolist at which he will sell his product but at the same time he cannot determine the amount of the commodity that the buyer will buy. His charging of the price is inversely proportional to the demand of his commodity. Hence, the price is fully under control of the monopolist but no control over the demand that is determined by the purchaser.

According to D. Salvatore, "Monopoly is the form of market organisation in which there is a single firm selling a commodity for which there are no close substitutes."

According to A. Koutsoyiannis, "Monopoly is a market situation in which there is a single seller, there are no close substitutes for commodity it produces, there are barriers to entry".

4.5.2. Features of Monopoly

The features of monopoly market are as follows:

1) **Single Seller:** A single producer of a particular commodity or service in a monopoly market gathering large number of buyers, who can be an individual, a group of partners, a joint stock company or a state, it is the alone source of supply for goods and services having no close substitute. The market is referred as 'pure monopoly' as this market structure where the firm is the industry. This is rather being a theoretical concept as sometimes there emerges close substitutes by manufacturers who hold important market share leading to an imperfect form of market which is extremely termed as monopolistic competition.
2) **Restricted Entry:** There is a restriction on the free entry of new organisations as new sellers are prohibited to enter the monopoly market. Following are the primary barriers restricting the entry of new sellers:
   i) Government licence or franchise,
   ii) Resource ownership,
   iii) Patents and copyrights,
   iv) High start-up cost, and
   v) Decreasing average total cost.
3) **Homogeneous Product:** The product which is produced by the monopoly firm has no close substitute and is a homogeneous product. The buyer is forced to purchase the product which is available at the labelled price.
4) **Full Control over Price:** The features of a monopoly market where entry is restricted of the new sellers or buyers making the competition restricted, market conditions are fully influenced by the monopolist thus a company under monopoly is free from price pressure and is independent to charge according to his advantage and with the intention of maximising of profit via predetermined choice of quantity. Therefore a monopolist is a 'price maker' instead of 'price taker' decides the price that is acceptable to the buyers. However, to side step the entry from new participants of market it is essential for the company to manage the set product or service price that is within the example of theorem of monopoly.

5) **Price Discrimination:** The practice of charging different prices from different buyers for the same goods or service is price discrimination. The monopolist has an advantage of being in the market by discriminating the prices as per his convenience.
6) **Increased Scope for Mergers:** Due to execution of control by a single entity under a monopoly, it increases the scope of vertical and horizontal mergers. These mergers maintain supply chain management as well as absorb competition efficiently.
7) **Price Elasticity:** The price increase and market demand dictate the price elasticity to complete value ratio with regards to the demand of the product or service being offered by Monopoly Company or individual. Surplus and loss which are categorised as 'burden' is very common to see within a monopoly. It has been referred recently as gain in hiding the consumer as well as the monopolist.
8) **Lack of Innovation:** There is an inclination of the monopolies towards losing efficiency over a period of time where designing new strategies and marketing skills become secondary due to single market domination.
9) **Lack of Competition:** The scope for 'perfect competition' diminishes when the design of the market serves a monopoly and absence of business competition as well single merchant of unusual good having no substitution and market demand is the demand for induced turnout conforming comprehensive market control. Monopoly turns to be fundamental market failure due to the incompetence as a result of market dominance.

4.5.3. Advantages of Monopoly

Following are the explanation of the advantages of monopoly:

1) **Research and Development:** Funding of high cost capital investment spending is done through supernormal profits which is useful in improving products and lowering of the costs in the long term by using successful research, e.g., Telecommunications and Pharmaceuticals.
2) **Economies of scale:** Monopolies are more efficient than smaller firms as they can produce at lower costs which increase the output and decreases average cost of production. Consumers receive this in the form of lower prices.
3) **Competition for Corporate Control:** Financial market discipline is the basic requirement for monopolies. If potentially low-cost monopolies fail to perform then it is liable to be taken over by a bid.
4) **Stability of Prices:** Mostly, the prices in the monopoly market are stable because there is only one firm involved in the market which fixes the price if and when it feels like. But in other structures of market the price are unstable and elastic due to the existence of competition but it does not exist in the monopoly market.
5) **Source of Revenue for the Government:** Monopoly firms give huge amount of money which is received by the government as tax.
6) **Massive Profits:** Absence of competitors resulting in large number of sales monopoly firms incline to receive heavy profits from their operations. Utilisation of such massive profits in launching of other products, carrying out of research and development which are many among those that are beneficial to firm.

4.5.4. Disadvantages of Monopoly

Following are the disadvantages of monopoly:

1) **Exploitation of Consumers:** A monopoly market is known for the exploitation of consumers where there are no products for competition and the consumer deals crudely in terms of quantity, quality and pricing. Inferior or sub-standard goods are easily produced by the firm depending on its choice as they will be purchased at the end of the day due to lack of competing products in the market that is already available.
2) **Dissatisfied Consumers:** Due to compromise in quality, consumers will be dealt rudely by a monopoly market. Hence, it is very obvious that the consumers who are unsatisfied complain about the products which are produced by the firms.
3) **Higher Prices:** Due to lack of competition in the market, there are no price war which benefits the consumer and as a result of this monopoly firms are free to charge high prices of goods and services.

4) **Price Discrimination:** Price discrimination is a practice in monopoly firms where different prices are charged for the same product from different consumers.

5) **Inferior Goods and Services:** Due to absence of competition the inferior goods and services are produced deliberately by the monopoly firms as they are aware of that the goods will be sold definitely.

4.5.5. Pricing under Imperfect Competition: Monopoly

Like a perfect competitive firm a monopoly firm also aims at earning maximum profits where a monopolist adjusts his output and attains maximum profits by fixing a price. This maximum profit earned by a monopoly firm at that level of output where marginal revenue is equal to marginal cost. If marginal revenue exceeds marginal cost the output will be expanded by the firm which will enhance its revenue more than its cost. Monopoly will keep on expanding its production till marginal revenue continues to exceed marginal cost. The monopoly firm finds equilibrium level of output at the point where marginal revenue is equal to marginal cost. At this level of output firm earns maximum profits. In order to study equilibrium under monopoly let us draw the demand and supply or cost curves of a monopolist.

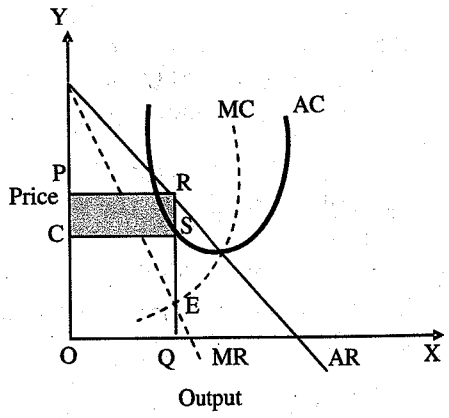


Figure 4.17: Pricing under Imperfect Competition: Monopoly

In **Figure 4.17,** AR and MR are the demand and marginal revenue curves of a monopolist. AC and MC are the respective cost and supply curves. The usual equilibrium of MR = MC is equally applicable to the monopolist. In the figure MR and MC have intersected at point E which is the equilibrium point. At this point the monopolist produces and supplies output quantity Q.

This is the only profit-maximising condition for the monopolist. Under the given demand-cost structure no other level of output can help to enhance its profit. In equilibrium the monopolist charges price OP which is determined by a corresponding point R on the average revenue curve. The total revenue of the monopolist is then,

$$TR = OQ \times OP = OQRP$$

Likewise the total cost of the monopolist is ruled by a point on the average cost curve. QS or OC is the average cost of producing output Q in which the total cost will be:

$$TC = OQ \times OC = OQSC$$

The profits of the monopolist due to the difference between TR and TC are:

$$\text{Profits} = TR - TC = OQRP - OQSC = CSRP$$

4.5.6. Pricing and Output Decisions under Monopoly

The demand curve of the firm is same as of market demand curve under monopoly the slope of the market demand curve is downward as downward sloping demand curves are faced by the monopolist. For profit maximisation the firms must operate at the level of output where marginal revenue and marginal cost are equal. Firms are price makers under monopoly setting market prices affect the individual production decision of the firms.

Under monopoly, price always exceeds marginal revenue as depicted by a downward-sloping monopoly demand curve. Here the price is average revenue because, and a downward sloping demand curve's requirement is that MR is less than AR. Thus, for determining the price-output under monopoly the following conditions must be fulfilled:

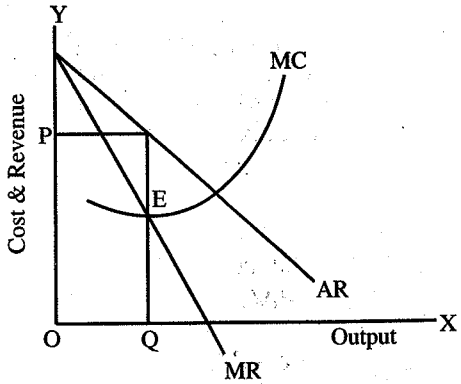


Figure 4.18: Equilibrium Position

Conditions for Equilibrium

Following are the two conditions for equilibrium in a monopoly market which are the basis of price-output determination:

1) MC = MR
2) MC curve must cut MR curve from below

The **figure 4.18** shows that at E, MC curve cuts MR curve that means, equilibrium price is OP and equilibrium output is OQ at E.

Assumptions of Monopoly

The following assumptions are the basis of the analysis for determining the price, output and profit under monopoly:

1) For a homogeneous product there is one seller or producer.
2) There are no close substitutes for the product.
3) Due to pure competition in the factor market the price of each input is given to him for buying the inputs.
4) It is the logical approach of the monopolist to maximise the profit with the minimum of costs.
5) Since, there are a large number of buyers on the demand side but no one is capable to affect the price of the product by one's individual actions. Hence the price of the product is stated for the consumer.
6) The treatment for all the consumers is same in the monopoly and a monopolist charges a uniform price for his product and does not charge varying prices.
7) The power of the monopolist is unrestricted and the price under monopoly is uncontrolled.
8) There is no threat of entry of other firms.

4.5.6.1. Equilibrium of the Monopoly Firm in Short-Run

The monopoly firm attains equilibrium in the short-run at the point where profit is maximum and losses are minimum.

1) **Super-Normal Profit:** In the short run, SAC and SMC are the short-run average and marginal revenue curves, and AVC is the average variable cost curve of the firm. Due to simple reasons, the AVC curve is not shown in **figure 4.19**. D = AR is the demand curve whose equivalent marginal revenue curve is MR.

   At point E, is the short-run monopoly equilibrium where the SMC curve cuts the MR curve from below. The monopolist sells OM output at MP Price. The price MP, being above the short-run average cost MA, the monopolist earns AP profit per unit of output.

   Thus total monopoly profits are:

   AP × CA = the area CAPB.

2) **Normal Profit:** When a monopolist earns only normal profit then it is shown by the short-run equilibrium of the monopolist in **figure 4.19**. OM output is determined by the equality of SMC curve and MR curve at point E which is sold at MP Price. At this level of output, the monopolist earns normal profit as SAC curve is tangent to the AR curve. The monopolist has the knowledge that any level of output other than OM would carry losses because the SAC curve will be higher than the AR curve.

3) **Minimum Loss:** The monopolist incurs losses in a short-run situation as shown in **Figure 4.19.** The equilibrium point E which is determined by the equation SMC = MR. But the monopoly price MP, established by demand conditions, does not cover the short-run average costs of production PA.

   It only covers the average variable costs MP, represented by the tangency of the demand curve D and the AVC curve at point P. PA is thus per unit loss which the monopolist incurs. Total losses are equal to BP × PA= BPAC.

   In this **figure 4.19**, P is the close-down point for this firm. If the price from the point MP is lowered downwards by the market conditions, the production has to be stopped temporarily by the monopolist. The firm will close down.

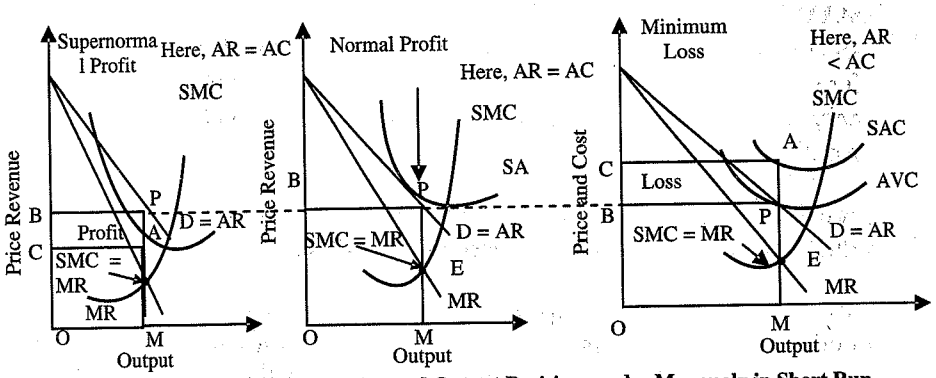


Figure 4.19: Price Determination and Output Decisions under Monopoly in Short Run

4.5.6.2. Equilibrium of the Monopoly Firm in Long-Run

When at that level of output, marginal cost equals marginal revenue the long run equilibrium of the firm under monopoly is achieved. Due to non-entry of the new firms in the market the profit of the monopoly firm is abnormal in the long run. Despite this, there is no loss to monopoly firm as in the long run all the costs are changeable and are to be recovered. Supposing, in the long run the variable cost is not recovered by the monopoly firm then it should abandon production and must leave the market.

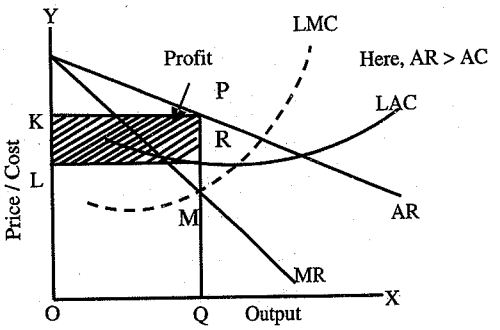


Figure 4.20: Equilibrium of Monopoly firm in Long

In the **figure 4.20**, at point M firm attains equilibrium where LMC of the firm intersects MR. OQ output is produced by the firm at the equilibrium price OK. The profit PR per unit is achieved, because at the output level OQ average revenue of the firm exceeds its average cost, the shaded area KLRP is equal to the total profit.

4.5.7. Comparison of Monopoly & Perfect Competition

The markets of monopoly and perfect competition have some similarities and dissimilarities where dissimilarities are more important than similarities.

4.5.7.1. Similarities between Monopoly and Perfect Competition

Following are the similarities of the two market situations:

1) **Goal:** The firms which are operating under perfect competition or monopoly have only one objective of maximising their profits and these types of firms are called rational firms.
2) **Manager:** The ownership being the same in both the markets, hence, manager is the owner of the firms in both the markets.
3) **Product:** Homogeneous product is produced in both the markets.
4) **AC and MC Curves:** The AC and MC curves in the short-run and in the long-run are U-shaped in both the markets.
5) **Equilibrium:** In both the markets equilibrium is achieved by equating MR with MC. Thus a firm operating under perfect competition and monopoly will be in equilibrium at that level of output where MR equals MC.
6) **Short Run Profit:** The firm in the short run equilibrium may earn normal, super normal, or sub normal profit operating in the market, either under perfect competition or monopoly.

4.5.7.2. Difference between Monopoly and Perfect Competition

Basis of Difference	Monopoly	Perfect Competition
1) **No. of Firms**	Existence of only one firm.	Existence of large number of firms.
2) **Nature of Product**	Under monopoly type of competition there is only one firm and only single product.	Homogeneous products are produced.
3) **Entry of Firms**	'No entry.'	Free entry and exit of firms.
4) **Degree of Monopoly Power**	Existence of full monopoly power.	Monopoly power is zero.
5) **Price Policy of Firm**	There is a single seller and he is the price-maker.	Seller is the price-taker.

4.5.8. Price Discrimination under Monopoly

Sometimes, the monopolist can charge different prices for the same commodity from different buyers, provided these buyers form different markets or belong to what are called non-competing groups. This is known as **discriminating monopoly** or **price discrimination.** Price discrimination refers charging different prices from different customers or for different units of the same product.

According to Mrs. Joan Robinson, "Discriminating monopoly or price discrimination is charging different prices for the same product or same price for the differentiated product".

The product may be differentiated by time, appearance, or place so that the purchasers are not able to shift to the low price commodity. The product is basically the same but may have slight or illusory difference. **For example,** a cinema theatre has different charges on the basis of the location of seats, though the same movie is shown at a particular time and the cost of its exhibition does not differ between customers. No doubt, the comforts differ between classes, but they are generally not in proportion to the charge differences.

4.5.8.1. Conditions for Price Discrimination

Price discrimination is possible only under the following conditions:

1) **Market Imperfections:** Price discrimination is possible when there is some degree of market imperfection. The individual seller is able to divide and keep his market into separate parts only if it is imperfect.
2) **Agreement among Rival Sellers:** Price discrimination also takes place when the seller of a commodity is a monopolist or when rivals enter into an agreement for the sale of the product at different prices to different customers.
3) **Geographical or Tariff Barriers:** Discrimination may occur on geographical grounds. The monopolist may discriminate between home and foreign buyers by selling at a lower price in the foreign market than in the domestic market.
4) **Differentiated Products:** Discrimination is possible when buyers need the same service in connection with differentiated products. Railways charge different rates for the transport of coal and copper.
5) **Ignorance of Buyers:** Discrimination also occurs when small manufacturers sell goods made to order. They charge different rates to different buyers depending upon the intensity of their demand for the product.

4.5.8.2. Objectives of Price Discrimination

There are following objectives of price discrimination:

1) **Maximisation of Revenue:** The basic objective of price discrimination is that, by setting different prices for the same product in different markets / segments, a business can increase its total sales revenues. Price discrimination is clearly in the interests of businesses who achieve higher profits. A discriminating monopoly is extracting consumer surplus and turning it into extra supernormal profit. Of course businesses may not be driven solely by the aim of maximising profit. A company will maximise its revenues if it can extract from each customer the maximum amount that person is willing to pay.
2) **Prevent Consumer Switching:** A dominant firm may "exploit" final consumers by means of price discrimination, with the result in the reduction of consumer welfare. It is a barrier to prevent consumers switching from one supplier to another. The business must be able to prevent "consumer switching" – a process whereby consumers who have purchased a product at a lower price are able to re-sell it to those consumers who would have otherwise paid the expensive price.
3) **Attain a Single Market:** It is sometimes a policy objective to attain a single market across the region. There being one seller of the product under monopoly, the monopoly firm is the industry itself. Therefore, the demand curve for its product is downward sloping to the right, given the tastes and incomes of its customers. It is a price-maker which can set the price to its maximum advantage.
4) **Reduction in Rivals:** Price discrimination can be used by a dominant firm to weaken actual or potential rivals. Price discrimination also might be used as a predatory pricing tactic – i.e. setting prices below cost to certain customers in order to harm competition at the supplier's level and thereby increase a firm's market power.

4.5.8.3. Types of Price Discrimination

There are three main types of price discrimination practiced by the monopolists:

1) **First Degree Price Discrimination:** The discriminatory pricing that attempts to take away the entire consumer surplus is called first degree price discrimination. First degree discrimination is possible only when a seller is in a position to know the price each buyer is willing to pay. That is, the monopolist knows buyer's demand curve for the product. What the seller does is that he first sets price at the highest possible level at which all those who are willing to buy, purchase atleast one unit each of the commodity. When the consumer surplus of this section of consumers is exhausted, he gradually lowers down the price so that the consumer surplus of the users of the subsequent units can be extracted. This procedure is continued until the whole consumer surplus available at the price where MR = MC is extracted. Also, consider the case of services of exclusive use, e.g., medical services. A doctor, who knows or can guess the paying capacity of his patients, can charge the highest possible fee from visibly the riches patient and the lowest fee from the poorest one. The first degree of price discrimination is the limit of discriminatory pricing.

2) **Second Degree Price Discrimination:** The second degree of discriminatory pricing is to charge different prices for the different quantities of purchase. The second degree price discrimination is also called 'block pricing system'. A different price is charged from different category of consumers. A monopolist adopting the second degree price discrimination intends to siphon-off only the major part of the consumer surplus, rather than the entire of it. The second degree price discrimination is feasible where:
   i) The number of consumers is large and price rationing can be effective, as in case of utilities like telephones and natural gas and also consumer durabilities;
   ii) Demand curves of all the consumers are identical; and
   iii) A single rate is applicable for a large number of buyers.

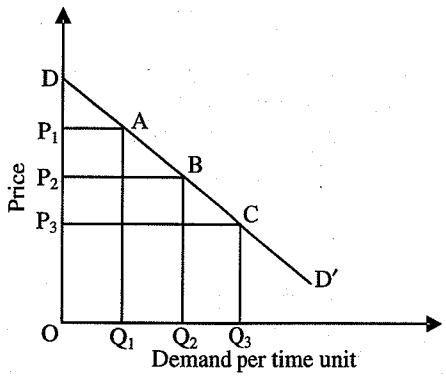


Figure 4.21: Second Degree Price Discrimination

As shown in **figure 4.21,** a monopolist using second degree price discrimination sets the price first at OP_1 and sells OQ_1. After selling OQ_1, he sets a lower price, OP_2, and sells Q_1Q_2 units. After the sale of Q_1Q_2 additional units, he sets still a lower price, OP_3, for the next additional purchase of Q_2Q_3 units. Thus, by adopting a block-pricing system, the monopolist maximises his Total Revenues (TR) as:

$$TR = (OQ_1 \cdot AQ_1) + (Q_1Q_2 \cdot BQ_2) + (Q_2Q_3 \cdot CQ_3)$$

If a monopolist is restrained from price discrimination and is forced to choose any one of the three prices, OP_1, OP_2 or OP_3 his total revenue will be much less.

3) **Third Degree Price Discrimination:** When a profit maximising monopoly firm sets different prices in different markets having demand curves with different elasticises, it is using third degree price discrimination. A monopolist is often faced with two or more markets, completely separated from each other – each having a demand curve with different elasticity. Therefore, a uniform price cannot be set for all the markets without loosing the possible profits. The monopolist is, therefore, required to allocate total output between the different markets so that profit can be maximised in each market. Profit in each market would be maximum only when MR = MC in each market. The monopolist, therefore, allocates its total output between the markets in such proportions that in all the market his MR = MC.

4.5.8.4. Price and Output Determination under Discriminating Monopoly

Suppose there are two markets to which a price discriminating monopolist has to sell his product – market A and market B. Both markets have different price elasticity – demand is more elastic in market B than in market A. **Figure 4.22** illustrates price and output determination under price discrimination. The monopolist sells his product in two markets A and B. Market A has high elastic demand for the product and market B has low elastic demand. Accordingly, the demand curve in market A is Da and its corresponding marginal revenue curve is MRa and in market B the corresponding curves are Db and MRb. Panel 3 in above figure shows AMR, the total marginal revenue curve drawn by the lateral summation of the MRa and MRb curves, and MC is the marginal cost curve. The point of intersection between the AMR and MC curves at E determines the

equilibrium level of output OM. The monopolist divides this output between the two markets by equating the marginal cost ME with the marginal revenue of each market. To equal the marginal cost ME with MRa and MRb draw a line EA parallel to the horizontal axis. It cuts MRa at E1 and MRb at E2 which become equilibrium points for the sale of output in each market. Thus, the quantity sold in market A is OM_1 and in market B it is OM_2 so that $OM_1 + OM_2$ equal the total output OM. The price in the highly elastic (foreign) market is M_1P_1 and in the less elastic (domestic) market M_2P_2, $M_2P_2 > M_1P_1$.

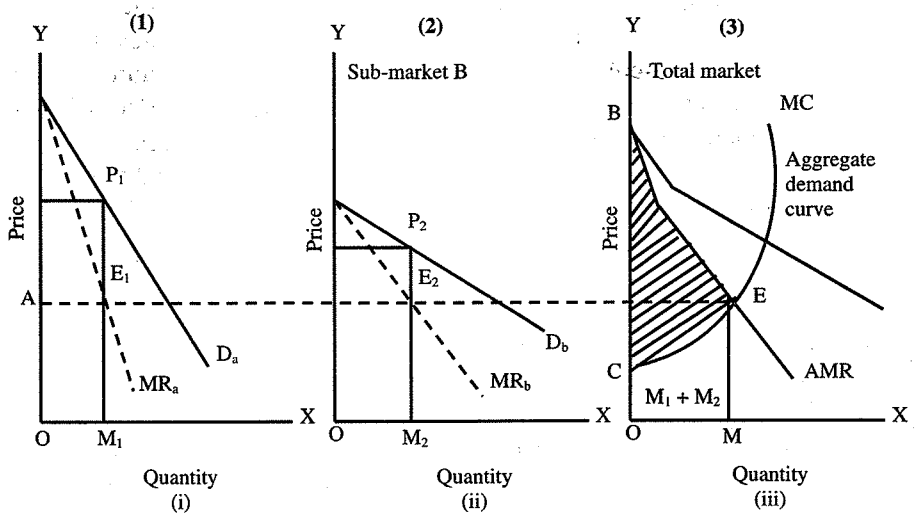


Figure 4.22: Price Discrimination in two Markets A and B

Total profits earned by the discriminating monopolist are MEC. We may conclude that under price discrimination the monopolist sells his product in two separate markets with different elasticity of demand so that he maximizes his profits when he sells more at a lower price in the foreign market with elastic demand and sells less at a higher price in domestic market with less elastic demand. It follows that when marginal revenues equal and prices differ in the two markets, price discrimination is possible and profitable.

4.6. BILATERAL MONOPOLY

4.6.1. Introduction

Bilateral monopoly is a market consisting of a single seller (monopolist) and a single buyer (monopsonist). In this type of market the seller tends to charge high prices on the buyer and the buyer will tend to pay a price that is as low as possible. Thus a transaction in a bilateral monopoly occurs only through a bargaining between the buyer and the seller and in this type of transaction there is no fear of a third person coming in. Thus, this type of market delays the transaction for the reason that the buyer and seller need to arrive at a better deal through their bargaining power.

For example, if a single firm produced all the copper in a country and if only one firm used this metal, the copper market would be a bilateral monopoly market. The equilibrium in such a market cannot be determined by the traditional tools of demand and supply. Economic analysis can only define the range within which the price will eventually be settled. The precise level of the price (and output), however, will ultimately be defined by non-economic factors, such as the bargaining power, skill and other strategies of the participant firms. Under conditions of bilateral monopoly economic analysis leads to indeterminacy which is finally resolved by exogenous factors.

4.6.2. Assumptions of Bilateral Monopoly

The assumptions of bilateral monopoly are given below:

1) There is a single commodity with no close substitutes.
2) The monopolist is its sole producer or seller.
3) The monopolist is its only buyer.
4) The monopolist and the monopolist are both free to maximise their own individual profits.

4.6.3. Price-Output Determination of Bilateral Monopoly

Given these assumptions, price and output determination under bilateral monopoly is illustrated in **Figure 4.23** where D is the demand curve of the monopolist's product and MR is its corresponding marginal revenue curve of the monopolist. The MC curve of the monopolist is the supply curve (S) facing the monopolist. The upward slope shows that if the monopolist wants to buy more, he will have to pay a higher price.

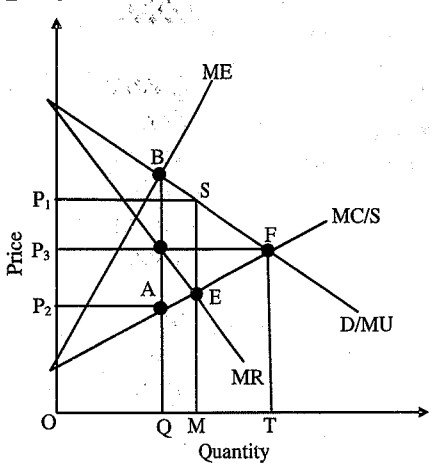


Figure 4.23: Price-Output Determination of Bilateral Monopoly

So when he buys more units of the product, his marginal outlay or marginal expenditure increases. This is shown by the upward sloping ME curve which is the marginal expenditure curve to the total supply curve MC/S. The curve D is the marginal utility (MU) curve of the monopolist.

Let us first take the equilibrium position of the monopolist. The monopolist is in equilibrium at point E where his MC curve cuts the MR curve from below. His profit maximising price is OP_1 (=MS) at which he will sell OM quantity of the product. The monopsonist is in equilibrium at point B where his marginal expenditure curve ME intersects the demand cure D/MU. He buys OQ units of the product at OP_2 (=QA) price, as determined by point A on the supply curve MC/S. So there is disagreement over price between the monopolist who wants to charge a higher price OP; and the monopsonist who wants to pay a lower piece OP_1 From a theoretical viewpoint, there is indeterminacy in the market.

In actuality, the actual quantity of the product sold and its price depends upon the relative bargaining strength of the two. The greater the relative bargaining strength of the monopolist, the closer will price be to OP_1 and the greater the relative strength of the monopsonist, the closer will price be to OP. Thus the price will settle somewhere between OP_1 and OP_2. If the monopoly and monopsony firms merge into a single firm with the monopsonist taking over the monopoly firm, the MC/S curve of the monopsonist becomes his marginal cost curve. The merged firm would thus maximise its profits at point F where its MC/S curve cuts the D/MU curve. It will supply and use OT output at OP_3 price. In this situation, the merged firm gets much larger output (OT) than the monopoly output (OM) at a lower price (OP_3) than the monopoly price (OP_1).

However, it may not be possible to merge the monopoly firm with the monopsony firm. Economists have suggested another solution to the problem of bilateral monopoly that of join profit maximisation. In this case, the monopolist and monopsonist agree on the quantity to be sold and bought to each other but disagree on the price to be charged. On this basis, they want to maximise joint profits because they feel that they have got information about each other's wants and aspirations.

This case is illustrated in **Figure 4.24** where the monopolist is in equilibrium at A when his MC/S curve = MR curve. He wants to sell OQ quantity at OP_1 (=QB) price. On the other hand, the monopsonist is in equilibrium at point B when his demand curves D/MU = ME curve.

He wants to buy OQ quantity at OP_2 price. Depending on the relative bargaining strength of each, the price can be anywhere between P_2 and P_1 and is thus indeterminate. But their joint profits are $P_1P_2 \times OQ$ that can be divided between the monopolist and the monopsonist.

The division of joint profits is also a theoretical possibility like the solution of the bilateral monopoly problem which is indeterminate.

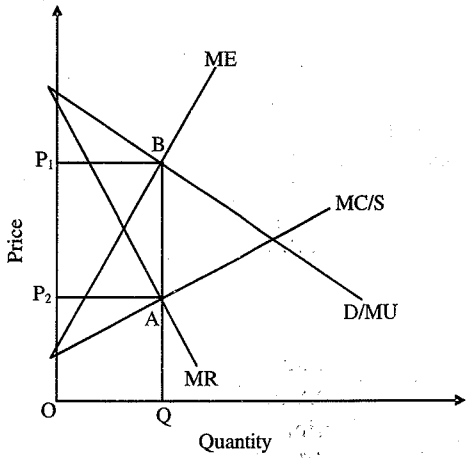


Figure 4.24: Price-Output Determination of Bilateral Monopoly

4.7. MONOPOLISTIC COMPETITION

4.7.1. Meaning and Definition of Monopolistic Competition

Monopolistic competition means a situation in market where differentiated products are sold in market by many firms. "There is a keen competition which is not perfect though among many firms producing very alike products". Firms cannot have any influence that can be perceptible on the other seller's price-output policies. Also it cannot be much influenced by their actions. Thus, monopolistic competition means the competition among huge number of sellers that produce close substitutes for each other which are not perfect substitutes.

According to Baumoul, "The term monopolistic competition refers to the market structure in which the sellers do have a monopoly (they are the only sellers) of their own product, but they are also subject to substantial competitive pressures from sellers of substitute product".

Imperfect competition is referred to that market situation in which one or more of the conditions of perfect competition are absent. The main form of imperfect competition is Monopolistic competition.

4.7.2. Features of Monopolistic Competition

Some of the features of monopolistic competition are as follows:

1) **Large Number of Sellers:** There are huge sellers in a monopolistically competitive market who have a small market share individually. These huge firms unlike perfect competition do not produce perfect substitutes. These firms produce and sell close substitute products. This makes the tough and real competition among firms.
2) **Product Differentiation:** The different seller's products are differentiated in a monopolistic competitive market on the basis of brands. There is mostly so much advertisement of such brands that brand association of customer with manufacturer and a kind of loyalty towards brand is developed. A monopoly element is aroused because of product differentiation to the producer over the product that is in competition.
3) **Freedom of Entry or Exit:** Market allows free entry of new firms and also allows exit of existing firms.
4) **Independent Behaviour:** Every firm possess an independent policy in monopolistic competition. Since, there are a large number of sellers; major portion of total output is not controlled by anyone. By updating price-output policy, no sellers can have any perceptible impact on other's sale and could not influence them.
5) **Product Groups:** Under monopolistic competition, there is not any 'industry' but 'group' of firms that exist which produce the similar product. Each firm is an industry in itself and produces a different product. Very closely related products are produced by the Chamberlain lumps together and these are called as product groups. **For example,** cigarettes, cars, etc.
6) **Selling Costs:** Sales are essentially pushed up by the selling costs because product differentiation exists in monopolistic competition. It includes salesman expenses, advertisement costs, seller's allowances for displaying windows, free sampling, premium gifts and coupons, free service, etc.
7) **Non-Price Competition:** A firm increases profits and sales of its product in monopolistic competition without any price cut. Products can be changed by a monopolistic competitor either by quality variation, packaging variation or by updating promotional programmes, etc.
8) **Control over Price:** Prices are controlled by the firms to some extent. **For example,** professionals such as solicitors, restaurants, etc., are price maker. It is also said that their prices are administered by them. A price which is set by the firm's conscious decision is termed as Administered Price. Also, firms must decide the price changing frequency.

4.7.3. Advantages of Monopolistic Competition

Following are some advantages of monopolistic competition:

1) **Promotion of Competition (Lack of Barriers to Entry):** One important feature of this market is lack of entry barrier which makes it relatively easy for other firms to exit and enter the market. A barrier to entry creates difficulty for new firms due to advertising, intellectual property rights, large start-up costs, etc. Therefore, in long-run this ensures that none of the firm will find themselves with power of monopoly (to exploit consumers) because of new firms entering in the market.

2) **Differentiation Brings Greater Consumer Choice and Variety:** One of the major advantage of monopolistic competition is that in order to be a firm that is competitive in marketplace; a firm has a main goal to differentiate it from other rivalry competitors to gain greater customer advantage that is much essential appealing to sovereignty of consumer. Consumer sovereignty means where goods to be produced are determined by the consumers within a market. Greater choice and variation of products and services is also with this for consumers to buy from rather than single choice; they have wider range of consumer choice. There may be either monopoly that is just one product, or perfectly competitive where all products are homogenous and generic.
3) **Product and Service Quality Development:** Monopolistic competition possesses an advantage of enhancing a firm's ability to improvise quality of a product with the help of its brand. Branding is defended by the economists as a way of enhancing consumer's trust and reliability. Brands help to raise the strength of maintaining high quality that is based on the financial stake of business in its reputation.
4) **Consumers become More Knowledgeable of Products:** Monopolistic competition provides a positive externality which is accompanied the intense marketing and advertising. The reason behind this is that firms try to differentiate their products that make the consumer more aware and informed about the options in products and services. This helps to gain understanding about the uniqueness, features and aspects that a particular product possess in comparison of other products. Hence, this leads to further competition because firms are able to recognise the consumer's wants and degree of demand.

4.7.4. Disadvantages of Monopolistic Competition

Following are the disadvantages of monopolistic competition:

1) **Liable of Excess Capacity:** In monopolistic competition, firms do not produce enough output which leads to lowering down of average cost and benefits attained from economies of scale. Due to less marginal revenue as compared to marginal cost, they are reducing their 'economic profits'. In fact, at some levels, the expense and funding going into marketing, packaging and advertising can be considered as extreme waste.
2) **Allocatively Inefficient:** As compared to perfect competition, it can be shown that firms undergoing monopolistic competition possess an allocation efficiency element as the price is above the marginal cost curve and less in long-run as competition is more. As demand curve slope downward, this shows that for a monopolistically competitive firm the price must be greater than the marginal cost. The product produced for society to benefit is not enough which shows it is allocatively inefficient because they want more, results the company to lose money.
3) **Higher Prices:** Monopolistic competition imparts another drawback that is they have tendency to extenuate a mark-up on the revenue's marginal cost because of firms having 'some market power' in comparison of a perfect competitive firm having prices equal to marginal cost. This arises difficult to governmental authority regarding regulation for following 2 reasons:
   i) Many firms are there, and
   ii) There would be loss made by these firms, hence this forces out such firms from business eventually.
4) **Advertising:** Marketing and Advertising is beneficial to customers to some extent because it provides information about the products and services. Also this leads to increase in the competition and thus can also impact negatively the consumer sovereignty. There has been argument about manipulation and distortion regarding consumer's desires along with obviously reducing competition. The reason of this being, consumers becoming captivated over the differentiation perception.

4.7.5. Pricing under Imperfect Competition: Monopolistic Competition

As there is product differentiation between firms in a monopolistic competitive market, a perfect elastic demand is not faced by the firm for its products. Each firm is capable to determine its own product's price and thus each firm is termed as a price maker. Generally, there is downward sloping of firm's demand curve for its products. Also, with lesser product differentiation as compared to competitors, there will be more elasticity in the curve.

Conditions for the Equilibrium of an Individual Firm

The price-output equilibrium and determination conditions of an individual firm may be described as follows:

1) Marginal Cost = Marginal Revenue, and
2) There must be intersection of MR curve by MC curve from below.

4.7.5.1. Short-Run Equilibrium of the Industry

The analysis in short-run of the firm under monopolistic competition is based on some assumptions. These assumptions are as follows:

1) There are large numbers of sellers who act independently of each other. These sellers are monopolist in their own sphere,
2) Each seller's product shows differentiation from other products,
3) A determinate elastic demand curve (AR) is attained by firm,
4) There is perfect elastic supply of factor-services for the production of the question product,
5) Each firm's short-run cost curves differ from each other, and
6) There is a restriction to entre new firms in the industry.

Explanation

Under these assumptions, a price and output is fixed by each firm so as to maximise the profits. There is an equilibrium point of price and output at which the Short-Run Marginal Cost (SMC) is equal to marginal revenue. Since in short-run, there is difference in cost, only normal profits are earned by the firm with lower unit costs. There will be loss in case while it is able to cover up the average variable cost.

1) **Super-Normal Profit:** In **figure 4.25** at point E, the SMC (short-run marginal cost) curve cuts the MR curve. This point E shows the output OQ and price QA (= OP). Resulting from this, the firm earns supernormal profit which is represented by the area PABC.
2) **Normal Profit: Figure 4.25** shows the same equilibrium of output and price. But the firm only covers the short-run average unit cost in this case as shown by the demand curve D's tangency and the short-run average unit cost curve that is SAC located at A. It earns normal profit.
3) **Minimum Loss: Figure 4.25** represents a condition where the firm is incapable of covering its short-run average unit cost and thus incurs losses. The equality of MR and SMC curves helps to set price at point E and the price is QA that occupies the average variable cost only. At point A, the tangency of AVC (the average variable cost curve) and the demand curve D makes it a shutdown point. When the price is lowered below QA by the firm, then firm will stop further production. However, there will be a loss to firm at this price which is equal to area CBAP during the short-run in the hope of cost lowering in the long-run.

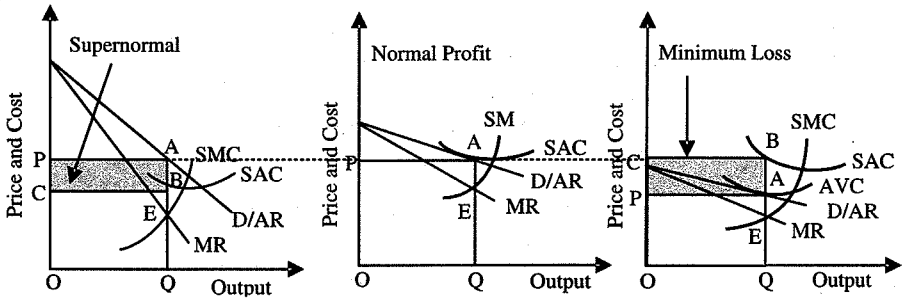


Figure 4.25: Pricing and Output Decisions under Monopolistic Competition in Short Run

4.7.5.2. Long-Run Equilibrium of the Industry

There is exit and entry of the firms in long-run in a competitive industry that is monopolistic. The process of adjustment will lead to existence of normal profits only. In the long run, this is a realistic assumption that no firm can incur either loses or earn super-normal profits. This is because of similar products in the market.

In a monopolistic competitive industry, if the firms in short-run earn super-normal profits, there will be an incentive to enter for new firms in the industry. Profits per firm will keep on decreasing with entry of more firms as

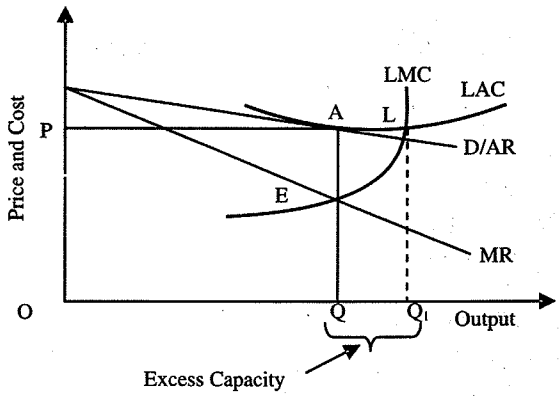


Figure 4.26: Long-Run Equilibrium of the Industry

there will be sharing of product demand among large number of firms. This happens till wiping of all profits and then only normal profits are earned by the all firms. Thus, it is clear that all firms will earn only normal profits in long run. In the **figure 4.26**, in long run all firms are at equilibrium at point E where (1) LMC = MR, and (2) MR is cut by LMC from below and the curve LAC is tangent at point A to the D/AR curve. Since at point A, price QA = LAC, normal profits are earned by each firms and there is no tendency of firms to enter or exit the industry.

4.7.6. Product Differentiation

Product differentiation is basically preference given by the customer to one product over the other. The reason of preference is different such as quality, looks, packing, advertisement, price, sale promotion policy of the company, expertise of salesman etc. The reason of preference may not necessarily be real but customers think it so because of above reasons. Due to preference shown by the consumers for a product, the producer also gets a little liberty to cover price of the product. The producer also takes advantage of the situation to certain extent. The producer also faces tough competition by close substitute products offered by other companies. The preference given to a particular product of a particular company by the customers due to their past experience makes them loyal to a particular brand which negatively affects the demand of similar products in the market. The preferential purchasing of product by consumer lies with a particular belief due to some logical reasons.

4.7.6.1. Aspects of Product Differentiation

Various aspects of product differentiation are as follows:

1) **Product Attributes:** Physical appearance, improved quality, customer friendly function i.e., easy to operate by one and all, easy to place in suitable location, improved material giving it more strength and less weight, design as per latest trends in the market and final finish of the product are some of the basic attributes for differentiation. **For example,** Personal computers vary from capacity, speed, software, graphic display as well as price range. Text books on different subjects also differ in content, presentation, organisation, readability and aids for teaching, graphics and design. The retail store for men and women has different varieties of style, material and quality. Furnitures have different varieties i.e., oak, maple, steel etc., and each has its positive and negative points as well as price range suiting to different segments of society.

2) **Some Control over Price:** Every product including the monopolistic product has to maintain a control over its price due to easy availability of different substitutes at competitive price in the market. The consumer preference for a particular product brand is also limited to certain payment limit. If the producer does not care for this limit, the customers are likely to switchover their loyalty to nearest available substitute.

3) **Service:** Service and surrounding atmosphere also create product differentiation. Some customers prefer to purchase product from a departmental store where they get good collection of goods, personal help, good manners of the staff as well as easy exchange of goods and personal help. However, some customers ignore all these aspects and prefer whole sale store where they get products at cheaper rates. Customers certainly prefer a drycleaner who returns the clothes in one day over the one who returns in three days.

4) **Location:** Products available at easily accessible locations are preferred by customers over those not so easily accessible. Small stores which do not match in facilities, infrastructure and variety of products compared to departmental stores easily compete due to their prominent location and they also charge higher prices. Motels situated on highways always charge more for their room accommodation than motels which are at less convenient location.

5) **Brand Name and Packaging:** Brand names, style of packaging and product association with a celebrity also make a difference in preference of customers. The brand name accompanies a history of proven quality and long period association with public. It carries trust of customers with it which makes them to go with it. Britania Biscuits, Godrej Furniture, Anacin, Disprin tablets for headache are **some examples** of brand products. Some companies hire celebrities like Amitabh Bachchan, Dhoni, Sachin Tendulkar, Aishwrya Rai etc., to promote their products through advertisements on TV channels. It certainly promotes preference of customers. The packaging with a print of gods and goddesses, shrines, historical forts or famous gates or famous natural sites etc., attract customers.

4.7.6.2. Advantages of Product Differentiation

The advantages of product differentiation are given below:

1) Due to product differentiation the production, sales promotion, inventory, as well as administrative costs are less. The cost of their brands goes up. Producing one brand rather than producing many is simply cheaper.

2) Concentration is made for single product, this helps to achieve consumer dominance as well as sales.

4.7.6.3. Disadvantages of Product Differentiation

The disadvantages of product differentiation are as follows:

1) Due to differentiation of product, the sale remains limited to substitutes and other products are not sold and go waste. The surety of sale of product due to differentiation brings lethargy on producers of other products who do not make efforts to further improve their product. The people involved in marketing of other products also do not bother to improve their efforts on sale promotion. Thus it brings waste and inefficiency in the market.

2) The expenditure on advertising also goes up due to endorsement by celebrities. It creates hurdles for other competitors and discourages competition.

3) Due to monopolistic nature of product in the market the variation of attributes is little. The customers are deprived of different varieties attributes which the other products may offer.

4.7.7. Difference between Monopolistic Competition and Perfect Competition

Points of Difference	Monopolistic Competition	Perfect Competition
1) **Number of Firms**	Many firms but in comparison to perfect competition there are less firms.	Large number of firms.
2) **Nature of Product**	Differentiation in product.	Product homogeneity.
3) **Taker or maker**	Price-maker.	Price-taker.
4) **MR and AR**	MR < AR, both are downward sloping.	MR = AR and parallel to the x-axis.
5) **Mobility**	Factors of production are not perfectly mobilised.	Factors of production are perfectly mobilised.
6) **MC Curve**	MC < AR (Price).	MC = AR (Price).
7) **Selling Cost**	Substantial selling costs.	No selling costs.
8) **Production**	Production is below optimum scale.	Production is on optimum scale.

4.7.8. Difference between Monopolistic Competition and Monopoly

Points of Difference	Monopolistic Competition	Monopoly
1) **Number of Sellers**	Large number of sellers.	Single seller.
2) **Nature of the Product**	Product differentiation is there.	No product differentiation.
3) **Degree of Elasticity**	More elasticity.	Less elasticity.
4) **Selling Costs**	Substantial amounts are spent in the form of selling costs.	Little amounts are spent to inform buyers. Merely informative cost is incurred.
5) **Nature of Long-run Profit**	In this case AR = LAC i.e., only earn normal profits.	In this case AR > LAC i.e., average revenue is equal to average cost in long run.
6) **Price Discrimination**	Price discrimination cannot be practiced by the seller.	Price discrimination is mostly practiced.

4.8. OLIGOPOLY

4.8.1. Meaning and Definition of Oligopoly

Oligopoly is defined as a condition where there is huge competition among a few large firms and there is an element of interdependence in firm's decision-making. This interdependence is recognisable by each oligopoly firm. Any decision made by one firm (be it on price, product or promotion) shows impact on competitor's trade resulting into countermoves. This results in dependence of one's competitor behaviour on one's own behaviour.

And thus, this needs to be considered while making decisions. A major change in policy by one firm will show immediate and obvious impacts on its competitors. In this way competitor generally react with their counter-policies. This leads to beginning of a game having moves and countermoves between the rivalry firms. The oligopolists have to use both weapons of defensive and aggressive marketing in order to play this game.

According to P.C. Dooley, "An oligopoly is a market of only a few sellers, offering either homogenous or differentiated products. There are so few sellers that they recognise their mutual dependence".

According to Grinols, "An oligopoly is a market situation in which each of small number of interdependent, competing producer influences but does not control the market".

In other words, an oligopoly is a market situation where number of firms are small (but more than two) which sell differentiated or homogenous products that are much close. Thus, oligopoly may be of following two kinds:

1) With product differentiation oligopoly, and
2) Without product differentiation oligopoly which is also called 'competition among a few'.

For example, automobile industry or cold-drinks industry. There is limited number of manufacturers for automobile and cold drink manufacturing industry in India.

4.8.2. Features of Oligopoly

Features of oligopoly market are as follows:

1) **Interdependence:** Decision-making inter-dependence among few firms of oligopoly market is an important feature of oligopoly. The reason behind this is that when the competitors are few then any change in output, price, product by one firm shows immediate effect on rival's fortune, who will be retaliate to change their own output, price or advertising technique according to the situation. Therefore, this is clear that an oligopolistic firm must take into consideration only demand for industry product in the market along with other firm's reactions in the industry to any major decision taken by it.
2) **Importance of Advertising and Selling Costs:** Oligopolists interdependence shows direct impact on the various firms to employ marketing weapons that are of both natures, aggressive and defensive so as to gain greater market share or to maintain their existing share. Many firms need to incur good deal on advertising and other sales promotion measures for this purpose. Hence, advertising and selling costs play a vital role in oligopoly market.
3) **Group Behaviour:** The oligopoly theory is a group behaviour theory and it is not a theory of individual or mass behaviour. On oligopolist's part, profit maximising behaviour may not be very much valid. Generally, there is no acceptance of group behaviour theory.
4) **Indeterminate Demand Curve:** Due to firm's interdependence in oligopoly and also because of a particular firm's inability to predict other firm's behaviour, determinateness and definiteness is lost by the demand curve facing an oligopolistic firm. Oligopoly's firm do not face a determinate demand curve on interdependence and uncertainty account. There is shift in the oligopolistic demand curve due to reaction of rival firms to changes made by this firm. When one firm changes the price, the product's demand will depend on the price change by rivals.
5) **Few Sellers:** There are a small number of sellers in the oligopoly market. Here, small number of sellers represents that a significant fraction of total industry output is produced by every firm and these can have a noticeable impact on the market conditions.
6) **Aggressive and Defensive Marketing Methods:** Various defensive and aggressive methods are used by the oligopoly firms so as to increase or maintain their market share. They use sales promotion and also extensive advertisement. According to **Prof. W.J. Bannal,** "Advertising can become a death and life matter under oligopoly". This is because every oligopolist tends to face a fierce competition.
7) **Competition and Combination:** The competition is not perfect in oligopoly. On one hand, there may be violent, fierce, cut throat and cruel competition but on the other hand, disadvantages of rivalry and competition are realised by oligopolists. Hence, some policy of collusion may be worked out by the oligopolist firms in order to avoid harmful competition.

8) **Identical or Differentiated Products:** Identical products are produced by some of the oligopolistic industries like perfect competition in this regard, while differentiated products are produced by others that are more like monopolistic competition. While considering both sides of issue of product differentiation, this characteristic seems to be little wishy-washy. It indicates that there are two main varieties of oligopolistic industries. These are as follows:
   i) **Identical Product Oligopoly:** In this type of oligopoly, raw materials are processes or intermediate goods are produced and they are used as inputs by other industries. **For example**, steel, aluminium and petroleum.
   ii) **Differentiate Product Oligopoly:** In this type of oligopoly, main focus is on goods sold for personal consumption. The key is that people possess different needs and wants which makes them enjoy variety. **For example**, household detergents, automobiles and computers.
9) **Small Number of Large Firms:** The main characteristic of oligopoly market is that it is an industry which is dominated by small number of large firms, each of which is relatively large in comparison to the overall market size. This specific characteristic gives a substantial market control to relatively large firms. Each firm does not show much control as monopoly but it certainly has some more as compared to a monopolistically competitive firm.

4.8.3. Advantages of Oligopoly

The advantages of oligopoly are as follows:

1) Huge profits are made by the large firms that have strong hold over the market because there are few market players.
2) Mostly, products of two varying competitive companies are derived out of one main large firm in oligopoly. Hence, whichever company earns profit; ultimately profit is earned by the parent company.
3) Prices can be decided by companies as per their choice because companies in oligopolistic market have full control over it. Though this practice is illegal, it works in favour of these businesses.
4) The companies generate high profits and this profit can be used for development and innovation of new processes and products.
5) The average production cost of goods can be lowered with the help of oligopoly. This is due to the reason that firms which produce similar goods can manufacture products with each other collaboration.

4.8.4. Disadvantages of Oligopoly

Some of the disadvantages of oligopoly are as follows:

1) Though price setting can prove to be advantageous for the firms, but if it's not done in realistic manner, it may tend to be disadvantageous to customers.
2) In the oligopolistic market, small businesses creative plans or ideas fail to realise because of they cannot overcome the major market player's control. Realisation of them is possible only when one of major player uses it.
3) In an oligopolistic market, small business fails to bring their establishment as a brand because most of market is captured by larger firms already.
4) Dominant companies do not think to improvise their products because of little competition. Independent decisions cannot be taken by the firms and they have to always consider the other dominant player's views in the market.
5) Due to presence of various entry barriers, new firms cannot easily enter the market. The goal of micro economy of fair wealth distribution is not fulfilled because major players make the maximum profit leaving small players with little benefits thereby.

4.8.5. Pricing Power/Models of Pricing and Output Determination under Oligopoly

Under oligopoly, there is no definite price-output determination theory. This is because of interdependence of oligopolistic firms to make decisions and uncertainty reaction patterns of rival firms. This leads to uncertainty of each firm's demand curve. Due to Interdependence of the firm's behaviour and uncertain reaction patterns, a variety of behaviour patterns can be obtained. Different assumptions are made by different economists

regarding oligopolist firm's aim and thus different behaviour patterns are assumed by firms accordingly. Different behaviour patterns may be:

1) Co-operation by the rivals in their objective pursuit,
2) Fight among rivals so as to increase market shares, and
3) Wide variety agreements.

Hence, under oligopoly, a large variety of price-output determination models have been developed by economists. This depends on the group behaviour assumptions of oligopolist firms.

The different models are includes:

1) Non-collusive oligopoly model of **Sweezy (Kinked Demand Curve)**, and
2) Collusive oligopoly model.

4.8.6. Kinked Demand Curve: Price Rigidity

This model which is termed as Kinked Demand Curve is a model of oligopoly that was developed by **Paul M. Sweezy and Hall and Hitch.** With the help of kinked demand curve analysis, he tried to show that output and price once determined tend to stabilise and not fluctuate under oligopolistic trend and conditions. Price cuts matches the price cut but price increases do not match price increase is the assumption on which this model is based. The reason behind this is simple that a price rise will lead to sharp sale reduction, but a reduction in price attracts some new customers. That means price established once remains unchanged at same level for longer duration. As per oligopoly pattern, only few firms are there in industry. The kinked demand curve represents a type of partial collusion and main motive of its proposal was to explain why such oligopolies might be characterised by stable prices unusually.

4.8.6.1. Assumptions of Kinked Demand Curve

Some assumptions are being considered for the kinked demand curve hypothesis of price rigidity. These assumptions are as follows:

1) Oligopolistic industry has only few firms.
2) Products produced by these firms are close substitutes of each other.
3) Product differentiation is absent and quality of the product is same.
4) Advertising expenditures are absent.
5) There is a prevailing or established product price in the market at which all the sellers are satisfied.
6) Attitude of each seller depends on rival's attitude.
7) Any attempt on seller's part to raise his sales by price reduction of his product will be counteracted by other following sellers.
8) Other sellers will not raise price if any one seller raises price. They will cater more customers by sticking to existing prices leaving the seller who raises price.
9) Changes in the marginal cost must not impact on the output, so the marginal cost curve tends to pass through the dotted part of marginal revenue curve.

4.8.6.2. Explanation of Kinked Demand Curve

As per observations, prices remain much inflexible for longer time in many oligopolistic industries. There is infrequent change in price even in declining costs faced by the firm. For this rigidity in price, many explanations have been given under oligopoly and out of these explanations, most popular is kinked demand curve hypothesis which was given by **American economist Sweezy.**

According to the kinked demand curve hypothesis, the demand curve that faces an oligopolist has a 'kink' at prevailing price level. The kink is observed at the existing level of price because the demand curve segment above this level tends to be highly elastic and below this level, segment tends to be inelastic. dD, a kinked demand curve with P, a kink point is shown in the **figure 4.27.**

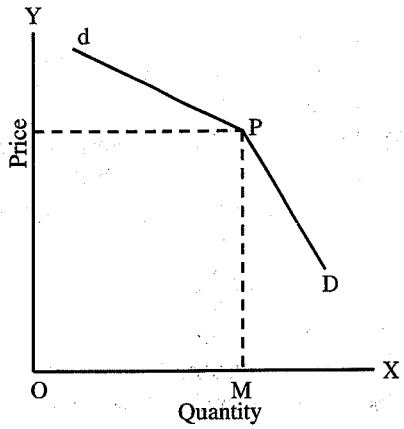


Figure 4.27: Kinked Demand Curve under Oligopoly

MP shows the prevailing price level and OM shows what firm produces and sells output. Here, it is visible that the upper segment dp is relatively elastic and the lower segment PD of the demand curve dD is relatively inelastic. This elasticities difference is because of the specific competitive reaction pattern that is assumed in the kinked demand curve hypothesis. As per belief of each oligopolist, lowering down the price below prevailing price level will make his competitors follow him and they will also reduce prices, whereas raising the price above prevailing price level will make his competitors not follow to increase its price. The reason behind this is that when an oligopolist lowers down its product price its competitors may feel that if they will not follow the pattern then it will make them loose their customers and customers will move to buy the products from lower price providing firm. So, to maintain their customers, they also lower down their prices. This makes upper part of demand curve price elastic. At other side, an increase in price by the firm will cause substantial sales reduction because rise in price will lead to withdrawing of customers and go to competitors who welcome new customers to gain much sales. Therefore, these happy competitors will not motivate to match the rise in price. A great deal is lost by the oligopolist who raises its price and thus refrain from price hike. This oligopolists behaviour explains the inelastic lower part of the demand curve. Thus, each oligopolist will stick to the prevailing price after analysing that no gain in changing it and formation of kink will be there at the prevailing price. Hence, **Keynes** demand curve theory helps to explain the sticky or rigid prices.

4.8.6.3. Criticisms of Kinked Demand Curve

Criticism of kinked demand curve is explained on the following grounds:

1) **Does not Explain how Prevailing Price is Determined:** Kinked demand curve explains simply why and how a price once set remains stable in the market. Therefore, it explains half-truth and not the full. Further, another version of kinked demand curve given by **Hall and Hitch** attempts to explain how determination of prevailing prices is done. But again that is not a valid explanation too.
2) **Does not Explain Price-Output Determination:** Firms behave in a concerted manner under oligopoly that is collusive. Under this oligopoly, there is no kink on the firm's demand curve.
3) **Kinked Demand Curve Theory is Applicable in Depression Only:** It is already stated that with decrease in demand, as per kinked demand curve hypothesis, the price remains stable most likely. But this does not happen in case of increasing demand i.e., price does not remain same there. Hence, we can say that this theory is applicable only in depression and not in boom. Kink in the demand curve may vary while the condition of boom is there.
4) **Explains Price Rigidity in Differentiated Oligopoly Only:** Price rigidity is explained by this theory in only differentiated oligopoly. In fact, its application is not certain in differentiated oligopoly as well. It is possible that the marginal curve's vertical position of a kinked demand curve may be not much wide for marginal cost curve to pass through it.

4.8.7. Collusive Oligopoly

Competing firms are termed as collude when they make some sort of agreement about output and pricing. These agreements may be either facet or formal. But open or formal agreements are not legal in most of the countries. The oligopolists agreement between two firms is generally kept secret or tacit. Collusive oligopoly comes in frame when firm enters the collusive agreement. Collusion can be of following two types:

1) Perfect/ Cartels collusion, and
2) Imperfect/ Price leadership collusion.

4.8.7.1. Cartels /Perfect Collusion

Market price is driven down to production cost level when rival oligopolists enter into competition over price with each other. Therefore, there is a strong incentive for oligopolists collusion, this leads to rise in price and restriction in output. Opposite of competition is collusion. It states that the firms cooperate with each other so as to take joint actions that keep their bargaining position stronger against the customer. There are cartel arrangements primarily in the perfect collusion. A Cartel is basically an explicit agreement that is carried out among independent firms on various subjects such as output, prices, market share etc. The firm's desire to earn large joint profits provides impulse so as to form cartels. But this kind of desire is short lived and thus, the cartel's formal arrangement cannot be a phenomenon of long duration. There are basically following two type of cartels:

1) Centralised cartels, and
2) Market-sharing cartels.

In case of centralised cartels, the output, pricing, distribution and sales of profits is done by a central cartel association of all the firms. In case of market sharing cartels, the firms that form cartel would agree on the market shares, with or without any common understanding on prices.

1) **Centralised Cartels:** The output and price decisions for the whole industry along with every member firm are taken by Central Cartel Board in order to gain maximum joint profits under the centralised cartel system. The output quota of each member firm is fixed by the Cartel Board. As per prior agreement, there is distribution of total profits among the firms. The total industrial output and price are determined in such a way that there is minimisation of total production cost. Let us assume only two firms are there. Further, assuming that the central cartel board knows the average revenue curve (demand curve) and corresponding total marginal revenue curve of the industry. The CMC (Combined Marginal Cost) curve for industry can be found out by horizontal summation of MC curves of two firms. The determination of the industrial profit maximising output is done by equating combined marginal revenue and combined marginal cost. The illustration of the price-output determination is given in **figure 4.28.**

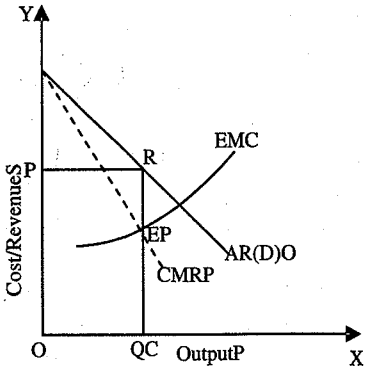


Figure 4.28: Centralised Cartels

In the **figure 4.28,** CMR and AR (D) are marginal revenue curve and demand curve of the industry. Where combined marginal cost curve is represented by EMC. At point E, industry is in equilibrium where EMC cuts CMR. QC is the Equilibrium output. Thus, OP is equilibrium price. Now, the main question arising that how to determine each firm's output quota. Here, each firm is asked to produce enough output at which MC of each firm becomes equal to MC of total output at equilibrium. Thus, each firm's marginal cost will be equal.

2) **Market Sharing Cartel:** Market sharing can be defined by the following two methods:
   i) **Market Sharing by Non-price Competition:** Firms do agree to sell goods at a uniform price that is already agreed by all. But there is freedom to member firms to produce and sell the quantity at which they can achieve maximum profits. Also there is freedom to change the product design and other method except any change in price so as to promote sales. While the different firm's cost is same, the agreed uniform price acts as the monopoly price. Low price will be fixed in the case when there is new firm entry threat. If the production cost differs, the price will be fixed by the process of bargaining between firms. Also one more thing is kept in view that even a high cost firm at agreed price gets some profits. But while the production cost differs, cartel is not stable due to reason that low cost firm will be tempted to lower down prices more in order to increase the profits. Even secret concessions may be given by low cost firms to buyers. Thus, cartel will not work and leads to breakdown.

   ii) **Market Sharing by Quota:** The oligopolistic firms may agree about the output quota produced by them as well as sell this quantity at uniform price. If costs and products of different firms are perfectly identical to each other then in this case the price and output quota of each firm can be determined such as to maximise the profits, i.e., emerging of monopoly solution will be there. In cases of production cost differences, the different firm's quota is decided by bargaining between firms which is based on the past sales of firms or productive firm's capacity. There can be another form of market sharing quota. Rather than determining uniform quota and price of each firm by cartel, geographical market division between cartel firms may be used. In this arrangement, there may be difference in the output and prices. However, it is noticeable that all types of cartels are temporary, where production cost is different for different firms in the oligopoly market. Cartels do collapse sooner or later in such cases.

4.8.7.2. Price Leadership/ Imperfect Collusion

There are few or more than two sellers in an oligopolistic situation who are capable to exercise monopolistic influence. We generally find existence of 'price leadership' in such a market situation. One assumes the price leader's role under price leadership and fixes the product price for entire industry. Other industrial firms just follow the price leader and accept the price which is fixed by him and also make adjustment regarding their output to this price. Generally, a price leader is a dominant or very large firm or the firm whose cost of production is lowest. As a result of price war, it often happens that there is establishment of price leadership where there is emerging of one firm as winner.

Price – Output Determination under Price Leadership

Various models are developed by the economists that are related to determination of price-output under price leadership (**figure 4.29**) on the basis of specific assumptions in context of price leader's behaviour and his followers.

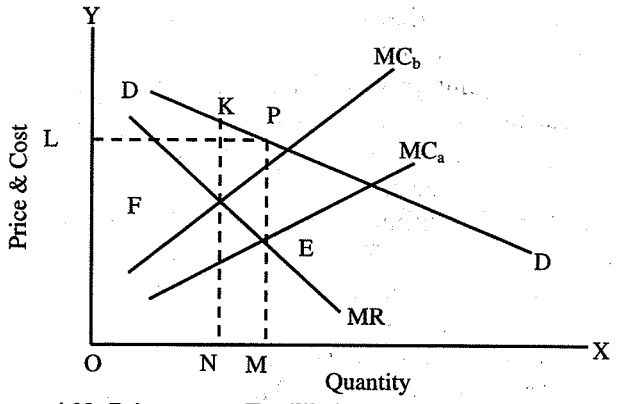


Figure 4.29: Price-output Equilibrium under Price Leadership

Assumptions under Price Leadership

1) Only two firms exists which are termed as A and B where A has lower production cost as compared to B;
2) The firm's product is identical or homogenous which means consumers are indifferent between the firms;
3) There is equal market share of both A and B firm, i.e., they have the same demand curve that will be half of total market demand curve.

The output and price decisions are illustrated in **figure 4.30**. It is assumed that there are identical revenue curves which are faced by all the firms are shown by MR and AR curves. But the cost curves are different: the low-cost or largest firm has its cost curves as shown by MC_1 and AC_1 whereas all other firms are smaller in size shown by MC_2 and AC_2.

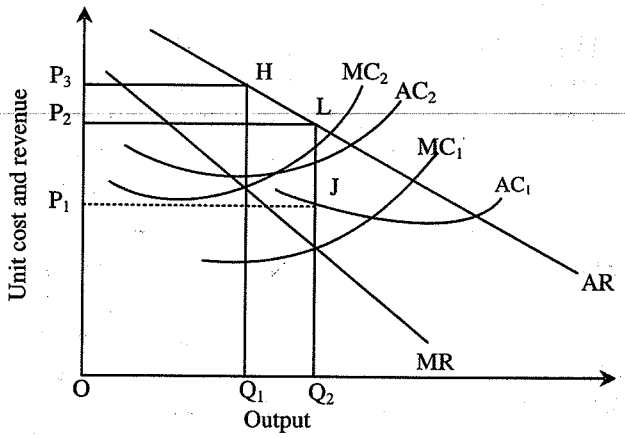


Figure 4.30: Price-output Equilibrium under Price Leadership

The reason behind this is that the economies of scale are shown by the largest firm and its production cost is lower as compared to other firms. Under given revenue and cost conditions, the firm with low cost would find it more profitable to fix its price at OP_2 (= LQ_2) and sell quantity OQ_2. Profit is maximum at this level, since at this output level, MC = MR. On other side, profit maximisation by the high cost firms would be at price OP_3 and quantity OQ_1. But, if higher price OP_3 is charged, they would lose their customers to the low-cost firm.

Therefore, high cost firms are forced to accept the OP_2 price and also recognition of price leadership of the low cost firm is done. Also note that elimination of other firms can be seen by the low cost firm to become monopolist by cutting the price to OP_1 (= JQ_2). The low cost firm can sell its all output OQ_2 at price OP but it tends to make nominal profit only. However, it may not do so for the fear of anti-monopoly laws.

4.8.8. Difference between Monopoly and Oligopoly

Points of Difference	Monopoly	Oligopoly
1) **Number of Firms**	Existence of one firm only.	Existence of few numbers of firms.
2) **Nature of Product**	Single product.	Differentiated or homogeneous products.
3) **Entry of Firms**	Concept of 'no entry' is there.	Concept of restricted entry and exit of firms.
4) **Degree of Monopoly Power**	Here full monopoly power exists.	Here limited degree of monopoly power exists due to differentiation of products.
5) **Control Over Price**	A considerable control over prices by the sellers can be seen.	Only some control exists over prices by the seller.

4.8.9. Difference between Monopolistic Competition & Oligopoly

Points of Difference	Monopolistic Competition	Oligopoly
1) **Number of Sellers**	Large.	Only Few sellers.
2) **Demand Curve**	Curve slopes Downward.	Demand curve is Kinky.
3) **Price Determination**	Fixed Price.	Other firm influences Price.
4) **Relation of Firms**	Independent firms.	Largely inter-dependent firms.
5) **Product-differentiation**	Product-differentiation is certainly found.	Product differentiation may or may not be found.

4.9. DUOPOLY

4.9.1. Meaning and Definition of Duopoly

Duopoly is a kind of oligopoly where only two sellers compete with each other. This kind of market is given a special reference in the economics as this theory was established around 1838, when theory of any other form of market was not developed. Even the theory of oligopoly market was developed during the decade of 1930.

The credit for the development of the concept of duopoly goes to a French economist **A.A. Cournot**. He propounded this theory in the year 1838. After him, there were various other economists who introduced their version of duopoly.

Duopoly is same as the monopoly market structure. In duopoly market structure there are only two suppliers in the market and these suppliers dominates the whole market to such a level that they jointly control the market price. This competition will continue, unless both the competitors start working together. Instead of being very similar to monopoly it is the simplest case of oligopoly market structure. In this type of market, competition is very limited because prices are usually high in comparison to other type of competitive market. A duopoly is a market situation where there are only two producers who sell almost similar products. Further, their pricing and market policies may vary. These firms can operate with cooperation among each other or can have cut-throat competition among them. They may fix their price as same or they can also change their price.

According to Dr. John, "Duopoly is that situation of a market in which there are two producers of a product, either perfectly identical or almost identical. They are not bound by an agreement regarding price or the quantity of production".

According to Webster, "Duopoly is an oligopoly limited to two sellers".

4.9.2. Features of Duopoly

The features of duopoly are as follows:

1) There are two sellers or producers in the market.
2) The products made by both the two sellers and producers may be exactly the same or almost the same.
3) Both producers can jointly agree over a single price or they may charge independent price.
4) The producers may adopt independent marketing and production strategies or they may collaborate to work together in this regard.
5) The producers can work in collaboration or they can indulge in competition with each other.

4.9.3. Price Determination under Duopoly

The condition of price determination is explained as follows:

1) **In Case of Agreement between the Firms:** In case there is an agreement between the producers and sellers then, the price determination will be same as under monopoly. The price will be determined at the point where both marginal revenue and marginal cost are equal. In order to attain this condition, demand must be elastic and cost structure of both the producer must be same. In case, the production cost of one producer is less than the other, the producer will fix the low price of the product. Hence, this producer will capture larger portion of the market.

2) **In Case of No Agreement between the Firms:** When producers under duopoly has no agreement and they are producing goods of similar nature, then there could be a struggle regarding the prices of the goods. Both the producer will try to capture the market through minimising the price of the goods, and this will continue till both price and marginal cost gets equal. There are chances that the price is not same in the duopoly form of market.

4.10. INDUSTRY ANALYSIS

4.10.1. Introduction

A number of environmental factors influence the organisations. It is up to managers to ensure that this influence is harnessed in a positive way, leading to organisational success. For the firm to make profit, it must create value for customers or buyers. Hence, the firm needs to understand its customers. While creating value, the firm has to obtain goods and services from suppliers.

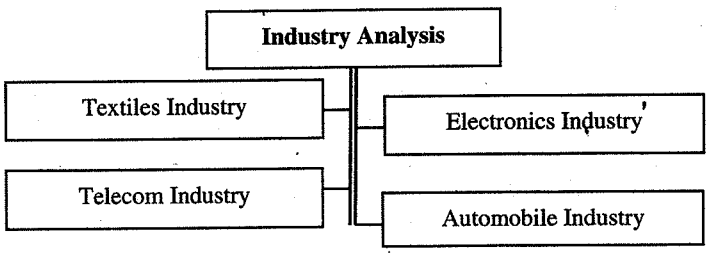
So, it must value its suppliers and form enduring business relationships with them. While creating value for its buyers, the firm must closely look at the rivals who are there in the arena competing for the same 'space'. Hence, the firm must understand competition. Thus, buyers, suppliers and competitors form the substance of a firm's industry environment.

The industry (or competitive) environment is typically characterised by a group of firms producing the same or related products or services. Forces from the industry environment directly affect the firm, and the amount of influence the firm has over its industry is dependent on the dominance of its competitive position. Most strategic management books utilise Michael Porter's Five Forces Model as a framework for analysing the competitive forces within the industry.

Like so many other models used to make strategic decisions today, the implicit assumption of this model is that the industry is operating within an economy closed to the greater society and ecosystem. From the view of the Five Forces Model, industry analysis is traditionally portrayed in strategic management books from the rather static perspective of "what is" within the industry.

This model suggests that strategic managers scan the product market segments in which they compete for opportunities and threats without much regard for context. Their primary focus is on increasing market share within defined industry boundaries, and the competition is defined as those competitors who directly compete with them in individual product or service categories. Cooperative relationships are typically limited to those with direct suppliers and buyers.

Capabilities to create value are viewed as residing in a single firm, and organisational performance is measured primarily in terms of how well the individual firm is managed with respect to its economic sustainability. Thus, within this traditional paradigm of industry analysis, strategic managers engage in adaptive learning within well-defined industry segments. Some of the major sectors of the business environment are



4.10.2. Textiles Industry

India's textiles sector is one of the oldest industries in Indian economy dating back several centuries. Even today, textiles sector is one of the largest contributors to India's exports with approximately 11 per cent of total exports. The textiles industry is also labour intensive and is one of the largest employers. The textile industry has two broad segments. First, the unorganised sector consists of handloom, handicrafts and sericulture, which are operated on a small scale and through traditional tools and methods.

The second is the organised sector consisting of spinning, apparel and garments segment which apply modern machinery and techniques such as economies of scale. The textile industry employs about 40 million workers and 60 million indirectly. India's overall textile exports during F.Y. 2015-16 stood at U.S. $40 billion.

The Indian textiles industry is extremely varied, with the hand-spun and handwoven textiles sectors at one end of the spectrum, while the capital intensive sophisticated mills sector at the other end of the spectrum. The decentralised power looms/hosiery and knitting sector form the largest component of the textiles sector. The close linkage of the textile industry to agriculture (for raw materials such as cotton) and the ancient culture and traditions of the country in terms of textiles make the Indian textiles sector unique in comparison to the industries of other countries. The Indian textile industry has the capacity to produce a wide variety of products suitable to different market segments, both within India and across the world.

4.10.3. Electronics Industry

The electronics sector produces electronic equipment for industries and consumer electronics products, such as computers, televisions and circuit boards. Electronics sector industries include telecommunications, equipment, electronic components, industrial electronics and consumer electronics. Electronics companies produce electrical equipment, manufacture electrical components and retail these products to make them available for consumers. The electronics industry, in particular, is emerging as one of the most important industry in the Indian market. The electronics industry in India dates back to the early 1960s.

Electronics was initially restricted to the development and maintenance of fundamental communication systems including radio broadcasting, telephonic and telegraphic communication, and augmentation of defence capabilities. Until 1984, the electronics sector was primarily government owned. The late 1980s witnessed a rapid growth of the electronics industry due to sweeping economic changes, resulting in the liberalisation and globalisation of the economy. The economic transformation was motivated by two compelling factors – the determination to boost economic growth, and to accelerate the development of export-oriented industries, like the electronics industry.

India is also an exporter of a vast range of electronic components and products for the following segments:

1) Display technologies,
2) Entertainment electronics,
3) Optical storage devices,
4) Passive components,
5) Electromechanical components,
6) Telecom equipment,
7) Transmission and signalling equipment,
8) Semiconductor designing, and
9) Electronic Manufacturing Services (EMS).

4.10.4. Automobile Industry

The Indian automotive industry has emerged as a 'sunrise sector' in the Indian economy. India is emerging as one of the world's fastest growing passenger car markets and second largest two wheeler manufacturer. It is also home for the largest motorcycle manufacturer and fifth largest commercial vehicle manufacturer. India is the largest base to export compact cars to Europe. Moreover, hybrid and electronic vehicles are new developments on the automobile canvas and India is one of the key markets for them. Global and Indian manufacturers are focussing their efforts to develop innovative products, technologies and supply chains.

The automobile industry, one of the core sectors, has undergone metamorphosis with the advent of new business and manufacturing practices in the light of liberalisation and globalisation. The sector seems to be optimistic of posting strong sales in the next couple of years in view of a reasonable surge in demand.

4.10.5. Telecom Industry

India is currently the world's second-largest telecommunications market and has registered strong growth in the past decade and half. The Indian mobile economy is growing rapidly and will contribute substantially to India's Gross Domestic Product (GDP), according to report prepared by GSM Association (GSMA) in collaboration with the Boston Consulting Group (BCG).

The liberal and reformist policies of the Government of India have been instrumental alongwith strong consumer demand in the rapid growth in the Indian telecom sector. The government has enabled easy market access to telecom equipment and a fair and proactive regulatory framework that has ensured availability of telecom services to consumer at affordable prices. The deregulation of Foreign Direct Investment (FDI) norms has made the sector one of the fastest growing and a top five employment opportunity generator in the country. The Indian telecom sector is expected to generate four million direct and indirect jobs over the next five years according to estimates by Randstad India. The employment opportunities are expected to be created due to combination of government's efforts to increase penetration in rural areas and the rapid increase in smartphone sales and rising internet usage.

International Data Corporation (IDC) predicts India to overtake U.S. as the second-largest smartphone market globally by 2017 and to maintain high growth rate over the next few years as people switch to smartphones and gradually upgrade to 4G.

4.11. PROFIT POLICY

4.11.1. Introduction

Generally it is believed that firms only have profit maximisation as its corporate goal. Firms do pursue other goals, besides profit. It must, however, be recognised that earning profits is one of the most important goals of any firm. A firm may follow many policies which may reduce short-run profit in order to establish better long-run situations. The studies conducted in recent years have pointed out that profit maximisation is often not the ultimate aim of management. There are two sets of problems which the firm faces in its profit policy decisions:

1) Profit-Limiting Factors
2) Setting Profit Standards

4.11.2. Profit-Limiting Factors

The objectives, other than profits, may be referred to as limitational factors. These limitational factors are supposed to reduce the profits of the firms below that level which a firm with profit maximisation as a sole objective would have earned. It must, nevertheless, be well understood that in practice these limiting factors are blended into the overall operating policies of a firm so as to achieve maximum long-run profits. There are several factors for limiting or controlling profits.

1) **Internal Factors:** The limiting factors which arise due to the firm's operations and indirectly result in reduction of profits are:
   i) **Management's Reluctance to Change:** In order to ensure the security of their job, managers normally do not wish to take high risk adventures, though these may promise a high rate of profit.
   ii) **Building a Reputation:** Some management try to build a reputation for their firms, irrespective of the level of profits they earn. They do so with an idea to make their firms as leaders in the industry.
   iii) **Preference for Liquidity:** Many a time, a firm needs to enter new areas of production. Some firms avoid entering into new ventures to maximise profits because that would require increased investment in fixed assets, thus reducing liquidity.
2) **External Factors:** Those factors which are beyond the control of the firm and force the firm to abandon the profit-maximising goals are:
   i) **Company's Goodwill:** A company may choose to limit profits in order to create an impression of charging "fair" prices.
   ii) **Aiming at Entry-Barrier:** In order to preserve its monopoly position a firm may prefer lower profits as that would deter entry of new firms into the industry.
   iii) **To Limit Demands of Labour Unions:** In industries having strong trade unions, a high profit rate would invite demands for higher wages on the ground that these demands do not adversely affect the firm.
   iv) **To Avoid Legal Problem:** Anti-trust legislations prohibit excessive profit earning by the firms. In order to avoid these anti-trust legislations, the firms may limit their profits.

4.11.3. Setting Profit Standards

Though, firms do not always seek to maximise profits, it cannot be denied that profit-earning is the only meaningful measure of corporate success and managerial effectiveness. Given this measure, it is desirable to employ some kind of profit yardstick to know what an "acceptable" performance of an enterprise from the point of view of shareholders and the management itself.

Various criteria are to be applied in deciding what an acceptable rate of profit is. These can be classified into two broad categories:

1) **Overall Standards:** The following profit standards for the company as a whole can be used as relevant criteria:
   i) **Profit Standard to Attract Capital:** Some companies view the 'rate necessary to attract equity capital' as a criterion for profit-earning.
   ii) **"Earning-by-the-Competitors" Standard:** Sometimes firms set their profit standards on the basis of comparative earnings. According to this standard, an acceptable rate of profit is the one which is earned by other firms in the same industry or by some selected firms in other industries operating under similar conditions.
   iii) **Historical Rate of Profit as a Standard:** In case the company's past earnings in 'normal' periods have been adequate enough to attract capital, to contain competition and keeping the shareholders satisfied, the company may treat the historical profit rate as the standard for profit-earning.
   iv) **'Retained Earnings' Standard:** Some managements consider adequate internal finance as a source of strength for the firm and, therefore, consider the rate sufficient to finance growth from internal sources as an acceptable rate of profit.
2) **Standards for Internal Use:** To determine profit, standards for internal use are even more difficult than overall standards. The problem of profit-making at different sub-centres within the firm is not only difficult but has received very little attention in literature.The need for profit standards for various profit-making sub-centres obviously arises in case of firms where management responsibility has been delegated to the heads of the divisional units/departments.

4.11.4. Role of Profit

Following are the important roles of profit in market system:

1) **Measure of Performance:** It measures the net effectiveness and soundness of a business effort. A higher profit is an indicator that the business is being run successfully and effectively. It is true that profit is far from being a perfect measure of business efficiency but it is probably the best indicator of the general efficiency of a firm. It is certainly the only one which allows quick and easy comparison of performance of various firms.
2) **Premium to Cover Costs of Staying in Business:** Profit is the premium that covers the costs of staying in business – replacement, obsolescence, market and technical risk and uncertainty. Seen from this point of view, it may be argued that there is no such thing as profit; these are only the costs of being and staying in business. The management of a business has to provide adequately for these costs by generating sufficient profit.
3) **Ensuring Supply of Future Capital:** Profit ensures the supply of future capital for innovation and expansion, either directly, by providing the means of self-financing out of retained profits, or indirectly, through providing sufficient inducement for new external capital which will optimise the company's capital structure and minimise its cost of capital.
4) **Motivating Entrepreneur:** The function of profit is that it not only creates a motivation in the entrepreneur to bear risk and uncertainty but also creates a capability in him to bear losses. If a firm is receiving continuous profits it can bear losses also some times. A firm which gets losses only, will have to be closed down.
5) **Provide Resources:** Profits not only motivate an entrepreneur for innovation but also provide resources for this purpose. In the absence of profits, programmes of technological improvement, innovation would be adversely affected because without resources, Research and Development (R & D) is not possible.

4.11.5. Profits and Control

The use of the profit incentive and profit accounting in the measurement and control of executive performance in large business enterprises is another managerial aspect of profits. Perhaps the major internal threat to vitality in the big corporation is bureaucratic deviationism, the tendency for middle and lower management to think in terms of security and routine or personal ambitions that conflict with the company's profit-making objectives.

When lower supervisory levels are rated in importance by simple size standards – sales volume, number of employees, production performance – they have no direct feel of the profit system, and no taste for the painful work of minimising costs. Companies that have become concerned about this problem have sought methods of measuring and rewarding executive performance that will guard against this hardening of the profit-seeking arteries. Such plans have two characteristics in common:

1) The initial one is realignment of the managerial organisation to shift the basic breakdown of operating responsibility from a functional basis (e.g., marketing *versus* production *versus* finance) to a commodity basis. The effect is to break the company up into several integral operating units (divisions) and give the executive authority over all functions in his division, and make him responsible for profits. The autonomy of division managers is, of course, not complete. They operate under general supervision and have the benefit of "advice" from a general staff, often at a higher executive echelon. But within the framework of the rules of the game imposed by broad company policies, they have authority and responsibility.
2) The other common trait is re-orientation of accounting reports to conform to the areas of executive responsibility. Each executive is given a profit goal for his operation and his performance is appraised on the basis of periodic profit and loss statements, as well as subordinate budgetary controls.

This kind of managerial decentralisation and control-by-profits has important advantages for the big company. The number of centres of initiative that are geared to profit objectives is multiplied. Greater scope can be given to division managers to exercise their own ingenuity in solving problems which can be better understood at the operating unit level, and top management can economise routine supervision and have more time for planning and for broad policy. By testing many executives on the firing line of experience with full-fledged integral operations, a screening device is provided for finding fireballs in the lower ranks.

This philosophy of managerial control raises many interesting economic issues that complicate its application to administrative problems. These issues centre around the methods for measuring divisional profits, and the levels at which profit goals are set.

4.11.6. Problems Involved in Profit Policy

Problems involved in profit policy are as follows:

1) **Determination of Selling Price:** The selling price of the product of an enterprise bears a direct relationship with its profit. An enterprise can fix selling price of its product or products only after considering the price policies of its competitors, otherwise it may be unsuccessful in its marketing efforts. Demand of its product may decrease substantially and its sales and profit may also decrease.
2) **Controlling Cost of Production:** Success of an enterprise depends upon its ability to control its cost of production, so that it may be minimised and all the normal and abnormal wastages may be checked. The techniques of budgetary control and standard costing may be adopted to control the cost of production. The less is the cost of production, the more will be the profit of an enterprise.
3) **Determination of Amount of Depreciation:** In the determination of Profit Policy the determination of amount of depreciation should be provided on fixed assets. Depreciation is a direct charge against profit of an enterprise though its determination is not very easy. As there are many methods of determining depreciation, it is very difficult to select a particular method.
4) **Valuation of Stock:** It bears direct relationship with the amount of profit of an enterprise. If the value of stock is high, profit of the enterprise will be more; conversely if the value of stock is low, profit of the enterprise will be less.

5) **Determination of the Target Rate of Return:** The objectives and aims of a business may be different. In fact, most business concerns like to earn a target rate of return on their investment. There are four criteria to judge the target rate of return:
   i) Rate adequate enough to attract equity capital.
   ii) Rate earned by other companies in the same industry.
   iii) Normal or historical profits rate of return.
   iv) Rate sufficient to finance growth from internal sources.

4.12. BREAK-EVEN ANALYSIS

4.12.1. Meaning and Definition of Break-Even Analysis

The study of cost-volume profit analysis is often referred to as **'Break-Even Analysis'** and the two terms are used interchangeably by many. This is so, because break-even analysis is the most widely known form of cost-volume-profit analysis. The term "Break-Even Analysis" is used in two senses–narrow sense and broad sense.

In its broad sense, break-even analysis refers to the study of relationship between costs, volume and profit at different levels of sales or production. In its narrow sense, it refers to a technique of determining that level of operations where total revenues equal total expenses, i.e., the point of no profit, no loss.

According to Matz, Curry and Frank, "A Break-even analysis indicates at what level of costs and revenue are in equilibrium".

4.12.2. Assumptions of Break-Even Analysis

The break-even analysis is based upon the following assumptions:

1) All elements of cost, i.e., production, administration and selling distribution can be segregated into fixed and variable components.
2) Variable cost remains constant per unit of output irrespective of the level of output and thus fluctuates directly in proportion to changes in the volume of output.
3) Fixed cost remains constant at all volumes of output.
4) Selling price per unit remains unchanged or constant at all levels of output.
5) Volume of production is the only factor that influences cost.
6) There will be no change in the general price-level.
7) There is only one product or in case of multi-products, the sales mix remains unchanged.
8) There is synchronization between production and sales.

4.12.3. Break-even Point (BEP)

It is a point where sales revenue equals the costs to make and sell the product and no profit or loss is reported. **Break-even point** is the level of sales at which profit is zero. At break-even point sales are equal to fixed cost plus variable. This concept is further explained by the following equation:

Break-even Sales = Fixed Cost + Variable Cost

Break-even point is also called **'Critical Point', 'Equilibrium Point', 'No-Profit No-Loss Point',** or **'Balancing Point'**.

According to Keller and Ferrara, "The break-even point of a company or a unit of a company is the level of sales income which will equal to the sum of its fixed costs and variable costs".

According to Charles T. Horngren, "The break-even point is that point of activity (sales volume) where total revenues and total expenses are equal; it is the point of zero profit and zero loss".

4.12.4. Computation of the Break-Even Point

The break-even point can be computed by the following methods:

1) Algebraic Formula Method
2) Graphic or Chart Method

4.12.4.1. Algebraic Formula Method for Computing the Break-Even Point

The break-even point can be computed in terms of:

1) **Break-Even Point in Units:** As the break-even point is the point of no profits no loss, it is that level of output at which the total contribution equals the total fixed costs. It can be calculated with the help of following formula:

$$\text{Break Even Point} = \frac{\text{Fixed Cost}}{\text{Selling Price per unit} - \text{Variable Cost per unit}} = \frac{\text{Fixed Cost}}{\text{Contribution per unit}}$$

2) **Break-Even Point in Terms of Budget-Total or Money Value**

At break-even point:

Total Sales = Total Fixed Cost + Total Variable Cost

Or $S = F + V$

(where S = Sales, F = Fixed Cost and V = Variable Cost)

Or $S - V = F$

Or $\frac{S-V}{S-V} = \frac{F}{S-V}$ (Dividing both Sides by S – V)

Or $1 = \frac{F}{S-V}$

Or $S \times 1 = \frac{F \times S}{S-V}$ (Multiplying both sides by S)

Hence, Break-Even Sales

$$= \frac{\text{Fixed Cost}}{\text{Selling Cost} - \text{Variable Cost}} \times \text{Sales} = \frac{\text{Fixed Cost}}{\text{Contribution}} \times \text{Sales}$$

With the use of P/V Ratio,

$$\text{B.E.P.} = \frac{\text{Fixed Cost}}{\text{P/V ratio}}$$

$$\left(\text{As, } \frac{\text{Contribution}}{\text{Sales}} = \text{P/V ratio}\right)$$

3) **Break-Even Point as a Percentage of Estimated Capacity:** Break-even point can also be computed as a percentage of the estimated sales or capacity by dividing the break-even sales by the capacity sales.

For example, if a firm has an estimated capacity of 1,00,000 units of products and its break-even point is reached at 50,000 units, then the break-even point is at 50% of capacity (1,00,000/50,000). If information as to total contribution at full capacity is available, the break-even point as a percentage of estimated capacity can be found as under:

$$\text{B.E.P (as \% age of capacity)} = \frac{\text{Fixed Cost}}{\text{Total Contribution}}$$

Example: From the following information, calculate the break-even point in units and in sales value:

Output	3,000 units
Selling price per unit	₹30
Variable cost per unit	₹20
Total fixed cost	₹20,000

Solution:

$$\text{Break Even Point (in units)} = \frac{\text{Fixed Cost}}{\text{Selling Price per unit} - \text{Variable Cost per unit}}$$

$$= \frac{20,000}{30-20} = \frac{20,000}{10} = 2,000 \text{ units.}$$

$$\text{Break Even Point (in sales value)} = \frac{\text{Fixed Cost} \times \text{Sales}}{\text{Sales} - \text{Variable Cost}}$$

Fixed Cost = ₹20,000 (given)
Sales = 3,000 × 30 = ₹90,000
Variable Cost = 3,000 × 20 = ₹60,000

Hence, B.E.P. (in sales value)

$$= \frac{20,000 \times 90,000}{90,000 - 60,000} = \frac{20,000 \times 90,000}{30,000} = ₹60,000$$

Otherwise, as the B.E.P. is 2000 units, break-even sale would be: 2,000 × 30 = ₹60,000

4.12.4.2. Graphical Method of Break-Even Analysis or Break-Even Chart/Graph

The break-even point can also be computed graphically. A break-even chart is a graphical representation of marginal costing. The break-even charts 'portrays a pictorial view of the relationships between costs, volume and profits'. It shows the break-even point and also indicates the estimated profit or loss at various levels of output. The break-even point as indicated in the chart is the point at which the total cost line and the total sales line intersect.

Construction of Break-Even Charts

Construction of break-even chart involves the drawing of fixed cost line, total cost line, and sales line as follows:

1) Select a scale for production on the horizontal axis and a scale for costs and sales on the vertical axis.
2) Plot the fixed cost on the vertical axis and draw fixed cost line passing through this point parallel to horizontal axis.
3) Plot the variable costs for some activity levels starting from the fixed cost line and join these points. This will give the total cost line. Alternatively, obtain total cost at different levels, plot the points starting from horizontal axis and draw the total cost line.
4) Plot the maximum or any other sales volume and draw the sales line by joining zero and the point so obtained.

There are three methods of drawing a break-even chart/ graph. These methods of drawing break-even chart have been explained with the help of the **example**.

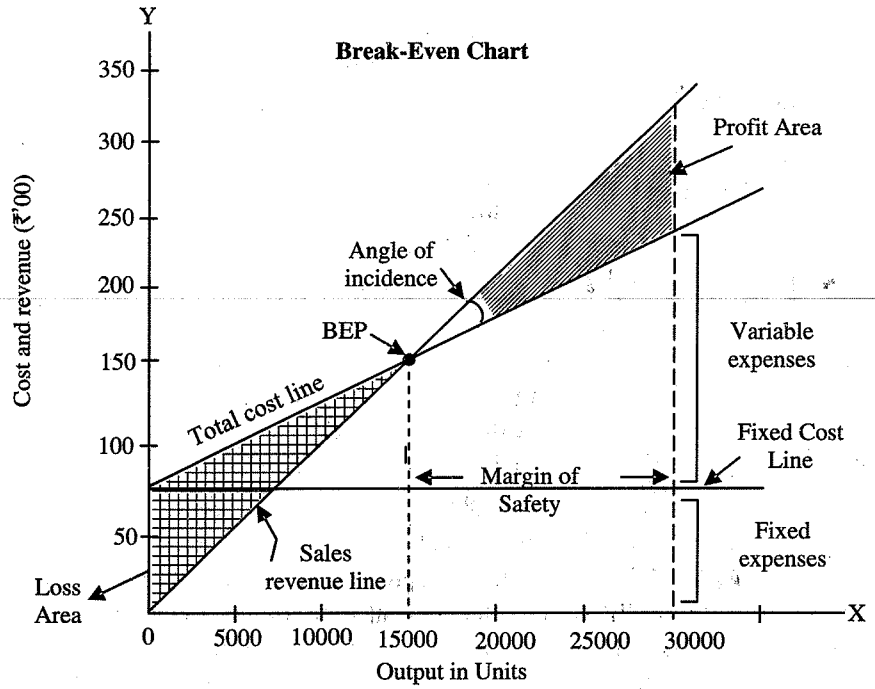
Example: Plot the following data on a graph (break-even chart) and determine:

1) Break-Even Point
2) Profit if the output is 25,000 units.

Output (Units)	Variable Cost Per unit (₹)	Total Variable cost (₹)	Fixed Expenses (₹)	Total Cost (₹)	Selling Price (per unit) (₹)	Total Sales (₹)
0	5	0	75,000	75,000	10	0
5,000	5	25,000	75,000	1,00,000	10	50,000
10,000	5	50,000	75,000	1,25,000	10	1,00,000
15,000	5	75,000	75,000	1,50,000	10	1,50,000
20,000	5	1,00,000	75,000	1,75,000	10	2,00,000
25,000	5	1,25,000	75,000	2,00,000	10	2,50,000
30,000	5	1,50,000	75,000	2,25,000	10	3,00,000

Solution:

1) **First Method:** Under this method following steps are taken to draw the break-even chart.



i) Volume of production/output or sales is plotted on horizontal axis, i.e., X-axis. The volume of sales or production may be expressed in terms of rupees, units or as a percentage of capacity.

ii) Costs and sales revenue are represented on vertical axis, i.e., Y-axis.

iii) Fixed cost line is drawn parallel to X-axis. The line indicates that fixed expenses remain constant at all levels of activity.

iv) The variable costs for different levels of activity are plotted over-the fixed cost line. The variable cost line is joined to fixed cost line at zero level of activity. As the variable cost line is drawn above the fixed cost line, it represents the total cost at various levels of output/sales.

v) Sales values at various levels of output are plotted and a line is drawn joining these plotted points. This line is called the sales (revenue) line.

vi) The point of intersection of total cost line and sales (revenue) line is called the break-even point.

vii) The number of units to be produced at break-even point can be determined by drawing a perpendicular to the X-axis from the point of intersection of cost and sales line.

viii) The sales revenue at break-even point can be determined by drawing a perpendicular to the Y-axis from the point of inter-section of cost and sales line.

ix) The area below the break-even point represents the loss area as the total sales are less than the total cost and the area above the break-even point indicates the area of profit as the sales revenue exceeds the total cost.

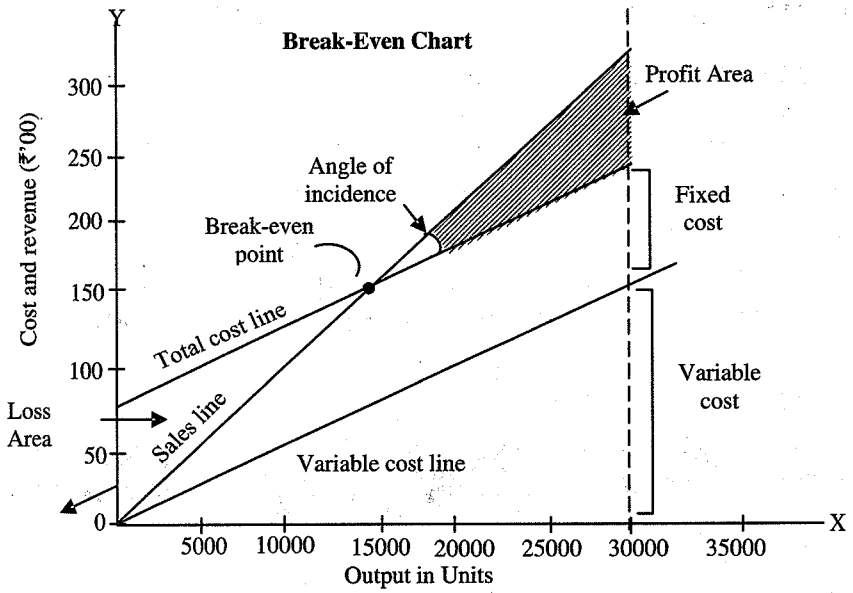
2) **Second Method:** The break-even chart can also be drawn by another method which is a variation of the first method. Under this method, the variable cost line is drawn first and then total cost line is drawn. The fixed cost line, so drawn, represents the total cost (Variable + Fixed) at various levels of output.. This method is useful to the management for decision making because it reveals additional information:

i) The variable costs are shown directly for various levels of output/sales.

ii) Marginal contribution at various levels of sales is indicated clearly by the difference between sales line and variable cost line.

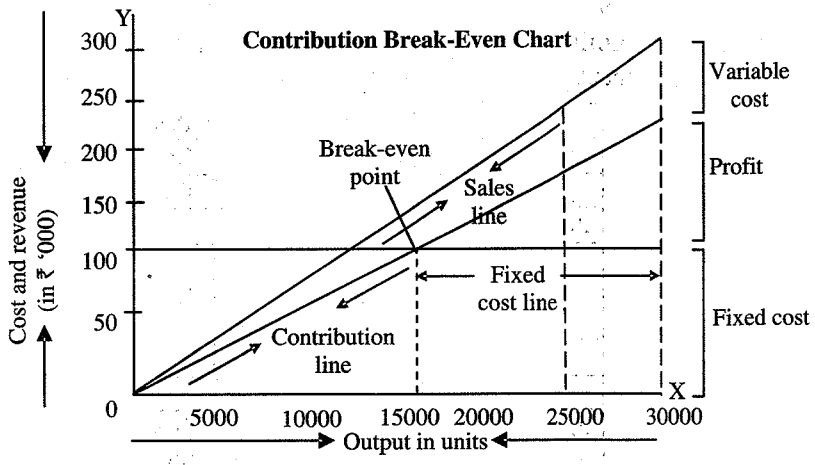
iii) It indicates the recovery of fixed costs at various levels of production.

A small variation of this method is that of showing the various elements of fixed and variable costs, **for example,** major cost elements such as direct material cost, labour cost, variable factory overhead cost, variable selling overhead and fixed costs.



3) **Third Method – Contribution Break-Even Chart:** This is a modified form of a simple break-even chart as shown in the first-two methods above. Under this method total cost line is not drawn, rather another line called contribution line is drawn from the origin and this line goes up with the increase in the level of output. The fixed cost line is drawn parallel to the x-axis as in the first method. The sales line is also drawn as usual. In this method, the question of intersection of sales line with the total cost line does not arise because there is no cost line. The break-even point is that point where the contribution line crosses the fixed cost line. At this point, total contribution is equal to the total fixed cost and hence there is no profit or loss.

As the contribution increases more than the fixed cost, profit shall arise to the organization and as contribution decreases from the fixed cost, there shall be loss to the organization. The contribution break-even chart shows clearly contribution at different levels of activity and indicates that all levels below the break-even point are unable to cover the fixed costs.



In the above example, at level of output/sales of 25,000 units, there is a profit of ₹50,000 as indicated by the break-even charts under the three methods.

4.12.5. Significance of Break-Even Analysis

The significance of break-even analysis are as follows:

1) To forecast profit accurately, it is absolutely essential to determine the relationship between costs and profits on one hand and on volume and profit on the other hand. It aims at measuring variations in cost with volume. Profit planning considers the projected level of output, optimum product combination, estimated revenue, total cost of production and is thus based on break-even analysis.
2) Break-even analysis is used in setting up flexible budgets which show costs at various levels of activities.
3) Break-even analysis helps management in the evaluation of performances for control purposes.
4) Break-even analysis may be helpful in formulating pricing policies by projecting the effect that various price structures have on costs and profits, especially when the demand for the product is elastic.
5) Break-even analysis helps to ascertain the amount of overhead costs that could be charged to product costs at different levels of operation.

4.12.6. Limitations of Break-Even Analysis

The limitations of break-even analysis are as follows:

1) Break-Even analysis can sometimes gives misleading results of production capacity changes.
2) All costs cannot be easily and accurately segregated into fixed and variable components.
3) Total fixed costs do not remain fixed beyond certain ranges of activity levels.
4) When a firm sells more than one product the sales mix changes continuously due to changes in demand. Moreover, it becomes difficult to forecast accurately the volume of sales mix which would yield maximum profit.
5) Sometimes changes in opening and closing stocks may be significant.

4.13. PROFIT FORECASTING

4.13.1. Introduction

Forecasting, particularly on a short-term basis (one year to three years), is essential to planning for business success. This process, estimating future business performance based on the actual results from prior periods, enables the business owner/manager to modify the operation of the business on a timely basis. This allows the business to avoid losses or major financial problems should some future results from operations not conform to reasonable expectations. Profit forecasting is an essential function of any management. It involves a careful analysis of the cost and revenue data in terms of the inter-relationship between sales volume, prices, fixed costs and variable costs. The necessity of profit forecasting is directly related to the extent of competition faced by the firm.

Profit forecasting relates to projection of future earnings and involves the analysis of actual and expected behaviour of firms (including its components), the sales volume, prices, competitors' strategies *vis-à-vis* firm's own strategies, etc. The real use of profit forecasting can be made by developing a profit-improvement plan related to the profit forecast for the period. A simplified version of the relation between profit forecast and profit-improvement plan can be given as below:

Profit Forecast	Profit Improvement Plan
1) Sales forecast	Change in sales
	Change in sales-mix
2) Cost budget:	
i) Production cost	Cost reduction policies
ii) Selling and distribution cost	
iii) Administration cost	
iv) R&D cost	
3) Capital expenditure budget	Developing a rational capital expenditure policy
4) Planned profit level	Enhancing return on investment

4.13.2. Factors of Profit Forecasting

The major factors of profit forecast are as follows:

1) **Turnover:** It must, however, be emphasised at the outset that the product is the starting point for all planning activities. To a manufacturer, the special aspect of a product is most relevant which earns good profit. A higher turnover indicates a healthier performance.

2) **Costs:** It is the costs that form the basis for many managerial decisions. It is the level of costs relative to revenue that determines the firm's overall profitability. In order to maximise profits, a firm tries to increase its revenue and lower its costs. The costs can be brought-down either by producing the optimum level of output, using the least cost combinations of inputs or increasing factor productivities, or by improving the organisational efficiency.

   The elements of costs are described as follows:

   i) **Sales Costs:** These costs consist of salesmen's compensation, sales promotion, market research and administration.

   ii) **Product Development:** In many organisations, this activity is part of R&D's responsibility. Product development involves R&D and production engineering.

      a) **R&D:** R&D must be used to help to close the gap between the required or desired profit and that anticipated, after all cost reduction and marketing plans have been made.

      b) **Production Engineering:** It coordinates search for knowledge in rational manner cutting across the entire spectrum of integrated management and processing activities to attain optimal economic objectives of sufficiency. It lays-down a disciplined use of strategies for increased productivity with ensured quality and quantity.

   iii) **Distribution:** When a product has been developed and made ready and its price also determined, the next task is distribution, to bring it to the market and reach it to the consumer.

   iv) **Inventories:** Holding large stocks will mean high inventory carrying charges and possible losses caused by price declines. Similarly, shortages in inventories interrupt production, making machines and men idle and causing sales loss. Hence, there is need for inventory control or what is sometimes termed as inventory planning.

   v) **Production:** It reflects the ability of the organisation to produce whatever is demanded by the environment. The measures of production include profits, sales, market share, students graduated, clients served and the like.

   vi) **General Administration:** Making policies is the function of administration. In all kinds of business, the function of administration is the same. In a manufacturing organisation, the administration plan should show the relationship between the cost and numbers of administration staff and those in other functions and activities.

   vii) **Depreciation:** Depreciation means a fall in the quality or value of an asset. The originally invested capital wears-out or becomes obsolete with the passage of time. It cannot be recovered when the usefulness of these assets is exhausted. Businessmen, therefore, have realised that in order to state business income properly, some provision should be made to recover that part of the original asset which eventually becomes worthless because of depreciation.

   viii) **Reserves:** It is an amount set aside out of profits and other surpluses. It is not designed to meet any existing liability, contingency or diminution of value of assets. They are available for distribution through the profit and loss account and used to write-off capital loss such as loss on sale of a fixed asset, discount allowed on shares or debentures, etc.

4.13.3. Advantages of Profit Forecasting

Profit forecasting serves a number of useful purposes:

1) **Planning:** The profit forecasting process ensures that managers do plan for future operations, and that they consider how conditions in the next year might change and what steps they should take now to respond to these changed conditions. This process encourages managers to anticipate problems before they arise, and hasty decisions that are made on the spur of the moment, based on expediency rather than reasoned judgment will be minimised.

2) **Coordination:** The profit forecasting serves as a vehicle through which the actions of different parts of an organisation can be brought together and reconciled into a common plan. Without any guidance, managers may make their own decisions, believing that they are working in the best interests of the organisation

3) **Communication:** Through profit forecasting, top management communicates its expectations to lower level management, so that all members of the organisation may understand these expectations and can coordinate their activities to attain them.

4) **Motivation:** The profit forecasting can be a useful device for influencing managerial behaviour and motivating managers to perform in line with the organisational objectives. Profit forecasting provides a standard that under certain circumstances; a manager may be motivated to strive and to achieve.

5) **Control:** A profit forecasting assists managers in managing and controlling the activities for which they are responsible. By comparing the actual results with the budgeted amounts for different categories of expenses, managers can ascertain which costs do not conform to the original plan and thus require their attention.

4.13.4. Methods of Profit Forecasting

Following methods are used for profit forecasting:

1) **Spot Projection:** Projecting the entire profit and loss statement for a specified future period by forecasting each important element separately; forecasts are made about sales volume and prices and costs of producing the anticipated sales. Since profits are residual resulting from the forces that shape demand for the company's products and govern the behaviour of its costs, their prediction is subject to wide margins of error, from cumulating of errors in forecasting revenues and costs, and from the inter-relation of the various components of the income statement.

2) **Break-Even Analysis:** Analysis identifying functional relations of both revenues and costs to output rate, with profits related to output as a residual; or alternatively, relating profits to output directly by the usual data used in break-even analysis.

3) **Environmental Analysis:** Analysis relating the company's profits to key variables in the economic environment, such as the general business activity and the general price level. These variables are external to the company.

4.14. GOVERNMENT INTERVENTION IN MARKET

4.14.1. Introduction

The term government intervention means the interference or regulation implemented by the government on different decisions or activities of individual or organisation within its legal authorities. The result of this type of intervention could be both favourable and unfavourable. In simple terms, government intervention is any action taken by the public sector organisation or the government with an aim to influence the economy and this influence can be beyond simple regulation of contract and laws related to public welfare.

The propagators of government intervention argue in favour of using various economic policies for managing the economic system drawbacks resulting in greater economic inequity. Further, they also argue that economic equilibriums cannot be obtained only through law of demand and law of supply. Thus, in order to correct the economic imbalance, government intervention must be used.

4.14.2. Reasons for Government Intervention

The government could interfere to improve the behaviour of individual or business in order to face the market or regulatory failure and also to attain the environmental and social benefits which would not have been delivered without the support of government. Some of the most important reasons why government intervention are important are as follows:

1) **Market Failure:** The market failure is a condition where market alone is unable to allocate the goods and services in the most effective way to its consumer. Hence, under this type of situation government intervention helps the organisation to overcome these issues.

2) **Regulatory Failure:** Usually organisation faces regulatory failure when the regulations are not properly designed and they deliver the undesirable consequences. Tax avoidances, higher cost of business, social disparity can be outcome of this in the overall economy and the society. Thus, an effective role can be played by the different regulatory tool or policy in order to deal with the problems and accomplishing the goals of the government in these situations.

3) **Unacceptable Risks:** This consists of human health and safety risks or any danger of loss to the physical environment. It also includes the emerging issues related to safety and environment, such risk which community is unable to deal with the deficiency in the current approach. Further, the chances of occurrences of any of such case will act as the factor for undertaking the government intervention.

4) **Equity Issues, Social Goals or Public Order and Protection:** The improvement in the efficiency is a driving force for the government intervention, whereas the equity and social objectives also play a major role in this concern. There can be justification for the government intervention for the promotion of fair results, facilitating the basic living standard to all in the society, availing the products and services which are valued by the society and safeguarding the society by securing the individuals from any kind of criminal or abusive conduct.

4.14.3. Arguments Against Government Intervention

The arguments against the government intervention are as follows:

1) The government is responsible for taking the wrong decisions. Often government takes wrong decisions under the pressure from the politics which results in investment in ineffective projects which deliver ineffective results.

2) Because of the government interventions, individual and business are not free to take decision, such as how to invest and how to operate. Hence, individual liberty is seized due to economic intervention.

3) The market has the competence for the decisions related to time and manner of production. Whereas, any type of government intervention affects this efficiency of the market.

4.14.4. Need for Government Intervention in Market/ Arguments for Government Intervention

Income, wealth and opportunity inequalities exist in the free market. Additionally, partiality exists in the private charities. Hence, for redistribution of the income in a society, it is important to have the government intervention.

1) **Regulation of Monopoly Power:** When a certain organisation controls whole or almost whole of the product or service market, it is said to be a monopoly. In case, monopoly exists in a *laissez faire* market, then most probably producer will charge high price where consumer with low income will not be able to buy the high price goods and services. The monopoly power can be capitalised by the firm while paying low wages to the manpower. The government can overcome these problems quite effectively. Either some tough laws or rival organisations can be formed by the government in order to manage the monopoly.

2) **Equitable Distribution of Wealth and Income:** If there is no government intervention, it results in increased monopoly power. The power of monopoly producer increases due to higher efficiency delivered by the employees and losses to the consumers. Thus, it increases inequality among the buyers. Competition is promoted in the market through government intervention and due to this competition goods are available at fair price to the customer and it promotes equality in distribution of income.

3) **Eliminate Market Failure:** Environmental exploitation is one of the most crucial problems in the contemporary world. Under a *laissez faire* economic situation, it is not possible to restrict any of the anti-environment activities performed by the industries. Therefore, if government intervenes, then it can help in saving the environment in much better way than alone by the industry. Thus, by using government intervention, it is feasible to manage an important market failure.

4) **Prevent Economic Swings:** The nature of the economy is dynamic and this dynamic nature is highly unpredictable. Additionally, it is not possible for any specific group or individual to prevent economy from this dynamic nature whereas, government possess much superior capacity which can be used by the market to prevent itself from such situation. Further, this involvement of economy will prevent market from undergoing losses and recession.

5) **Improve Market Infrastructure:** In order to perform the business activities effectively and efficiently, the electricity, water, roads, rail, communication, etc., acts as a crucial resources. Whereas, in the contemporary world, without the help of these infrastructure resources, no business can perform any business activity effectively. In case of *laissez faire* economic model, full liberty is given to the organisations and individuals to make any decision pertaining to the business. But, the decision related to the infrastructure development and improvement cannot be paid by the individuals from their own funds. Therefore, for the growth of national market infrastructure, it is important to have the interference from the government.

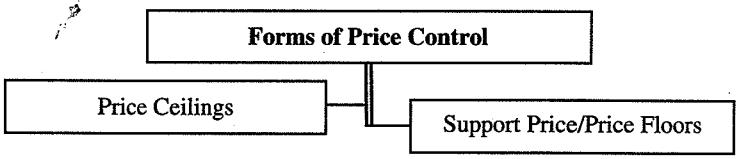
4.14.5. Price Control

The price regulation imposed by the government on the products and services in a certain market is termed as price control. The main objective of implementing these kinds of control is to ensure the affordability of staple foods and goods, to regulate the prices in case of shortages, to manage inflation, or to facilitate the least income for the specific product facilitators. Price control is mainly of two types, price ceiling which is the maximum charged price, and price floor indicating the least chargeable price.

The price control is mainly used as a mode of regulation to control or regulate the price level or the structure which an organisation charges. Price is seen as a critical factor. For addressing the competition, it is also utilised as a strategic factor. It also serves as a source of direct earning for the organisations. Hence, along with the cost incurred on production and sale, it also includes some margin of profit. Further, this price must be normal so that customers should not get frightened. Price is also a very important component for the customer because it directly affects their buying decisions. Generally, the quantity which will be sold in the market is determined by the price of that good. This is the reason behind the selling of fans at lower price during the winter.

4.14.5.1. Forms of Price Control

The basic forms of price Control are given as following:



4.14.5.2. Price Ceilings

Price ceiling is a form of government intervention which is in the favour of the buyers. Under this, an upper limit of the price is fixed by the government and hence producers cannot sell goods above that price. Further, there are various ill-effects of this intervention, like it reduces the motivation to producers and sometime quality of goods also suffers.

4.14.5.3. Support Price/Price Floors

Price floor is a form of government intervention which is in the favour of the producers. Under this, price floor buyer have to pay atleast the minimum price to the producer whereas, producer is free to charge price even above this level. Further, this price floor also possesses some of the ill-effects, like it raises the level of price above the level of equilibrium. If the prices are more, then the quantity supplied will be more than the quantity demanded, and hence it will result in creating surplus. As a layman this surplus seems to be good for the economy and the consumer, but actually this surplus results in generating such quantity which cannot be sold without offering high discount and as a result the price floor become ineffective.

This can be seen that both the price floor and price ceiling has positive and negative effect. In some situations, price control are desired. But, due considerations must be taken while implementing price control, particularly in case of their negative consequences.

4.14.6. Prevention and Control of Monopolies

The main objective of **MRTP Act, 1969 is** to avoid any kind of power accumulation for preventing the losses. This act also deals with the unfair, monopolistic and restrictive trade practices. Further, this law restricts monopoly and it protects the rights of the consumer. Monopolies and Restrictive Trade Practices Act, 1969

(MRTP) can be seen as one of the most critical economic laws which has been formed to make sure that no accumulation of economic power results from the economic system operations, causing overall losses. This act was passed on 1st June, 1970, and some amendments were made in 1974, 1980, 1982, 1984, and 1991. Apart from the State of Jammu & Kashmir, this act is applied to the entire country.

Areas of Control under MRTP ACT

The areas of control under MRTP Act are as follows:

1) **Monopolistic Trade Practice:** It is such a trade practice which prevents the competition on production, supply and distribution of goods and services. It restricts the capital investment and technical development to the shared detriment which allow the quality of goods and services to diminish. The monopolistic trade practice will be considered harmful for the welfare of public interest, if it gives significance to the existing economic condition of the nation and other factors which are significant to the certain situation. These trade practices will have the following outcomes:
   i) It enhances the cost of the production, distribution and supply of goods and services.
   ii) It increases the price at which goods are sold and as a result it increases the level of profit for the producer.
   iii) It minimises the level of competition in production, distribution and supply of goods and services.
   iv) It restricts the supply of goods and services to the consumers.
   v) It reduces the quality of goods and services.
   vi) It prevents the restricted competition through unfair means.

2) **Restrictive Trade Practices (RTPs):** In a broader sense, restrictive trade practice is such a practice which restricts the competition in the market. **According the [Section 2(O)] of MRTP Act 1969**, a restrictive trade practice means , a trade practice which has or may have , the effect of preventing, distorting, or restricting competition in any manner, and in particular, (i) which tends to disrupt the flow of capital or resources into the stream of production, or (ii) which tends to bring about the manipulation of prices, or condition of delivery, or to affect the flow of supplies in the market relating to goods or services in such a manner as to impose on the consumers unjustified costs or restrictions.

   Section 33(1) of the MRTP Act described the different RTPs as follows:
   i) Price Discrimination
   ii) Area Restriction
   iii) Concert or Collusion Cartels
   iv) Tie-up Sales
   v) Exclusive Dealing
   vi) Predatory Pricing
   vii) Full-Line Forcing
   viii) Resale Price Maintenance

3) **Unfair Trade Practices (UTPs):** An amendment was made in the MRTP act in the year 1984 and under this amendment; UTP was introduced as a part of MRTP act. This amendment was made on the recommendation of Sacher Committee Report. This committee introduced this amendment with an aim to prevent consumer from different types of unfair trade practices which were exploiting the consumer.

   Any trade activity which is performed to increase the sales, product supply or consumption, or any service terms by using any unfair or misleading activities along with the practices mentioned below are termed as unfair trade practices:
   i) Such advertisements which are misleading and presents a false image,
   ii) Bargain Sale,
   iii) Offering free Gift's, Prizes, etc., and initiating Contests or Lottery,
   iv) Not following the prescribed standards,
   v) Production of Spurious Goods,
   vi) Product Hoarding or Destruction.

4.14.7. Government Intervention and Pricing

Governments in different countries have played the various roles at different points of time. Even in countries where regulatory structure has been liberalized in recent years, certain controls have been retained in the large special interest. Promotion of business by the government is normally accepted by the private sector. Role of government in pricing is explained under following heads:

1) **Price Regulation:** It is usual for the Government to regulate prices charged by public utilities like gas and electric companies. The underlying object is to call forth the maximum output consistent with the monopolists' cost and consumer demand. This is shown in **figure 4.31**. In the absence of price fixation by the Government, the monopolist would produce OM output and charge OP (= MQ price, because here MR = MC and Q is a point on the demand curve and shows what the consumers are prepared to pay.

   Now let us suppose the Government fixes a lower price OP′ (= M′L). At this price, the monopolist produces a large output OM′, since this price cuts the demand curve DD at L. Although the monopolist has been compelled to charge a lower price, yet the consuming public has taken a larger quantity. This compensates the monopolist. OM′ is the new profit-maximising output and his profit will be OM′ output ×P′N′. The consumers are benefited by a larger amount being made available to them at a lower price.

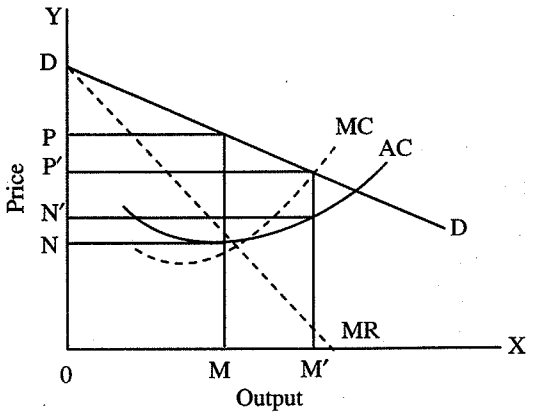


Figure 4.31: Price Regulation

2) **Taxation:** Taxation is regarded as a very suitable device for regulating monopolies so that they are not able to exploit the consuming public by misusing their monopolistic power. Such taxes may be of two types:
   i) Lump sum tax irrespective of the quantity of the output, and
   ii) A fixed tax per unit of the output.

   The lump sum tax has to be borne entirely by the monopolist since it cannot be shifted to the consumers by way of the price rise. Already, the monopolist is supposed to have fixed a price which maximises his profit. If he could raise the price further he would have done so. If he could decrease the output to increase his profit, he would have done so. Since he has already fixed price and output which brings him maximum profit, he cannot now touch them. Hence, by imposing a lumpsum tax, the government can take away all or any proportion of his profit without adversely affecting the general welfare.

   This is, therefore, a very effective way of controlling the monopolist's behaviour. As for the tax per unit of output, the monopolist will be induced to reduce output and raise the price in order to maximise his profit after paying the per unit tax. The price rise affects the consumer. The profits after the tax too are smaller than profits before the tax; since the tax is a variable cost. The total costs of the monopolist at the various levels of output are increased, but the total revenue remains the same. Hence, prices being higher and the output smaller, the general welfare is adversely affected. Such a tax is a wholesome check on the misuse of monopolist's power.

3) **Indirect Taxes:** These taxes can be used to raise the price of de-merit goods and products with negative externalities designed to increase the opportunity cost of consumption and thereby reduce consumer demand towards a socially optimal level.

4) **Subsidies:** Subsidies to consumers will lower the price of merit goods. They are designed to boost consumption and output of products with positive externalities – remember that a subsidy causes an increase in market supply and leads to a lower equilibrium price.

5) **Price Controls:** Controls on prices are usually enacted when policy-makers believe the market price is unfair to buyers or sellers. Government controls prices by following manners:
   i) **Price Ceiling:** A legal maximum on the price at which a good can be sold.
   ii) **Price Floor:** A legal minimum on the price at which a good can be sold.

iii) **Dual Pricing:** Dual pricing is selling the same or identical product at different prices in different markets. It may indicate an intention to engage in dumping. Dual pricing is a method of pricing of a product under which the product may have two prices. One is the controlled price fixed by the government and the other one is the market price usually determined on the basis of cost production and a reasonable margin of profit. **For example,** pricing of sugar and cement in India is a good example of dual pricing. Dual pricing is thus a price control of price regulation system.

4.14.8. System of Dual Price

Selling of similar or almost similar goods at different price in different market is termed as dual pricing. This may result in a purposeful dumping. Under dual price system, a particular product has two different prices. One of the prices will be the controlled price determined by the government and the second one will be the price determined by the market depending on the profit margin and production cost.

4.14.8.1. Objectives of Dual Pricing

The various objectives of dual pricing are as follows:

1) New market development along with export.
2) To effectively utilise the capacity.
3) To face the market competition.
4) To share the consumer's surplus rather than letting the customer decide.
5) To improve the market shares.

4.14.8.2. Tools of Dual Pricing

The various tools of dual pricing are given a following:

1) **Two Step Pricing:** This type of pricing includes charging prices for such products which are being transferred twice. In the **first** stage, the price of the product is charged on the basis of variable cost undergone for producing goods and services. In the **second** stage, fixed cost which was incurred for adding special facilities, are included. Hence, the product transfer price is formed by the addition of these two charges.

2) **Profit Sharing:** This is the method in which the transferring of product is done at the standard variable cost to the marketing unit. The manner in which profit sharing is done between the two profit centres may result in disputes. Therefore, top executives or senior managers participate to resolve this issue. As the profit distribution is carried out arbitrarily, the profitability of different segment is not depicted accurately. The producing units can seen as the unfair due to the contribution of producing unit which is dependent on the selling capability and real selling price.

3) **Two Sets of Prices:** In this method, the revenue earned will be transferred to the manufacturing unit at the rate of selling price, whereas purchasing unit is charged for standard cost. The parent company's account will be charged for the variation in the outside sales price and the standard cost. During the formation of combined financial statements, these charges will not be considered. This method is commonly implemented when the buying units and the selling units have occasional disputes and no method can be useful to resolve these issues.

4.15. EXERCISE

4.15.1. Very Short Answer Type Questions

1) What is revenue analysis?
2) Explain cost plus price.
3) What is market and market structure?
4) Explain the meaning of monopoly.
5) Write the features of duopoly.
6) What are the roles of profit?
7) What is price skimming?
8) What is price penetration?
9) Explain the concept of government intervention.
10) What is dual price system?

4.15.2. Short Answer Type Questions

1) What is total, average and marginal revenue?
2) Explain the relationship between revenue and price elasticity of demand.
3) Discuss marginal cost pricing, breakeven point and cyclical pricing.
4) Explain different types of market structure.
5) What are the advantages and disadvantages of perfect competition?
6) Differentiate between perfect competition and imperfect competition.
7) What are the features of monopoly?
8) Explain the bilateral monopoly.
9) Discuss the monopolistic competition.
10) What role does profit plays in Market System?
11) What is break even analysis and breakeven point?

4.15.3. Long Answer Type Questions

1) What is pricing? Discuss its importance and objectives.
2) What is pricing practices? Highlight penetrating pricing, price leadership, price skimming and transfer pricing.
3) Explain different types of market structure.
4) Discuss the meaning of is perfect competition? Explain its features and assumptions.
5) Highlight the price-output determination under perfect competition.
6) Write the advantages and disadvantages of monopoly.
7) Explain the equilibrium of Monopoly firm under different cost situations.
8) What is price discrimination? Explain with the help of suitable diagram and examples.
9) What are the features, advantages and disadvantages of monopolistic competition?
10) State various features of oligopoly and explain price rigidity with the help of kinked demand curve.
11) Write its feature and advantages of oligopoly.
12) Describe price rigidity. Also highlight collusive oligopoly.
13) What is industry analysis? Highlight industry analysis in textile and electronic industry.
14) Explain Break-even Analysis with assumptions, advantages and disadvantages.
15) Explain profit forecasting. Explain its significance and methods to forecast.
16) Discuss the concept of price control and their forms of control.
17) What do you mean by government intervention? Do you think government intervention is needed? Why? Why not?

UNIT 5

Consumption and Investment Function & Business Cycle

5.1. CONSUMPTION FUNCTION

5.1.1. Introduction

Consumption function was given by the famous economist **John Maynard Keynes.** It is a mathematical formula. Consumption function establishes the relation between the real disposable income and consumer spending. This variable is regarded as the essential component of short-term demand in an economy by Keynes.

The formula for consumption function can be expressed as follows:

$$C = a + c\,Yd$$

There are three essential items on which Keynes consumption function is based and shows the level of consumer spending:

1) Yd – disposable income
2) a – autonomous consumption (consumption when income is 0. (e.g. even with no income, you may borrow to be able to buy food))
3) c – Marginal propensity to consume (the % of extra income that is spent) Also known as induced consumption.

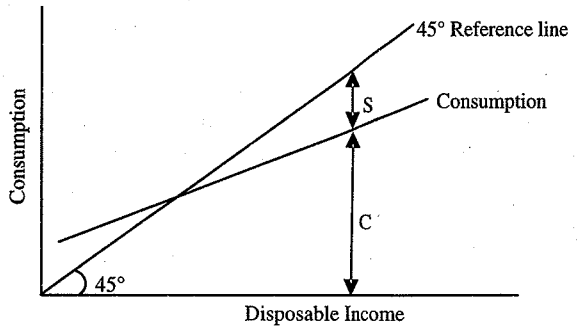


Figure 5.1: Consumption Function

The consumption is straightforwardly linked to income as per the previous data of several years and spending of the household is more as compared to savings. This relationship is well described in **figure 5.1.**

In the **figure 5.1,** the 45° line is drawn for measuring the level of consumption and saving. The every point drawn on this line is at equal distance from both the axis i.e. X-axis and Y-axis. The consumption or income is shown by the vertical distance from any point on the horizontal X-axis to this line. The saving during the specified year is shown by the amount with which the actual consumption reduces of the 45° line. In the above figure, it is clearly depicted that the consumption and saving is straightforwardly linked to the income and consumption level of the household is more than savings by the household. Hence, it proposes that consumption is mainly determined by the level of disposable income (Yd) meaning the more is the disposable income, the more will be the spending by the consumers. As per consumption function, the consumer spending increases with the increase in the level of income. On the other hand, the rise in spending will be considerably at lesser rate as compared to the income.

1) The major portion of the consumer's income is spent when the income is less. Thus, the average propensity to consume can be one or greater than one indicating that the consumer spends the each and every part of the income and no amount is left for saving. If the income of the consumer is less than the total income of the consumer is spent on fulfilling the basic essential needs of the consumer, therefore, no amount is left for saving.
2) On the contrary, if the income of the consumer increases then they are able to save some of the portion of their income. Hence, the rise in consumer spending is at a diminishing rate with the increase in income as compared to the disposable income. The average propensity to spend is less when the income of the consumer is more.

5.1.2. Determinants of Consumption Function

The determinants of the consumption function are as follows:

1) **Money Income of the Community:** This is one of the main determinants in consumption function. It takes into consideration that with the rise in income of the consumer or community, the expenditure on consumption also rises and vice-versa.
2) **Distribution of Income in the Community:** The level of consumption is less when there is inequality in the distribution of income between the rich section and the poor section of the community. On the contrary, if there is equality in distribution of income between the rich class and poor class of the community then the level of consumption is high.
3) **Changes in the Price Level:** The propensity to consume is low with increase in the level of price taking into account the other things remains constant and while the propensity to consume is high when the level of price decreases taking into consideration that the other things remain unchanged.
4) **Changes in the Wage Level:** The propensity to consume also rises with the increase in the level of wages considering other things remains unchanged whereas propensity to consume reduces with the fall in level of wages taking into account the other things remains constant.
5) **Change in Consumers' Tastes and Fashions:** The propensity to consume is also impacted by the change in taste and preferences or fashion of the consumer.

5.1.3. Nature of Consumption Function/Propensity to Consume

Propensity to consume is a schedule depicting the consumption level at various possible level of the income of the consumer. In other words, it is the ratio between C and Y.

According to Dillard, "A schedule showing various amounts of consumption which correspond to different levels of income is known as propensity to consume".

The following are the two most essential component of propensity to consume:

1) Average Propensity to Consume (APC)
2) Marginal Propensity to Consume (MPC)

5.1.3.1. Average Propensity to Consume (APC)

Average propensity to consume (APC) is defined as the ratio obtained by dividing the total consumption with the total income of the consumer at the given level of income. In other words, average propensity to consume is said as the ratio of consumption expenditure at any income level.

$$APC = \frac{C}{Y}$$

For example, if income (Y), is ₹100 crore and consumption (C) is ₹90 crore, then

$$APC = \frac{C}{Y} = \frac{90}{100} = 0.9 \text{ or } 90\%$$

This denotes that consumption expenditure forms 90% of the income in the economy.

5.1.3.2. Marginal Propensity to Consume (MPC)

Marginal propensity to consume is defined as the ratio of change in the consumption (ΔC) to change in the level of income (ΔY). It tries to establish the relation between the changes in the level of consumption and income.

$$\text{Marginal Propensity to Consume} = \frac{\text{Change in Consumption}}{\text{Change in Income}} \quad \text{or,} \quad MPC = \frac{\Delta C}{\Delta Y}$$

For example, if income rises from ₹200 crore to ₹240 crore and the consumption rises from ₹160 crore to ₹180 crore, then

$$MPC = \frac{\Delta C}{\Delta Y} \qquad \Delta C = ₹40 \text{ crore}$$

$$\Delta Y = ₹20 \text{ crore} \qquad MPC = \frac{20}{40} = 0.5 \text{ or } 50\%$$

5.1.4. Measures to Raise Propensity to Consume

The following are the measures to raise propensity to consume:

1) **Increased Wages:** The consumption function moves upward as the wages of the consumer increases whereas the employment level of the economy is severely affected with the increased wages. In short-run, it is impossible to increase the marginal revenue productivity of labour because in case the wages are increased then the costs will also increase with it, at the time when the rise in marginal revenue productivity of labour is not present. Hence, the situation of unemployment exists in the economy.
   Thus, in long-run the policy of wages must be framed in such a manner that the wages rises in the same proportion as the rise in labour productivity increasing the consumption level in the economy.

2) **Social Security Measures:** In long-run, the social security measures lead to increase in the consumption function. The forthcoming uncertain situation can be considerably reduced when the safety measures such as provision for unemployment relief, medical, facilities, old age pension, etc., are taken and the propensity to save will be decreased by the consumer. Hence, in order to increase the propensity to consume, the state must provide greater social security measures. During the times of depression, the unemployment relief and old age pension will lead to an increase in consumption expenditure and thereby assists the economy to recover from the period of depression. Therefore, the social security measure is likely to increase the consumption function whether in times of prosperity or during the period of depression

3) **Credit Facilities:** The consumption function also moves upwards due to the availability of inexpensive and simple credit facilities. The propensity to consume is increased when the purchases of durable goods by the consumer such as scooters, televisions, refrigerators etc., increases due to the easy and inexpensive availability of loans. The purchase of durable goods gives the equal effect whether the consumer purchases goods on instalment basis or on the basis of hire-purchase system. Therefore, the propensity to consume the durable goods is increased with the help of various credit facilitates.

4) **Advertisement:** In today's era, an advertisement is considered as one of the most prominent method to increase the propensity to consume. There are various forms of advertisement like radio, television cinema, newspaper, etc., through which the consumers are able to get the information and knowledge regarding the products. With the help of advertising only, the consumers get aware about the usage of product. As the consumers get influenced towards the product by appealing advertisement adopted by the seller so that they buy the product i.e. increases the propensity to consume.

5) **Development of the Means of Transport:** The propensity to consume also rises by the proper development of the means of transport. The shifting of product from the place of manufacturing to the place of final consumption or to the various parts of the country becomes very easy and also helps in the expansion of the market. The prices of the product can be reduced because of the reduction in transport cost. There is easy availability of goods to the consumer at their own towns. Hence, these factors lead to the movement in consumption function towards upwards.

6) **Urbanisation:** The urbanisation also contributes in the increase in propensity to consume. During the period of urbanisation, the peoples are so much attracted and are inclined towards the product by the presentation of new articles and demonstration effect that they tend to shift from rural areas to urban areas. Thus, leading to a movement in consumption function towards upward. Therefore, in order to increase the consumption function, the concept of urbanisation must also be taken into consideration.

5.1.5. Determinants of Propensity to Consume

The determinant of propensity to consume or the factors which affect the consumption function is classified into two broad categories viz., subjective factors and objective factors. These factors are elaborated as follows:

1) **Subjective Factors:** The subjective factors take into account only the factors which helps in persuading the consumer to save their some portion of income. Some of the factors helps in saving their portion of income are as follows:
   i) **Unanticipated Events:** The people save their part of income in order to meet their future unexpected happening or events which may or may not arise like illness, unemployment, accidents, etc.
   ii) **Fulfilment of Future Requirements:** The people are also induced to save their part of the income so that they are able to fulfil their future needs and wants like education of the children, marriages of their children, etc.

iii) **Increased Returns:** The people might also save their portion of income to make further investment for generation of additional return to the consumer by way of increasing the profits and interest on the investment. Hence, this results in raising the future income of consumers.

2) **Objective Factors:** The following are the objective factors which had a great impact on the consumption function as described by Keynes:
   i) **Changes in the General Price Level – Real Balance Effect:** The general price level is considered as an essential factor affecting the level of consumption in the economy. The propensity to consume reduces with the rise in general price level or in the situation of inflation in economy.
   ii) **Fiscal Policy:** The consumption function is also affected by the fiscal policy of the government mainly the taxation policy. The government reduces the level of consumption when the excise duties, sales tax, etc., are imposed and rises the savings of the economy. Similarly, there is rise in level of consumption of the people when the taxes are reduced by the government increasing the propensity to consume.
   iii) **Rate of Interest:** The propensity to consume and propensity to save is also impacted by the rate of interest. Normally, propensity to save is increased with the rise in rate of interest thereby leading to decrease in propensity to consume.

5.2. INVESTMENT FUNCTION

5.2.1. Introduction

Investment means providing the fund for present duration of the time period, the benefit of which will be obtained in the future period of time by generating additional funds. It will help the investor indemnifying for the forgone consumption, reduction in purchasing power and rise in ambiguity in future flow of funds. In terms of economics, the investment is the increase in the total physical stock of capital indicating the new expenditure on account of the rise in capital goods and stock.

The income and employment level in the economy is increased with the increase in investment level of aggregate demand. In other words, investment is also referred to as the purchase of securities like shares, debentures/bonds, mutual funds units, etc., at a given price with the aim of earning future returns on the purchased securities. This definition is approximately similar to the definition given in economics sense. However, the point of difference lies to the fact that some of the economist observes the investment from the viewpoint of secondary investor's to whom the funds is provided by the primary investor.

5.2.2. Types of Investment

The following are the types of investments:

1) **Real Investment and Financial Investment: Real investment** refers to the investment which results in the formation of further units of production. **For example,** investment in new factory buildings, new plants and machines, etc. **Financial investment** refers to the transmission of ownership from one person to the other and it does not increase the stock of real capital of an economy. **For example,** the purchases of stock and shares, debentures, government bonds, other securities, etc.

2) **Gross Investment and Net Investment: Gross investments** means the total amount of expenditure spend on the new fixed capital assets such as houses, factories, machinery or expenses done on the surplus inventory like raw materials and unsold consumption articles. **Net investment** means the amount of expenditure spend on the capital asset after deducting the amount of depreciation. In the Keynesian theory of employment and income, investment means net investment and net gross investment.

3) **Private Investment and Net Investment: Private investment** means the amount spend on investment by the private individual on different projects of the public sector with a view to earn maximum profit. Hence, this form of investment is profit-oriented. The peoples in this form of investment are motivated by marginal efficiency of capital and the rate of interest. **Public investment** means the form of investment done by the public authorities or government such as state government or local government (or bodies) on the various governmental projects. The social welfare is one of the main objective of this form of investment.

4) ***Ex-Ante* Investment and *Ex-Post* Investment:** The other name of **Ex-ante investment** is planned or intended investment and **Ex-post investment** is also called as unplanned or unintended investment. The summation of planned and unplanned investment helps to derive the total actual investment.

5) **Induced Investment and Autonomous Investment:** Induced investment is the form of investment is started due to the changes in income level. Generally, **induced investment** refers to the investment in which with the increase in the level of income the investment also rises. This form of investment is profit oriented. **Autonomous investment** refers to the investment in which there is no change in the level of income. The main objective of this form of investment is social welfare. It is completely indifferent with the change in income.

5.2.3. Investment Demand Curve

The **investment demand curve** refers to the graphical representation of the cumulative investment data of entire economy obtained by the summation of the investment spending by each firm. When the important investment projects are arranged in descending order of their expected rate of net profitability then investment demand curve arises. The investment is started at a point where the real interest rate becomes equal to the expected rate of net profit. There is opposite relationship between the real interest rate and demand for the quantity of capital goods indicated by the downward slope of investment demand curve as represented in **figure 5.2.**

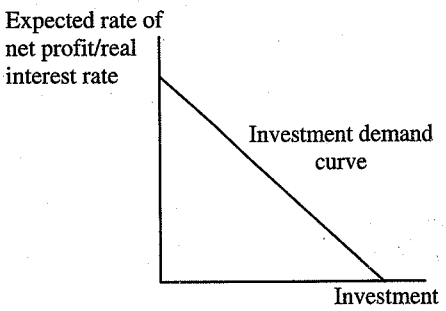


Figure 5.2: Investment Demand Curve

Moreover, there are other factors also which affect the investment spending are mentioned below:

1) **Acquisition and Operating Costs:** The investment demand curve moves leftward because of the reduction in expected rate of net profit with the increase in acquisition and operating costs for the machines. **For example,** the increase in wages or electricity bill will tend the investment demand curve to move towards left. On the contrary, the investment demand curve shows rightward trend due to the increase in expected rate of net profit with the reduction in acquisition and operating costs for the machines.

2) **Corporate Taxes:** The investment decisions are also affected by the corporate taxes due to the reason that the expected rate of net profits is taken into consideration by the businesses before making any investment into the company. The investment demand curve moves leftwards when the business taxes rises leading to the reduction in expected net profits while the investment demand curve moves towards right with the fall in corporate taxes.

3) **Technological Change:** Normally, the investments are encouraged by the improvement in technology. The more is improvement in technology, the greater is the quality product provided at reduced cost increasing the profits of the firm. If a profit arises in new product then it influences huge investment to the firm because the firm are already prepared to expand the production of such a product. Thus, the faster advancement in the technology leads to the movement of investment demand curve towards right.

4) **Expectations:** The investment is done with the expectation to derive benefit of such an investment in future period of time. The firms make investment in a particular venture with a view to have an increased return i.e. profit on the investment. The profitability of any investment is based on the expectation of future sales and profits generated with the help of investment. The investment demand curve will move towards right when the firm is confident regarding the future business conditions and vice-versa.

5.2.4. Investment and Income

In case when the investment plans is given in terms of level of disposable income then the investment decisions of a firm can be associated with the consumption plans of the households. The relationship depicts the amount of investment by the businesses at each possible output level i.e. income is called as **investment schedule**. Just as similar to the consumption and saving schedule showing the consumption and saving plans of the household similar to the investment schedule showing the investment spending of the firms. The business investment does not depend on the level of current disposable income due to the reason that they prospects to have long-term profits. Hence, it is supposed that an equal amount will be invested by the firms irrespective of the level of income.

Table 5.1 demonstrates the investment done by the firms with the given level of income:

Table 5.1: Investment Schedule Calculation

Level of output (Income), GDP = N	50	60	70	80	90	100
Investment, I	10	10	10	10	10	10

In **figure 5.3,** the X-axis represents the output (income) and Y-axis represents the investment. The slope of the investment schedule may be upward to some extent but it is assumed that the investment schedule is horizontal as shown in the given figure.

The investment schedule has more variability in comparison to the consumption schedule as it frequently shows upward or downwards trend. Hence, the investment is considered as one of the most unstable component of total spending. There are certain factors resulting in the change in investment like stability of capital goods, inconsistency in innovation, variability of profits, variability of expectations, etc.

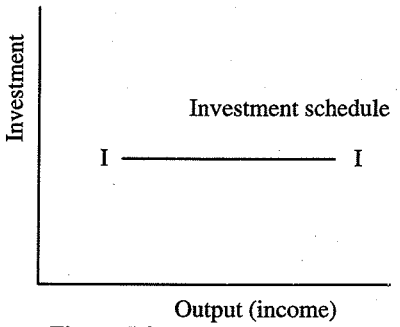
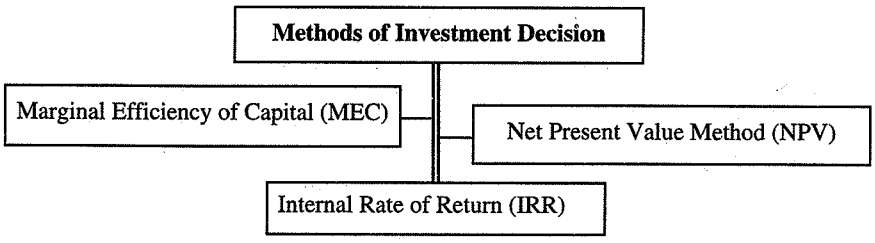


Figure 5.3: Investment Schedule

5.2.5. Methods of Investment Decision

The theory of investment is fundamentally the theory of demand for capital. The capital is productive and it yields a return over time. Due to this, it is demanded. Some interests are paid on money spent on capital acquisition and hence the return on capital is subject to a cost. Since every firm wants to maximise its profit, therefore with the given cost each firms demands capital stock that maximises its profit.

The theory of investment explains the concept that how firms decide the level of capital stock which maximises the profit. Theoretically as well as practically, there are different methods of making investment decisions. However, there are three most important methods of investment decision-making, as shown in **figure below:**



5.2.5.1. Marginal Efficiency of Capital (MEC) and Business Expectations

Marginal Efficiency of Capital depends on the expectations of businessmen. Expectation of a business increases due to invention whereas it goes down due to any threat to the returns on investment. It is also affected by the annual spirit of the entrepreneur. The expected annual rate of return on an additional unit of a capital good is called the marginal efficiency of capital. If some money invested in a newly produced asset then the marginal efficiency of capital is described as the expected rate of return to be received on that money.

According to Kurihara, "It is the ratio between prospective yield of additional capital goods and their supply price".

According to J.M. Keynes, "The marginal efficiency of capital is the rate of discount which makes the present value of the prospective yield from the capital asset equal to its supply price. The marginal efficiency of capital will progressively diminish as investment in the asset increases. The Marginal Efficiency of Capital (MEC) curve is, therefore negatively sloped".

Keynes assumed that an entrepreneur would expand his investment at the level where the marginal efficiency of capital fell to the level of the rate of interest. But it is an impractical assumption. According to Keynes an

entrepreneur is using only borrowed capital. In reality, it is required that the entrepreneur must have his own capital for the access to borrowed capital. Hence, the rate of interest is the secondary important thing for the entrepreneur. **Keynes** accepted the law of diminishing returns on capital that is widely held in conventional political economy. According to the law of diminishing returns on capital, when the investment increases then each additional unit of capital brings a decline in the productivity or efficiency of capital. **Keynes** does not explain that with an increase of capital applied to production why the rate of profit should decline or why, in the final analysis, it must decline to the level of the interest rate.

Calculation of MEC

The marginal efficiency of capital depends upon two factors:

1) **Expected Net Return on an Asset:** The income which a person receives after selling the product produced by the machine or asset is referred as the expected net return on that asset. The investment will increase when the expected rate of return is greater than the expenses made on a particular asset.
2) **Replacement Cost or Supply Price:** Assume that the other things remaining the same, the marginal efficiency of capital will increase if the supply price or the price of the machines falls. On the other hand it falls with the increase in the supply price.

$$MEC = \frac{\text{Expected Yearly Income}}{\text{Supply Price}} \times 100$$

Due to increase in consumption there is an increase in marginal efficiency of capital and investment. The amount of investment will increases at that level where the marginal efficiency of capital is more than the principal rate of interest. In other words, investment will increase upto that point where marginal efficiency of capital is equal to the rate of interest.

We can explain it by the following **table 5.2**:

Table 5.2: Schedule of Marginal Efficiency of Capital

Investment	MEC	Rate of Interest
₹60 crore	12%	8%
₹80 crore	10%	8%
₹100 crore	8%	8%
₹120 crore	6%	8%
₹140 crore	4%	8%

According to this schedule, we assume that rate of interest in the market is 8% which is fixed. The **table 5.2** shows that as the volume of investment increases the marginal efficiency of capital falls. The rate of interest and MEC both are equal to each other when the investment volume is ₹100 crore.

This diagram shows that the marginal efficiency of capital and rate of interest both are equal to each other (8%) when investment is ₹100 crore. So it is clear that the amounts of investment are determined by the rate of interest and marginal efficiency of capital.

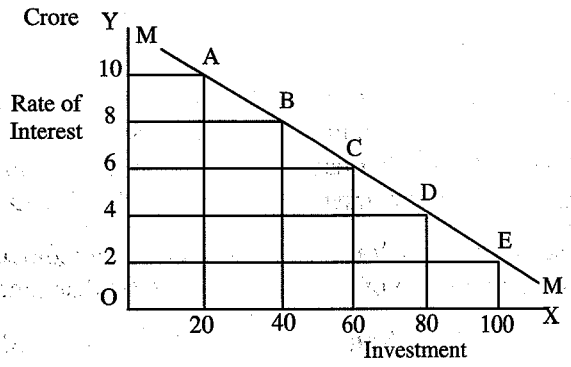


Figure 5.4: MEC Curve

Decision Rule

When we compare MEC of capital with market rate of interest, the following would be the rule for making investment decision:

1) If MEC > r, the investment project is accepted.
2) If MEC = r, the investment project is acceptable on non-profit.
3) If MEC < r, the project is rejected.

5.2.5.2. Net Present Value (NPV) Method

The net present value method is referred as a modern method of evaluating investment proposals. The time value of money and attempts to calculate the return on investments by introducing the factor of time element are considered in this method. This method considered the fact that the value of money which is earned today is more valuable in comparison to the same amount of money earned tomorrow. The net present values of all inflows and outflows which are arising during the entire life of the project is determined separately for each year by discounting these flows by the firm's cost of capital or a pre-determined rate.

Calculation of NPV

The following are the required steps to be followed for adopting the net present value method or evaluating investment proposals:

1) First of all selecting an appropriate rate of interest that should be selected as the minimum required rate of return is called 'cut-off rate of discount rate'.
2) The investor did not invest below the level of selected rate of return. In the market, the discount rate should be either the actual rate of interest on long-term loans or it should reflect the opportunity cost of capital of the investor.
3) Calculate the present value of total investment outlay, i.e. cash outflows at the determined discount rate. The present value shall be the same as the cost of investment if in the initial year the total investment is to be made.
4) Calculate the present values of total investment proceeds such as inflows (profit before depreciation and after tax) at the predetermined discount rate.
5) Net present value of each project is calculated by subtracting the present value of cash inflows from the present value of cash outflows for each project.
6) When present value of cash inflows either exceeds or is equal to the present values of cash outflows then the net present value is positive or zero and hence the proposal may be accepted. But in the situation when the present value of inflows is less than the present value of cash outflows, the proposal should be rejected.
7) When there are two mutually exclusive projects then the given projects should be ranked in order of net present values. In other words, the first preference should be given to the project having the maximum positive net present value.

The present value of ₹1 due in any number of years can be found with the use of the following mathematical formula:

$$PV = \frac{1}{(1+r)^n}$$

Where, PV = Present Value
r = Rate of interest/discount rate
n = Number of years

Net Present Value (NPV) is considered a sophisticated capital budgeting technique and gives clear consideration to the time value of money. All such techniques in one way or another, discount the firm's cash flows at a specified rate. The minimum return that is often called discount rate must be earned on a project to leave the firm's market value unchanged included required return, cost of capital, or opportunity cost.

The Net Present Value (NPV) is found by subtracting the present value of project's cash outflows (CF_0) from the present value of its cash inflows (CF_t) discounted at a rate equal to the firm's cost of capital (k).

NPV = Present Value of Cash Inflows – Present Value of Cash Outflows

$$NPV = \sum_{t=1}^{n} \frac{CF_t}{(1+k)^t} - CF_0 = \sum_{t=1}^{n} (CF_t \times PVIF_{k,t}) - CF_0$$

Accept-Reject Decision

When NPV is used to make accept-reject decisions, the decision criteria are as follows:

NPV > Zero	Accept the proposal
NPV < Zero	Reject the proposal
NPV=Zero	Indifference

5.2.5.3. Internal Rate of Return (IRR) Method

The internal rate of return (IRR) method is the other discounted cash flow (DCF) or time-adjusted method for appraising capital investment decisions. The yield on investment, marginal efficiency of capital, marginal productivity of capital, rate of return, and time-adjusted rate of return are also known as internal rate of return (IRR) method. The IRR method also considers the time value of money by discounting the cash streams as same in the present value method.

The basis of the discount factor, however, is different in both cases. In the case of the net present value method, the discount rate is the required predetermined rate of return which determinants are external to the proposal under consideration and usually the cost of capital. On the other hand the internal rate of return is based on facts which are internal to the proposal. In other words, the cash inflows as well as outflows are not considered for arriving at the required rate of return for finding out present values. But the internal rate of return (IRR) method depends completely on the initial amount and the cash proceeds of the project which is being evaluated for acceptance or rejection. It is, therefore, appropriately referred to as internal rate of return.

The rate of return that a project earns is referred to as internal rate of return. The discount rate (r) which equates the aggregate present value of the net cash inflows (CFAT) with the aggregate present value of cash outflows of a project is called internal rate of return. In other words, it is that rate which gives the project net present value (NPV) of zero.

Calculation of IRR

Calculation of IRR is done in the following two manners:

1) **When the Annual Cash Inflows are Equal:** The internal rate of return of the projects which results in equal cash inflows can be computed by determining present value factor in the following way:

$$\text{Present Value Factor} = \frac{\text{Initial Investment}}{\text{Annual Cash Inflow}}$$

When present value factor is known then it is found in the Annuity table on the line which represents number of years corresponding to economic life of the project. When exact present value factor is not available the internal rate of return will be in between two factors and nearest factor is taken as internal rate of return. It can be computed by the following formula:

$$IRR = X + \frac{P_X - I}{P_X - P_Y}(Y - X)$$

Where, IRR = Internal rate of return,
Y = Higher discount rate
X = Lower discount rate,
Px = Present value of cash inflows at X
Py = Present value of cash inflows at Y,
I = Initial investment

For clear understanding, a portion of the Present Value Annuity Table is below:

Present Value Annuity Table

Present Value of an Annuity of Re.1							
Year	6%	7%	8%	9%	10%	11%	12%
01	0,9434	0.9346	0.9929	0.9174	0.9091	0.9009	0.2989
02	1.8334	1.8010	1.7833	1.7591	1.7355	1.7125	1.6900
03	2.6730	2.6243	2.5771	2.5313	2.4864	2.4437	2.4018
04	3.4651	3.3872	3.3121	3.2397	3.1699	3,1024	3.0373
05	4.2123	4.1002	3.9927	3.8896	3.7909	3.6959	3.6048

06	4.9173	4.7655	4.6229	4.4859	4.3553	4.2305	4.1114
07	5.5824	5,3839	5.2064	5.0429	4.8664	4.7122	4.5638
08	6.2098	5.9713	5.7466	5.5348	5.3349	5.1461	4.9676
09	6.8117	6.5152	6.2469	5.9852	5.7590	5.5370	5.3282
10	7.3601	7.0236	6.7101	6.4176	6.1446	5.8892	5.6502

2) **When the Annual Cash Inflows are not Equal:** The internal rate of return of the projects which results in unequal cash inflows can be computed by making trial calculations in order to compute accurate internal rate of return (IRR) which associates the present value of cash inflows and cash outflows. In this process of calculation of IRR, the following steps are required:

i) **Determination of First Trial Rate:** It is calculated on the basis of present value factor which is as follows:

$$\text{Present Value Factor} = \frac{\text{Initial Investment}}{\text{Annual Cash Inflow}}$$

$$\text{Average Annual Cash Inflow} = \frac{\text{Total Cash Inflows}}{\text{Economic Life of the Project}}$$

After this calculation, annuity table is used to find out the IRR.

ii) **Application of Second Trial Rate:** If the NPV gives positive value, we apply the higher rate of discount, and if still it gives positive net present value, we increase the discount rate until NPV becomes negative. If NPV becomes negative then IRR lies between these two rates.

Accept-Reject Decision

When IRR is used to make accept-reject decisions, the decision criteria are as follows:

IRR > k (Required Rate of Return)	Accept the proposal
IRR < k	Reject the proposal
IRR = k	Indifference

Example: Project X and Project Y costs ₹ 50,000 and ₹ 25,000 respectively. Their cash flows are given below. You are required to find out the internal rate of return for each project and decide on that basis which project is more profitable.

Year	Cash Inflows	
	Project X (₹)	Project Y(₹)
1	5,000	10,000
2	15,000	10,000
3	30,000	10,000
4	20,000	10,000
5	10,000	

Solution: First, calculate the present value factor for both the projects.

$$\text{PV factor} = \frac{\text{Initial Cost of Project}}{\text{Average Annual Cash Inflow}}$$

$$\text{PV factor for Project X} = \frac{₹\ 50,000}{₹\ 16,000} = 3.125$$

$$\text{PV factor for Project Y} = \frac{₹\ 25,000}{₹10,000} = 2.5$$

Present value for Project X for 5 yrs. comes to 18% (present value for 5 yrs. is 3.127, which is nearly equal to 3.125).

Similarly, for Project Y it is 22% for 4 yrs.

Take 18% and 22% as discounting rate for Projects X and Y. In this case, the present value of Project X and Y is as follows:

Computation of Present Value of Cash Inflow

Year	Project X			Project Y		
	P.V. Factor at 18%	Cash Inflow (₹)	Present Value (₹)	PV Factor at 22%	Cash Inflow (₹)	Present Value (₹)
1	0.847	5,000	4,235	0.820	10,000	8,200
2	0.718	15,000	10,770	0.672	10,000	6,720
3	0.609	30,000	18,270	0.551	10,000	5,510
4	0.518	20,000	10,360	0.451	10,000	4,510
5	0.437	10,000	4,370		–	–
		80,000	**48,005**		**40,000**	**24,940**

In order to equate the present values with the cash outlays of ₹ 50,000 and ₹ 25,000 for project X and Y respectively, the IRR will have to be lower than 18% for project X and 22% for project Y because the NPV is negative in both the cases. So, we must consider second trial rate at a discount rate of 16% for X and 20% for Y. The present values at these rates are as follows:

Computation of Present Value of Cash Inflows

Year	Project X			Project Y		
	P.V. Factor at 16%	Cash Inflow (₹)	Present Value (₹)	PV Factor at 20%	Cash Inflow (₹)	Present Value (₹)
1	0.862	5,000	4,310	0.833	10,000	8,330
2	0.743	15,000	11,145	0.694	10,000	6,940
3	0.641	30,000	19,230	0.579	10,000	5,790
4	0.552	20,000	11,040	0.482	10,000	4,820
5	0.476	10,000	4,760		–	–
		80,000	**50,485**		**40,000**	**25,880**

Since the NPV of both the projects is positive, now apply the formula to calculate the IRR.

$$IRR = X + \frac{Px - I}{Px - Py}(Y - X)$$

IRR for Project X

$$= 16 + \frac{₹\ 50,485 - ₹\ 50,000}{₹\ 50,485 - ₹\ 48,005}(18 - 16) = 16 + \frac{₹\ 970}{₹\ 2,480} = 16.39\%$$

IRR for Project Y

$$= 20 + \frac{₹\ 25,880 - ₹\ 25,000}{₹\ 25,880 - ₹\ 24,940}(22 - 20)$$

$$= 20 + \frac{₹\ 880}{₹\ 940} \times 2 = 20 + \frac{₹\ 1,760}{₹\ 940} = 21.87\%$$

Project Y is more profitable than project X because it shows a higher IRR.

5.3. MULTIPLIER

5.3.1. Meaning and Definition of Multiplier

Keyne's theory of income, output and employment embodies the concept of multiplier. This is an important concept as it helps in determining the change in income as a result of change in investment. In economic terms, multiplier simply implies that change observed in a factor of production at final level will not be same as

change at the initial level, rather it would be a multiplied value. **For example,** though income increases with increase in investment, yet percentage increase in income is not same as percentage increase in investment. Rather, it is much more as compared to increase in investment. The number of times increment in the income to the increment in investment is called multiplier. Since, Keneysian theory of macroeconomics establishes a relationship between investment and income, the above stated phenomena is termed as investment multiplier. So, multiplier seeks to establish a relationship between final changes in income as a result of initial changes in investment. It represents the ratio of change in income due to change in investment.

$$K=\frac{\Delta Y}{\Delta I}$$

Where, K = Multiplier,
ΔY = Change in income,
ΔI = Change in investment.

Thus, it can be deduced that initially there is an increase in consumption with the increase in investment and finally many more times increase in income takes place. **For example**, if by increasing investment by ₹50 crore, ultimate increase in income is ₹100 crores, then multiplier will be $\frac{100}{50}=2$. So, the multiplier is the number which when multiplied with the amount of investment reveals us the increase in income.

The **figure 5.5** shown below explains how change in investment causes change in income, which further leads to change in consumption, thus having a multiplier effect.

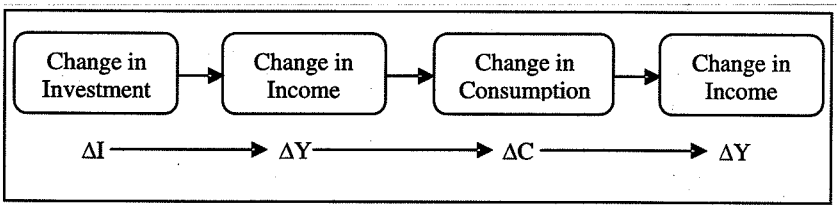


Figure 5.5: Multiplier

Expenditure on consumption made by one person becomes the income of other. In this way, change in consumption leads to change in income. This chain continues till ΔC falls to zero.

According to Keynes, "Investment multiplier tells us that when there is an increment of aggregate investment; income will increase by an amount which is K times the increment of investment."

According to Kurihara, "The multiplier is the ratio of change in income to the change in investment."
Because of this multiplier effect, Governments need to be cautious while making any changes in government spending or taxation, as they would indirectly affect the whole economy. So, e.g., if a government wants to raise real GDP by ₹10000 crore and the multiplier has been estimated at 4, it would have to raise its spending by ₹2500 crore through social sector schemes or other available instruments.

5.3.2. Assumptions of Multiplier

Keyne's theory of the multiplier works under certain assumptions which limit the operation of the multiplier. They are as follows:

1) Induced investment remains absent while change takes place in autonomous investment.
2) The marginal propensity to consume is considered constant.
3) Consumption is a function of present income.
4) Time lags are absent in multiplier process. Therefore, any change in investment instantaneously leads to a multiple times change in income.
5) Until the multiplier process gets completed, the new level of investment is maintained steady.
6) It is considered that there is net increase in investment.
7) Consumer goods are available corresponding to effective demand for them.

8) Consumer goods industries have surplus capacity to meet the increased demand for consumer goods in response to a rise in income following increased investment.
9) Other factors of production are conveniently available within the economy.
10) The multiplier process operates in an economy which is an industrialised economy.
11) The economy is closed and thus is devoid of any foreign influences.
12) Prices of products are constant.
13) The effect of consumption on investment due to accelerator is ignored.
14) The economy is not a full employment economy, i.e., has less than full employment level.

5.3.3. Average Propensity to Consume (APC)

At a given level of income in the economy, the ratio of total consumption to total income gives the Average Propensity to Consume (APC). It is the ratio of consumption expenditure to any particular level of income:

$$APC = \frac{C}{Y}$$

If income (Y), is ₹50 crore and consumption (C) is ₹40 crore, then

$$APC = \frac{C}{Y} = \frac{40}{50} = 0.8 \text{ or } 80\%$$

This implies that consumption expenditure forms 80 per cent of the income in the economy.

5.3.4. Marginal Propensity to Consume (MPC)

Marginal propensity to consume gives the relationship between change in consumption (ΔC) and change in income (ΔY). It represents the ratio of the two.

$$\text{Marginal Propensity to Consume} = \frac{\text{Change in Consumption}}{\text{Change in Income}}$$

or, $$MPC = \frac{\Delta C}{\Delta Y}$$

If income increases from ₹100 crore to ₹120 crore and consumption increases from ₹80 crore to ₹90 crore, it means that change in income by ₹20 crore has caused a change in consumption by ₹10 crore.

$$MPC = \frac{\Delta C}{\Delta Y} = \frac{10}{20} = 0.5$$

5.3.5. Diagrammatic Representation of Multiplier

Equilibrium between aggregate demand and aggregate supply determines the level of national income. This means that the national income level is fixed at the level where C + I curve intersects the 45° income curve. With this curve, multiplier can be explained. It is explained in **figure 5.6**.

In this **figure 5.6**, marginal propensity to consume is represented by C. Marginal propensity to consume has been here assumed to be equal to $^1/_2$ i.e., 0.5.

As shown in figure, the slope of MPC (marginal propensity to consume) curve has been taken to be 0.5 C + I. This curve represents the aggregate demand. The aggregate demand curve C + I which intersects the 45° line at point E so that the level of income equal to OY_1 is determined. We can find-out how much increment in income will occur as a result of EH

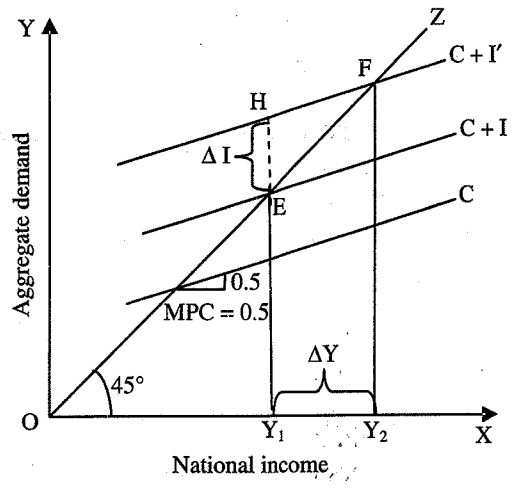


Figure 5.6: Keynes's Income Multiplier

amount increment in investment. Because of this increase in investment by EH amount, the aggregate demand curve shifts upward to the new position C + I′. This new aggregate demand curve C + I′ intersects the 45° income line at point F so that the equilibrium level of income increases to OY_2. So, increment in income is Y_1Y_2. Figure clearly shows that Y_1Y_2 is greater than EH. On measuring, it is found that Y_1Y_2 is twice the length of EH. So, the marginal propensity to consume is here equal to ½ and therefore the size of multiplier will be equal to 2.

5.3.6. Importance of Multiplier

The concept of multiplier is one of the important contributions of Keynes's to the income and employment theory. Its importance lies in the following:

1) **Investment:** The effect of investment in income and employment theory is highlighted by the concept of multiplier. Due to multiplier process, a decline in income and employment takes place whenever investment falls and vice-versa. Since the consumption function is stable during the short-run fluctuations in income and employment are due to fluctuations in the rate of investment. In this way, multiplier explains the process of income propagation and importance of investment.

2) **Trade Cycle:** Whenever there are fluctuations in the level of income and employment due to variations in the rate of investment, the multiplier process brings out variations in different phases of the trade cycle. There is a observed cumulative decline in income and employment when there is a fall in investment. This leads to recession and ultimately to depression. Opposed to this, an enhanced investment causes revival, and ultimately to a boom. Hence, multiplier is among the most significant tools used for the analysis of trade cycles.

3) **Saving-Investment Equality:** Multiplier also brings equality between saving and investment. A divergence between saving and investment leads to a rise in income via multiplier process if there is an increase in investment. This increment in income is more than the increase in initial investment. Therefore, saving also increases and equals investment.

4) **Formulation of Economic Policies:** The multiplier helps modern states in formulating economic policies. Thus, this principle assumes state intervention in economic affairs. It consists the following:
   i) **To Achieve Full Employment:** To remove unemployment and achieve full employment, state takes decision regarding the amount of investment to be injected into the economy. Income and employment gets increased by multiplier times the initial increase in investment. If one time investment is not sufficient to bring full employment, then regular doses of investments can be injected by state till the full employment level is reached.

   ii) **Deficit Financing:** Importance of deficit financing is also highlighted by the multiplier. In a state of depression, traditional monetary policy tool of lowering the rate of interest is not helpful because low rate of interest fails to encourage private investment as the marginal efficiency of capital is quiet low. In this scenario, increased public expenditure through public investment programmes by creating a budget deficit is helpful. It increases income and employment by multiplier time the increase in investment.

   iii) **Public Investment:** Public investment helps in creating or controlling income and employment via multiplier process. If public investment is increased during the investment, it can have greatest multiplier effect on income and employment. This is because during a depression the MPC is high (or the MPS is low). On the contrary, in periods of overfull employment, income and employment would be seriously affected due to decline in investment where the MPS is high (or MPC is low).

   The most appropriate policy would be to reduce investment where the MPC is low (or MPS is high), to have gradual decline in income and employment. The crucial thing is the timing of investment which has to be made in such a manner that the multiplier is able to work with full force and there is little scope for the income stream to peter out. Moreover, public investment should not replace the private investment but supplement it so that it could be enhanced during depression and reduced during inflation. Thus, multiplier's forward and backward operations will help in two different situations.

iv) **To Control Trade Cycles:** The expansion, boom, slowdown and depression in a trade cycle can be controlled by state on the basis of the multiplier effect on income and employment. When the economy is experiencing inflationary pressures, the state can control them by a reduction in investment which leads to a cumulative decline in income and employment via multiplier process. Besides, an increase in investment in a deflationary situation often helps an increase the level of income and employment. This becomes possible because of multiplier process.

5.3.7. Criticisms of Multiplier

The post-Keynesian economists have severely criticised the multiplier theory on the following grounds:

1) **Tautological: Prof. Haberler** has criticised Keynes' multiplier as – tautological. This implies that it may be a redundant concept. It is a truism which defines the multiplier as necessarily true as:

$K = 1/1 - (\Delta C/\Delta Y)$

According to Professor Hansen, "Such a coefficient is a mere arithmetic multiplier (i.e., a truism) and not a true behaviour multiplier based on a behaviour pattern which establishes a verifiable relation between consumption and income. A mere arithmetic multiplier, $1/1 - (\Delta C/\Delta Y)$ is tautological".

2) **Mechanical Relationship: According to Hazlitt,** "The Keynesian multiplier is a strange concept about which some Keynesians make more fuss than about anything else in the Keynesian system". It is a myth for there can never by any precise, predictable or mechanical relationship between investment and income. Thus, he regards it is a needless theoretical toy, making the study of macroeconomics unnecessarily complex.

3) **Effect of Acceleration is Ignored:** The concept of multiplier has also been criticised by many scholars on the ground that it does not take into consideration the change in investment as a result of change in consumption. The fact is that multiplier is influenced not only by investment but also by consumption expenditure. As per this, even if no investment is undertaken and consumption expenditure is increased, then also the multiplier effect will continue working.

4) **Equality between Saving and Investment:** Since **Keynes** is of the opinion, that ex-post (realised) saving and ex-post (realised) investment are always equal to each other, his concept of multiplier contradicts with the concept of natural equality between ex-post saving and ex-post investment. The basis of concept of multiplier is the fact that a part of the total income which is not spent on consumption is hoarded. Income is not fully invested. However, this is not valid as per the concept of equality between saving and investment because a part of income which is not spent is invested and as such realised saving is equal to realised investment.

5) **Relation between Income and Consumption:** A linear relation between consumption and income is established by Keynes's multiplier theory with the assumption that the value of MPC lies between 0 and 1. However, practically the behaviour of consumption in relation to income is complicated and non-linear.

According to Gardner Ackley, "The relationship does not run simply from current income to current consumption, but rather involves some complex average of past and expected income and consumption. There are other factors than income to consider".

6) **Perfectly Elastic Supply of Consumer Goods:** The assumption behind the concept of multiplier is that the supply of consumer goods is perfectly elastic. However, **Keynes** himself has refuted this possibility in his book. He has taken into consideration such situations in which production process/supply process is hurdled due to some constraints in the factors of production like mobility constraints. These hurdles are observed even when the factors of production are unemployed and are willing to work.

5.4. ACCELERATOR

5.4.1. Concept of Accelerator

Accelerator is an economic term being acquired from mathematics. In mathematics, 'acceleration' is defined as **'rate of change in velocity'**. The principle of acceleration in economics states that evaluation of net investment is based on the rate of change in the final output. This is also known as the principle of induced investment.

For example, let us assume that "X" has a carpet manufacturing unit and he is utilising full production capacity of his unit. Suppose, the demand for carpet is increased by two carpets per day, there will be an additional investment on new looms by 'X'. If the price of one carpet is ₹10, then the price of two carpets will be ₹20. Assuming the price of one loom is ₹100 that is a weaving capacity of two carpets in one day; therefore to meet the production of ₹20, further investment of ₹100 is required.

The investment-output ratio will be 100:20 or 5:1. Hence, with investing capital worth five rupees we can produce an output worth rupee one. This ratio is termed as accelerator or capital output ratio. Thus, accelerator is the rate of change in capital ΔK with respect to change in income or output ΔY. In economics, accelerator is technically an investment made for the sake of increasing the production which has to be many times more than the value of the production.

Principle of acceleration analyses the change in net investment that is the result of change in income, because accelerator is the rate of change in basic change in income. Thus;

$$W=\frac{\Delta K}{\Delta Y}$$

Where, W = Accelerator
ΔK = Change in capital
ΔY = Change in income

The principle of acceleration states that the level of income or output is solely responsible for the need for capital stock in the society. For the continuation of the induced investments, it is mandatory that the consumption should increase endlessly. Thus, the principle of acceleration is said to be the principle of induced investment. According to this principle, it requires to keep in mind that the net investment does not depend on the absolute amount of final output. Net investment depends on the rate of change in final output. Hence, the change in the rate of output results in the change in the amount of net investment.

5.4.2. Assumptions of Acceleration

Possibly acceleration is based on the following assumptions:

1) **No Excess Capacity in Consumer Goods Industries:** In the economy all the firms are producing with full capacity utilisation and are potentially sound to fulfil the current demand only. If there exists any fresh demand that cannot be met out with the available capacity of production as there is no extra production capacity. Therefore, net investment is needed.
2) **Fixed Accelerator of Fixed Capital-Output Ratio:** Due to technical reasons, the assumption on which the acceleration principle is based is that the accelerator or capital output ratio is always fixed.
3) **Permanent Increase in Demand:** Principle of acceleration is based on the assumption that there should be a fixed increase in demand. Based on this assumption the firms will then tend towards new investment in order to get their capital stock increased.
4) **Firm will Increase Capacity:** To counter additional rise in demand the firm will continue to increase its capacity of production as and when required.
5) **Capital Equipment is Indivisible:** Investment in capital equipment is to be executed only once. Suppose a machine has to operate for a period of five years, it has to be installed in the beginning of the plan period. The investment pertaining to the installation of the machinery cannot be covered for five years.
6) **Surplus Capacity in Producer's Goods Industry:** Acceleration principle assumes that due to excess in capacity of producer's goods industry, any new demand of machinery can be immediately fulfilled.
7) **Availability of Credit:** Availability of credit for the firms to increase their production capacity.
8) **Law of Constant Returns:** Working of the capital goods industries is been covered by the law of constant returns.
9) **Elastic Supply of Resources:** There should be elasticity in the supply of factors of production; then only production of capital goods will be increased by the capital goods industries as per the requirement.

5.4.3. Explanation of Acceleration

The following formula and schedule helps to explain the principle of acceleration:

'Principle of acceleration' clearly states that the increase in the level of production is directly proportional to the increase in capital goods. Accelerator or capital-output ratio is responsible for the rise in capital goods. Thus, to enhance the production worth one rupee, capital value ₹3 is required, the capital-output ratio or accelerator will be 3:1 or 3. This states that, if there is a requirement of additional capital production of value ₹10 then the amount of capital required will be equivalent to 3 × 10 = ₹30.

Acceleration is calculated with the help of the following formula:

$$W = \frac{\Delta K}{\Delta Y} \qquad (\Delta K = I)$$

or, $$W = \frac{I}{\Delta Y} (\Delta Y = Y_t - Y_{t-1})$$

Where, W = Accelerator, ΔK = Change in capital
I = Net investment, ΔY = Change in income
Y_t = Income in 't' time-period, Y_{t-1} = Income in 't – 1' time-period

Briefly, it has been proved by the principle of acceleration that the 'net investment is affected by acceleration on the basis of the rate of change of income or output' but, not on the complete change. The rate of change in income remains fixed until net investment remains fixed. Thus, with this rate there will be an increase and decrease in the investment.

5.4.4. Importance of the Acceleration

The following facts clearly state the importance of acceleration principle:

1) **Explanation of Violent Fluctuations in Investment:** Net investment depending upon the rate of change of production of goods is responsible for greater fluctuation in the production of capital equipments or net investment in the production of goods that can be well explained by acceleration principle. With the help of the principle of acceleration, it can be explained why the production of capital equipments or net investment in the production of goods fluctuates. This is because there is a dependence of net investment on the rate of change of production of goods.

2) **Explanation of Free Market Economy:** Every free market economy is characterised by short-term cyclical fluctuation that has been proved by the combination of the principle of acceleration and the principle of multiplier. These cyclical fluctuations have been explained by **Prof. Samuelson** in different ways by the reciprocal action of acceleration and multiplier.

3) **Trade Cycle:** The famous theory of trade cycle as discussed by **Prof. Hicks,** which is based on the principle of acceleration, is an important part of the exposition of trade cycle. **According to Robertson,** "It deserves a place of pride in any analysis of trade cycle". It presents scientific description of trade cycles.

4) **Basis of the Theory of Growth:** The theory of economic development is immensely signified by the principle of acceleration. In this theory of economic development, **Prof. Harrod** has made broad utilisation of acceleration. This theory depicts that in order to attain stable rate of growth, the rate of investment of an economy must be equal to the multiple of accelerator and multiplier.

5.4.5. Limitations of the Acceleration

Principle of acceleration has been mainly criticised by **Knox, Hutt, Tinbergen,** etc., on the following numbers:

1) **Unrealistic Assumption of Constant Capital-Output Ratio:** The principle of acceleration depends on the assumption of constant capital-output ratio. But in reality, it is never remain constant. It changes in the long-run due to technological variations. Whereas, in the short-run despite there is no change in technological factors, there is a possibility of change in capital-output ratio. This is due to the reason that sometime demand goes through changes for those goods having more capital-output ratio contrary to some other time when demand for these goods may increase having less capital-output ratio. Hence this assumption is not realistic.

2) **Impractical: According to Tinbergen** the Principle of acceleration is an impractical concept. It is based on this assumption that the economy lacks excess capacity. But in real life, majority of the firms operate at less production capacity. Therefore, as demand increases investment is not necessarily increased by these firms. Those firms which are operating at full production capacity, the production is enhanced with the existing capacity which is according to the increase in demand. When it is ascertained that the demand has increased constantly, the net investment is undertaken.

3) **Inaccurate:** The principle of acceleration being inaccurate states that a producer will undertake only the required net investment which is necessary for capital-output ratio in response to increase in demand. In reality, it has been observed that the producer purchase capital goods in bulk that exceeds their existing requirement to meet further increase in demand. This proves to be more economical. Therefore, on one side, net investment is more than accelerator and, on the other side, due to excessive capacity no further increase in investment despite of increase in demand. Thus, the principle of acceleration is not applicable.

4) **Inelastic Supply of Factors:** The impractical assumption of the principle of acceleration is that in capital goods industry, the supply of factors is elastic, therefore when there is an increase in their demand, consequently the supply also increases. In reality, for the production of capital goods supply of many essential factors is inelastic in nature. Hence, sometimes there is long delay between demand for and supply of capital goods.

5) **No Explanation of Actual Behaviour of Investors:** According to the principal of acceleration with increase in demand, investors decide mechanically to increase their investment. This not being a suitable explanation of the real behaviour of the investors. Factually, these investors do take investment decisions after proper considerations and consultations. According to the principle of acceleration, an entrepreneur is treated like a thermostat to examine when the production capacity has been over utilised and to balance it by attempting additional investment. This principle fails to explain how the correct decisions regarding increase in net investment in response to increase in demand, is taken by the entrepreneur class.

6) **Simple Truism:** Simple truism is being viewed by many economists as the principle of acceleration in the short-run. It demonstrates that as demand increases the investment will also increase. Though this fact is evident to all but, it is unable to analyse the economic problems. It is possible that this principle is applicable in the long-run economic analysis as it states the changes in investment depending upon the rate of change in consumption, not on the level of consumption because accelerator is not constant in the long-run, this principal is not feasible.

7) **Limited Use:** Due to its limited utility the principle of acceleration is applicable only in the increase in demand but not when there is a decrease in demand. Hence, this principle is unable to explain the lower turning point of the trade cycle.

5.5. BUSINESS CYCLE

5.5.1. Meaning and Definition of Business Cycle

'Business cycle' or 'trade cycle' is a term that describes economic activity variations occurring in an almost regular time series in all the capitalist societies. Various indicators like, volume of employment, price-level, output and income demonstrates the amount of economic activity in a community. The graph portrays like a wave when these indicators are plotted on it. Thus, it is a regular phenomenon of rising and falling of the economic activity, where each and every movement of rise and fall taken collectively describes a **'Trade Cycle' or a 'Business Cycle'.**

According to Keynes, "A trade cycle is composed of periods of good trade characterise by rising prices and low unemployment percentages altering with periods of bad trait characterised by falling prices and high unemployment percentages."

According to W.C. Mitchell, "Business cycles are a species of fluctuations in the economic activities of organised communities. The objective 'business' restricts the concept to fluctuations in activities which are systematically conducted on a commercial basis. The noun 'cycles' bars out fluctuations which do not occur with a measure of regularity."

Generally, business cycles are defined as alternate periods of prosperity and depression of good and bad trade varying from 3-4 years.

5.5.2. Features of Business Cycle

The essence and features of business cycle are as follows:

1) **Recurring Fluctuations:** Business cycles are represented by variations which occur regularly in an unhindered pattern. This states that the repetition of expansion and contraction does not have a uniform or regular period.

2) **Period of Business Cycle is no Longer than a Year:** The period of completion of a particular business cycle is around 3 to 4 years. But, in some cases the duration of the business cycle is either shorter or longer than that of the duration of a normal business cycle. Those causes that are associated directly or indirectly with the physical season and are due to seasonal fluctuations in economic activity taking place within the period of one calendar year makes it distinctly different. Usually, the period of business cycle is not less than one year.

3) **Presence of the Alternating Forces of Expansion and Contraction:** The interchangeable forces of prosperity and depression in an economy that are in-built in the system represent a business cycle. The force of expansion phase thrust the economy to a high level of activity over a period of time. Initially the force is of low intensity but finally regained by a counter force which results in contraction, ending up the process with depression.

4) **Phenomenon of the Crisis: According to Keynes** business cycle is characterised by the event of crisis. It means that the peak and trough are uneven as the expansion phase of the business cycle ends up suddenly and depression slows down gradually.

5) **Synchronic:** Existence of business cycle is contemporary. They have an enfolding attribute of causing any changes in an individual industry or sector. **For example,** there is a parallel occurrence of depression or contraction in all the industries or sectors of the economy. Recession due to the chain reaction overcomes the whole economy from one industry to another.

5.5.3. Types of Business Cycle

Business cycles have been classified under the following heads:

1) **Major and Minor Cycles:** Major cycles are the variations of business activity that occurs between successive 'break-downs' or crises of failures. The term, 'crisis' refers to an obstruction from time-to-time that is a character of the economic activity. Therefore, major cycles are the intervals that emerge between larger break-downs in economic activity or between major recessions. Another term for major cycle is 'Juglar Cycles' coined after a French economist, Clement Juglar during 19th century. On the basis of his findings he established the cyclical character of business fluctuations.

   Facts arising from the variations in business states that each major cycle comprises of two or three minor cycles. In business, upswings in the major cycles are frequently interrupted by minor downswings. Similarly, the downswings in major cycles are obstructed by minor downswings. Minor cycles are referred to 'shorter cycles' in major cycles. The duration of minor cycles on an average is approximately forty months.

2) **Building Cycles:** The cycles of building construction refers to building cycles. Time span of building cycles are longer than that of business cycles. In a building industry fluctuations affect in regular durations. These are upswings and downswings, where the duration of the building cycles varies from fifteen to twenty years.

3) **Kondratieff Cycles:** The span of these cycles is of 50 or 60 years and is referred as 'long waves'. The term long waves in economic activity was discovered by Russian economist, **'Kondratieff'**. Thus, these long waves are termed as Kondratieff cycles. During the period 1780-1920, Kondratieff on the basis of the statistical data could establish 21/2 cycles in England and France. Each full cycle was of the duration of 60 years.

5.5.4. Phases of Business Cycle

The distinct and closely related phases of business cycle are as follows:

1) **Prosperity or Expansion:** In this phase, there exist an optimistic feeling between businessmen and industrialists that is represented by enhanced production, high capital investment in fundamental industries. Bank credit expansion, high prices and high profits. Boom exists more than the normal expansion of

aggregate activities in an economy. The level of economic activity rises which is not being the nature of 'X' increases at cost of 'Y'; the whole system escalates. The entrepreneurs are motivated in general to expand their activities and invest more resources in the process of production.

The whole process being initiated by these stimulants is named as 'starters' or 'originating forces'. When this process gets impetus, economic variations like output, employment, investment, consumption, profits and prices soar collectively.

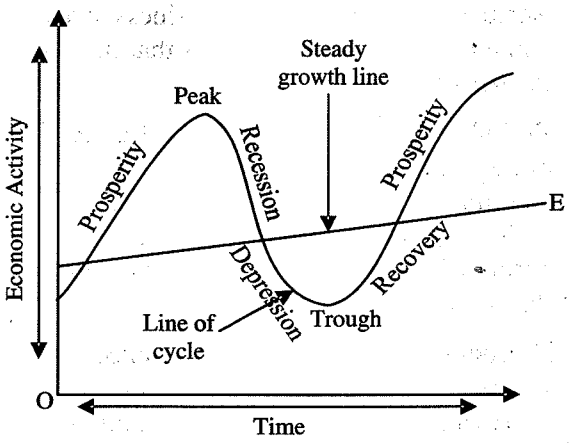


Figure 5.7: Phases of Business Cycle

2) **Boom or Peak:** Prosperity attains the highest level and then the recession of the economic activity from the peak phase, that is generally represented by retardation in the rate of expansion resulting in slowing of the growth and then reaching the peak.

3) **Recession:** The beginning of recession means that boom is over signifying liquidation in stock markets, burden in the banking system, clearance of some bank loans, fall in prices, acute decline in the demand for capital equipment and abandonment of new projects by the norms characterised by recession. Full unemployment in income, expenditure, prices and profits are the overall effects of recession.

As these signals are observed by entrepreneurs they become suspicious about the future, that results in the fall of psychological factors along with real economic factors pertaining to excess investment, reduced liquidity, escalated costs leading to steep fall in prices indicates the period of recession. The after effects of this phase are the prevalence of negative sentiments, reduction in fresh investments, debt recovery by banks, stocks liquidation and fall in prices. The spill-over effect is on the consumers who defer their purchases causing slump in the economy. Ultimately recession gains momentum resulting into depression altogether.

4) **Depression:** Depression is the protective period where business activities in the country are below normal. Sharp decline in production, mass employment, low employment, declining prices, reduction in profits and wages, credit reduction high rate of business failures, creating an atmosphere of total negativity and hopelessness. Deadlock in the construction activities. However, the consumer goods industry is not much affected during depression.

5) **Trough:** The period of acute stress and despair in the economy describes Trough. This phase of economy experiences downward or downturn shift as the economic activities are retarded and ultimately come to a halt. Then once again these activities record an upward swing in the economy.

6) **Recovery:** When the lowest point of depression has attained the gradual increase in business activity is restored. Where the entrepreneur feels positive for the existing economic situation resulting in improvement in business activities. Every economic activity like industrial production, volume of employment gathers pace. Prices and profits raise slowly, wages increase, banks expand their credit base as well as starting of new investment in capital goods industry. Pessimistic environment is succeeded by an atmosphere of hope.

5.5.5. Causes of Business Cycle

The main causes of Trade/ Business cycle are:

1) **Banking Operations:** Role of banking operations is very important in its contribution to the cyclical phenomenon. These operations pertain to expansion and contraction of credit, change in discount rates, and variations in the ratio of deposits to cash reserves. The banks are capable to change the volume of money supply in the economy.

2) **Capital Goods and Consumer Goods:** The cause of business cycles is due to the proportional changes between the production of capital goods and consumer goods in the economy that leads to deficit and excess in supply of commodity in the short run.

3) **Purchasing Power:** Market maladjustments resulting in cyclical fluctuations are due to disagreement of the purchasing power with the expansion and contraction of production.

4) **Profit Mania:** Profit cravings of producer are another notable cause of the business cycles. Being very optimistic, he is under the delusion of the volume and accurate description of demand. If the retail trade is fast then the result is that the tendency of the producer intensified by significantly increasing the volume of production and resulting into a small boom in the raw materials and labour markets. If there is a decrease in the retail trade, the producer becomes suspicious and immediately resorts to output reduction and termination of some orders for raw materials, plant, etc., thus the processes of rise and fall in prices is intensified.

5) **Human Psychology:** The human psychology is another contributor to the occurrence of business cycles. Likewise human psychology is to withstand frequent changes in a cyclical manner – from excitement to despair. 'Optimism and pessimism are giving birth to one another in an endless chain. At the level of boom psychological turn at the peak into the opposite direction take place. At the bottom of the depression there occurs a turn in the opposite direction. There is no generalised explanation of these psychological changes.

6) **Cyclical Changes in Weather:** Weather changes also contribute to the evolution of trade cycle. Agricultural production and the prices of those basic goods being consumed by the working class in a society are affected by changes in weather conditions. It further affects the wage rate, cost of raw material, etc., that contributes to the fluctuations in the economic activity.

5.5.6. Theories of Business Cycle

Some of the important theories of business cycle are as follows:

1) **Sunspot or Climatic Theory: William Herschel** and **Stanley Jevors** are associated with the oldest theory known as Sunspot theory or Climatic theory. They connected business cycles with the sunspot activity. This theory explains that climatic conditions are affected by the appearance of the spot on the sun affecting rainfall and agricultural crops. Depression is the result of crop failure. Conversely, it leads to prosperity when there is no spot on the sun.

   Hence, meteorological conditions that are closely linked, represented by cycles in sunspots and business activity through the changes in agricultural production. Due to regular variations in climatic conditions, the prosperity follows depression. According to this theory, business cycles are natural phenomenon but today, this theory is fully unaccepted. As trade cycle being a complex fact it cannot be associated with climatic conditions.

2) **Psychological Theory: Prof. Arthur. C. Pigou,** an English economist has stated the psychological theory of business cycle and has been affirmed by J.M Keynes that trade cycle is generated by psychological factors. **Keynes** specifies that excess of optimism and excess of pessimism among people is the cause for trade cycle. When entrepreneurs are optimistic during good trade conditions they would results in increased production thus, spreading this feeling to others. This results in enhanced production beyond limits leading to excessive production resulting in losses. When pessimism prevails among these entrepreneurs it decreases their investment and production.

   Finally, these variations due to optimism and pessimism cause depression. Having an element of truth in this theory it is unable to explain the occurrence of boom and its revival. Eventually, this theory unsuccessful to explain the revolution of trade cycle. Monetary factors that are reflected in the changes in interest rates are not considered.

3) **Hayek's Monetary Over-Investment Theory: Hayek** stated that full employment and state of equilibrium persist in the economy until and unless natural rate of interest equals market rate of interest. Inequality between market and natural rate of interest causes trade cycles in the economy. Thus, when market rate of interest is below natural rate of interest prosperity exists and vice-versa leads to depression in the economy. The rate at which supply of voluntary savings equals the demand for funds taken on loan is the natural rate of interest. Whereas, the demand and supply of money determines the market rate of interest being the money rate existing in the market.

According to this theory:

i) When the market rate of interest is below the natural rate of interest then prosperity begins. Thus, in this situation the demand for investment funds exceeds the supply of available saving.

ii) Due to the continuous rise in the prices of factors, increasing the cost of production and reducing the profits of producers of capital goods their investment declines under the anticipation of loss in the future, resulting in the fall in the rate of natural interest. Hence, a decline in the rate of natural interest.

iii) The course of revival in the economy begins, investments are determined, banks start increasing the supply of money reducing the market interest rate below the natural interest rate due to the fall in prices coming to an end during depression.

4) **Under Consumption Theory: J.A. Hobson** who being the chief exponent of this theory stated that the appearance of trade cycles is due to improper distribution of national income. National income is under-distributed due to boom, these businessmen and entrepreneurs accumulate income from business activities and grew richer as they cannot consume it wholly they intend to save. During the period of boom there is increased savings the level of consumption declines resulting in the fall in the demand of consumer goods due to no consumption. The supply of consumer goods will be in excess of the demand of these goods enhances the reduction of the prices of these goods creating a pessimistic outlook. Depression is the outcome of this continuous downward trend.

5) **Keyne's Saving and Investment Theory:** In "General theory of Employment" Keynes has given an explanation of business or trade cycle but never in detail. The main subject matter of his theory is the balanced fluctuations of business cycle in cumulative income, output and employment. He further states that business cycle appear due to fluctuations in the rate of investment which is due to Marginal Efficiency of Capital (MEC). **According to Keynes,** Marginal Efficiency of Capital (MEC) is defined as the relation between the expected yield of that type of capital and the cost incurred in producing the units. Hence, in simpler terms the expected rate of profit on new investment is defined as marginal efficiency of capital.

6) **Hick's Theory of Trade Cycle: Prof. Hicks** elaborates the reality of trade cycles by linking the principles of multiplier and acceleration. According to Hicks, investment is of two types:
   i) Autonomous investment, and
   ii) Induced investment.

   Autonomous investment is also called free investment which is independent of the variations in income, output and consumption, while **Induced investment** is driven by the fluctuations in income, output and consumption. Multiplier effect signifies the force of autonomous investment, whereas the force of induced investment is expressed in acceleration. Hence, **according to Hicks,** cyclical fluctuations in economic activity *via* multiplier and accelerator are caused by autonomous investment and induced investment respectively.

7) **Haw trey's Monetary Theory: Prof. R.G. Hawtrey** stated that trade cycle are truly a monetary aspect of economic activity. Prosperity and depression in the economy is the result of the changes in the demand of the monetary flow on the part of the businessmen. Partial depression, not general depression is due to non-monetary factors like strikes, floods, earthquakes, droughts, wars, etc. In reality, cyclical variations are the result of expansion and contraction of bank credit leading to fluctuations in the flow of monetary demand of producers and traders. Presently, the principal means of payment is credit floated by banks. This credit is expanded or contracted either by lowering or raising the rate of interest, or by purchasing or selling securities to traders. Thus, prosperity of depression is brought by increasing or decreasing the flow of money in the economy.

8) **Innovations Theory:** An innovative theory of business cycle was propounded by **Joseph Schumpeter,** an impressive treatise on 'Business Cycle' in 1939. Schumpeter considered business cycle as a unit of the dynamic process of growth and development which made his approach distinct from other approaches. He further states that no business cycle can exist in a static state. His analysis is based on the assumption that an economy is in a state of full employment equilibrium. This point is considered to be the point of reference that can measure all the changes in economic activity. This innovative

theory of Schumpeter is to affect disturbance in equilibrium causing entrepreneurs to expand their level and rate of activity. This results in the process of aggregate demand for productive machinery as technological changes create increased demand for capital. Thus, after some starting hesitations, majority of entrepreneurs gradually incline to adopt these changes. This tremendously enhances cumulative investment in the economy.

These innovations comprises of:

i) Introduction of a new product,
ii) Introduction of a new method,
iii) Introduction or opening of a new market,
iv) A new source of raw material, or
v) A new form of business organisation.

Schumpeter explains that modifications are not inventions. Inventions are scientific actions undertaken by a scientist inside a lab while innovations are done by an entrepreneur and are applied in a market. That is why, business cycle are caused by innovative changes, but not due to inventions.

5.5.7. Impact of Business Cycle

The impacts or effects of business cycle on business firm may sometimes be good or sometimes bad. Impacts of business cycle can be explained as follows:

1) **Effect during Expansion:** During the concluding phase of prosperity inputs start declining in comparison to their demand. There is shortage of additional workers that are available by offering higher wages in contrast to the existing wage rate. Labour market becomes seller's market. Thus, similar situation prevails in other input markets resulting into speedy rise in input prices thereby, increasing output and employment. Therefore, cost of living comparatively exceeds more than the income of the household.

   Whereupon, those wage earners and households depending on fixed income revise their expenditure on consumption. There is a thrust in the resistance of the consumers leading to decline or slowing of the actual demand. The foremost and distinct blow is on the demand for new houses, flats and apartments which is followed by the termination of the demand for cement, iron and steel, construction and labour. The after effect of this trend is visible in other durable goods industries of automobiles, refrigerators, furniture, etc.

   Inflation in its beginning stage is somewhat beneficial and establishes global expansion of business activity where some groups gain while others lose. Inflation is appreciated till the stage of full employment:

   i) It boosts production and investment and by that employment, income and demand increases.
   ii) Businessmen gain during inflation as rise in cost is not so fast with respect to price rise. This result in loss to the investing class as well as the loss for wage-earning class (fixed income) as rise in wages is disproportionate to rise in prices.
   iii) During inflation, debtors (borrowers) gain while creditors (lenders) lose.
   iv) Agriculturists gain.
   v) Since government being the largest borrower, it gains and the burden of debt declines.
   vi) Social effects of inflation include:
      a) Rich become still richer, poor become still poorer;
      b) People lose confidence in the currency;
      c) Sometimes inflation results in revolution;
      d) The political party in power will lose in election; and
      e) Speculation, hoarding, profiteering and corruption.

2) **Effect during Recession:** The following effects of recession are:
   i) Demand in recession is finally caused by a parallel and continuous decline in the income of wage and interest earners, as well as reduction in the demand for inputs; prices of inputs, wages, interest, etc. Again, producers lower their prices to abolish their inventories and also to fulfil their financial

commitments. Furthermore, consumers postpone their purchases in the hope of further decrease in price. This results in the growing disparity between demand and supply. When this process accelerates, it settles into permanent recession. Reduction in investments leading to decline in income and consumption. This process is just opposite to expansion.

Further deterioration in the demand for both consumer and capital goods is due to the reduction in production and employment, and slashing of investment. Despite a fall in wage rates, investment borrowings decrease, shrinking of bank credit, stock prices decreases and unemployment increases.

ii) Fast decline in the level of national income and expenditure. Regular reduction in the prices of consumer and capital goods. Jobs are lost by workers. Difficult to pay-off their debts by the debtors. Demand for bank credit backflows resulting in the escalation of bank's cash balances. No attraction and gain in investing in stocks.

5.5.8. Measures to Control Business Cycles

Economic development of an economy is negatively affected by periodic fluctuations in business and economic activities. Therefore, the government should take protective and corrective measures to maintain balance in economic system. Following two types of measures are chosen to control business cycles:

1) **Preventive Measures:** The aim of preventive measures is to avoid the happening of business cycles, which include all measures that are chosen during the period of expansion to regulate purchases, protection of assets and avoiding irresponsible credit expansion. The adoption of these measures is viable only if the managers of the firm have devoted knowledge:
   i) The existence of accurate connection between cyclical changes in common business activity and changes in the business of the given firm.
   ii) The quality of the cyclical price movements during the purchases and sales of the distinct business in respect to the general business cycle, which has been secured by a business manager who has initiated the foremost crucial step in chalking out a creative programmes that is adaptable to the business cycle.

   The various protective measures against the effects of business cycles are as follows:
   i) Preservation of assets, and avoidance of unnecessary increase in plant and equipment and in dividends during the expansion phase;
   ii) Too much inventories and purchase commitments outside the available financial resources and excessive sales promotion must be avoided resulting in the cancellation of orders;
   iii) Standard of credit must be adjustable and application of terms and conditions must be rigid during expansion and liberal during contraction;
   iv) Curtailing the dependence of agriculture on nature;
   v) Equilibrium between demand and supply;
   vi) Control on experimental activities and,
   vii) Nationalisation of basic industries.

2) **Corrective Measures:** The main corrective measures are:
   i) **Monetary Policy:** The control over the supply of money and the cost of credit in the economy refers to monetary policy. Monetary policy is the utilisation of various method of credit in the economy and various method of credit control by the central bank. Two types of credit control used by the central bank are- Quantitative measures and selective measures of credit control. The method through which total credit in the economy is controlled is known as quantitative measures.

   The main methods of quantitative measures are Bank rate, Open market operations, changes in Cash Reserve Ratio (CRR) and Statutory Liquidity Ratio (SLR). The motive of the selective measures of credit control is to control the availability of credit for specific purposes and business activities. The main measures of selective credit control are Rationing, direct action and moral persuasion.

ii) **Fiscal Policy:** Fiscal policy caters to the achievement of certain objectives by managing public revenue, expenditure, and public debt. Taxes are the principal source of revenue. Government's fiscal policy affects the business activities and encourages more investment. Therefore fiscal policy of the government is capable to control cyclical fluctuations by inducing suitable changes in taxation, public expenditure and public debt policies.

iii) **Direct Controls:** For fast and effective control of business cycle, government applies direct physical controls that includes licensing, rationing of scarce and essential commodities, control of prices and wages, export-import control, control over exchange, hoardings and black marketing, and control of Monopolistic and Restrictive Trade Practices (MRTP), etc.

5.6. BUSINESS DECISIONS

5.6.1. Introduction

Decision-making is an integral part of today's business management. Making a decision is one of the most difficult tasks faced by a professional manager. A manager has to take several decisions in the management of business. The life of a manager is filled with making decisions after decisions. Decision-making is a process and a decision is the product of such a process. Managerial decision-making serves as a link between traditional economics and the decision-making sciences for business decision-making.

5.6.2. Features of Business Decisions

The features of business decisions can be clarified as follows:

1) **Scientific Decisions:** These decisions are taken after systematic study and quantitative evaluation of the data. The present decisions are based on data from past performance and establish a course of action for the future. Although scientific decisions are time consuming yet these are accurate.

2) **Strategic Decisions:** These decisions are taken keeping in view the targets and objectives to be achieved in the long-period. The success of these decisions depends on effectiveness and efficient implementation of policies.

3) **Tactical Decisions:** These decisions are short-period decisions and are taken to support the strategic decisions. These are more flexible and can be easily changed according to the circumstances and needs.

4) **Decisions Under Certainty:** The business decisions may be taken under certainty which means a state of knowledge in which decision-maker knows in advance the specific outcome to which each alternative will invariably lead. In other words, the decision maker has perfect knowledge of the environment and the result of whatever decision he might make. In such a decision the conditions of a business problem are taken to be known with certainty. However, the real life is dynamic and only few things remain certain. Such decisions may be very short period decisions.

5) **Decisions Under Risks:** Generally in every business risk element is involved. Risk is defined as a state of knowledge in which each alternative leads to one of a set of specific outcomes. Each outcome accruing with a probability that is known to the decision maker.

   Risk may be regarded as the quantitative measurement of a decision such as a loss or a gain. In other words, under conditions of risks the decision-maker possesses some objective knowledge of the business and can deduce the probability of profit under different types of decisions. It may be pointed out that some of the risks can be insured and easily avoided.

6) **Decisions Under Uncertainty:** Since real world is very uncertain, most of the business decisions are undertaken in a state of uncertainty. It is a situation in which a set of possible results are expected from one or more alternatives. Their probabilities are either not known or are meaningless.

   Unlike risk, therefore, uncertainty is a subjective phenomenon. Uncertainty may relate to profit, demand, production, cost, pricing and investment. The presence of uncertainty in business is a great obstacle in decision-making.

5.6.3. Types of Business Decisions

The major kinds of economic decision are as follows:

1) **What to Produce:** The first major economic decision of any economy relates to the type and the range of goods to be produced. Since resources are limited, one must choose between different alternative combinations of goods and services that may be produced. Allocation of resources between the different types of goods, e.g., consumer goods and capital goods, is another major concern to any economy. At firm level, this decision would involve review of market demand and availability of raw materials and technology. This can also be referred to as the problem of choice.

2) **How to Produce:** Having decided on what to produce, the economy must determine the techniques of production to be used. This can also be viewed as the problem of efficiency; efficiency is maximised when the limited stock of resources yields the maximum possible volume of goods and services, or renders the maximum benefit to the society.

3) **For Whom to Produce:** This means how the national product should be distributed. This is essentially the problem of distribution. Once the goods are produced, they need to be distributed among the various economic agents. In a market economy such a distribution is done on the basis of "ability to pay" principle; this implies that those who have more in terms of wealth and income would have more of the commodities than those who have less. However, in a command economy such a distribution is done on the basis of "according to need" principle; this implies that people would be rewarded according to their needs and not their ability to pay.

5.6.4. Business Cycle and Business Decisions

The business decision of any organisation is greatly affected by its business cycle. The different strategic decisions are taken by the managers and entrepreneurs depending on the business cycle stage. A business never remains same, hence, in order to meet the future needs it must keep itself updated from time to time. Hence, different phase of the business cycle seeks demand of different actions from the organisation.

Thus, in case economy is going through its expansion phase then management must take such moves which will be suitable as per the need of time like organisation will increase its level of output. On the other hand, when business is facing hard time then excess of expenses can result in its exit from the market. Therefore, organisation must take decisions accordingly. Management may decide to shut its some of the weak units for either temporary basis or permanent basis. Hence, there are various stages in the life span of the business which seeks the management to take suitable decisions as per the need of the time.

The different phases of business cycle and its connection with the different business decision are stated below:

1) **Boom:** This stage represents the fast growth of economy. In this stage, there is high demand and there is an increase in the customer expenditure. Thus, the decision related to the increase in production has to be taken by the management. In order to accomplish this, sophisticated technology can be adopted by the management, use of manpower can be increased or different policy changes can be implemented.

2) **Downturn:** This is the stage where there is a decline in the growth. There will be a decrease in the consumer expenditure. The decisions about the level of production and the use of additional manpower have to be taken by the management.

3) **Recession:** This is the stage in which recession can be resulted from the ongoing decline in the economic activities. There will be decrease in demand and business will have a tough time to sustain. Thus, the organisation takes decisions related to the seizing of expansion by following the strategies such as rightsizing or downsizing, producing lesser number of products either for the time being or permanently have to be taken by the management.

4) **Recovery:** This is the stage where GDP begins to increase once again with the increases in customer expenditure. Thus, the decisions related to increased production of goods and services are to be taken by the management.

5.7. EXERCISE

5.7.1. Very Short Answer Type Questions

1) Explain the meaning of consumption function.
2) What is average propensity to consume?
3) What is multiplier?
4) Write the meaning of accelerator.
5) Name the types of business cycle.

5.7.2. Short Answer Type Questions

1) Highlight the determinants of consumption function.
2) What is investment function?
3) Write few importance of multiplier.
4) Explain business cycle and business decisions.

5.7.3. Long Answer Type Questions

1) What are the types of investments? Highlight the investment demand curve.
2) Explain the marginal efficiency of capital, net present value method and internal rate of return method.
3) Highlight the meaning of multiplier. Also, discuss the APC and MPC.
4) What are the importance and limitations of accelerators?
5) Explain business cycle. Highlight its phases and causes.
6) Discuss the theories of business cycle. Also, suggest measures to control business cycle.
7) What is business decisions? Also, mention its features and types.

SAVITRIBAI PHULE PUNE UNIVERSITY

MBA - FIRST SEMESTER EXAMINATION, 2013

ECONOMIC ANALYSIS FOR BUSINESS DECISION

Time: $2\frac{1}{2}$ Hours **Max. Marks: 50**

Note: 1) All questions are compulsory.
2) Each question has an internal option.
3) Each question carries 10 marks.
4) Figures to the right indicate marks for that question/sub-question.
5) Your answers should be specific and to the point.
6) Support your answers with suitable live examples.
7) Draw neat diagrams and illustrations supportive to your answer.
8) Use of calculators is permitted (as applicable).

Ques 1 a) Explain how profit or value maximisation objectives predict the behaviour of a firm more accurately than other objectives. **(05)**

Ans: According to traditional economic theory, profit maximisation is the sole objective of business firms. Profit maximisation means the largest absolute amount of money profits in given demand and supply conditions. Conventional price theory is based upon profit maximisation hypothesis. Profit maximisation hypothesis helps not only in predicting the behaviour of business firms but also the price-output behaviour under different market conditions. No other hypothesis can explain and forecast the behaviours of firms better than this hypothesis.

Refer Unit-1, Page No. 27

Ques 1 b) What might be the objective(s) of the following non-profit organisations: **(05)**
i) A Museume

Ans: Objectives of a Museum

1) **Social Responsibility of Business:** In real sense the assumption of social responsibilities implies recognition and understanding of the aspirations of society and determination to contribute to its achievement. Thus, museum depends on society for existence, sustenance and encouragement.
2) **Fund Raising:** A key economic goal of museums is to raise funds for fulfilling their charitable purposes. The process of fundraising goes beyond holding events or sending out mailings. A complete development effort includes creating a database of regular donors, applying for grants, seeking individual and corporate donations and holding events such as balls, auctions, raffles and sporting events. The cost of fundraising efforts can outweigh the total money spent on a museum's charitable purpose at new or smaller organizations.

Ques 1 b) ii) Critical Care Unit of a Hospital

Ans: Objectives of a Critical Care Unit of a Hospital

1) **Satisfactory Level of Profits:** Every management determines a 'satisfactory aspiration level' on the basis of its past experience and judgment about future uncertainty. If the 'aspiration level' is attained easily it is revised upwards for future operations, while if it remains unachieved or proves highly difficult to be achieved it is revised downwards. This, in case of Critical Care Unit of a Hospital; would be to maintain its operating costs, so that quality products, latest technology equipments and medicines can be used.
2) **Welfare:** The Critical Care Unit of a hospital has to keep welfare of different groups of people as its objective. First and foremost, the welfare of the workers and staffs has to be considered. They should be provided good working conditions, fair wages and other benefits to increase their involvement in the hospital. It is very important as it can improve efficiency and productivity of the staff because in a critical care unit the well behaved and well practiced staffs are required. So, if the hospital pay attention to their welfare, they woodwork more effectively.

Or

Ques 1 c) Explain how firms and individuals participate and interact in the product market and in factor market. (10)

Ans: Participation and Interaction between Firms and Individuals in Product Market
Refer Unit-1, Page No.39

Participation and Interaction between Firms and Individuals in Factor Market
Refer Unit-1, Page No.39

Ques 2 a) State various features of oligopoly and explain price rigidity with the help of kinked demand curve. **(10)**

Ans: Features of Oligopoly
Refer Unit-4, Page No. 162

Kinked Demand Curve
Refer Unit-4, Page No. 164

Or

Ques 2 b) Explain income elasticity of demand with suitable examples. **(05)**

Ans: Income Elasticity of Demand
Refer Unit-2, Page No. 75

Types of Income Elasticity of Demand
Refer Unit-2, Page No. 75

Ques 2 c) Explain market experiment method and regression technique of demand forecasting. **(05)**

Ans: Market Experiments/Econometric Method
Refer Unit-2, Page No.83

Regression Method
Refer Unit-2, Page No.83

Ques 3 a) How is marginal cost different from incremental cost? **(05)**

Ans: Out of Syllabus

Ques 3 b) How are sunk costs treated in business decision-making? **(05)**

Ans: Out of Syllabus

Or

Ques 3 c) In what areas can a firm use 'insurance' to reduce risk. **(04)**

Ans: Out of Syllabus

Ques 3 d) Define risk, uncertainty and expected value. **(06)**

Ans: Out of Syllabus

Ques 4 a) Capital is a '*sine qua non*' for economic development. Briefly summarise the importance of capital market in the process of economic development. **(10)**

Ans: Out of Syllabus

Or

Ques 4 b) What are the fiuctions of a Central Bank of India? How do they differ from the functions of other banks? **(10)**

Ans: Out of Syllabus

Ques 5 a) Discuss the WTO agreement embodying the results of the GATT Uruguay Round. **(10)**

Ans: Out of Syllabus

Or

Ques 5 b) What is the importance of national income estimates to a country? **(5)**

Ans: Out of Syllabus

Ques 5 c) Explain PPP in International trade. **(05)**

Ans: Out of Syllabus

SAVITRIBAI PHULE PUNE UNIVERSITY

MBA - FIRST SEMESTER EXAMINATION, 2014

ECONOMIC ANALYSIS FOR BUSINESS DECISIONS

Time: 2½ Hours **Max. Marks: 50**

Note: 1) All questions are compulsory.
2) Each question carries 10 marks.
3) Figures to the right indicate marks for that question/sub-question.
4) Your answers should be specific and to the point.
5) Draw neat diagrams and illustrations supportive to your answer.

Ques 1 a) What are the steps involved in a scientific approach of managerial economic analysis in solving business problems. **(10)**

Ans: Out of Syllabus

Or

Ques 1 b) Explain circular flow of economic activity in Four Sector Model. **(10)**

Ans: Out of Syllabus

Ques 2 a) State various features of oligopoly and explain the concept of 'Price-Rigidity' with the help of Kinked demand curve. **(10)**

Ans: Features of Oligopoly
Refer Unit-4, Page No. 162

Kinked Demand Curve
Refer Unit-4, Page No. 164

Or

Ques 2 b) What do you mean by the term 'Market' in Economics? Explain different types of market structure. **(10)**

Ans: Market
Refer Unit-1, Page No. 36

Different Market Structures
Refer Unit-4, Page No. 142

Ques 3 a) i) Explain the concept of Average Fixed Cost, Average Variable Cost, Marginal Cost and explain their relationship. **(07)**

Ans: Concept of Average Fixed Cost, Average Variable Cost, Marginal Cost/ Types of Cost
Refer Unit-3, Page No. 105

Interrelationship between Average Cost and Marginal Cost
Refer Unit-3, Page No. 117

Ques 3 a) ii) What is the economic significance of opportunity cost? **(03)**

Ans: Refer Unit-3, Page No. 106

Or

Ques 3 b) Explain Break-even Analysis with assumptions, advantages and disadvantages. **(10)**

Ans: Break-Even Analysis
Refer Unit-4, Page No. 174

Assumptions of Break-Even Analysis
Refer Unit-4, Page No. 174

Advantages/Significance of Break-Even Analysis
Refer Unit-4, Page No. 179

Disadvantages/Limitation of Break-Even Analysis
Refer Unit-4, Page No. 179

Ques 4 a) Explain the regulatory role of RBI with regard to currency, credit, balance of payment and monetory policy. **(10)**

Ans: Out of Syllabus

Or

Ques 4 b) Explain the role and functions of 'Capital Market'. **(10)**

Ans: Out of Syllabus

Ques 5 a) Explain the following terms that are part of Union Budget such as Revenue Account, Capital Account, Revenue Deficit, Fiscal Deficit, Plan and Non-plan expenditure. **(10)**

Ans: Out of Syllabus

Or

Ques 5 b) Explain the various methods of calculating GDP (Gross Domestic Product) in India. **(10)**

Ans: Out of Syllabus

SAVITRIBAI PHULE PUNE UNIVERSITY

MBA - FIRST SEMESTER EXAMINATION, OCTOBER - 2015

ECONOMIC ANALYSIS FOR BUSINESS DECISIONS

Time: 2½ Hours **Max. Marks: 50**

Note: 1) Answer Five questions.
2) All questions carry equal marks.

Ques l a) Define 'business economics'. Explain its role to take business decisions.

Ans: Business Economics/Managerial Economics
Refer Unit-1, Page No. 16

Role of Business Economics in Decision-Making/Significance of Managerial Economics in Decision-Making
Refer Unit-1, Page No. 23

Or

Ques l b) Explain the contribution of Adam Smith in business economics with his invisible hand theory.

Ans: Out of Syllabus

Ques 2 a) What is price elasticity of demand? Explain various kinds of price elasticity with suitable examples.

Ans: Price Elasticity of Demand
Refer Unit-2, Page No. 71

Types of Price Elasticity/ Types/Degrees of Price Elasticity
Refer Unit-2, Page No. 71

Or

Ques 2 b) What is the concept of 'Oligopoly'? Explain the concept of price rigidity with the help of kinked demand curve.

Ans: Oligopoly
Refer Unit-4, Page No. 161

Price Rigidity/ Kinked Demand Curve
Refer Unit-4, Page No. 164

Ques 3 a) What is the concept of cost volume profit analysis? Discuss its merits and demerits.

Ans: Out of Syllabus

Or

Ques 3 b) Write short notes on (any Two):
i) Decision Tree Analysis

Ans: Out of Syllabus

ii) Hedging

Ans: Out of Syllabus

iii) Opportunity Cost

Ans: Opportunity Cost
Refer Unit-3, Page No. 106

iv) Break-even Point Analysis

Ans: Break-even Point Analysis
Refer Unit-4, Page No. 174

Ques 4 a) What is the concept of 'capital market'? Explain its role and functions in Economics.

Ans: Out of Syllabus

Or

Ques 4 b) What is RBI? Explain its role and functions in Indian economy.

Ans: Out of Syllabus

Ques 5 a) What is the concept of foreign trade? Explain the role of WTO in foreign trade.

Ans: Foreign Trade
Ans: Out of Syllabus

Or

Ques 5 b) Differentiate between the following terms (any Two):
i) Capital account and Revenue account

Ans: Out of Syllabus

Ques 5 b) ii) Planned expenditure and Non-planned expenditure

Ans: Out of Syllabus

Ques 5 b) iii) Revenue deficit and Fiscal deficit

Ans: Out of Syllabus

SAVITRIBAI PHULE PUNE UNIVERSITY

MBA - FIRST SEMESTER EXAMINATION - 2016

ECONOMIC ANALYSIS FOR BUSINESS DECISIONS

Time: $2^1/_2$ Hours **Max. Marks: 50**

Note: 1) Solve any one from each question.
2) All questions are compulsory.
3) All questions carry equal marks,

Ques 1 a) Explain how firms and individuals participate and interact in the product market and in the factor market.

Ans: Way in which Firms and Individuals Participate and Interact in the Product Market and Factor Market
Refer Unit-1, Page No.38

Or

Ques 1 b) What role does profit plays in Market System?

Ans: Role of Profit
Refer Unit-4, Page No. 173

Ques 2 a) Explain the concept and various determinants of market demand.

Ans: Concept of Market Demand/Concept of Demand
Refer Unit-2, Page No. 61

Determinants of Market Demand/Demand
Refer Unit-2, Page No. 62

Or

Ques 2 b) Write a detailed note on – Price output decisions in multi-plant firms.

Ans: Out of Syllabus

Ques 3 a) Elaborate the meaning and various types of cost in detail.

Ans: Meaning of Cost
Refer Unit-3, Page No. 105

Type of Cost
Refer Unit-3, Page No.105

Or

Ques 3 b) Discuss meaning of risk. Explain the decision-making under risk in detail.

Ans: Out of Syllabus

Ques 4 a) Explain composition and functions of money market in India.

Ans: Out of Syllabus

Or

Ques 4 b) Discuss role of Securities and Exchange Board of India (SEBI) in monitoring and regulating capital market in India.

Ans: Out of Syllabus

Ques 5 a) Write note on:
i) Difference between WTO and GATT

Ans: Out of Syllabus

Ques 5 a) ii) GDP and PPP

Ans: Out of Syllabus

Or

Ques 5 b) Define following terms in relation with Union Budget:

i) Revenue account

Ans: Out of Syllabus

Ques 5 b) ii) Capital account

Ans: Out of Syllabus

Ques 5 b) iii) Revenue deficit

Ans: Out of Syllabus

Ques 5 b) iv) Capital deficit

Ans: Out of Syllabus

Ques 5 b) v) Plan and non-plan expenditure

Ans: Out of Syllabus

SAVITRIBAI PHULE PUNE UNIVERSITY

MBA - FIRST SEMESTER EXAMINATION - 2017

ECONOMIC ANALYSIS FOR BUSINESS DECISIONS

Time: 2¼ Hours **Max. Mark: 50**

Note: 1) Solve any one from each question.
2) All questions are compulsory.
3) All questions carry equal marks.

Ques 1 a) What do you understand by "Basic Economic Problem"? Why it is necessary to understands the Basic Economic Problem" in the context of Managerial Decision-Making? (10)

Ans: Out of Syllabus

Or

Ques 1 b) What did Adam Smith mean by The Invisible Hand? Explain with suitable example. (10)

Ans: Out of Syllabus

Ques 2 a) Explain the concept of "Elasticity of Demand". Differentiate between Price and Cross Elasticity of Demand. (10)

Ans: Concept/Meaning and Definition of Elasticity of Demand
Refer Unit-2, Page No. 70

Difference between Price and Cross Elasticity of Demand
Refer Unit-2, Page No. 76

Or

Ques 2 b) What do you understand by the term "Market Equilibrium"? Expound the equilibrium of Monopoly firm under different cost situations. (10)

Ans: Market Equilibrium
Refer Unit-3, Page No. 94

Equilibrium of Monopoly Firm under Different Cost Situations/Equilibrium of Monopoly Firm
Refer Unit-4, Page No. 151

Ques 3 a) "Describe "Cost Volume Profit Analysis". What are the advantages and disadvantages of Cost Volume Profit Analysis? (10)

Ans: Out of Syllabus

Or

Ques 3 b) Why does an exporter face a foreign exchange risk? How can the exporter hedge tis foreign exchange risk? (10)

Ans: Out of Syllabus

Ques 4 a) Discuss the Roles and Functions of RBI with respect to: (10)
i) Currency

Ans: Out of Syllabus

ii) Credit Control

Ans: Out of Syllabus

iii) Balance of Payment

Ans: Out of Syllabus

Or

Ques 4 b) What is "Capital Market"? Explain the Role of SEBI in Indian Capital Market. **(10)**

Ans: Out of Syllabus

Ques 5 a) Enlighten the following terms and their significance in foreign trade: **(2×5=10)**

i) GATT

Ans: Out of Syllabus

ii) WTO

Ans: Out of Syllabus

Or

Ques 5 b) Illuminate the following terms: **(2×5=10)**

i) Fiscal Deficit

Ans: Out of Syllabus

Ques 5 b ii) Capital Account *versus* Revenue Account

Ans: Out of Syllabus

Solved Paper (2017)

SAVITRIBAI PHULE PUNE UNIVERSITY

MBA - FIRST SEMESTER EXAMINATION - 2018

ECONOMIC ANALYSIS FOR BUSINESS DECISIONS

Time: 2¼ Hours **Max. Mark: 50**

Note: 1) Solve any One from each question.
2) All questions are compulsory.
3) All questions carry equal marks.

Ques 1 a) Discuss the circular flow of income and output in open economy. **(10)**

Ans: Out of Syllabus

Or

Ques 1 b) "Is profit the only objective of a firm?" Explain. **(10)**

Ans: No, a firm does not have only an objective to earn profit. Profit is one of the objectives of the firm.

Objectives of the Firm
Refer Unit-1, Page No. 27

Ques 2 a) What is price discrimination? Explain with the help of suitable diagram and examples. **(10)**

Ans: Price Discrimination/ Price Discrimination under Monopoly
Refer Unit-4, Page No. 153

Diagram and Example of Price Discrimination/ Price and Output Determination under Discriminating Monopoly
Refer Unit-4, Page No. 154

Or

Ques 2 b) Discuss the features of oligopoly? Explain price rigidity. **(10)**

Ans: Features of Oligopoly
Refer Unit-4, Page No. 162

Kinked Demand Curve: Price Rigidity
Refer Unit-4, Page No. 164

Ques 3 a) With the help of suitable diagram and example explain cost-output relationship in short-run. **(10)**

Ans: Cost-Output Relationship in Short-Run
Refer Unit-3, Page No. 114

Or

Ques 3 b) Write a note on decision tree approach. **(10)**

Ans: Out of Syllabus

Ques 4 a) Explain the components of money market. **(10)**

Ans: Out of Syllabus

Or

Ques 4 b) What are the objectives of monetary policy? Explain the quantitative methods of Monetary Policy. **(10)**

Ans: Out of Syllabus

Ques 5 a) Who implements fiscal policy? Explain the structure and deficits of Union Budget. **(10)**

Ans: Out of Syllabus

Or

Ques 5 b) Write short note on: **(10)**
i) GDP (Gross Domestic Product).

Ans: Out of Syllabus

Ques 5 b) ii) Purchasing power parity.

Ans: Out of Syllabus

UNIT-WISE CLASSIFICATION OF PREVIOUS YEAR QUESTION PAPERS

UNIT I

First Semester Examination, 2013

1 a) Explain how profit or value maximisation objectives predict the behaviour of a firm more accurately than other objectives.

1 b) What might be the objective(s) of the following non-profit organisations:

i) A Museume

ii) Critical Care Unit of a Hospital

1 c) Explain how firms and individuals participate and interact in the product market and in factor market.

First Semester Examination, 2014

2 b) What do you mean by the term 'Market' in Economics?

First Semester Examination, 2015

1 a) Define 'business economics'. Explain its role to take business decisions.

First Semester Examination, 2016

1 a) Explain how firms and individuals participate and interact in the product market and in the factor market.

First Semester Examination, 2017

First Semester Examination, 2018

1 b) "Is profit the only objective of a firm?" Explain.

UNIT II

First Semester Examination, 2013

2 b) Explain income elasticity of demand with suitable examples.

2 c) Explain market experiment method and regression technique of demand forecasting.

First Semester Examination, 2014

First Semester Examination, 2015

2 a) What is price elasticity of demand? Explain various kinds of price elasticity with suitable examples.

First Semester Examination, 2016

2 a) Explain the concept and various determinants of market demand.

First Semester Examination, 2017

2 a) Explain the concept of "Elasticity of Demand". Differentiate between Price and Cross Elasticity of Demand.

First Semester Examination, 2018

UNIT III

First Semester Examination, 2013

First Semester Examination, 2014

3 a) i) Explain the concept of Average Fixed Cost, Average Variable Cost, Marginal Cost and explain their relationship.

3 a) ii) What is the economic significance of opportunity cost?

First Semester Examination, 2015

3 b) Write short notes on:
i) Opportunity Cost

First Semester Examination, 2016

3 a) Elaborate the meaning and various types of cost in detail.

First Semester Examination, 2017

2 b) What do you understand by the term "Market Equilibrium"?

First Semester Examination, 2018

3 a) With the help of suitable diagram and example explain cost-output relationship in short-run.

UNIT IV

First Semester Examination, 2013

2 a) State various features of oligopoly and explain price rigidity with the help of kinked demand curve.
2 b) Explain different types of market structure.

First Semester Examination, 2014

2 a) State various features of oligopoly and explain the concept of 'Price-Rigidity' with the help of Kinked demand curve.
3 b) Explain Break-even Analysis with assumptions, advantages and disadvantages.

First Semester Examination, 2015

2 b) What is the concept of 'Oligopoly'? Explain the concept of price rigidity with the help of kinked demand curve.
3 b) Write short notes on:
i) Break-even Point Analysis

First Semester Examination, 2016

1 b) What role does profit plays in Market System?

First Semester Examination, 2017

2 b) Expound the equilibrium of Monopoly firm under different cost situations.

First Semester Examination, 2018

2 a) What is price discrimination? Explain with the help of suitable diagram and examples.
2 b) Discuss the features of oligopoly? Explain price rigidity.

UNIT V

First Semester Examination, 2013

First Semester Examination, 2014

First Semester Examination, 2015

First Semester Examination, 2016

First Semester Examination, 2017

First Semester Examination, 2018

MODEL PAPER - I

ECONOMIC ANALYSIS FOR BUSINESS DECISIONS

MBA – FIRST SEMESTER EXAMINATION

Time: 2½ Hours **Max. Marks: 50**

Note: 1) All questions shall be compulsory with internal choice within the questions.
2) FIVE questions each carries 10 marks.
3) Figures to the right indicate marks for that question/sub-question.

Q. 1 Answer any 5 out of 8: **(2 marks each)**

i) State the concept of economy.
ii) What do you mean by the term 'Market' in Economics?
iii) State the meaning of utility analysis.
iv) What is income elasticity?
v) State the law of supply.
vi) Give the meaning of opportunity cost.
vii) What is bilateral monopoly?
viii) What is investment function?

Q. 2 Answer any 2 out of 3: **(5 marks each)**

i) Is economic an art or science?
ii) Explain market experiment method and regression technique of demand forecasting.
iii) Explain the change in market equilibrium.

Q. 3 (a) Explain how firms and individuals participate and interact in the product market and in the factor market. **(10 marks)**

Or

Q. 3 (b) Explain the concept of "Elasticity of Demand". Differentiate between Price and Cross Elasticity of Demand. **(10 marks)**

Q. 4 (a) What is price elasticity of demand? Explain various kinds of price elasticity with suitable examples. **(10 marks)**

Or

Q. 4 (b) Elaborate the meaning and various types of cost in detail. **(10 marks)**

Q. 5 (a) What is the concept of 'Oligopoly'? Explain the concept of price rigidity with the help of kinked demand curve. **(10 marks)**

Or

Q. 5 (b) Explain business cycle. Highlight its phases and causes. **(10 marks)**

MODEL PAPER - II

ECONOMIC ANALYSIS FOR BUSINESS DECISIONS

MBA – FIRST SEMESTER EXAMINATION

Time: 2½ Hours **Max. Marks: 50**

Note: 1) All questions shall be compulsory with internal choice within the questions.
2) FIVE questions each carries 10 marks.
3) Figures to the right indicate marks for that question/sub-question.

Q. 1 Answer any 5 out of 8: **(2 marks each)**

i) Highlight the concept of firm.
ii) What is consumer surplus?
iii) What is elasticity of supply?
iv) State the meaning of economics of scale.
v) What is revenue?
vi) State the meaning of price leadership.
vii) Give the meaning of consumption function
viii) What is multiplier?

Q. 2 Answer any 2 out of 3: **(5 marks each)**

i) Explain different types of market structure.
ii) Write short notes on Break-even Point Analysis.
iii) What is accelerator? State the assumption of accelerator.

Q. 3 (a) Define 'business economics'. Explain its role to take business decisions. **(10 marks)**

Or

Q. 3 (b) Explain the concept and various determinants of market demand. **(10 marks)**

Q. 4 (a) With the help of suitable diagram and example explain cost-output relationship in short run and long run. **(10 marks)**

Or

Q. 4 (b) Discuss the meaning of is perfect competition. Highlight the price-output determination under perfect competition. **(10 marks)**

Q. 5 (a) What is price discrimination? Explain with the help of suitable diagram and examples. **(10 marks)**

Or

Q. 5 (b) What is business decision? And also mention its nature and types. **(10 marks)**

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